



SE301

Fundamentals of Engineering Design and Systems Management



SE301 FUNDAMENTALS OF ENG DESIGN AND SYSTEMS MGMT

ABET eCOURSEBOOK AY2026

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UNITED STATES MILITARY ACADEMY
WEST POINT.

SECTION I

COURSE ASSESSMENTS



UNITED STATES MILITARY ACADEMY
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TAB A

AY2026 COURSE ASSESSMENT



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996



MADN-SE

14 May 2026

MEMORANDUM FOR LTC John Caddell, Systems and Decision Sciences Program Director,
Department of Systems Engineering, West Point, NY 10996

SUBJECT: Academic Term (AT) 26-2 Course Assessment – SE301 Fundamentals of Engineering Design
and Systems Management.

1. The attached report summarizes the SE301 assessment and curriculum change recommendations from AT 26-2.
2. EXSUM: The course focused on the entire SDP, with four lessons on Systems Thinking and twenty-five lessons on each of the phases of the SDP. Overall performance was in line with past academic year semesters. Based on the AY26-1 SE301 course assessment, a few course adjustments were implemented for this semester. These improvements focused on timing of graded events and incorporating workshop lessons to allow cadets to work with their project teams during class time.
3. The point of contact for this action is the undersigned.

Encls

1. Course Assessment
2. Course Instructional Memorandum
3. Course Syllabus
4. Course Objective Assessment Table
5. Course End Feedback
6. Cadet Course End Feedback Comments

JOSEPH P. KEARNEY
MAJ, EN/AC
Course Director

Enclosure 1: Course Assessment

Section I – Course Information and History

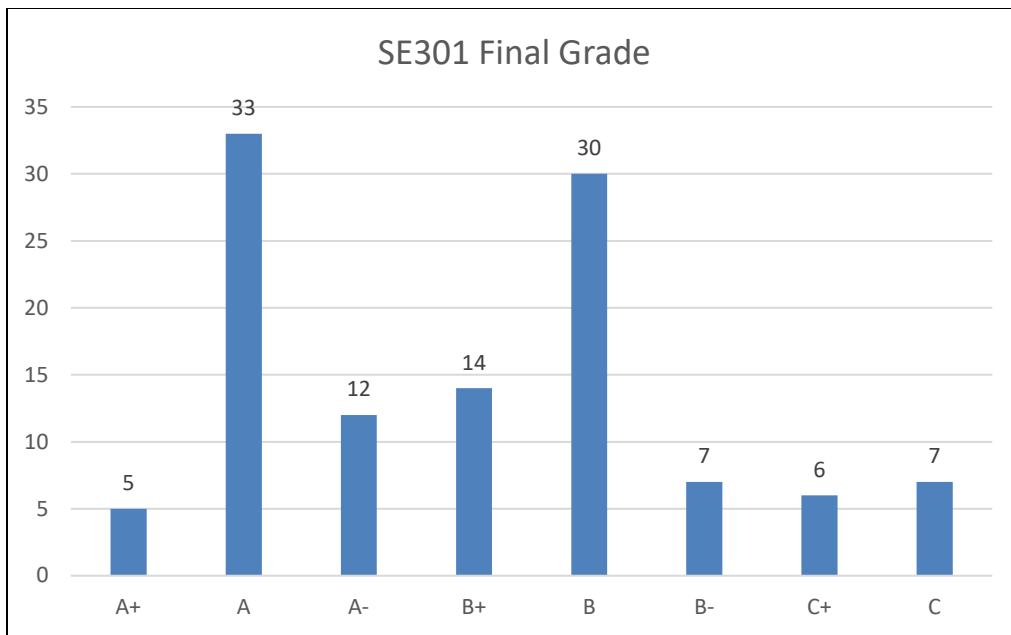
1. Historic Course Offerings and Enrollments.

a. Offered.

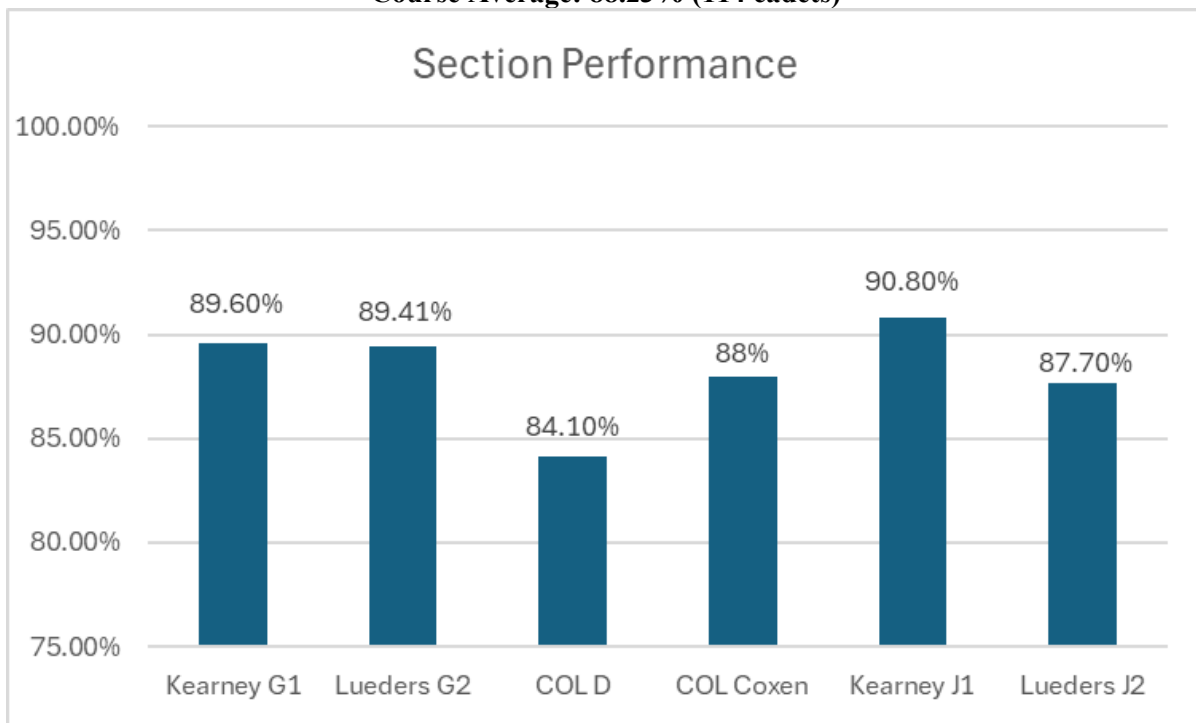
	AY22		AY23		AY24			AY25			AY26
	22-1	22-2	23-1	23-2	24-1	24-2	24-3	25-1	25-2	25-5	22-1
SE301	X	X	X	X	X	X	X	X	X	X	X

b. Historical Course Final Grade Averages.

Academic Term	SE301 Average
AYT 26-2	88.23
AYT 26-1	87.7%
AYT 25-2	91.5%
AYT 25-2	92.1%
AYT 25-1	90.8%
AYT 24-2	89.7%
AYT 24-1	88.3%
AYT 23-2	84.6%
AYT 23-1	85.3%
AYT 22-2	83.5%
AYT 22-1	86.8%
AYT 21-2	83.5%
AYT 21-1	85.9%
AYT 20-2	93.0% ¹
AYT 20-1	85.9%



Course Average: 88.23% (114 cadets)



2. Course Strategy.

SE301 provides the baseline approach to the fundamentals of systems thinking, systems engineering, and systems decision making. The course is designed to allow cadets the opportunity to learn engineering design, systems engineering, and engineering management processes on an individual and project team level so that each cadet will

have the experience necessary to succeed in future Department of Systems Engineering courses. Cadets learn and apply the SDP throughout the 30-lesson course. The course includes four homework assignments that focus on applying systems thinking and the SDP to a complex engineering problem, three Written Progress Reviews (WPRs) focused on course topics, a final project presentation, a final project report, and one Term End Exam (TEE). Individual work accounts for 65% of the course, and the remaining 35% is group work.

Section II – Course Assessment

1. Achievement of Course Objectives.

- a. **Course Objective Assessment Table:** Enclosure 4
- b. **Results of Course End Feedback:** Enclosure 5
- c. **Results of Cadet Free-Form Comments:** Enclosure 6

2. Course Content.

- a. **Body of Knowledge.** The body of knowledge addressed across the five blocks of this course adequately cover the course objectives. The first block covers systems engineering and systems thinking. The remaining four blocks cover the four phases of the systems decision process in more detail. Each phase includes between four-five individual lessons.
- b. **Effective Assessment.** The department's major-graded-event content approval process provides an adequate assessment tool for the assessment of the course objectives.

3. Course Processes.

- a. **Textbooks.**
 - i. Parnell, Gregory S., Driscoll, Patrick J., Henderson, Dale L., Decision Making in Systems Engineering and Management, 2nd Edition, John Wiley & Sons, Inc., 2011.
 - ii. Cabrera, Derek and Cabrera, Laura, Systems Thinking Made Simple: New Hope for Solving Wicked Problems, 2nd Edition, Odyssean, 2015.
- b. **Software.** Cadets utilize Microsoft Excel, PowerPoint, and Word. SE301 introduces Plectica.
- c. **Graded Requirements.** The graded requirements in SE301 are appropriate for gauging student comprehension, extending fundamental ideas, and demonstrating the real-world relevance of the material. Across the five blocks of instruction, there were a total of ten graded events.

4. Course Pedagogy.

- a. **Lecture Materials:** The course balanced lectures with practical applications for each of the phases of the SDP. Generally, for each phase there were three – four lectures on the material followed by a “lab” type experience to build value models, implementation plans, cost analysis, etc. For some of the phases, the cadets were allowed one class to apply the tools to

their homework assignments in class. Lectures were consistent with previous versions of SE301.

- b. **Homework Assignments:** Homework Assignments (1x Individual, 75 points and 3x Group Homeworks, 70 points each). The homework assignments will reinforce the phases of the SDP to include Systems Thinking (75 points), Problem Definition (70 points), Solution Design (70 points), and Decision Making (70 points). These separate homework assignments will support the project to apply the SDP to a complex engineering problem.
- c. **Final Project:** Project Presentation (Group, 1 x 100 points). The group project aims to assess each cadet's ability to function as a contributing member of a team and apply the Systems Decision Process (SDP) to a complex engineering problem and present a recommended solution and implementation plan to the client / decision-maker. The project will build upon the homework assignments that frame and define the problem, develop candidate solutions, and make a recommended decision on the candidate solutions. h

Section III – Assessment of Recent Changes and Proposed Future Changes.

1. Assessment of Changes to Current Academic Term (based on previous Course Assessments).

- a. **Course Objectives.** No changes.
- b. **Textbook.** No changes.
- c. **Curriculum.**
 - i. Increase the Systems thinking block to a total a of Systems Thinking Classes. In doing so add more practical exercises that enable the Cadets to achieve the lesson objective of visualizing the system.
 - ii. Remove the 'Intro to' classes for start of each phase for the SDP. The lessons do not generate enough good material / critical thinking for 75 minute lessons.
 - iii. Add a Cost Estimation class.
 - iv. Add a decision making graded lab. The decision making example of the car problem is great, however in the group project it tends to be a single cadet working on the excel sheet and the rest of the group does not attempt to learn / help. A graded lab would ensure all cadets leave the course with decision modeling skills.
 - v. Continue to allow cadets to choose their own project, but have presentations occur in person. Because of the scheduling and only having two lessons after Thanksgiving break, presentations were recorded and submitted via Teams. Cadets put in varying levels of effort and preparation into these recorded presentations. The pressure of standing in front of peers to present, as well as the experience with public speaking, is valuable for cadets. We recommend having Lessons 28 and 29 of SE301 be dedicated solely to in-person presentations.

d. **Graded Requirements.**

- i. Last semester, all graded events, including WPRs and the TEE were submitted on Canvas / Gradescope. However, the WPRs and TEE were administered in-person and on paper. For WPR 1-3 and the TEE, cadets were allowed to bring one 3"x5" notecard to reference, however they were given a 20 minute time limit for utilization.
- ii. Last semester, cadets were allowed to develop and pitch and topic of their own choosing for the course project. We sustained this approach during AY25 and recommend continuing to do so. Cadets were much more motivated to work on their project because they had agency over the topic. The main challenge cadets faced was finding data to support their projects during the performance modeling portion of the SDP. A lot of these issues with data collection could have been avoided with better time management by cadets.

2. **Proposed Change for Next Academic Term.**

a. **Course Objectives.** None.

- b. **Textbook.** Cadets should not have to purchase the entire Cabrera text. Only three chapters of the text were required reading for the course this semester. I recommend trying to purchase only those chapters via an e-book at a reduced rate. Another option would be to have cadets watch the professionally produced YouTube videos that Dr. Cabrera has created, and not purchase the Cabrera text at all.

c. **Curriculum.**

- i. Less Graded Events (HW1-3, Presentation, WPR1, WPR2, WPR3/Lab, TEE)
 - ii. Tie HW1 into group project topic, scope out group projects more.
- d. **Graded Requirements.** Significant updates to WPR 1 – 3, new problems to prevent old tests being passed around.



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996



MADN-SE

02 January 2026

INSTRUCTIONAL MEMORANDUM
NO. 301-26-2

FUNDAMENTALS OF ENGINEERING DESIGN & SYSTEMS MANAGEMENT

1. **PURPOSE.** To provide cadets with administrative details necessary to successfully complete SE301, Fundamentals of Engineering Design and Systems Management.
2. **REFERENCES.**
 - a. Department of Systems Engineering Administrative Memo 1-1, "General Administrative Instructions," August 2025.
 - b. Department of Systems Engineering Administrative Memo 1-2, "Standards for Written Requirements," August 2025.
 - c. Department of Systems Engineering Administrative Memo 1-3, "Academic Documentation Policy," August 2025.
 - d. USMA publication. "Documentation and Acknowledgement of Academic Work," Office of the Dean, June 2025.
 - e. "Guidelines for Technical Communication Department Guidance," Department of Systems Engineering & Department of Civil & Mechanical Engineering, August 2020.
 - f. Course Website: Canvas— "SE301 FNDTN ENGIN DSGN & SYS MGMT (2026-2)"
3. **GENERAL.** This memorandum provides each cadet with the course description, course objectives, course texts, lesson assignments, grading plan, and criteria for passing the course. Your Instructor will inform you in advance of any deviation from the guidance provided in this memorandum.
4. **SPECIFIC.**
 - a. Course Scope. SE301 serves as a foundational course for all cadets taking the Engineering Management, Systems Engineering, or Systems and Decision Sciences majors and the Core Engineering Sequence. This course presents the methodological framework and techniques for designing, implementing, managing, and re-engineering

complex systems or processes. Cadets learn engineering design and engineering management processes and gain an appreciation for future environments and system life cycles. Cadets analyze real-world examples and complete practice problems to illustrate mastery of course topics. Cadets also use spreadsheet and application design software to model and analyze design alternatives. SE301 introduces the Systems Decision Process (SDP), incorporates material from courses in the USMA core curriculum, and previews the modeling and decision-making tools cadets will learn in follow-on Department of Systems Engineering courses. The course is designed to allow cadets to learn engineering design and engineering management processes on an individual level so that each cadet will have the experience necessary to succeed in future Systems Engineering courses.

- b. **Course Strategy.** SE301 provides the baseline approach to the fundamentals of systems thinking, systems engineering, and systems decision making. The course is designed to allow cadets the opportunity to learn engineering design, systems engineering, and engineering management processes on an individual and project team level so that each cadet will have the experience necessary to succeed in future Department of Systems Engineering courses. Cadets learn and apply the SDP throughout the 30-lesson course. The course includes four homework assignments that focus on applying systems thinking and the SDP to a complex engineering problem, an in-class graded lab, three Written Progress Reviews (WPRs) focused on course topics, a final project presentation, and one Term End Exam (TEE). Individual work accounts for 69% of the course, and the remaining 31% is group work.

- c. **Course Objectives.**

- 1) Apply systems thinking and the SDP to design an engineered solution to a large-scale, complex, multi-disciplinary engineering problem.
- 2) Understand the four phases of the SDP and perform the individual tasks in each phase to address an engineering problem.
- 3) Perform stakeholder and requirement analysis to ensure alternatives achieve the desired outcome and account for the political, social, and economic dimensions of an engineering problem.
- 4) Demonstrate creativity in the formulation of alternative solutions to an engineering problem.
- 5) Apply Life Cycle Cost Analysis methods in comparing alternatives.
- 6) Apply mathematics, basic science, and engineering to analyze a physical system or process and apply the results to the solution of an engineering problem.
- 7) Understand the role of modeling and simulation in the development and analysis of an engineered solution to meet stakeholder requirements.
- 8) Communicate technical findings effectively, both orally and in writing.

d. Course Texts.

- 1) Driscoll, Patrick J., Parnell, Gregory S., Henderson, Dale L., Decision Making In Systems Engineering and Management, 3rd Edition, John Wiley & Sons, Inc., 2022.
- 2) Cabrera, Derek and Cabrera, Laura, Systems Thinking Made Simple: New Hope for Solving Wicked Problems, 2nd Edition, Odyssean, 2015.

e. Additional References. Additional reading and course material will be posted on the course Canvas page.

f. Lesson Assignments. The syllabus is posted on the course Canvas page.

g. Performance Assessment.

- 1) The course consists of 1,000 points. All graded work is individual work unless otherwise noted. The following is a breakdown of the graded work:
 - a) Homework Assignments (1x Individual, 75 points and 3x Group Homeworks, 70 points each). The homework assignments will reinforce the phases of the SDP to include Systems Thinking (75 points), Problem Definition (70 points), Solution Design (70 points), and Decision Making (70 points). These separate homework assignments will support the project to apply the SDP to a complex engineering problem.
 - b) Written Partial Reviews (WPRs). (Individual, 1 x 125 points and 2 x 90 points) The purpose is to reinforce concepts taught throughout the course. The WPRs will consist of interpretation, application, and explanation of concepts given scenario questions.
 - c) Project Presentation (Group, 1 x 100 points). The group project aims to assess each cadet's ability to function as a contributing member of a team and apply the Systems Decision Process (SDP) to a complex engineering problem and present a recommended solution and implementation plan to the client / decision-maker. The project will build upon the homework assignments that frame and define the problem, develop candidate solutions, and make a recommended decision on the candidate solutions. The group project will expand on these recommended solutions to develop an implementation plan for the decision maker.
 - d) Peer evaluations: cadets will anonymously grade the level of effort made by their teammates during group throughout the semester. (10 pts)
 - e) Graded Lab: The graded lab is to reinforce the performance modeling and decision making process on an individual basis (60 points).
 - f) Term End Examination (TEE). (Individual, 200 points). The purpose of this exercise is to reinforce concepts taught during the entire course.

- g) Instructor Points (IP). (Individual, 40 points). The purpose of Instructor Points is to ensure cadets are prepared for class each day, complete assigned problems, and evaluate performance on in-class quizzes. Instructors will provide you with a written statement about how they will assign instructor points.

2) Student performance assessment summary:

Graded Event	Format	Lesson	Points	% of Course
HW#1: Systems Thinking Homework	Individual	9	75	7.50%
WPR 1	Individual	13	125	12.50%
HW#2: Problem Definition Homework	Group	14	70	7.00%
HW#3: Solution Design Homework	Group	21	70	7.00%
WPR 2	Individual	19	90	9.00%
Decision Making Graded Lab	Individual	24	60	6.00%
WPR 3	Individual	25	90	9.00%
HW#4: Decision Making Homework	Group	26	70	7.00%
Project Presentation	Group	28/29	100	10.00%
Peer evaluation	Individual	30	10	1.00%
Instructor points	Individual	30	40	4.00%
Term End Examination	Individual	TEE	200	20.00%
Course Total			1000	

5. ADMINISTRATIVE GUIDANCE.

- Minimum Requirements. Score at least 50% on the TEE and complete the course with at least 65%.
- Graded Events. Unless there are exceptional circumstances, cadets will take all WPRs and the TEE and complete all graded events during their primary scheduled times IAW the DPOM. Cadets who will miss an event must contact the instructor as soon as they know about the conflict. Cadets are responsible for coordinating with the Instructor to make up any material/requirements missed.
- Late Submissions. Failure to submit a graded event on time for other than legitimate, compelling reasons will result in a reduced grade (5% per 24-hour period) and may result in disciplinary action. Cadets must submit all major graded events ($\geq 10\%$ of course grade) regardless of lateness. Any graded event not submitted prior to lesson 30 will receive a grade of zero. All graded events must be submitted to pass the course.
- Peer evaluations of participation. Cadets must submit an assessment of the contributions of each team member to any submission of group work. Cadets should contact their instructor at the earliest opportunity if there are problems with teamwork in their group.
- Final Grades. Final grades will be awarded IAW reference 2.a, Department of Systems Engineering Admin Memo 1-1.

- f. Documentation. Cadets may use any available reference material in preparing coursework, but the use of any references **MUST BE PROPERLY DOCUMENTED**. Documentation procedures are outlined in references 2b through 2d. Generative AI tools should be used cautiously and must always be explicitly documented. AI tools should not be used as sources demonstrating the truth of a statement. Use of large language models (LLMs) for assignments will be treated similarly to assistance from another cadet (e.g. general discussion of ideas is acceptable but having a cadet or AI tool write text for you will result in point deductions). Cadets should consult with their instructor if they have any uncertainty about what is an appropriate use of these tools.
- g. Additional Instruction (AI). AI is available at mutually agreed upon times between the cadet and their Instructor. Before visiting an instructor for additional instruction, cadets are expected to make a legitimate attempt to complete the daily assignments and practice problems to standard and bring those assignments to the AI session. Cadets are expected to have **specific questions for any AI session**. Cadets will bring their notes and course textbooks to any AI session. Cadets should primarily use their instructor for AI. However, they can seek AI from other instructors currently teaching the course. No AI is given after the first version of a WPR has been provided, whether that version is the primary or a special make-ahead exam. The TEE's last AI session is before 1600 on the last day of classes.
- h. Class. If you plan or anticipate being absent from class, let your instructor and section marcher know *prior to the class*, in person or via email. If you cannot attend class, you must inform your Instructor NLT 24hrs prior to the missed lesson.

JOSEPH P. KEARNEY
MAJ, EN/AC
SE301 Course Director

Enclosure 3: SE301 Syllabus

Block	Lsn	Day 1	Day 2	Title	Assinged Reading (Complete Before Class)	Graded Events
Systems Thinking	1	1/7/2026	1//626	Course Introduction, Systems Thinking, Systems Engineering and SDP	Read: Driscoll (3e) 1.1, 1.2, 1.5	
	2	1/9/2026	1/8/2025	Systems Thinking I	Read: Driscoll (3e) 2.4-2.8	
	3	1/12/2026	1/13/2026	Systems Thinking II		
	4	1/14/2026	1/15/2026	Systems Thinking III		
Problem Definition	No scheduled lesson		1/16/2026			
	5	1/20/2026	1/21/2026	Introduction to the Systems Decision Process (SDP)	Read: Driscoll (3e) Ch 4	
	6	1/22/2026	1/23/2026	Research and Stakeholder Engagement	Read: Driscoll (3e) Ch 1.7, 5.2, 5.3	
	No scheduled lesson		1/26/2026	CHANGE DUE TO WEATHER CANCELLATION		
	7	1/28/2026	1/27/2026	Stakeholder analysis	Read: Driscoll (3e) 5.2, 5.3	
	8	1/30/2026	1/29/2026	Functional analysis	Read: Driscoll (3e) 5.5, 5.6	
	9	2/2/2026	2/3/2026	Requirements analysis	Read: Driscoll (3e) 5.4	
	10	2/4/2026	2/5/2026	Value Modeling (Qualitative)	Read: Driscoll (3e) Ch 6.1, 6.2 Read HW # 2 Guidance	HW # 1 DUE 03 2359 FEB
	No scheduled lesson		2/6/2026			
	11	2/9/2026	2/12/2026	Value Modeling (Quantitative) I	Read: Driscoll (3e) Ch 6.3	
	No scheduled lesson		2/10/2026	CHANGE DUE TO POWER OUTAGE		
	12	2/11/2026	2/17/2026	Value Modeling (Quantitative) II	Read: Saaty - Decision Making with AHP Read: Driscoll (3e) Ch 5.6	
	No scheduled lesson		2/13/2026			
	No scheduled lesson		2/20/2026			HW # 2 DUE 22 2359 FEB
	13		2/18/2026	2/19/2026	WPR 1	
No scheduled lesson		2/23/2026	2/24/2026	CHANGE DUE TO BLIZZARD		
Solution Design	14	2/25/2026	2/26/2026	Solution Design Intoduction, Alternative Generation	Read: Driscoll (3e) 7.2.8, 7.2.9, 7.2.10	
	No scheduled lesson		2/27/2026			
	15	3/2/2026	3/3/2026	Modeling and feasibility screening	Read: Driscoll (3e) Ch 3.2, 7.3, 9.3	
	16	3/4/2026	3/5/2026	Cost Analysis I: Present Value		
	No scheduled lesson		3/6/2026			
	17	3/9/2026	3/10/2026	Cost Analysis II: Cost Estimation		
	18	3/11/2026	3/12/2026	Cost Analysis III		
	No scheduled lesson		3/13/2026			
19		3/16/2026	3/17/2026	WPR 2		
Decision Making	20	3/18/2026	3/19/2026	Intro to Solution Design & Performance Modeling I	Read: Driscoll (3e) Ch 9.1-9.4	
	No scheduled lesson		3/20/2026			
	21	3/23/2026	3/24/2026	Performance Modeling II		HW # 3 DUE 22 2359 MAR
	22	3/25/2026	3/26/2026	Sensitivity Analysis	Read: Driscoll (3e) Ch 5.7, 9.8.1	
	No scheduled lesson		3/27/2026			
	23	4/7/2026	4/8/2026	Risk Analysis		
	No scheduled lesson		4/9/2026	4/10/2026		
	24	4/13/2026	4/14/2026	Decision Making Graded Lab & WPR 3 Review		
25		4/15/2026	4/16/2026	WPR 3		
No scheduled lesson		4/17/2026				
Solution Implementation	26	4/20/2026	4/21/2026	Introduction to Solution Implementation	Read: Driscoll (3e) Ch 12.1-12.4	HW # 4 DUE 19 2359 APR
	No scheduled lesson		4/22/2026	4/24/2026		
	27	4/27/2026	4/28/2026	Monitoring and control	Read: Driscoll (3e) 12.8	
	28	4/29/2026	4/30/2026	Project Presentations (3 Groups)		
	29	5/4/2026	5/5/2026	Project Presentations (3 Groups)		
	30	5/6/2026	5/7/2026	Semester review		
	No scheduled lesson		5/8/2026	5/11/2026		
5/12/26-5/16/26		TEEs Week		Review Lesson 1- 30		

Enclosure 4: SE301 Course Objective Assessment Table

SE301 Course Objectives		Cadet Assessment	Direct Assessment	Course Director's Assessment	Applicable Graded Events
1.	Apply systems thinking and the Systems Decision Process (SDP) to design an engineered solution to a large-scale, complex, multi-disciplinary engineering problem.		3.3	4.0	1) Homework 1 2) WPR 1 & 2 3) Project Pres. & Lab 4) TEE
2.	Understand the four phases of the Systems Decision Process and perform the individual tasks in each phase to address an engineering problem.		3.0	4.0	1) Homework 2,3,4 2) WPR 1 & 2 3) Project Pres. & Lab 4) TEE
3.	Perform stakeholder and requirement analysis to ensure alternatives achieve the desired outcome and account for the political, social, and economic dimensions of an engineering problem.		3.3	3.3	1) Homework 2 2) WPR 1 3) Project Pres. & Lab 4) TEE
4.	Demonstrate creativity in the formulation of alternative solutions to an engineering problem.		3.0	3.5	1) Homework 3 2) WPR 1 & 2 3) Project Pres. & Lab 4) TEE
5.	Apply Life Cycle Cost Analysis methods in comparing alternatives		3.3	2.5	1) Homework 4 2) WPR 1 3) Project Pres 4) TEE
6.	Apply mathematics, basic science, and engineering to analyze a physical system or process and apply the results to the solution of an engineering problem		3.3	3.5	1) Homework 3,4 2) WPR 2 3) Project Pres. & Lab 4) TEE
7.	Understand the role of modeling and simulation in the development and analysis of an engineered solution to meet stakeholder requirements		3.0	2.5	1) Homework 4 2) WPR 2 3) TEE
8.	Communicate technical findings effectively, both orally and in writing		3.3	3.3	1) Homework 1-4 2) Project Presentation
Overall Assessment			3.21	3.32	
NOTES: 1) Graded Events included: 2xWPR, 4xHomeworks, TEE, Project 2) Cadet Assessment is based on the End-of-Course Survey which is currently unavailable. 3) Course Director Assessment is: -- *Based on course average for "All graded events" -- All others based on individual questions on graded events Range: 1 = strongly disagree to 4 = strongly agree					

Enclosure 5: Course End Feedback
Tabular report aggregated for Course – SE301

Student surveys are unavailable at the time of course closeout. These will be added as soon as they become available.

Enclosure 6: Cadet Comments

Student comments are unavailable at the time of course closeout. These will be added as soon as they become available.



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996



MADN-SE

20 December 2025

MEMORANDUM FOR LTC John Caddell, Systems and Decision Sciences Program Director,
Department of Systems Engineering, West Point, NY 10996

SUBJECT: Academic Term (AT) 26-1 Course Assessment – SE301 Fundamentals of Engineering Design
and Systems Management.

1. The attached report summarizes the SE301 assessment and curriculum change recommendations from AT 26-1.
2. EXSUM: The course focused on the entire SDP, with four lessons on Systems Thinking and twenty-five lessons on each of the phases of the SDP. Overall performance was in line with past academic year semesters. Based on the AY25-2 SE301 course assessment, a few course adjustments were implemented for this semester. These improvements focused on timing of graded events and incorporating workshop lessons to allow cadets to work with their project teams during class time.
3. The point of contact for this action is the undersigned.

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JOSEPH P. KEARNEY
MAJ, EN/AC
Course Director

Enclosure 1: Course Assessment

Section I – Course Information and History

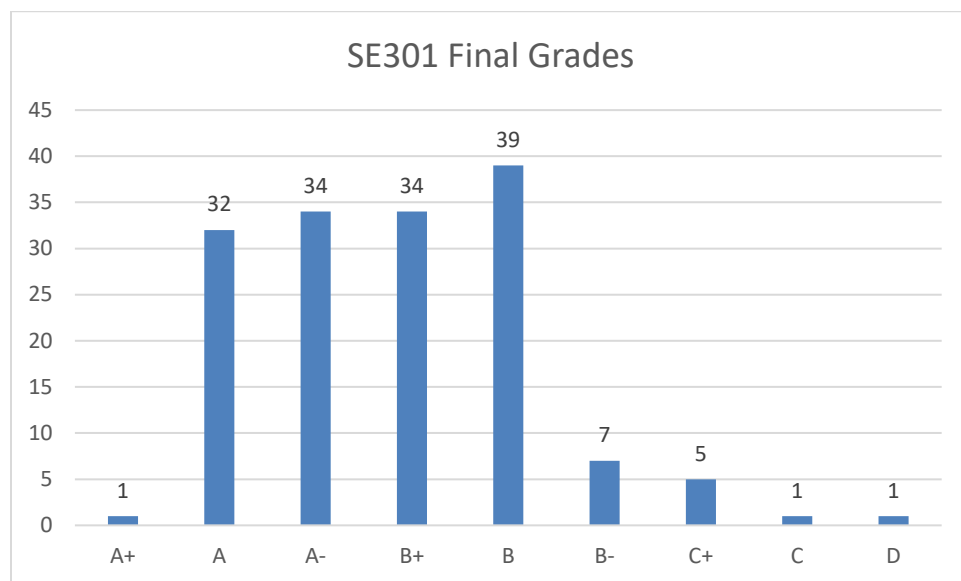
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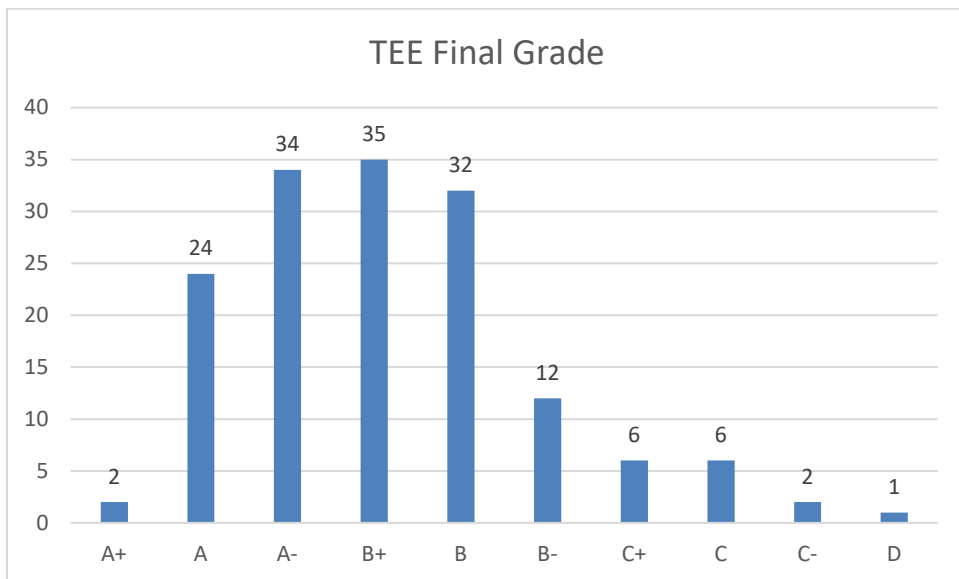
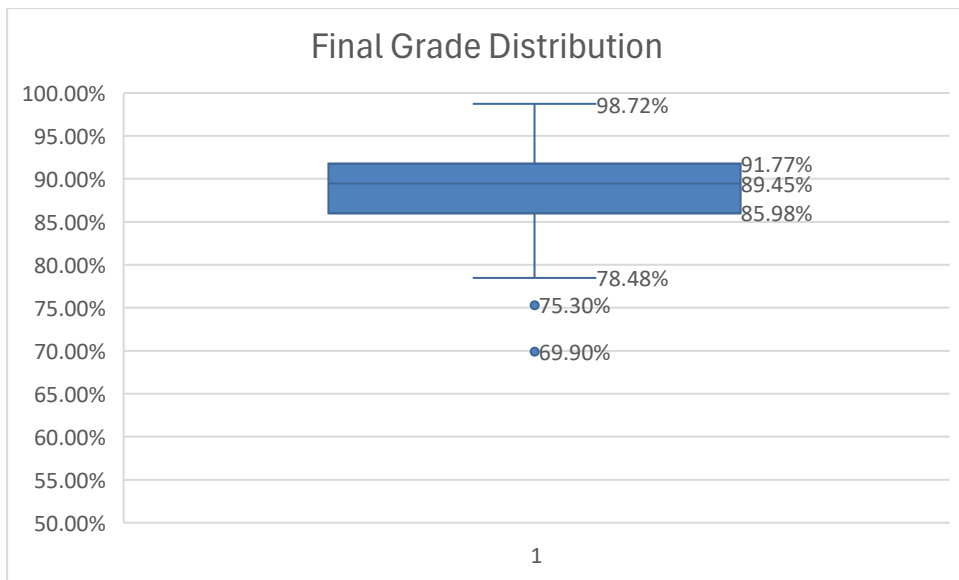
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	22-1	22-2	23-1	23-2	24-1	24-2	24-3	25-1	25-2	25-5	22-1
SE301	X	X	X	X	X	X	X	X	X	X	X

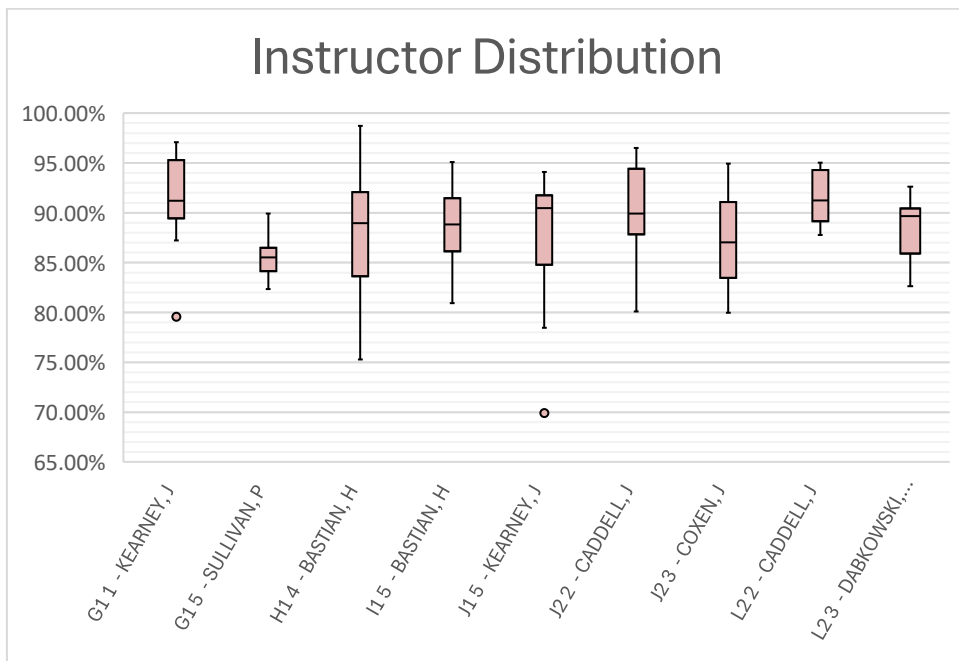
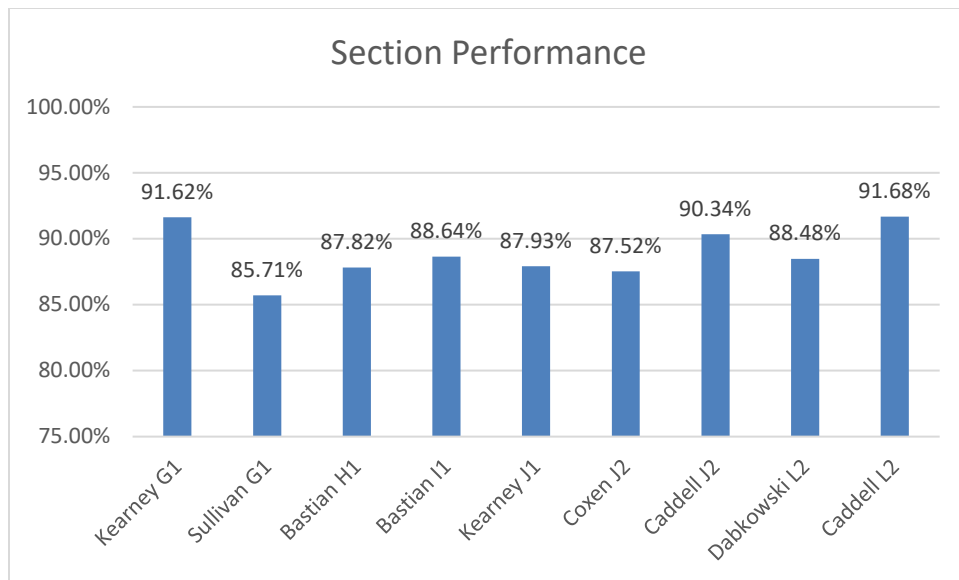
b. Historical Course Final Grade Averages.

Academic Term	SE301 Average
AYT 26-1	87.7%
AYT 25-2	91.5%
AYT 25-2	92.1%
AYT 25-1	90.8%
AYT 24-2	89.7%
AYT 24-1	88.3%
AYT 23-2	84.6%
AYT 23-1	85.3%
AYT 22-2	83.5%
AYT 22-1	86.8%
AYT 21-2	83.5%
AYT 21-1	85.9%
AYT 20-2	93.0% ¹
AYT 20-1	85.9%



Course Average: 87.7% (159 cadets)





2. Course Strategy.

SE301 provides the baseline approach to the fundamentals of systems thinking, systems engineering, and systems decision making. The course is designed to allow cadets the opportunity to learn engineering design, systems engineering, and engineering management processes on an individual and project team level so that each cadet will have the experience necessary to succeed in future Department of Systems Engineering courses. Cadets learn and apply the SDP throughout the 30-lesson course. The course includes four homework assignments that focus on applying systems thinking and the SDP to a complex engineering problem, three Written Progress Reviews (WPRs) focused on course topics, a final project presentation, a final project report, and one Term End

Exam (TEE). Individual work accounts for 65% of the course, and the remaining 35% is group work.

Section II – Course Assessment

1. Achievement of Course Objectives.

- a. **Course Objective Assessment Table:** Enclosure 4
- b. **Results of Course End Feedback:** Enclosure 5
- c. **Results of Cadet Free-Form Comments:** Enclosure 6

2. Course Content.

- a. **Body of Knowledge.** The body of knowledge addressed across the five blocks of this course adequately cover the course objectives. The first block covers systems engineering and systems thinking. The remaining four blocks cover the four phases of the systems decision process in more detail. Each phase includes between four-five individual lessons.
- b. **Effective Assessment.** The department's major-graded-event content approval process provides an adequate assessment tool for the assessment of the course objectives.

3. Course Processes.

- a. **Textbooks.**
 - i. Parnell, Gregory S., Driscoll, Patrick J., Henderson, Dale L., Decision Making in Systems Engineering and Management, 2nd Edition, John Wiley & Sons, Inc., 2011.
 - ii. Cabrera, Derek and Cabrera, Laura, Systems Thinking Made Simple: New Hope for Solving Wicked Problems, 2nd Edition, Odyssean, 2015.
- b. **Software.** Cadets utilize Microsoft Excel, PowerPoint, and Word. SE301 introduces Plectica.
- c. **Graded Requirements.** The graded requirements in SE301 are appropriate for gauging student comprehension, extending fundamental ideas, and demonstrating the real-world relevance of the material. Across the five blocks of instruction, there were a total of ten graded events.

4. Course Pedagogy.

- a. **Lecture Materials:** The course balanced lectures with practical applications for each of the phases of the SDP. Generally, for each phase there were three – four lectures on the material followed by a “lab” type experience to build value models, implementation plans, cost analysis, etc. For some of the phases, the cadets were allowed one class to apply the tools to their homework assignments in class. Lectures were consistent with previous versions of SE301.
- b. **Homework Assignments:** Homework Assignments (1x Individual, 75 points and 3x Group Homeworks, 70 points each). The homework assignments will reinforce the

phases of the SDP to include Systems Thinking (75 points), Problem Definition (70 points), Solution Design (70 points), and Decision Making (70 points). These separate homework assignments will support the project to apply the SDP to a complex engineering problem.

- c. **Final Project:** Project Report (Group, 1x 90 points) and Presentation (Group, 1 x 50 points). The group project aims to assess each cadet's ability to function as a contributing member of a team and apply the Systems Decision Process (SDP) to a complex engineering problem and present a recommended solution and implementation plan to the client / decision-maker. The project will build upon the homework assignments that frame and define the problem, develop candidate solutions, and make a recommended decision on the candidate solutions. The group project will expand on these recommended solutions to develop an implementation plan for the decision maker. The report and presentation will evaluate the cadet's ability to communicate effectively in an oral and written manner.

Section III – Assessment of Recent Changes and Proposed Future Changes.

1. Assessment of Changes to Current Academic Term (based on previous Course Assessments).

- a. **Course Objectives.** No changes.
- b. **Textbook.** No changes.
- c. **Curriculum.** No changes.
- d. **Graded Requirements.**
 - i. Last semester, all graded events, including WPRs and the TEE were submitted on Canvas / Gradescope. However, the WPRs and TEE were administered in-person and on paper. For WPR 1-3 and the TEE, cadets were allowed to bring one 3"x5" notecard to reference, however they were given a 20 minute time limit for utilization.
 - ii. Last semester, cadets were allowed to develop and pitch and topic of their own choosing for the course project. We sustained this approach during AY25 and recommend continuing to do so. Cadets were much more motivated to work on their project because they had agency over the topic. The main challenge cadets faced was finding data to support their projects during the performance modeling portion of the SDP. A lot of these issues with data collection could have been avoided with better time management by cadets.

2. Proposed Change for Next Academic Term.

- a. **Course Objectives.** None.
- b. **Textbook.** Cadets should not have to purchase the entire Cabrera text. Only three chapters of the text were required reading for the course this semester. I recommend trying to purchase only those chapters via an e-book at a reduced rate. Another option would be to have cadets watch the professionally produced YouTube videos that Dr. Cabrera has created, and not purchase the Cabrera text at all.
- c. **Curriculum.**

- i. Increase the Systems thinking block to a total a of Systems Thinking Classes. In doing so add more practical exercises that enable the Cadets to achieve the lesson objective of visualizing the system.
 - ii. Remove the 'Intro to' classes for start of each phase for the SDP. The lessons do not generate enough good material / critical thinking for 75 minute lessons.
 - iii. Add a Cost Estimation class.
 - iv. Add a decision making graded lab. The decision making example of the car problem is great, however in the group project it tends to be a single cadet working on the excel sheet and the rest of the group does not attempt to learn / help. A graded lab would ensure all cadets leave the course with decision modeling skills.
 - v. Continue to allow cadets to choose their own project, but have presentations occur in person. Because of the scheduling and only having two lessons after Thanksgiving break, presentations were recorded and submitted via Teams. Cadets put in varying levels of effort and preparation into these recorded presentations. The pressure of standing in front of peers to present, as well as the experience with public speaking, is valuable for cadets. We recommend having Lessons 28 and 29 of SE301 be dedicated solely to in-person presentations.
 - vi. Provide more guidance on project topics. Many cadets proposed topics that led to challenges with finding appropriate data. While instructors attempted to provide input and recommended ideas, a set list of projects or successful past projects in SE301 would be very beneficial to the cadets. Since the semester project proposals are created early in the semester, cadets do not yet have a good understanding of the SDP.
- d. **Graded Requirements.** Significant updates to WPR 1 – 3, new problems to prevent old tests being passed around.



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996



MADN-SE

15 August 2025

INSTRUCTIONAL MEMORANDUM
NO. 301-26-1

FUNDAMENTALS OF ENGINEERING DESIGN & SYSTEMS MANAGEMENT

1. **PURPOSE.** To provide cadets with administrative details necessary to successfully complete SE301, Fundamentals of Engineering Design and Systems Management.
2. **REFERENCES.**
 - a. Department of Systems Engineering Administrative Memo 1-1, "General Administrative Instructions," August 2025.
 - b. Department of Systems Engineering Administrative Memo 1-2, "Standards for Written Requirements," August 2025.
 - c. Department of Systems Engineering Administrative Memo 1-3, "Academic Documentation Policy," August 2025.
 - d. USMA publication. "Documentation and Acknowledgement of Academic Work," Office of the Dean, June 2025.
 - e. "Guidelines for Technical Communication Department Guidance," Department of Systems Engineering & Department of Civil & Mechanical Engineering, August 2020.
 - f. Course Website: Canvas— "SE301 FNDTN ENGIN DSGN & SYS MGMT (2026-1)"
3. **GENERAL.** This memorandum provides each cadet with the course description, course objectives, course texts, lesson assignments, grading plan, and criteria for passing the course. Your Instructor will inform you in advance of any deviation from the guidance provided in this memorandum.
4. **SPECIFIC.**
 - a. Course Scope. SE301 serves as a foundational course for all cadets taking the Engineering Management, Systems Engineering, or Systems and Decision Sciences majors and the Core Engineering Sequence. This course presents the methodological framework and techniques for designing, implementing, managing, and re-engineering

complex systems or processes. Cadets learn engineering design and engineering management processes and gain an appreciation for future environments and system life cycles. Cadets analyze real-world examples and complete practice problems to illustrate mastery of course topics. Cadets also use spreadsheet and application design software to model and analyze design alternatives. SE301 introduces the Systems Decision Process (SDP), incorporates material from courses in the USMA core curriculum, and previews the modeling and decision-making tools cadets will learn in follow-on Department of Systems Engineering courses. The course is designed to allow cadets to learn engineering design and engineering management processes on an individual level so that each cadet will have the experience necessary to succeed in future Systems Engineering courses.

- b. Course Strategy. SE301 provides the baseline approach to the fundamentals of systems thinking, systems engineering, and systems decision making. The course is designed to allow cadets the opportunity to learn engineering design, systems engineering, and engineering management processes on an individual and project team level so that each cadet will have the experience necessary to succeed in future Department of Systems Engineering courses. Cadets learn and apply the SDP throughout the 30-lesson course. The course includes four homework assignments that focus on applying systems thinking and the SDP to a complex engineering problem, three Written Progress Reviews (WPRs) focused on course topics, a final project presentation, a final project report, and one Term End Exam (TEE). Individual work accounts for 65% of the course, and the remaining 35% is group work.

- c. Course Objectives.

- 1) Apply systems thinking and the SDP to design an engineered solution to a large-scale, complex, multi-disciplinary engineering problem.
- 2) Understand the four phases of the SDP and perform the individual tasks in each phase to address an engineering problem.
- 3) Perform stakeholder and requirement analysis to ensure alternatives achieve the desired outcome and account for the political, social, and economic dimensions of an engineering problem.
- 4) Demonstrate creativity in the formulation of alternative solutions to an engineering problem.
- 5) Apply Life Cycle Cost Analysis methods in comparing alternatives.
- 6) Apply mathematics, basic science, and engineering to analyze a physical system or process and apply the results to the solution of an engineering problem.
- 7) Understand the role of modeling and simulation in the development and analysis of an engineered solution to meet stakeholder requirements.
- 8) Communicate technical findings effectively, both orally and in writing.

d. Course Texts.

- 1) Driscoll, Patrick J., Parnell, Gregory S., Henderson, Dale L., Decision Making In Systems Engineering and Management, 3rd Edition, John Wiley & Sons, Inc., 2022.
- 2) Cabrera, Derek and Cabrera, Laura, Systems Thinking Made Simple: New Hope for Solving Wicked Problems, 2nd Edition, Odyssean, 2015.

e. Additional References. Additional reading and course material will be posted on the course Canvas page.

f. Lesson Assignments. The syllabus is posted on the course Canvas page.

g. Performance Assessment.

- 1) The course consists of 1,000 points. All graded work is individual work unless otherwise noted. The following is a breakdown of the graded work:
 - a) Homework Assignments (1x Individual, 75 points and 3x Group Homeworks, 70 points each). The homework assignments will reinforce the phases of the SDP to include Systems Thinking (75 points), Problem Definition (70 points), Solution Design (70 points), and Decision Making (70 points). These separate homework assignments will support the project to apply the SDP to a complex engineering problem.
 - b) Written Partial Reviews (WPRs). (Individual, 1 x 125 points and 2 x 90 points) The purpose is to reinforce concepts taught throughout the course. The WPRs will consist of interpretation, application, and explanation of concepts given scenario questions.
 - c) Project Report (Group, 1x 90 points) and Presentation (Group, 1 x 50 points). The group project aims to assess each cadet's ability to function as a contributing member of a team and apply the Systems Decision Process (SDP) to a complex engineering problem and present a recommended solution and implementation plan to the client / decision-maker. The project will build upon the homework assignments that frame and define the problem, develop candidate solutions, and make a recommended decision on the candidate solutions. The group project will expand on these recommended solutions to develop an implementation plan for the decision maker. The report and presentation will evaluate the cadet's ability to communicate effectively in an oral and written manner.
 - d) Peer evaluations: cadets will anonymously grade the level of effort made by their teammates during group work twice during the semester. The first evaluation will be after the submission of HW2 (10 points) and the final evaluation will be after the submission of the project report (20 points).

- e) Term End Examination (TEE). (Individual, 200 points). The purpose of this exercise is to reinforce concepts taught during the entire course.
- f) Instructor Points (IP). (Individual, 40 points). The purpose of Instructor Points is to ensure cadets are prepared for class each day, complete assigned problems, and evaluate performance on in-class quizzes. Instructors will provide you with a written statement about how they will assign instructor points.

2) Student performance assessment summary:

Graded Event	Format	Lesson	Points	% of Course
HW#1: Systems Thinking Homework	Individual	9	75	7.50%
WPR 1	Individual	13	125	12.50%
HW#2: Problem Definition Homework	Group	16	70	7.00%
Peer evaluation 1	Individual	16	10	1.00%
Project Presentation	Group	18	50	5.00%
HW#3: Solution Design Homework	Group	22	70	7.00%
WPR 2	Individual	21	90	9.00%
WPR 3	Individual	27	90	9.00%
HW#4: Decision Making Homework	Group	28	70	7.00%
Project Report	Group	30	90	9.00%
Peer evaluation 2	Individual	30	20	2.00%
Instructor points	Individual	30	40	4.00%
Term End Examination	Individual	TEE	200	20.00%
Course Total			1000	

5. ADMINISTRATIVE GUIDANCE.

- a. Minimum Requirements. Score at least 50% on the TEE and complete the course with at least 65%.
- b. Graded Events. Unless there are exceptional circumstances, cadets will take all WPRs and the TEE and complete all graded events during their primary scheduled times IAW the DPOM. Cadets who will miss an event must contact the instructor as soon as they know about the conflict. Cadets are responsible for coordinating with the Instructor to make up any material/requirements missed.
- c. Late Submissions. Failure to submit a graded event on time for other than legitimate, compelling reasons will result in a reduced grade (5% per 24-hour period) and may result in disciplinary action. Cadets must submit all major graded events ($\geq 10\%$ of course grade) regardless of lateness. Any graded event not submitted prior to lesson 30 will receive a grade of zero. All graded events must be submitted to pass the course.
- d. Peer evaluations of participation. Cadets must submit an assessment of the contributions of each team member to any submission of group work. Low scores from other team members will result in reduced grades for that piece of work. Cadets should

contact their instructor at the earliest opportunity if there are problems with teamwork in their group.

- e. Final Grades. Final grades will be awarded IAW reference 2.a, Department of Systems Engineering Admin Memo 1-1.
- f. Documentation. Cadets may use any available reference material in preparing coursework, but the use of any references **MUST BE PROPERLY DOCUMENTED**. Documentation procedures are outlined in references 2b through 2d. Generative AI tools should be used cautiously and must always be explicitly documented. AI tools should not be used as sources demonstrating the truth of a statement. Use of large language models (LLMs) for assignments will be treated similarly to assistance from another cadet (e.g. general discussion of ideas is acceptable but having a cadet or AI tool write text for you will result in point deductions). Cadets should consult with their instructor if they have any uncertainty about what is an appropriate use of these tools.
- g. Additional Instruction (AI). AI is available at mutually agreed upon times between the cadet and their Instructor. Before visiting an instructor for additional instruction, cadets are expected to make a legitimate attempt to complete the daily assignments and practice problems to standard and bring those assignments to the AI session. Cadets are expected to have **specific questions for any AI session**. Cadets will bring their notes and course textbooks to any AI session. Cadets should primarily use their instructor for AI. However, they can seek AI from other instructors currently teaching the course. No AI is given after the first version of a WPR has been provided, whether that version is the primary or a special make-ahead exam. The TEE's last AI session is before 1600 on the last day of classes.
- h. Practice Problems. Practice Problems and reading assignments are assigned to enhance the understanding of text material and assist the cadets in preparing for graded events. Instructors may choose to allocate some of their Instructor Points toward practice problems.
- i. Class. You are expected to bring your texts and laptop to each class. If you plan or anticipate being absent from class, let your instructor and section marcher know *prior to the class*, in person or via email. If you cannot attend class, you must inform your Instructor NLT 24hrs prior to the missed lesson.

As mentioned above, the Canvas Page "SE301 FNDTN ENGIN DSGN & SYS MGMT (2026-1)" will be this course's primary means of communication. The course instructor will post all announcements and assignments on it.

John Caddell
CPT(P)
Course Director

Joe Kearney

LTC
Program Director

DISTRIBUTION: 1 each Instructor, SE301
 1 each cadet, SE301

Enclosure 3: SE301 Syllabus

SE301: Fundamentals of Engineering Design and Systems Management

Block	Lsn	Day 1	Day 2	Title	Assignment (Complete Prior to Class)	Supplementary Reading (Optional)	Graded event
Problem Definition	1	8/18/2025	8/19/2025	Course Introduction, Systems Thinking, and SDP	Read: Driscoll (3e) 2.1, 2.4-2.8	The Value of Systems Thinking - Bing video	
	2	8/20/2025	8/21/2025	Systems Engineering in the Military and Industry	Read: Driscoll (3e) 1.1, 1.2, 1.5, 13.5	"Thinking about thinking": Soldiers Have a Better Way to Solve Problems I AUSA	
	No scheduled lesson		8/22/2025				
	3	8/25/2025	8/26/2025	Systems Thinking and the DSRP Methodology	Read: Cabrera Ch. 3 (p. 52-69)		
	4	8/27/2025	8/28/2025	Introduction to the Systems Decision Process (SDP)	Read: Driscoll (3e) Ch 4		
	No scheduled lesson		8/29/2025				
	5	9/3/2025	9/2/2025	Introduction to Problem Definition	Read: Driscoll (3e) Ch 5.1	Curtis - Doing the Right Problem	
	6	9/5/2025	9/4/2025	Research and Stakeholder Engagement	Read: Driscoll (3e) Ch 1.7, 5.2, 5.3		
	7	9/8/2025	9/9/2025	Stakeholder analysis	Read: Driscoll (3e) 5.2, 5.3	Watch: How Thinking Works	
	8	9/10/2025	9/11/2025	Functional analysis	Read: Driscoll (3e) 5.5, 5.6		
	No scheduled lesson		9/12/2025				
	9	9/15/2025	9/16/2025	Requirements analysis	Read: Carson - Implementing Structured Requirements to Improve Requirements Quality Driscoll (3e) 5.4		HW #1 Due (75 pts) Issue HW #2 Guidance
	10	9/17/2025	9/18/2025	Value Modeling (Qualitative)	Read: Driscoll (3e) Ch 6.1, 6.2 Read HW #2 Guidance		
Solution Design	No scheduled lesson		9/19/2025				
	11	9/22/2025	9/23/2025	Value Modeling (Quantitative) I	Read: Driscoll (3e) Ch 6.3		
	12	9/24/2025	9/25/2025	Value Modeling (Quantitative) II	Read: Saaty - Decision Making with AHP Read: Driscoll (3e) Ch 5.6		
	No scheduled lesson		9/26/2025				
	13	9/29/2025	9/30/2025	WPR 1	Review: Lessons 1-12		WPR 1 (125 pts)
	14	10/1/2025	10/2/2025	Introduction to Solution Design	Read: Driscoll (3e) Ch 7.1-7.2		
	15	10/6/2025	10/7/2025	Alternative Generation	Read: Driscoll (3e) 7.2.8, 7.2.9, 7.2.10		
	16	10/8/2025	10/9/2025	Modeling and feasibility screening	Read: Driscoll (3e) Ch 3.2, 7.3, 9.3		HW #2 Due (70 pts) Issue HW #3 and Presentation Guidance
	No scheduled lesson		10/10/2025				
	17	10/14/2025	10/15/2025	Project Presentation Working Session			
	No scheduled lesson		10/16/2025				
	18	10/20/2025	10/21/2025	Project Presentations	To Do: Continue writing Project Briefing Turn-In: Project Presentation slides prior to class Read:		Project Presentations (70 pts)
	19	10/22/2025	10/23/2025	Cost Analysis I	Read: Driscoll (3e) Ch 8.1, 8.2, 8.4		
Decision Making	No scheduled lesson		10/24/2025				
	20	10/27/2025	10/28/2025	Cost Analysis II	In-Class PE and Project Working Day		
	21	10/29/2025	10/30/2025	WPR 2			90 points
	No scheduled lesson		10/31/2025				
	22	11/3/2025	11/4/2025	Introduction to Decision Making	Read: Driscoll (3e) Ch 9.1-9.4 Turn-In: Homework 3 Due NLT 2200	Keeney - Creativity in Decision Making	
	23	11/5/2025	11/6/2025	Performance Modeling I		Hakes - An Economic Evaluation of the Moneyball Hypothesis	
	No scheduled lesson		11/7/2025				HW #3 Due (70 pts) Issue HW #4 Guidance
	No scheduled lesson		11/10/2025				
	24	11/13/2025	11/12/2025	Performance Modeling II	Read: SE301 HW 4 Guidance		Issue final project Guidance
	25	11/17/2025	11/18/2025	Sensitivity	Read: Driscoll (3e) Ch 5.7, 9.8.1	-	
	26	11/19/2025	11/20/2025	Risk Analysis	Read: Driscoll (3e) Ch 5.7, 9.8.1		
	No scheduled lesson		11/24/2025				
	No scheduled lesson		11/26/2025				
Solution Implementation	27	12/2/2025	12/3/2025	WPR 3			90 pts
	28	12/4/2025	12/5/2025	Introduction to Solution Implementation	Read: Driscoll (3e) Ch 12.1-12.4 Turn-In: Homework 4 Due NLT 2200		HW #4 Due (70 pts)
	29	12/8/2025	12/9/2025	Monitoring and control	Read: Driscoll (3e) 12.8		
Course Summary	30	12/10/2025	12/11/2025	Semester review and Project Report Working Session	To Do: Continue writing Project Report To Do: Watch assigned Project Group Presentation To Do: Come to Feedback Session with Questions Turn-In: Project Report Due NLT 2200		Project Report (90 pts)
	No scheduled lesson		12/12/2025				
	12/16/2025-12/20/2025			TEEs Week	Review Lesson 1 - 30		TEE (200 pts)

Enclosure 4: SE301 Course Objective Assessment Table

SE301 Course Objectives		Cadet Assessment	Direct Assessment	Course Director's Assessment	Applicable Graded Events
1.	Apply systems thinking and the Systems Decision Process (SDP) to design an engineered solution to a large-scale, complex, multi-disciplinary engineering problem.		3.3	4.0	1) Homework 1 2) WPR 1 & 2 3) Project Pres. & Report 4) TEE
2.	Understand the four phases of the Systems Decision Process and perform the individual tasks in each phase to address an engineering problem.		3.0	4.0	1) Homework 2,3,4 2) WPR 1 & 2 3) Project Pres. & Report 4) TEE
3.	Perform stakeholder and requirement analysis to ensure alternatives achieve the desired outcome and account for the political, social, and economic dimensions of an engineering problem.		3.3	3.3	1) Homework 2 2) WPR 1 3) Project Pres. & Report 4) TEE
4.	Demonstrate creativity in the formulation of alternative solutions to an engineering problem.		3.0	3.5	1) Homework 3 2) WPR 1 & 2 3) Project Pres. & Report 4) TEE
5.	Apply Life Cycle Cost Analysis methods in comparing alternatives		3.3	2.5	1) Homework 4 2) WPR 1 3) Project Pres. & Report 4) TEE
6.	Apply mathematics, basic science, and engineering to analyze a physical system or process and apply the results to the solution of an engineering problem		3.3	3.5	1) Homework 3,4 2) WPR 2 3) Project Pres. & Report 4) TEE
7.	Understand the role of modeling and simulation in the development and analysis of an engineered solution to meet stakeholder requirements		3.0	2.5	1) Homework 4 2) WPR 2 3) TEE
8.	Communicate technical findings effectively, both orally and in writing		3.3	3.3	1) Homework 1-4 2) Project Presentation 3) Project Report
Overall Assessment			3.21	3.32	
NOTES: 1) Graded Events included: 2xWPR, 4xHomeworks, TEE, Project 2) Cadet Assessment is based on the End-of-Course Survey which is currently unavailable. 3) Course Director Assessment is: -- *Based on course average for "All graded events" -- All others based on individual questions on graded events Range: 1 = strongly disagree to 4 = strongly agree					

Enclosure 5: Course End Feedback
Tabular report aggregated for Course – SE301

Student surveys are unavailable at the time of course closeout. These will be added as soon as they become available.

Enclosure 6: Cadet Comments

Student comments are unavailable at the time of course closeout. These will be added as soon as they become available.



UNITED STATES MILITARY ACADEMY
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TAB B

AY2025 COURSE ASSESSMENT

NO COURSE ASSESSMENT AVAILABLE



UNITED STATES MILITARY ACADEMY
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TAB C

AY2024 COURSE ASSESSMENT

NO COURSE ASSESSMENT AVAILABLE



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB D

AY2023 COURSE ASSESSMENT



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996



MADN-SE

15 May 2023

MEMORANDUM FOR COL Julia Coxen, Professor and Department Head, Department of Systems Engineering, United States Military Academy, West Point, New York 10996.

THRU Dr. Vikram Mittal, Systems and Decision Sciences Program Director, Department of Systems Engineering, West Point, NY 10996

SUBJECT: AT23-2 Course Assessment – SE301: Fundamentals of Engineering Design & Systems Management

1. The attached report summarizes the SE301 assessment and curriculum change actions for Academic Term (AT) 23-2.
2. EXSUM: This semester is the sixth consecutive semester that SE301 integrated CES cadets and DSE majors into a single course, which continues to be a success. Additionally, the course focused on the entire SDP, with five lessons on Systems Thinking and five lessons on each of the phases of the SDP. Overall performance is lower compared to previous semesters. Based on the AY 21-1 SE301 course assessment and requirements to deliver course content in a hybrid learning environment, several course improvement recommendations were implemented from the previous semester. These improvements focused on leveraging technology and pedagogical efficiencies which enabled flexible and redundant course material delivery and assessment.
3. The point of contact for this memorandum is the undersigned at x4698 or Ahmed.Bahabry@westpoint.edu.

AHMED BAHABRY, Ph.D.
Assistant Professor
SE301 Course Director

Encls

1. Course Assessment
2. Course Instructional Memorandum
3. Course Syllabus
4. Course Objective Assessment Table
5. Course End Feedback
6. Cadet Course End Feedback Comments

2. Course Assessment
3. Course Instructional Memorandum
4. Course Syllabus
5. WPR 1 Analysis
6. WPR 2 Analysis
7. Final Grades and OML
8. Course End Feedback

Enclosure 1: Course Assessment

Section I – Course Information and History

1. Historic Course Offerings and Enrollments.

a. Offered.

	AY17		AY18		AY19		AY20			AY 21-1	
	17-1	17-2	18-1	18-2	19-1	19-2	20-1	20-2	20-3	21-1	21-2
SE300	X	X	X	X	X						
SE301	X	X	X	X	X	X	X	X	X	X	X

b. Enrollments.

	Enrollment	
Academic Term	SE300	SE301
21-2	94	
21-1	121	
20-3	107	
20-2	91	
20-1	227	
19-2	111	
19-1	106	132
18-2	55	95
18-1	89	91
17-5		14
17-2	97	37
17-1	100	107

c. Historical Course Final Grade Averages.

	Course Average	
Academic Term	SE300	SE301
21-2	86.26	
21-1	86.26	
20-3	91.5%	
20-2	90.2%	
20-1	86.1%	
19-2	88.4%	
19-1	81.4%	86.5%
18-2	83.0%	88.0%
18-1	83.9%	89.0%

17-5		89.8%
17-2	82.8%	90.5%
17-1	84.5%	87.8%

2. Course Strategy.

SE301 provides the baseline approach to the fundamentals of systems thinking, systems engineering, and systems decision making. The course is designed to allow cadets the opportunity to learn engineering design, systems engineering, and engineering management processes on an individual and project team level so that each cadet will have the experience necessary to succeed in future Department of Systems Engineering courses. Cadets learn and apply the Systems Decision Process (SDP) throughout the 30-lesson course. The course includes one individual homework assignment that focuses on applying systems thinking to an emerging real-world initiative to integrate virtual reality in the Army's rotary-wing training curriculum. The course includes three group homework assignments where cadet teams implement the SDP to devise a strategy for various projects of their choosing. Lastly, the course includes two WPRs focused on course topics, a final project presentation, a final project report, and one Term End Exam (TEE). Individual work accounts for 67.5% of the course and the remaining 32.5% is group work.

Section II – Course Assessment

1. Achievement of Course Objectives.

- a. **Course Objective Assessment Table:** Enclosure 4
- b. **Results of Course End Feedback.** Enclosure 5
- c. **Results of Cadet Free-Form Comments.** Enclosure 6

2. Course Content.

- a. **Body of Knowledge.** The body of knowledge addressed across the five blocks of this course adequately cover the course objectives. The first block covers systems engineering and systems thinking and introduces the systems decision process. The remaining four blocks cover the four phases of the systems decision process in more detail. Each phase includes between 5-7 individual lessons.
- b. **Effective Assessment.** The department's major-graded-event content approval process provides an adequate assessment tool for the assessment of the course objectives.

3. Course Processes.

- a. **Textbooks.**
 - i. Parnell, Gregory S., Driscoll, Patrick J., Henderson, Dale L., Decision Making in Systems Engineering and Management, 2nd Edition, John Wiley & Sons, Inc., 2011.

- ii. Cabrera, Derek and Cabrera, Laura, Systems Thinking Made Simple: New Hope for Solving Wicked Problems, 2nd Edition, Odyssean, 2015.
- b. **Software.** Cadets utilize Microsoft Excel, PowerPoint, and Word. SE301 provides an introduction to Microsoft Project and Vensim as well.
- c. **Graded Requirements.** The graded requirements in SE301 are appropriate for gauging student comprehension, extending fundamental ideas, and demonstrating the real-world relevance of the material. Across the five blocks of instruction, there were a total of nine graded events. The course consisted of 67.5% individual work 32.5% group work with 4 to 5-person teams.

4. Course Pedagogy.

- a. **Lecture Materials:** The course balanced lectures with practical applications for each of the phases of the SDP. Generally, for each phase there were two - three lectures on the material followed by a “lab” type experience to build value models, implementation plans, cost analysis, etc. For some of the phases, the cadets were allowed one class to apply the tools to their homework assignments in class. Lectures were consistent with previous versions of SE301.
- b. **SE Case Studies:** This semester was the sixth time that SE Case Studies were introduced at the end of almost every class to reinforce topics in the lecture with real world applications of systems engineering. Cadets appeared to enjoy the discussions around the case studies as they provided relevance to the materials. Example case studies include: The Apollo Program, Ford Class Aircraft Carrier, and the Mars Climate Orbiter. These cases described both successes and failures in systems engineering. Consistent with each of the last four semesters, these case studies were often not discussed due to instructors running out of time. These case studies are too robust to add to the daily lesson prep. They should be truncated to focus on specific lesson objectives or removed from the course completely.
- c. **Homework Assignments:** Homework assignments followed a similar structure to past semesters, where the first homework assignment focuses on systems thinking and the subsequent homework assignments address each phase of the SDP. HW 1 focused on the applications of systems thinking, specifically, the DSRP methodology, to frame a problem/system of the cadet team’s choosing. Homework 2-4 focused on the application of each phase of the SDP to a group project. Each cadet presented a group project topic to their respective instructor immediately following the submission of Homework 1. Instructors selected 4-5 topics per section and assigned a team of 4-5 cadets to each of the selected topics. These topics remained the focus of each of the remaining homework assignments.
- d. **Integration of SE301/SE300:** This semester was the sixth semester in which SE301 cadets from D/SE majors and SE300 cadets from the CES were integrated into the same course and sections. Course evaluations are not available at this time, but cadets were asked to evaluate how they felt this combination of D/SE and CES contributed to their learning. Based on informal feedback received from instructors and cadets, it seems that integrating the course continues to be beneficial to everyone. Performance of CES versus D/SE cadets was tracked for every graded event except for the course end project, which included teams that were composed of CES and D/SE cadets.

Section III – Assessment of Recent Changes and Proposed Future Changes.

1. Assessment of Changes to Current Academic Term (based on previous Course Assessments).

- a. **Course Objectives.** No changes.
- b. **Textbook.** No changes.
- c. **Curriculum.** No changes.
- d. **Graded Requirements.**
 - i. Last semester, all graded events, including WPRs and the TEE were submitted on Blackboard. This semester, we continued with a similar strategy for all graded events, except for WPRs and the TEE, using Microsoft Teams. Regardless of the medium, using a digital submission process allows for instructors to easily use a standardized rubric, view each other's grades, and ensure equity across the course. The only downside to Teams is that it does not have SafeAssign. However, this semester's projects were substantively different from previous offerings, decreasing the likelihood of an honor issue.
 - ii. Last semester, cadets were allowed to develop and pitch and topic of their own choosing for the course project. We sustained this approach during AY21-2 and recommend continuing to do so. Cadets were much more motivated to work on their project because they had agency over the topic.
 - iii. Last semester, SE301 was taught using a hybrid approach. This semester, due to COVID19 protocols, we were forced to continue this practice. While Teams is an effective medium for online learning, I recommend SE301 go back to a live modality.
 - iv. Last semester, due to the academic schedule adjustments, SE301 was unable to implement the Infantry Warrior Simulation (IWARS) lab. This semester, we changed the simulation lab to focus on System Dynamics. This change allowed cadets to experience a different approach to simulation. I recommend maintaining the use of System Dynamics, as it's applicability to social and technical problems surpasses that of iWARS.
 - v. Last semester, the Written Partial Reviews and Term End Exam were administered and graded through Blackboard. This semester, all WPRs and the TEE were administered in-person, using paper-copies of the exams. If SE301 is taught in person, I recommend continuing with in-person, hand-written exams.

2. Proposed Change for Next Academic Term.

- a. **Course Objectives.** None.
- b. **Textbook.** Continue to make sure that the registrar's office knows that the Cabrera textbook should be the second edition. Also, cadets should not have to purchase the entire Cabrera text. Only three chapters of the text were required reading for the course this semester. I recommend trying to purchase only those chapters via an e-book at a reduced rate. Lastly, ensure the Green Book is the correct edition.
- c. **Curriculum.**
 - i. Reduce the number of Performance Modeling lessons to one in order to make room for an additional Quantitative Value Modeling lesson. During the semester, we found that while cadets conceptually understand value modeling, they need additional repetitions with it, especially when using Excel. As SE301 includes majors and non-majors, it is vitally

important they understand how to conduct this analysis, especially for their homework and course project.

- ii. Reduce the number of Alternative Generation lessons to one in order to make room for an additional Cost Analysis. During the semester, we found that cadets can calculate Net Present Value; however, they struggle with including costs from across the system's lifecycle. This second lesson should focus on helping the cadets identify important costs during each phase of a system's lifecycle so that their cost profiles for the project are more complete.
- iii. Continue to allow cadets to choose their own project. This was something they cadets greatly enjoyed and allows them to see the universal applicability of the SDP. Additionally, continue to have the cadets record their presentations. This allows the instructor to view them at their leisure, forces the cadets to rehearse, and practice their public speaking skills. More importantly, this allows for the feedback sessions during lessons 28 and 29 to be more fruitful. A recommended COA is to have two groups per session in class, with the other two working remotely. For those in-person, print out the rubric and have them grade each other. Leave time for the cadets to provide each other with feedback. This is a great way to allow cadets to see other projects and give feedback to their peers.

d. **Graded Requirements.** Consider revising Homework #4 so that it is less similar to the project presentation and report. The sequencing of this year's homework 4 resulted in a short window between the assignment's submission and that of the project presentation. If feedback was not provided on time, this led to cadets submitting the same work twice. Consider rewriting homework #4 or increasing the time between assignment submissions.



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996



MADN-SE

6 August 2020

INSTRUCTIONAL MEMORANDUM
NO. 301-21-1

FUNDAMENTALS OF ENGINEERING DESIGN & SYSTEMS MANAGEMENT

1. **PURPOSE.** To provide each cadet with the administrative details necessary to successfully complete SE301, Fundamentals of Engineering Design and Systems Management.
2. **REFERENCES.**
 - a. Department of Systems Engineering Administrative Memo 1-1, "General Administrative Instructions," 8 July 2020.
 - b. Department of Systems Engineering Administrative Memo 1-2, "Standards for Written Requirements," 8 July 2020.
 - c. Department of Systems Engineering Administrative Memo 1-3, "Academic Documentation Policy," 8 July 2020.
 - d. USMA publication. "Documentation of Academic Work," Office of the Dean, July 2018.
 - e. "Communication Skills Handbook, A Basic Guide to Oral and Written Communications," Department of Systems Engineering, April 2000.
 - f. Course Website: Microsoft Teams – "SE AY21-2 SE301 – Engineering Design and Systems Management"
3. **GENERAL.** This memorandum provides each cadet with the course description, course objectives, course texts, lesson assignments, grading plan, and criteria for passing the course. Your instructor will inform you in advance of any deviation from the guidance provided in this memorandum.
4. **SPECIFIC.**
 - a. Course Scope. SE301 serves as the foundational course for all cadets taking the Engineering Management, Systems Engineering, or Systems and Decision Sciences

majors. This course presents the methodological framework and techniques for designing, implementing, managing, and re-engineering complex systems or processes. Cadets learn engineering design and engineering management processes and gain an appreciation for future environments and system life cycles. Cadets analyze real-world examples and complete practice problems to illustrate mastery of course topics. Cadets also use spreadsheet software and application design software for modeling and analyzing design alternatives. SE301 introduces the Systems Decision Process (SDP), incorporates material from courses in the USMA core curriculum, and previews the modeling and decision making tools that cadets will learn in follow-on Department of Systems Engineering courses. The course is designed to allow cadets the opportunity to learn engineering design and engineering management processes on an individual level so that each cadet will have the experience necessary to succeed in future Systems Engineering courses. Cadets will spend one to two lessons in a computer lab environment.

- b. **Course Strategy.** SE301 provides the baseline approach to the fundamentals of systems thinking, systems engineering, and systems decision making. The course is designed to allow cadets the opportunity to learn engineering design, systems engineering, and engineering management processes on an individual and project team level so that each cadet will have the experience necessary to succeed in future Department of Systems Engineering courses. Cadets learn and apply the Systems Decision Process (SDP) throughout the 30-lesson course. The course includes four homework assignments that focus on applying systems thinking and the SDP to a complex, engineering problem, two WPRs focused on course topics, a final project presentation, final project report, and one Term End Exam (TEE). Individual work accounts for 67.5% of the course and the remaining 32.5% is group work.
- c. **Course Objectives.**
 - 1) Apply systems thinking and the Systems Decision Process (SDP) to design an engineered solution to a large-scale, complex, multi-disciplinary engineering problem.
 - 2) Understand the four phases of the Systems Decision Process and perform the individual tasks in each phase to address an engineering problem.
 - 3) Perform stakeholder and requirement analysis to ensure alternatives achieve the desired outcome and account for the political, social, and economic dimensions of an engineering problem.
 - 4) Demonstrate creativity in the formulation of alternative solutions to an engineering problem.
 - 5) Apply Life Cycle Cost Analysis methods in comparing alternatives.
 - 6) Apply mathematics, basic science, and engineering to analyze a physical system or process and apply the results to the solution of an engineering problem.

- 7) Understand the role of modeling and simulation in the development and analysis of an engineered solution to meet stakeholder requirements.
 - 8) Communicate technical findings effectively, both orally and in writing.
- d. Course Texts.
- 1) Parnell, Gregory S., Driscoll, Patrick J., Henderson, Dale L., Decision Making In Systems Engineering and Management, 2nd Edition, John Wiley & Sons, Inc., 2011.
 - 2) Cabrera, Derek and Cabrera, Laura, Systems Thinking Made Simple: New Hope for Solving Wicked Problems, 2nd Edition, Odyssean, 2015.
- e. Additional References. Additional reading and course material will be posted on the course Teams page.
- f. Lesson Assignments. Syllabus is posted on the course Teams page.
- g. Performance Assessment.
- 1) The course consists of 1000 points. All graded work is individual work unless otherwise noted. The following is a breakdown of the graded work:
 - a) Homework Assignments (1x Individual, 50 points and 3x Group, 75 points each). The homework assignments will reinforce the phases of the SDP to include Systems Thinking (50 points), Problem Definition (75 points), Solution Design (75 points) and Decision Making (75 points). These individual homework assignments will support the semester long project to apply the SDP to a complex, engineering problem.
 - b) Written Partial Reviews (WPRs). (Individual, 2 x 125 points) The purpose is to reinforce concepts taught throughout the course. The WPRs will consist of interpretation, application, and explanation of concepts given scenario questions.
 - c) Project Report (Group, 1x 100 points) and Presentation (Group, 1 x 75 points). The purpose of the group project is to assess each cadet's ability to function as a contributing member of a team and apply the Systems Decision Process (SDP) to a complex, engineering problem and present a recommend solution and implementation plan to the client / decision-maker. The project will build upon the individual homework assignments that frame and define the problem, develop candidate solutions, and make a recommended decision on the candidate solutions. The group project will expand on these recommended solutions to develop an implementation plan for the decision maker. The report and presentation will evaluate the cadet's ability to communicate effectively in an oral and written manner.

- d) Term End Examination (TEE). (Individual, 175 points). The purpose of this exercise is to reinforce concepts taught during the entire course.
- e) Instructor Points (IP). (Individual, 75 points). The purpose of Instructor Points is to ensure cadets are prepared for class each day, complete assigned problems, and/or evaluate performance on in-class quizzes.

2) Student performance assessment summary:

Graded Event	Format	Lesson	Points	% of Course
HW#1: Systems Thinking Homework	Individual	6	50	5%
WPR 1	Individual	11	125	12.5%
HW#2: Problem Definition Homework	Individual/ Group	12	75	7.5%
HW#3: Solution Design Homework	Group	17	75	7.5%
WPR 2	Individual	22	125	12.5%
Project Presentation	Group	23	75	7.5%
HW#4: Decision Making Homework	Group	25	75	7.5%
Systems Thinking Presentation	Individual	30	50	5%
Instructor Points	Individual	1-30	75	7.5%
Term End Examination	Individual	TEE	175	17.5%
Project Report	Group	5 May	100	10%
Course Total			1000	

5. ADMINISTRATIVE GUIDANCE.

- a. Minimum Requirements. Score at least 50% on the TEE and complete the course with at least 65%.
- b. Graded Events. Unless there are exceptional circumstances, cadets will take all WPRs and the TEE and complete all graded events during their primary scheduled times IAW the DPOM. Cadets who will miss an event must contact the instructor as soon as they know about the conflict. Cadets have the responsibility to coordinate with the instructor to make up any material/requirements missed.
- c. Late Submissions. Failure to submit a graded event on time for other than legitimate, compelling reasons will result in a reduced grade (5% per 24-hour period) and may result in disciplinary action. Cadets are required to submit all major graded events ($\geq 10\%$ of course grade) regardless of lateness. Any graded event not submitted prior to lesson 30 will receive a grade of zero. All graded events must be submitted to pass the course.
- d. Final Grades. Final grades will be awarded IAW reference 2.a, Department of Systems Engineering Admin Memo 1-1.

- e. Documentation. Cadets may use any available reference material in preparing course work, but use of any references **MUST BE PROPERLY DOCUMENTED**. Documentation procedures are outlined in references 2b through 2d.
- f. Additional Instruction (AI). AI is available at mutually agreed upon times between the cadet and their instructor. Prior to visiting an instructor for additional instruction, cadets are expected to make a legitimate attempt to complete the daily assignments and practice problems to standard and bring those assignments with them to the AI session. Cadets are expected to have **specific questions for any AI session**. Cadets will bring their notes and course text books to any AI session. Cadets should primarily use their own instructor for AI, however they are authorized to seek AI from other instructors teaching the course this semester. No AI is given after the first version of a WPR has been given, whether that version is the primary or a special make-ahead exam. The last AI session for the TEE is prior to 1600 on the last day of classes.
- g. Practice Problems. Practice Problems and reading assignments are assigned with the intent of enhancing the understanding of text material and assist the cadets in preparing for graded events. Instructors may choose to allocate some of their Instructor Points towards practice problems.
- h. Class. You are expected to bring your texts and laptop to each class. If you plan or anticipate being absent from class, let your instructor and section marcher know *prior to the class* in person or via email. In order to comply with USMA and Department policy, no more than 12 Cadets will be allowed to attend class in person. Your Instructor will assign you specific lessons in which you will attend class in person. IAW COVID-19 mitigation measures, lessons 1-4 will be conducted remotely. Beginning with lesson 5, classes will be conducted using the hybrid approach, with some cadets in-person and the remaining attending virtually. You are responsible for ensuring you attend class in the appropriate manner. When attending class in a remote status, you must turn on your camera for the duration of the lesson. If you are quarantined due to COVID-19, you will (1) inform your Instructor and (2) attend class via the online platform. If you are unable to attend class, you must inform your Instructor NLT 24hrs prior to the missed lesson.
- i. Teams. As mentioned above, the Microsoft Teams Page “SE AY21-2 SE301 – Engineering Design and Systems Management” will be the primary means of communication for this course. The course director will post all announcements and assignments to this page. It is your responsibility to consistently check this page for updated content and assignments. After you have read this sentence, send an email to your instructor acknowledging your understanding of the course standards.

FOR THE HEAD OF THE DEPARTMENT:

// ORIGINAL SIGNED //

ARTHUR J. MIDDLEBROOKS
MAJ, AR/49
Course Director

// ORIGINAL SIGNED //

JAMES H. SCHREINER
LTC, EN / FA47
Program Director

DISTRIBUTION: 1 each instructor, SE301
 1 each cadet, SE301

Enclosure 3: SE301 Syllabus

Lsn	Day 1	Title	Assignment (Complete Prior to Class)	Supplementary Reading (Optional)	Graded Event	Significant Event
Systems Thinking	1/19/2021	Introduction to the Systems Decision Process (SDP)	Read: Parnell Ch 6 (p 185-193) Read: Parnell Ch 9.1-9.10 (p 275-290) To Do: Exercises 9.1, 9.3, & 9.9			Remote
	1/20/2021		DROP			Remote
	1/22/2021	Systems Thinking	Read: Parnell Ch 2 (p 27-40; 54-59) Read: Senge - Systems Thinking and Org. Learning To Do: Exercise 2.1		Issue HW #1 Guidance	Remote
	1/26/2021	DSRP Methodology	Read: Cabrera Ch. 3 (p. 52-69)	Watch: How Thinking Works (https://www.youtube.com/watch?v=dUqRTWCd504)		Remote
	1/27/2021		DROP			Remote
Problem Definition	1/29/2021	Systems Thinking Workshop	Read: Systems Thinking Case Study			Remote
	2/2/2021	Introduction to Problem Definition	Read: Parnell Ch 10.1 (p 297-300) To Do: Exercise 10.1	Curtis - Doing the Right Problem	Issue Systems Thinking Presentation Guidance	Study Day/Majors Open Hour 3 February
	2/4/2021	Stakeholder Analysis	Read: Parnell Ch 1.3 (p 3-7) & Ch 10.2 (p 300-314) To Do: Exercise 10.2 Turn-In: Homework 1 Due NLT 1700 4FEB21	Reynolds - Stakeholder Theory and Managerial Decision-Making	HW #1 Due (50pts)	
	2/8/2021	Functional & Requirements Analysis	Read: Parnell Ch 10.3 (p 314-325) To Do: Exercise 10.3	Carson - Implementing Structured Requirements to Improve Requirements Quality	Issue HW #2 Guidance	
	2/10/2021	Value Modeling (Qualitative)	Read: Parnell Ch 10.4 (p 326-331; 333-337) Read: Saaty - Decision Making with AHP To Do: Exercise 10.6			
	2/12/2021	Value Modeling (Quantitative)	Read: Parnell Ch 10.4.3 (p 331-341) Read: SE301 HW 2 Guidance			500th Night (12-14 Feb) Pres Day 15 Feb (No Class)
	2/17/2021	Cabrera Systems Thinking Lecture (DROP)	Read: https://blog.cabreraesearch.org/systems-thinking-in-a-7-images Watch: https://www.youtube.com/watch?v=-sfReUj3o0 Lecture: 1955-2105 17 February 2021, Teams			
Solution Design	2/19/2021		DROP			
	2/23/2021	WPR 1	Review: Lessons 1-10		WPR 1 (150pts)	
	2/25/2021	Introduction to Solution Design	Read: Parnell Ch 11.1-11.2 (p 353-365) To Do: Exercise 11.1 Turn-In: Homework 2 Due NLT 1700 25FEB21	Frey - Engineering Design Thinking	HW #2 Due (75pts)	Study Day 24 February
	3/1/2021	Alternative Generation I	Read: Parnell Ch 11.3-11.6 (p.365-388) To Do: Exercises 11.2 & 11.3	Kelly - Creative Confidence Excerpt	Issue HW #3 Guidance	
	3/3/2021	Alternative Generation II	Read: SE301 HW 3 Guidance			
	3/5/2021		DROP			No Classes 8 March
	3/11/2021	Design Methods	Read: Parnell Ch 3.3 (p 74-77) & Ch 7.5 (p 212-221) Read: Axiomatic Design Technology		Issue Project Presentation and Report Guidance	Study Day 10 March
Decision Making	3/15/2021	Cost Analysis	Read: Parnell Ch 5.1-5.4 (p 137-172) To Do: Exercise 5.4 & 5.7			
	3/17/2021	Introduction to Decision Making	Read: Parnell Ch 12.1-12.4 (p 395-409) To Do: Exercise 12.1 Turn-In: Homework 3 Due NLT 1700 17MAR21	Keeney - Creativity in Decision Making	HW 3 (75pts)	
	3/19/2021	Performance Modeling I	Read: Parnell Ch 4.1-4.5 (p 95-121) To Do: Exercises 4.5, 4.6, 4.7, & 4.8	Hakes - An Economic Evaluation of the Moneyball Hypothesis	Issue HW #4 Guidance	
	3/23/2021	Performance Modeling II	Read: SE301 HW 4 Guidance			Founder's Day
	3/25/2021	Sensitivity and Risk Analysis	Read: Parnell Ch 3.4-3.6 and 12.5-12.6 (p 409-417) To Do: Exercise 12.7	McManus - A Framework for Understanding Uncertainty		Study Day 24 March
	3/29/2021	Tradespace Analysis	Read: Parnell Ch 12.7-12.10 (p 417-432) To Do: Exercises 12.3, 12.4, 12.9 & 12.10	Ross - Multi-Attribute Trade space Exploration		Honorable Living Day 30 March
	4/1/2021		DROP			
Solution Implementation	4/6/2021	WPR 2	Review: Lessons 12-21		WPR 2 (125pts)	No Classes 5 April
	4/8/2021	Introduction to Solution Implementation	Read: Parnell Ch 13.1-13.3 (p 447-453) To Do: 13.1 & 13.6 Turn In: Project Presentation Due NLT 1700 8APR21	Sauser - Why Projects Fail	Project Presentation Due (100pts)	Study Day 7 April Mission Commnd Conference 8 April (Modified Day)
	4/12/2021	Project Planning	Read: Parnell Ch 13.4 (p 453-456) To Do: 13.8 & 13.9	Lewis - One World Trade		
	4/14/2021	Execution, Monitoring, and Control	Read: Parnell Ch 13.5-13.8 (p 457-474) To Do: 13.13 & 13.14 Turn-In: Homework 4 Due NLT 1700 14APR21	Sharon Project Management vs Systems Engineering Management	HW 4 (75pts)	
Course Summary	4/16/2021	Introduction to Systems Thinking and Simulation	Read: Parnell Ch 4.6 (p 121-128) & Ch 4.1-4.5 (p 95-121) Read: System Dynamics (Bala et al) p. 3-22			Modified Day 16 April Sandhurst 17 April
	4/20/2021	Systems Thinking and Simulation Lab	Read: 7 Case Studies in Carbon and Climate			Study Day 21 April
	4/23/2021	Project Presentation Feedback Session I	To Do: Continue writing Project Report To Do: Come to Feedback Session with Questions			
	4/28/2021	Project Presentation Feedback Session II	To Do: Continue writing Project Report To Do: Come to Feedback Session with Questions			Study Day 27 April Projects Day 29 April
	5/3/2021	Course Summary	Read: Parnell Ch 14 (p 477-483) Turn In: Systems Thinking Presentation Due NLT 1700 3MAY21		Systems Thinking Presentation Due (50pts)	
	5/5/2021		Turn In: Project Report Due NLT 1700 5MAY21		Project Report (150pts)	Study Day 6 May
	5/7/2021					
	5/11/2021					

Enclosure 4: SE301 Course Objective Assessment Table

SE301 Course Objectives		Cadet Assessment	Direct Assessment	Course Director's Assessment	Applicable Graded Events
1.	Apply systems thinking and the Systems Decision Process (SDP) to design an engineered solution to a large-scale, complex, multi-disciplinary engineering problem.		3.3	4.0	1) Homework 1 2) WPR 1 & 2 3) Project Pres. & Report 4) TEE
2.	Understand the four phases of the Systems Decision Process and perform the individual tasks in each phase to address an engineering problem.		3.0	4.0	1) Homework 2,3,4 2) WPR 1 & 2 3) Project Pres. & Report 4) TEE
3.	Perform stakeholder and requirement analysis to ensure alternatives achieve the desired outcome and account for the political, social, and economic dimensions of an engineering problem.		3.0	3.5	1) Homework 2 2) WPR 1 3) Project Pres. & Report 4) TEE
4.	Demonstrate creativity in the formulation of alternative solutions to an engineering problem.		3.0	3.5	1) Homework 3 2) WPR 1 & 2 3) Project Pres. & Report 4) TEE
5.	Apply Life Cycle Cost Analysis methods in comparing alternatives		3.7	2.5	1) Homework 4 2) WPR 1 3) Project Pres. & Report 4) TEE
6.	Apply mathematics, basic science, and engineering to analyze a physical system or process and apply the results to the solution of an engineering problem		3.3	3.0	1) Homework 3,4 2) WPR 2 3) Project Pres. & Report 4) TEE
7.	Understand the role of modeling and simulation in the development and analysis of an engineered solution to meet stakeholder requirements		2.3	2.5	1) Homework 4 2) WPR 2 3) TEE
8.	Communicate technical findings effectively, both orally and in writing		3.3	3.0	1) Homework 1-4 2) Project Presentation 3) Project Report
Overall Assessment			3.12	3.3125	
NOTES: 1) Graded Events included: 2xWPR, 4xHomeworks, TEE, Project 2) Cadet Assessment is based on the End-of-Course Survey which is currently unavailable. 3) Course Director Assessment is: -- *Based on course average for "All graded events" -- All others based on individual questions on graded events Range: 1 = strongly disagree to 5 = strongly agree					

Enclosure 5: Course End Feedback
Tabular report aggregated for Course – SE301

Student surveys are unavailable at the time of course closeout. These will be added as soon as they become available.

Enclosure 6: Cadet Comments

Student comments are unavailable at the time of course closeout. These will be added as soon as they become available.



UNITED STATES MILITARY ACADEMY
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TAB E AY2022 COURSE ASSESSMENT



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996



MADN-SE

12 May 2022

MEMORANDUM FOR COL Rich Morales, Professor and Department Head, Department of Systems Engineering, United States Military Academy, West Point, New York 10996.

THRU Dr. Vikram Mittal, Systems & Decision Sciences Program Director, Department of Systems Engineering, United States Military Academy, West Point, New York 10996.

SUBJECT: Academic Term (AT) 22-2 Course Assessment – SE301 Fundamentals of Engineering Design and Systems Management.

1. The attached report summarizes the SE301 assessment and curriculum change recommendations from AT 22-1.
2. EXSUM: This semester is the ninth consecutive semester that SE301 integrated CES Cadets and DSE majors into a single course, which continues to be a success. Additionally, the course focused on the entire SDP, with four lessons on Systems Thinking and four-six lessons on each of the phases of the SDP. Overall performance was in line with past academic year semesters. Based on the AY22-1 SE301 course assessment, a few course adjustments were implemented for this semester. These improvements focused on dedicating more lessons to value modeling and cost analysis, and incorporating workshop lessons to allow Cadets to work with their project teams during class time.
2. The point of contact for this action is the undersigned.

Encls

1. Course Assessment
2. Course Instructional Memorandum
3. Course Syllabus
4. Course Objective Assessment Table
5. Course End Feedback
6. Cadet Course End Feedback Comments

SAM M. YOO
MAJ, AV
Course Director

Enclosure 1: Course Assessment

Section I – Course Information and History

1. Historic Course Offerings and Enrollments.

a. Offered.

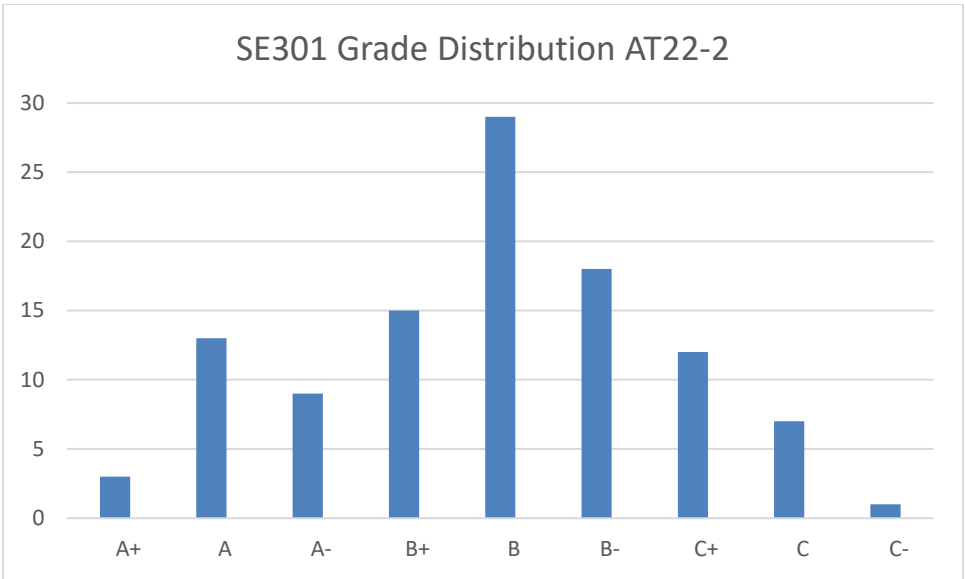
	AY18		AY19		AY20			AY21			AY22	
	18-1	18-2	19-1	19-2	20-1	20-2	20-3	21-1	21-2	21-5	22-1	22-2
SE300	X	X	X									
SE301	X	X	X	X	X	X	X	X	X	X	X	X

b. Enrollments.

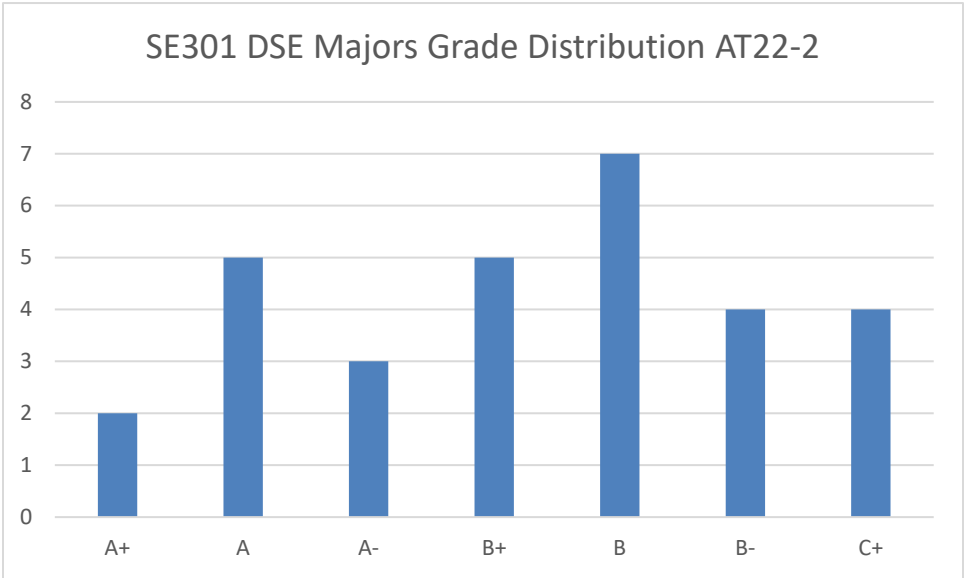
Academic Term	Enrollment	
	SE300	SE301
22-2	107	
22-1	148	
21-5	26	
21-2	94	
21-1	121	
20-3	107	
20-2	91	
20-1	227	
19-2	111	
19-1	106	132
18-2	55	95
18-1	89	91
17-5		14
17-2	97	37

c. Historical Course Final Grade Averages.

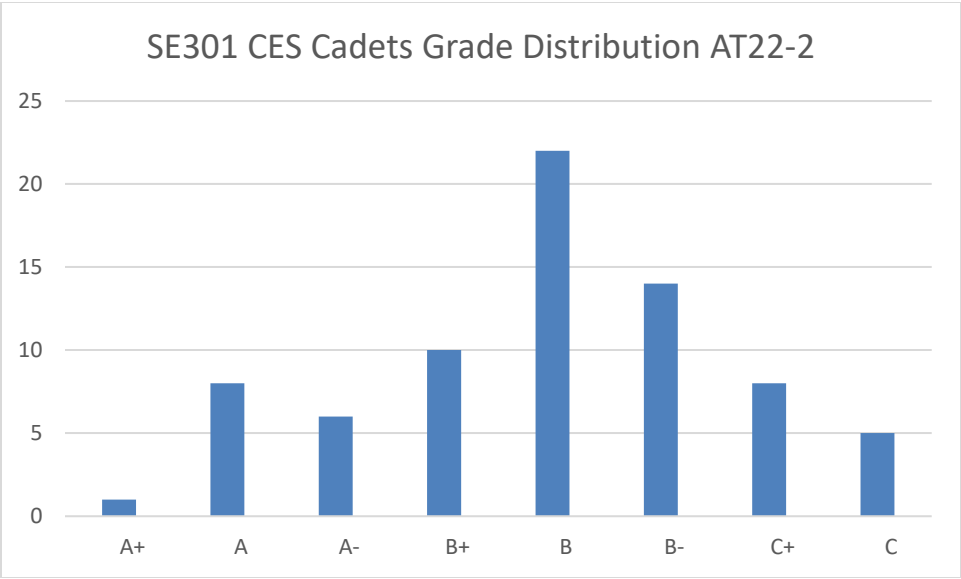
Academic Term	Course Average	
	SE300	SE301
22-2	85.49%	
22-1	87.03%	
21-5	90.75%	
21-2	86.26%	
21-1	85.43%	
20-3	91.5%	
20-2	90.2%	
20-1	86.1%	
19-2	88.4%	
19-1	81.4%	86.5%
18-2	83.0%	88.0%
18-1	83.9%	89.0%
17-5		89.8%
17-2	82.8%	90.5%



Course Average: 85.49% (107 Cadets)



Course Average for DSE Majors: 85.42% (33 Cadets)



Course Average for CES Cadets: 85.49% (74 Cadets)

2. Course Strategy.

SE301 provides the baseline approach to the fundamentals of systems thinking, systems engineering, and systems decision making. The course is designed to allow Cadets the opportunity to learn engineering design, systems engineering, and engineering management processes on an individual and project team level so that each cadet will have the experience necessary to succeed in future Department of Systems Engineering courses. Cadets learn and apply the Systems Decision Process (SDP) throughout the 30-lesson course. The course includes two individual homework assignments, the first of which focuses on applying systems thinking to an emerging real-world initiative to develop the future airport of the 2050s. The second individual assignment allows Cadets to apply the Problem Definition phase to the airport system. The course includes three group homework assignments where cadet teams implement the SDP to devise a strategy for the future airport system. Lastly, the course includes two WPRs focused on course topics, a final project presentation, a final project report, and a Term End Exam (TEE).

Section II – Course Assessment

1. Achievement of Course Objectives.

- a. **Course Objective Assessment Table:** Enclosure 4
- b. **Results of Course End Feedback:** Enclosure 5
- c. **Results of Cadet Free-Form Comments:** Enclosure 6

2. Course Content.

- a. **Body of Knowledge.** The body of knowledge addressed across the five blocks of this course adequately cover the course objectives. The first block covers systems engineering and systems thinking. The remaining four blocks cover the four phases of the systems decision process in more detail. Each phase includes between four-six individual lessons.
- b. **Effective Assessment.** The department's major-graded-event content approval process provides an adequate assessment tool for the assessment of the course objectives.

3. Course Processes.

- a. **Textbooks.**
 - i. Parnell, Gregory S., Driscoll, Patrick J., Henderson, Dale L., Decision Making in Systems Engineering and Management, 2nd Edition, John Wiley & Sons, Inc., 2011.
 - ii. Cabrera, Derek and Cabrera, Laura, Systems Thinking Made Simple: New Hope for Solving Wicked Problems, 2nd Edition, Odyssean, 2015.
- b. **Software.** Cadets utilize Microsoft Excel, PowerPoint, and Word. SE301 introduces Microsoft Project as well.
- c. **Graded Requirements.** The graded requirements in SE301 are appropriate for gauging student comprehension, extending fundamental ideas, and demonstrating the real-world relevance of the material. Across the five blocks of instruction, there were a total of nine graded events.

4. Course Pedagogy.

- a. **Lecture Materials:** The course balanced lectures with practical applications for each of the phases of the SDP. Generally, for each phase there were three – four lectures on the material followed by a “lab” type experience to build value models, implementation plans, cost analysis, etc. For some of the phases, the Cadets were allowed one class to apply the tools to their homework assignments in class. Lectures were consistent with previous versions of SE301.
- b. **SE Case Studies:** This semester was the ninth time that SE Case Studies were introduced at the end of almost every class to reinforce topics in the lecture with real world applications of systems engineering. Cadets appeared to enjoy the discussions around the case studies as they provided relevance to the materials. Example case studies include: The Apollo Program, Ford Class Aircraft Carrier, and the Mars Climate Orbiter. These cases described both successes and failures in systems engineering. Consistent with past semesters, these case studies were sometimes not discussed because instructors ran out of time. These case studies are too robust to add to the daily lesson prep. They should be truncated to focus on specific lesson objectives or removed from the course completely.
- c. **Homework Assignments:** Homework assignments followed a similar structure to past semesters, where the first homework assignment focuses on systems thinking and the subsequent homework assignments address each phase of the SDP. HW 1 focused on the applications of systems thinking, specifically, the DSRP methodology, to frame the development of NASA's 2050s Airports of Tomorrow project. Homework 2 required Cadets to individually apply the Problem Definition phase to the 2050s airport system. Homework 3-4 focused on the remaining phases of the SDP to a group project. Homework 3-4, the Project Presentation, and Final Project Report were all group assignments.
- d. **Integration of SE301/SE300:** This semester was the ninth semester in which SE301 Cadets from D/SE majors and SE300 Cadets from the CES were integrated into the same course and sections. Course evaluations are not available at this time, but Cadets were asked to evaluate how they felt this combination of D/SE and CES contributed to their learning. Based on informal feedback received from instructors and Cadets, it seems that integrating the course continues to be beneficial to everyone.

Section III – Assessment of Recent Changes and Proposed Future Changes.

1. Assessment of Changes to Current Academic Term (based on previous Course Assessments).

- a. **Course Objectives.** No changes.
- b. **Textbook.** No changes.
- c. **Curriculum.** No changes.
- d. **Graded Requirements.**
 - i. Last semester, all graded events, including WPRs and the TEE were submitted on Microsoft Teams. This semester, we used Microsoft Teams to collect graded HW assignments and the project submissions (presentation slides and technical report).

However, the WPRs and TEE were administered in-person and on paper. For WPR 1/2 and the TEE, Cadets were allowed to bring one 3"x5" notecard to reference. Using a digital submission process allows for instructors to easily use a standardized rubric, view each other's grades, and ensure equity across the course. The only downside to Teams is that it does not have SafeAssign. However, this semester's project was completely new from previous offerings, decreasing the likelihood of an honor issue.

- ii. Last semester, Cadets were allowed to develop and pitch and topic of their own choosing for the course project. We adjusted this approach during AY22-2 to apply the SDP to the NASA Blue Skies of Tomorrow project across the course. It is recommended to revert the course project to allow Cadets to select a topic of their choosing. Cadets seemed to struggle more with applying systems engineering to an area they were unfamiliar with. In AY 22-1, Cadets were much more motivated to work on their project because they had agency over the topic. The main challenge Cadets faced was conducting realistic Stakeholder Analysis and finding data to support their projects during the performance modeling portion of the SDP. A lot of these issues with inadequate stakeholder analysis and data collection could have been avoided with better time management by Cadets.

2. **Proposed Change for Next Academic Term.**

- a. **Course Objectives.** None.
- b. **Textbook.** Cadets should not have to purchase the entire Cabrera text. Only three chapters of the text were required reading for the course this semester. I recommend trying to purchase only those chapters via an e-book at a reduced rate. Another option would be to have Cadets watch the professionally produced YouTube videos that Dr. Cabrera has created, and not purchase the Cabrera text at all.
- c. **Curriculum.**
 - i. Allow Cadets to return to pitching project topics at the beginning of the semester rather than selecting a project for everyone, and select the top 4-5 topics to apply throughout the course. While requiring the same project for all Cadets helped with greater shared understanding and opportunities to enhance learning in class for all, the positives may have been outweighed by the negatives of a poor understanding of the aviation industry at large. Allowing Cadets to choose a topic area of interest motivates them to dive in deeper than in a topic area they are uninterested in.
 - ii. Require Cadets to submit a pre-recorded project presentation video rather than presenting live. Time and scheduling are the major limitations as conducting presentations in class takes two lessons to complete, without adequate time for feedback. By removing the in-class presentation requirement, two lessons could be added to expand the Solution Implementation block by a lesson, and/or incorporate a Systems Thinking workshop lesson back into the first block.
 - iii. Provide more guidance and clarity within homework assignment prompts and rubrics. Cadets had issues understanding exactly what was expected of them for assignments, which could be more clearly communicated in writing via the assignment prompts, and more detailed rubrics. While enhancements were made this semester to improving each homework assignments explicit requirements, it was noted that even more clarity is possible.
 - iv. Sustain standardized grading across the course by leveraging "Homework Calibration" meetings with all instructors to establish common standards. Data in previous semesters revealed a difference in the grading based on instructor. By calibrating together prior to each

assignment's return, a more fair and equitable assessment is possible across the course that is independent of which instructor is grading.

- d. **Graded Requirements.** This year's Homework 2 on Problem Definition was an individual assignment to stress the importance of understanding value modeling. While Cadets were able to independently apply their in-class learning to the assignment, the requirements were too large and required too much time for Cadets to complete. Recommend truncating the requirements for the Qualitative/Quantitative Value modeling down from ten total value measures to six.



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996



MADN-SE

06 December 2021

INSTRUCTIONAL MEMORANDUM
NO. 301-22-2

FUNDAMENTALS OF ENGINEERING DESIGN & SYSTEMS MANAGEMENT

1. **PURPOSE.** To provide each cadet with the administrative details necessary to successfully complete SE301, Fundamentals of Engineering Design and Systems Management.
2. **REFERENCES.**
 - a. Department of Systems Engineering Administrative Memo 1-1, "General Administrative Instructions," 10 July 2021.
 - b. Department of Systems Engineering Administrative Memo 1-2, "Standards for Written Requirements," 10 July 2021.
 - c. Department of Systems Engineering Administrative Memo 1-3, "Academic Documentation Policy," 10 July 2021.
 - d. USMA publication. "Documentation of Academic Work," Office of the Dean, July 2017.
 - e. "Guidelines for Technical Communication Department Guidance," Department of Systems Engineering & Department of Civil & Mechanical Engineering, August 2020.
 - f. Course Website: Microsoft Teams – "SE AY22-2 SE301 *XX Section and Instructor*"
3. **GENERAL.** This memorandum provides each cadet with the course description, course objectives, course texts, lesson assignments, grading plan, and criteria for passing the course. Your instructor will inform you in advance of any deviation from the guidance provided in this memorandum.
4. **SPECIFIC.**
 - a. Course Scope. SE301 serves as the foundational course for all cadets taking the Engineering Management, Systems Engineering, or Systems and Decision Sciences majors. This course presents the methodological framework and techniques for

designing, implementing, managing, and re-engineering complex systems or processes. Cadets learn engineering design and engineering management processes and gain an appreciation for future environments and system life cycles. Cadets analyze real-world examples and complete practice problems to illustrate mastery of course topics. Cadets also use spreadsheet software and application design software for modeling and analyzing design alternatives. SE301 introduces the Systems Decision Process (SDP), incorporates material from courses in the USMA core curriculum, and previews the modeling and decision making tools that cadets will learn in follow-on Department of Systems Engineering courses. The course is designed to allow cadets the opportunity to learn engineering design and engineering management processes on an individual level so that each cadet will have the experience necessary to succeed in future Systems Engineering courses. Cadets will spend one to two lessons in a computer lab environment.

- b. **Course Strategy.** SE301 provides the baseline approach to the fundamentals of systems thinking, systems engineering, and systems decision making. The course is designed to allow cadets the opportunity to learn engineering design, systems engineering, and engineering management processes on an individual and project team level so that each cadet will have the experience necessary to succeed in future Department of Systems Engineering courses. Cadets learn and apply the Systems Decision Process (SDP) throughout the 30-lesson course. The course includes four homework assignments that focus on applying systems thinking and the SDP to a complex, engineering problem, two WPRs focused on course topics, a final project presentation, final project report, and one Term End Exam (TEE). Individual work accounts for 65.0% of the course and the remaining 35.0% is group work.
- c. **Course Objectives.**
 - 1) Apply systems thinking and the Systems Decision Process (SDP) to design an engineered solution to a large-scale, complex, multi-disciplinary engineering problem.
 - 2) Understand the four phases of the Systems Decision Process and perform the individual tasks in each phase to address an engineering problem.
 - 3) Perform stakeholder and requirement analysis to ensure alternatives achieve the desired outcome and account for the political, social, and economic dimensions of an engineering problem.
 - 4) Demonstrate creativity in the formulation of alternative solutions to an engineering problem.
 - 5) Apply Life Cycle Cost Analysis methods in comparing alternatives.
 - 6) Apply mathematics, basic science, and engineering to analyze a physical system or process and apply the results to the solution of an engineering problem.

- 7) Understand the role of modeling and simulation in the development and analysis of an engineered solution to meet stakeholder requirements.
 - 8) Communicate technical findings effectively, both orally and in writing.
- d. Course Texts.
- 1) Parnell, Gregory S., Driscoll, Patrick J., Henderson, Dale L., Decision Making In Systems Engineering and Management, 2nd Edition, John Wiley & Sons, Inc., 2011.
 - 2) Cabrera, Derek and Cabrera, Laura, Systems Thinking Made Simple: New Hope for Solving Wicked Problems, 2nd Edition, Odyssean, 2015.
- e. Additional References. Additional reading and course material will be posted on the course Teams page.
- f. Lesson Assignments. Syllabus is posted on the course Teams page.
- g. Performance Assessment.
- 1) The course consists of 1000 points. All graded work is individual work unless otherwise noted. The following is a breakdown of the graded work:
 - a) Homework Assignments (2x Individual, 80 and 90 points, and 2x Group, 90 and 80 points each). The homework assignments will reinforce the phases of the SDP to include Systems Thinking & Stakeholder Analysis (80 points), Problem Definition (90 points), Solution Design (90 points) and Decision Making (80 points). These homework assignments will support the semester long project to apply the SDP to a complex, engineering problem.
 - b) Written Partial Reviews (WPRs). (Individual, 2 x 125 points) The purpose is to reinforce concepts taught throughout the course. The WPRs will consist of interpretation, application, and explanation of concepts given scenario questions.
 - c) Project Report (Group, 1x 100 points) and Presentation (Group, 1 x 80 points). The purpose of the group project is to assess each cadet's ability to function as a contributing member of a team and apply the Systems Decision Process (SDP) to a complex, engineering problem and present a recommended solution and implementation plan to the client / decision-maker. The project will build upon the homework assignments that frame and define the problem, develop candidate solutions, and make a recommended decision on the candidate solutions. The group project will expand on these recommended solutions to develop an implementation plan for the decision maker. The report and presentation will evaluate the cadet's ability to communicate effectively in an oral and written manner.

- d) Term End Examination (TEE). (Individual, 150 points). The purpose of this exercise is to reinforce concepts taught during the entire course.
- e) Instructor Points (IP). (Individual, 80 points). The purpose of Instructor Points is to ensure cadets are prepared for class each day, complete assigned problems, and/or evaluate performance on in-class quizzes.

2) Student performance assessment summary:

Graded Event	Format	Lesson	Points	% of Course
HW#1: Systems Thinking & SH Analysis	Individual	7	80	8%
WPR 1	Individual	11	125	12.5%
HW#2: Problem Definition Homework	Individual	12	90	9%
HW#3: Solution Design Homework	Group	17	90	9%
WPR 2	Individual	22	125	12.5%
HW#4: Decision Making Homework	Group	23	80	8%
Project Presentation	Group	27	80	8%
Instructor Points	Individual	1-30	80	8%
Project Report	Group	30	100	10%
Term End Examination	Individual	TEE	150	15%
Course Total			1000	

5. ADMINISTRATIVE GUIDANCE.

- a. Minimum Requirements. Score at least 50% on the TEE and complete the course with at least 65%.
- b. Graded Events. Unless there are exceptional circumstances, cadets will take all WPRs and the TEE and complete all graded events during their primary scheduled times IAW the DPOM. Cadets who will miss an event must contact the instructor as soon as they know about the conflict. Cadets have the responsibility to coordinate with the instructor to make up any material/requirements missed.
- c. Late Submissions. Failure to submit a graded event on time for other than legitimate, compelling reasons will result in a reduced grade (5% per 24-hour period) and may result in disciplinary action. Cadets are required to submit all major graded events ($\geq 10\%$ of course grade) regardless of lateness. Any graded event not submitted prior to lesson 30 will receive a grade of zero. All graded events must be submitted to pass the course.
- d. Final Grades. Final grades will be awarded IAW reference 2.a, Department of Systems Engineering Admin Memo 1-1.
- e. Documentation. Cadets may use any available reference material in preparing course work, but use of any references **MUST BE PROPERLY DOCUMENTED**. Documentation procedures are outlined in references 2b through 2d.

- f. Additional Instruction (AI). AI is available at mutually agreed upon times between the cadet and their instructor. Prior to visiting an instructor for additional instruction, cadets are expected to make a legitimate attempt to complete the daily assignments and practice problems to standard and bring those assignments with them to the AI session. Cadets are expected to have **specific questions for any AI session**. Cadets will bring their notes and course text books to any AI session. Cadets should primarily use their own instructor for AI, however they are authorized to seek AI from other instructors teaching the course this semester. No AI is given after the first version of a WPR has been given, whether that version is the primary or a special make-ahead exam. The last AI session for the TEE is prior to 1600 on the last day of classes.
- g. Practice Problems. Practice Problems and reading assignments are assigned with the intent of enhancing the understanding of text material and assist the cadets in preparing for graded events. Instructors may choose to allocate some of their Instructor Points towards practice problems.
- h. Class. You are expected to bring your texts and laptop to each class. If you plan or anticipate being absent from class, let your instructor and section marcher know *prior to the class* in person or via email. If you are unable to attend class, you must inform your Instructor NLT 24hrs prior to the missed lesson.
- i. Teams. As mentioned above, the Microsoft Teams Page “SE AY22-2 SE301 XX *Section and Instructor*” will be the primary means of communication for this course. The course instructor will post all announcements and assignments to this page. It is your responsibility to consistently check this page for updated content and assignments. After you have read this sentence, send an email to your instructor acknowledging your understanding of the course standards.

FOR THE HEAD OF THE DEPARTMENT:

SAM M. YOO
MAJ, AV
Course Director

VIKRAM MITTAL, PhD
Associate Professor
Program Director

DISTRIBUTION: 1 each instructor, SE301
 1 each cadet, SE301



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Enclosure 3: SE301 AY22-2 Syllabus

Lsn	Day 2	Title	Assignment (Complete Prior to Class)	Supplementary Reading (Optional)	Graded Event
Systems Thinking	1	1/5/2022	Introduction to the Systems Decision Process (SDP) Read: Pamell Ch 6 (p 185-193) Read: Pamell Ch 9.1-9.10 (p 275-290) To Do: Exercises 9.1, 9.3, & 9.9		
	2	1/7/2022	Systems Thinking Read: Pamell Ch 2 (p 27-40, 54-59) To Do: Exercise 2.1	Senge - Systems Thinking and Org. Learning	
	3	1/11/2022	DSRP Methodology Read: Cabrera Ch. 3 (p. 52-66) Read: 2022 Blue Skies Competition Q&A Summary Document	Watch: How Thinking Works (https://www.youtube.com/watch?v=gUgRTWCdXt4)	Issue HW #1 Guidance
	4	1/14/2022	Cabrera Systems Thinking Lecture (DROP) Read: https://blog.cabrera-research.org/systems-thinking-in-a-7-images Watch: https://www.youtube.com/watch?v=-sfReUu3o0 Lecture: 1940-2040 12 JAN 22 Zoom		
Problem Definition	5	1/19/2022	Introduction to Problem Definition Read: Pamell Ch 10.1 (p 297-300) Read: SE301 HW 2 Guidance To Do: Exercise 10.1	Curtis - Doing the Right Problem	Issue HW#2 Guidance
	6	1/21/2022	Stakeholder Analysis Read: Pamell Ch 13 (p 3-7) & Ch 10.2 (p 300-314) To Do: Exercise 10.2	Reynolds - Stakeholder Theory and Managerial Decision-Making	
	7	1/25/2022	Functional & Requirements Analysis Read: Pamell Ch 10.3 (p 314-325) To Do: Exercise 10.3 Turn-In: Homework 1 Due NLT 2200 27JAN22	Carson - Implementing Structured Requirements to Improve Requirements Quality	HW #1 Due (80pts)
	8	1/28/2022	Value Modeling (Qualitative) Read: Pamell Ch 10.4 (p 326-331; 333-337) Read: Saaty - Decision Making with AHP To Do: Exercise 10.6		
	9	2/1/2022	Value Modeling (Quantitative) I Read: Pamell Ch 10.4.3 (p 331-341)		
	10	2/3/2022	Value Modeling (Quantitative) II Read: Pamell Ch 10.6 (p.341-349) In-Class PE and Project Working Day		
	11	2/8/2022	WPR 1 Review: Lessons 1-10		WPR 1 (125pts)
Solution Design	12	2/10/2022	Introduction to Solution Design Read: Pamell Ch 11.1-11.2 (p 353-365) Read: SE301 HW3 Guidance To Do: Exercise 11.1 Turn-In: Homework 2 Due NLT 2200 10FEB22	Frey - Engineering Design Thinking	HW #2 Due (90pts) Issue HW #3 Guidance
	13	2/16/2022	Alternative Generation Read: Pamell Ch 11.3-11.5.2 & 11.5.5-11.6 (p.365-370 & p.386-388) To Do: Exercises 11.2 & 11.3	Kelly - Creative Confidence Excerpt	
	14	2/18/2022	Design Methods Read: Pamell Ch 3.3 (p 74-77) & Ch 7.5 (p 212-221) Read: Axiomatic Design Technology		
	15	2/23/2022	Cost Analysis Read: Pamell Ch 5.1-5.2, 5.4 (p 137-145, 167-172) To Do: Exercise 5.4 & 5.7		
	16	2/25/2022	Cost Analysis II In-Class PE and Project Working Day Read: Project Presentation and Report Assignment Turn-In: Homework 3 Due NLT 2200 27FEB22		HW #3 Due (90pts) Issue Project Presentation and Report Guidance
Decision Making	17	3/1/2022	Introduction to Decision Making Read: Pamell Ch 12.1-12.4 (p 395-409) Read: SE301 HW4 Guidance To Do: Exercise 12.1	Keeney - Creativity in Decision Making	Issue HW #4 Guidance
	18	3/3/2022	Performance Modeling I Read: Pamell Ch 4.1-4.5 (p 95-121) To Do: Exercises 4.5, 4.6, 4.7, & 4.8	Hakes - An Economic Evaluation of the Moneyball Hypothesis	
	19	3/15/2022	Performance Modeling II and Simulation Read: Pamell Ch 4.6 (p 121-128)		
	20	3/17/2022	Sensitivity and Risk Analysis Read: Pamell Ch 3.4-3.6 and 12.5-12.6 (p 409-417) To Do: Exercise 12.7	McManus - A Framework for Understanding Uncertainty	
	21	3/21/2022	Tradespace Analysis Read: Pamell Ch 12.7-12.10 (p 417-432) To Do: Exercises 12.3, 12.4, 12.9 & 12.10	Ross - Multi-Attribute Trade space Exploration	
Solution Implementation	22	3/24/2022	WPR 2 Review: Lessons 12-21		WPR 2 (125pts)
	23	3/29/2022	Introduction to Solution Implementation Read: Pamell Ch 13.1-13.3 (p 447-453) To Do: 13.1 & 13.6 Turn-In: Homework 4 Due NLT 2200 29MAR22	Sauser - Why Projects Fail	HW #4 Due (80pts)
	24	3/31/2022	Project Planning Read: Pamell Ch 13.4 (p 453-456) To Do: 13.8 & 13.9	Lewis - One World Trade	
	25	4/5/2022	Execution, Monitoring, and Control Read: Pamell Ch 13.5-13.8 (p 457-474) To Do: 13.13 & 13.14	Sharon Project Management vs Systems Engineering Management	
Course Summary	26	4/12/2022	Project Working Day To Do: Continue working on project presentation and writing Project Report		
	27	4/14/2022	Project Presentation Day 1 To Do: Continue writing Project Report Turn-In: Project Presentation Slides NLT 2200 13APR22		Project Presentation Slides Due (80pts)
	28	4/18/2022	Project Presentation Day 2 To Do: Continue writing Project Report		
	29	4/21/2022	USAACE Project Tour To Do: Continue writing Project Report		
	30	5/3/2022	Course Summary Read: Pamell Ch 14 (p 477-483) To Do: Course Evaluation in Blackboard Turn-In: Project Report Due NLT 2200 02MAY22		Project Report (100pts)

Enclosure 4: SE301 Course Objective Assessment Table

SE301 Course Objectives		Cadet Assessment	Direct Assessment	Course Director's Assessment	Applicable Graded Events
1.	Apply systems thinking and the Systems Decision Process (SDP) to design an engineered solution to a large-scale, complex, multi-disciplinary engineering problem.		3.0	3.5	1) Homework 1 2) WPR 1 & 2 3) Project Pres. & Report 4) TEE
2.	Understand the four phases of the Systems Decision Process and perform the individual tasks in each phase to address an engineering problem.		3.0	3.3	1) Homework 2,3,4 2) WPR 1 & 2 3) Project Pres. & Report 4) TEE
3.	Perform stakeholder and requirement analysis to ensure alternatives achieve the desired outcome and account for the political, social, and economic dimensions of an engineering problem.		3.0	3.0	1) Homework 1,2 2) WPR 1 3) Project Pres. & Report 4) TEE
4.	Demonstrate creativity in the formulation of alternative solutions to an engineering problem.		3.0	3.5	1) Homework 3 2) WPR 1 & 2 3) Project Pres. & Report 4) TEE
5.	Apply Life Cycle Cost Analysis methods in comparing alternatives		3.3	3.0	1) Homework 4 2) WPR 1 3) Project Pres. & Report 4) TEE
6.	Apply mathematics, basic science, and engineering to analyze a physical system or process and apply the results to the solution of an engineering problem		3.0	3.0	1) Homework 3,4 2) WPR 2 3) Project Pres. & Report 4) TEE
7.	Understand the role of modeling and simulation in the development and analysis of an engineered solution to meet stakeholder requirements		3.0	2.5	1) Homework 4 2) WPR 2 3) TEE
8.	Communicate technical findings effectively, both orally and in writing		3.0	3.0	1) Homework 1-4 2) Project Presentation 3) Project Report
Overall Assessment			3.04	3.10	
NOTES: 1) Graded Events included: 2xWPR, 4xHomeworks, TEE, Project Presentation and Report 2) Cadet Assessment is based on the End-of-Course Survey which is currently unavailable. 3) Course Director Assessment is: -- *Based on course average for "All graded events" -- All others based on individual questions on graded events Range: 1 = strongly disagree to 4 = strongly agree					

Enclosure 5: Course End Feedback
Tabular report aggregated for Course – SE301

Student surveys are unavailable at the time of course closeout. These will be added as soon as they become available.

Enclosure 6: Cadet Comments

Student comments are unavailable at the time of course closeout. These will be added as soon as they become available.



DEPARTMENT OF THE ARMY
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WEST POINT, NEW YORK 10996



MADN-SE

17 December 2021

MEMORANDUM FOR COL Rich Morales, Professor and Department Head, Department of Systems Engineering, United States Military Academy, West Point, New York 10996.

THRU Dr. Vikram Mittal, Systems & Decision Sciences Program Director, Department of Systems Engineering, United States Military Academy, West Point, New York 10996.

SUBJECT: Academic Term (AT) 22-1 Course Assessment – SE301 Fundamentals of Engineering Design and Systems Management.

1. The attached report summarizes the SE301 assessment and curriculum change recommendations from AT 21-2.
2. EXSUM: This semester is the ninth consecutive semester that SE301 integrated CES cadets and DSE majors into a single course, which continues to be a success. Additionally, the course focused on the entire SDP, with four lessons on Systems Thinking and four-five lessons on each of the phases of the SDP. Overall performance was in line with past academic year semesters. Based on the AY21-2 SE301 course assessment, a few course adjustments were implemented for this semester. These improvements focused on timing of graded events and incorporating workshop lessons to allow cadets to work with their project teams during class time.
3. The point of contact for this action is the undersigned.

Encls

1. Course Assessment
2. Course Instructional Memorandum
3. Course Syllabus
4. Course Objective Assessment Table
5. Course End Feedback
6. Cadet Course End Feedback Comments

RAYMOND T. VETTER
MAJ, EN/FA49
Course Director

Enclosure 1: Course Assessment

Section I – Course Information and History

1. Historic Course Offerings and Enrollments.

a. Offered.

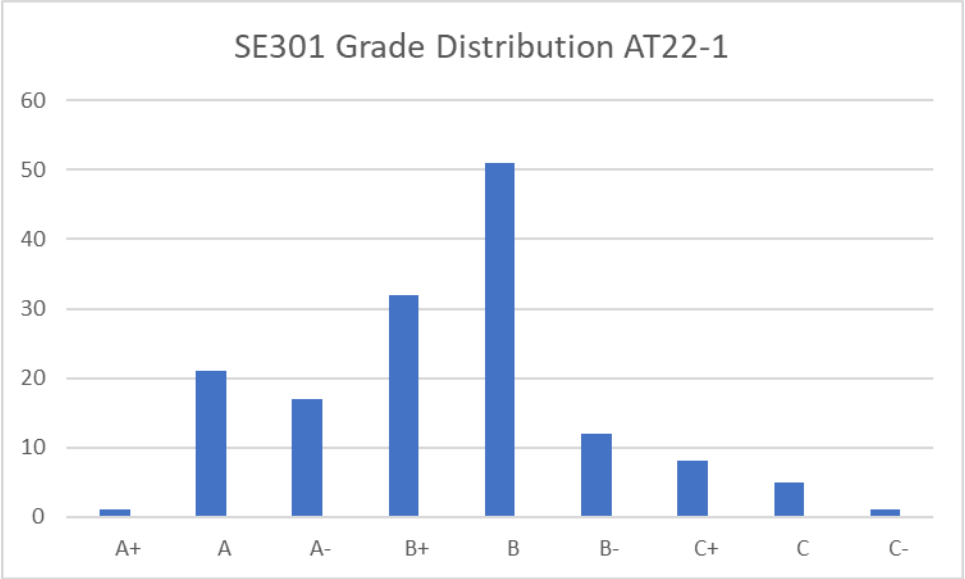
	AY18		AY19		AY20			AY21			AY22
	18-1	18-2	19-1	19-2	20-1	20-2	20-3	21-1	21-2	21-5	22-1
SE300	X	X	X								
SE301	X	X	X	X	X	X	X	X	X	X	X

b. Enrollments.

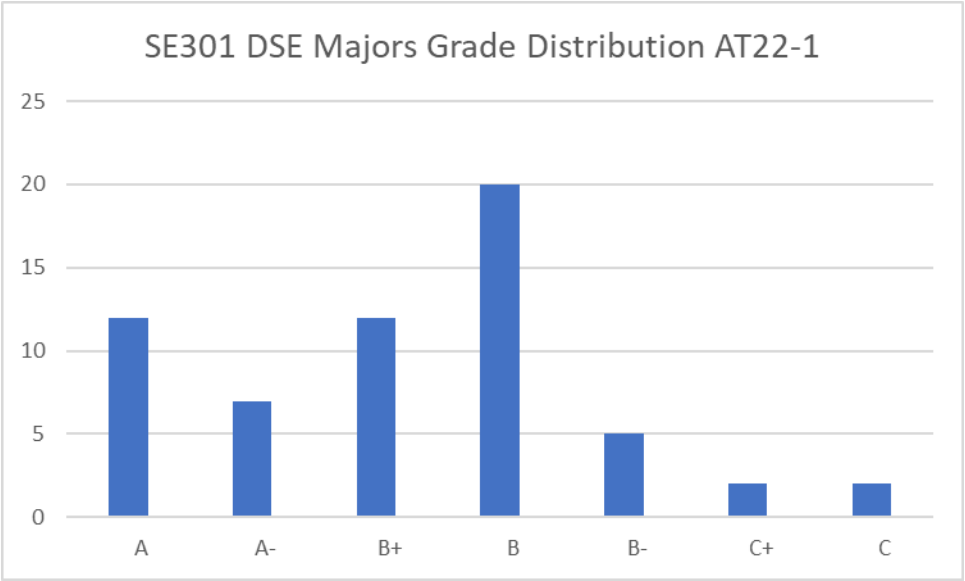
Academic Term	Enrollment	
	SE300	SE301
22-1	148	
21-5	26	
21-2	94	
21-1	121	
20-3	107	
20-2	91	
20-1	227	
19-2	111	
19-1	106	132
18-2	55	95
18-1	89	91
17-5		14
17-2	97	37

c. Historical Course Final Grade Averages.

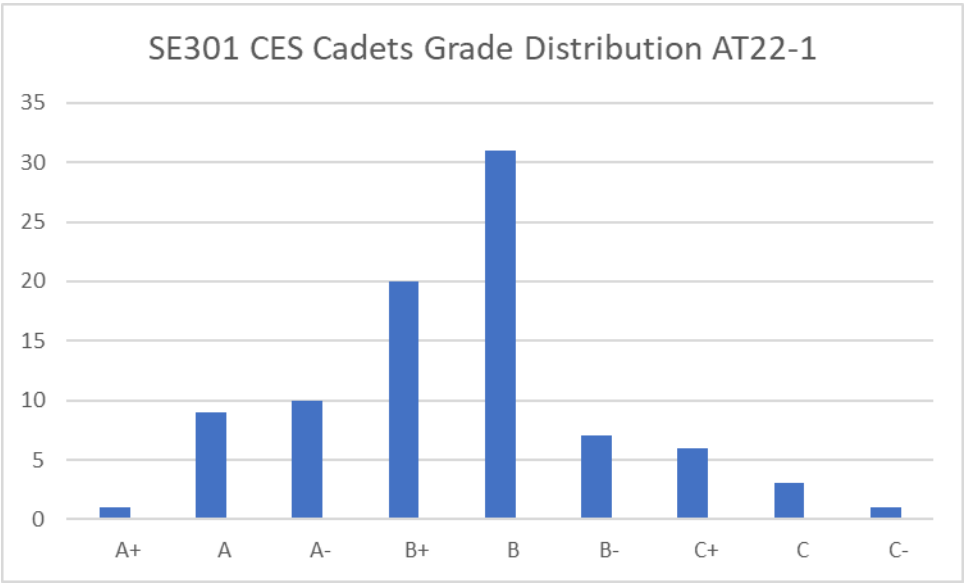
Academic Term	Course Average	
	SE300	SE301
22-1	87.03%	
21-5	90.75%	
21-2	86.26%	
21-1	85.43%	
20-3	91.5%	
20-2	90.2%	
20-1	86.1%	
19-2	88.4%	
19-1	81.4%	86.5%
18-2	83.0%	88.0%
18-1	83.9%	89.0%
17-5		89.8%
17-2	82.8%	90.5%



Course Average: 87.0% (148 cadets)



Course Average for DSE Majors: 87.5% (60 cadets)



Course Average for CES Cadets: 86.7% (88 cadets)

2. Course Strategy.

SE301 provides the baseline approach to the fundamentals of systems thinking, systems engineering, and systems decision making. The course is designed to allow cadets the opportunity to learn engineering design, systems engineering, and engineering management processes on an individual and project team level so that each cadet will have the experience necessary to succeed in future Department of Systems Engineering courses. Cadets learn and apply the Systems Decision Process (SDP) throughout the 30-lesson course. The course includes one individual homework assignment that focuses on applying systems thinking to an emerging real-world initiative to develop the Optionally Manned Fighting Vehicle. The course includes three group homework assignments where cadet teams implement the SDP to devise a strategy for various projects of their choosing. Lastly, the course includes two WPRs focused on course topics, a final project presentation, a final project report, and a Term End Exam (TEE).

Section II – Course Assessment

1. Achievement of Course Objectives.

- a. **Course Objective Assessment Table:** Enclosure 4
- b. **Results of Course End Feedback:** Enclosure 5
- c. **Results of Cadet Free-Form Comments:** Enclosure 6

2. Course Content.

- a. **Body of Knowledge.** The body of knowledge addressed across the five blocks of this course adequately cover the course objectives. The first block covers systems engineering and systems thinking. The remaining four blocks cover the four phases of the systems decision process in more detail. Each phase includes between four-five individual lessons.
- b. **Effective Assessment.** The department's major-graded-event content approval process provides an adequate assessment tool for the assessment of the course objectives.

3. Course Processes.

- a. **Textbooks.**
 - i. Parnell, Gregory S., Driscoll, Patrick J., Henderson, Dale L., Decision Making in Systems Engineering and Management, 2nd Edition, John Wiley & Sons, Inc., 2011.
 - ii. Cabrera, Derek and Cabrera, Laura, Systems Thinking Made Simple: New Hope for Solving Wicked Problems, 2nd Edition, Odyssean, 2015.
- b. **Software.** Cadets utilize Microsoft Excel, PowerPoint, and Word. SE301 introduces Microsoft Project and Vensim as well.
- c. **Graded Requirements.** The graded requirements in SE301 are appropriate for gauging student comprehension, extending fundamental ideas, and demonstrating the real-world relevance of the material. Across the five blocks of instruction, there were a total of nine graded events.

4. Course Pedagogy.

- a. **Lecture Materials:** The course balanced lectures with practical applications for each of the phases of the SDP. Generally, for each phase there were three – four lectures on the material followed by a “lab” type experience to build value models, implementation plans, cost analysis, etc. For some of the phases, the cadets were allowed one class to apply the tools to their homework assignments in class. Lectures were consistent with previous versions of SE301.
- b. **SE Case Studies:** This semester was the eighth time that SE Case Studies were introduced at the end of almost every class to reinforce topics in the lecture with real world applications of systems engineering. Cadets appeared to enjoy the discussions around the case studies as they provided relevance to the materials. Example case studies include: The Apollo Program, Ford Class Aircraft Carrier, and the Mars Climate Orbiter. These cases described both successes and failures in systems engineering. Consistent with past semesters, these case studies were sometimes not discussed because instructors ran out of time. These case studies are too robust to add to the daily lesson prep. They should be truncated to focus on specific lesson objectives or removed from the course completely.
- c. **Homework Assignments:** Homework assignments followed a similar structure to past semesters, where the first homework assignment focuses on systems thinking and the subsequent homework assignments address each phase of the SDP. HW 1 focused on the applications of systems thinking, specifically, the DSRP methodology, to frame the development of the Optionally Manned Fighting Vehicle. Homework 2-4 focused on the application of each phase of the SDP to a group project. Instructors selected 4-5 topics per section and assigned a team of 4-5 cadets to each of the selected topics. These topics remained the focus of each of the remaining homework assignments. Homework 2-4 were group assignments.
- d. **Integration of SE301/SE300:** This semester was the ninth semester in which SE301 cadets from D/SE majors and SE300 cadets from the CES were integrated into the same course and sections. Course evaluations are not available at this time, but cadets were asked to evaluate how they felt this combination of D/SE and CES contributed to their learning. Based on informal feedback received from instructors and cadets, it seems that integrating the course continues to be beneficial to everyone.

Section III – Assessment of Recent Changes and Proposed Future Changes.

1. Assessment of Changes to Current Academic Term (based on previous Course Assessments).

- a. **Course Objectives.** No changes.
- b. **Textbook.** No changes.
- c. **Curriculum.** No changes.
- d. **Graded Requirements.**
 - i. Last semester, all graded events, including WPRs and the TEE were submitted on Microsoft Teams. This semester, we used Microsoft Teams to collect graded HW assignments and the project submissions (presentation slides and technical report).

However, the WPRs and TEE were administered in-person and on paper. For WPR 2 and the TEE, cadets were allowed to bring one 3"x5" notecard to reference. Using a digital submission process allows for instructors to easily use a standardized rubric, view each other's grades, and ensure equity across the course. The only downside to Teams is that it does not have SafeAssign. However, this semester's projects were substantively different from previous offerings, decreasing the likelihood of an honor issue.

- ii. Last semester, cadets were allowed to develop and pitch and topic of their own choosing for the course project. We sustained this approach during AY22-1 and recommend continuing to do so. Cadets were much more motivated to work on their project because they had agency over the topic. The main challenge cadets faced was finding data to support their projects during the performance modeling portion of the SDP. A lot of these issues with data collection could have been avoided with better time management by cadets.

2. Proposed Change for Next Academic Term.

- a. **Course Objectives.** None.
- b. **Textbook.** Cadets should not have to purchase the entire Cabrera text. Only three chapters of the text were required reading for the course this semester. I recommend trying to purchase only those chapters via an e-book at a reduced rate. Another option would be to have cadets watch the professionally produced YouTube videos that Dr. Cabrera has created, and not purchase the Cabrera text at all.
- c. **Curriculum.**
 - i. Reduce the number of Performance Modeling lessons to one in order to make room for an additional Quantitative Value Modeling lesson. During the semester, we found that while cadets conceptually understand value modeling, they need additional repetitions with it, especially when using Excel. As SE301 includes majors and non-majors, it is vitally important they understand how to conduct this analysis, especially for their homework and course project.
 - ii. Reduce the number of Alternative Generation lessons to one in order to make room for an additional Cost Analysis. During the semester, we found that cadets can calculate Net Present Value; however, they struggle with including costs from across the system's lifecycle. This second lesson should focus on helping the cadets identify important costs during each phase of a system's lifecycle so that their cost profiles for the project are more complete. Cadets also do not understand the concept of discount rate unless they have taken EM381.
 - iii. Continue to allow cadets to choose their own project, but have presentations occur in person. Because of the scheduling and only having two lessons after Thanksgiving break, presentations were recorded and submitted via Teams. Cadets put in varying levels of effort and preparation into these recorded presentations. The pressure of standing in front of peers to present, as well as the experience with public speaking, is valuable for cadets. We recommend having Lessons 28 and 29 of SE301 be dedicated solely to in-person presentations.
 - iv. Provide more guidance on project topics. Many cadets proposed topics that led to challenges with finding appropriate data. While instructors attempted to provide input and recommended ideas, a set list of projects or successful past projects in SE301 would be very beneficial to the cadets. Since the semester project proposals are created early in the semester, cadets do not yet have a good understanding of the SDP.

- d. **Graded Requirements.** The sequencing of this year's Homework #4 resulted in a short window after the project presentation. Homework #4 should be due soon after the Decision-Making block is complete. Feedback from Homework #4 should help inform the final project presentation and final project report. Consider rewriting homework #4 or increasing the time between assignment submissions.



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996



MADN-SE

2 August 2021

INSTRUCTIONAL MEMORANDUM
NO. 301-22-1

FUNDAMENTALS OF ENGINEERING DESIGN & SYSTEMS MANAGEMENT

1. **PURPOSE.** To provide each cadet with the administrative details necessary to successfully complete SE301, Fundamentals of Engineering Design and Systems Management.
2. **REFERENCES.**
 - a. Department of Systems Engineering Administrative Memo 1-1, "General Administrative Instructions," 10 July 2021.
 - b. Department of Systems Engineering Administrative Memo 1-2, "Standards for Written Requirements," 10 July 2021.
 - c. Department of Systems Engineering Administrative Memo 1-3, "Academic Documentation Policy," 10 July 2021.
 - d. USMA publication. "Documentation of Academic Work," Office of the Dean, July 2017.
 - e. "Guidelines for Technical Communication Department Guidance," Department of Systems Engineering & Department of Civil & Mechanical Engineering, August 2020.
 - f. Course Website: Microsoft Teams – "SE AY22-1 SE301 – Engineering Design and Systems Management"
3. **GENERAL.** This memorandum provides each cadet with the course description, course objectives, course texts, lesson assignments, grading plan, and criteria for passing the course. Your instructor will inform you in advance of any deviation from the guidance provided in this memorandum.
4. **SPECIFIC.**
 - a. Course Scope. SE301 serves as the foundational course for all cadets taking the Engineering Management, Systems Engineering, or Systems and Decision Sciences majors. This course presents the methodological framework and techniques for

designing, implementing, managing, and re-engineering complex systems or processes. Cadets learn engineering design and engineering management processes and gain an appreciation for future environments and system life cycles. Cadets analyze real-world examples and complete practice problems to illustrate mastery of course topics. Cadets also use spreadsheet software and application design software for modeling and analyzing design alternatives. SE301 introduces the Systems Decision Process (SDP), incorporates material from courses in the USMA core curriculum, and previews the modeling and decision making tools that cadets will learn in follow-on Department of Systems Engineering courses. The course is designed to allow cadets the opportunity to learn engineering design and engineering management processes on an individual level so that each cadet will have the experience necessary to succeed in future Systems Engineering courses. Cadets will spend one to two lessons in a computer lab environment.

- b. **Course Strategy.** SE301 provides the baseline approach to the fundamentals of systems thinking, systems engineering, and systems decision making. The course is designed to allow cadets the opportunity to learn engineering design, systems engineering, and engineering management processes on an individual and project team level so that each cadet will have the experience necessary to succeed in future Department of Systems Engineering courses. Cadets learn and apply the Systems Decision Process (SDP) throughout the 30-lesson course. The course includes four homework assignments that focus on applying systems thinking and the SDP to a complex, engineering problem, two WPRs focused on course topics, a final project presentation, final project report, and one Term End Exam (TEE). Individual work accounts for 55.0% of the course and the remaining 45.0% is group work.
- c. **Course Objectives.**
 - 1) Apply systems thinking and the Systems Decision Process (SDP) to design an engineered solution to a large-scale, complex, multi-disciplinary engineering problem.
 - 2) Understand the four phases of the Systems Decision Process and perform the individual tasks in each phase to address an engineering problem.
 - 3) Perform stakeholder and requirement analysis to ensure alternatives achieve the desired outcome and account for the political, social, and economic dimensions of an engineering problem.
 - 4) Demonstrate creativity in the formulation of alternative solutions to an engineering problem.
 - 5) Apply Life Cycle Cost Analysis methods in comparing alternatives.
 - 6) Apply mathematics, basic science, and engineering to analyze a physical system or process and apply the results to the solution of an engineering problem.

- 7) Understand the role of modeling and simulation in the development and analysis of an engineered solution to meet stakeholder requirements.
 - 8) Communicate technical findings effectively, both orally and in writing.
- d. Course Texts.
- 1) Parnell, Gregory S., Driscoll, Patrick J., Henderson, Dale L., Decision Making In Systems Engineering and Management, 2nd Edition, John Wiley & Sons, Inc., 2011.
 - 2) Cabrera, Derek and Cabrera, Laura, Systems Thinking Made Simple: New Hope for Solving Wicked Problems, 2nd Edition, Odyssean, 2015.
- e. Additional References. Additional reading and course material will be posted on the course Teams page.
- f. Lesson Assignments. Syllabus is posted on the course Teams page.
- g. Performance Assessment.
- 1) The course consists of 1000 points. All graded work is individual work unless otherwise noted. The following is a breakdown of the graded work:
 - a) Homework Assignments (1x Individual, 50 points and 3x Group, 75 points each). The homework assignments will reinforce the phases of the SDP to include Systems Thinking (50 points), Problem Definition (75 points), Solution Design (75 points) and Decision Making (75 points). These individual homework assignments will support the semester long project to apply the SDP to a complex, engineering problem.
 - b) Written Partial Reviews (WPRs). (Individual, 2 x 125 points) The purpose is to reinforce concepts taught throughout the course. The WPRs will consist of interpretation, application, and explanation of concepts given scenario questions.
 - c) Project Report (Group, 1x 125 points) and Presentation (Group, 1 x 75 points). The purpose of the group project is to assess each cadet's ability to function as a contributing member of a team and apply the Systems Decision Process (SDP) to a complex, engineering problem and present a recommended solution and implementation plan to the client / decision-maker. The project will build upon the individual homework assignments that frame and define the problem, develop candidate solutions, and make a recommended decision on the candidate solutions. The group project will expand on these recommended solutions to develop an implementation plan for the decision maker. The report and presentation will evaluate the cadet's ability to communicate effectively in an oral and written manner.

- d) Term End Examination (TEE). (Individual, 200 points). The purpose of this exercise is to reinforce concepts taught during the entire course.
- e) Instructor Points (IP). (Individual, 75 points). The purpose of Instructor Points is to ensure cadets are prepared for class each day, complete assigned problems, and/or evaluate performance on in-class quizzes.

2) Student performance assessment summary:

Graded Event	Format	Lesson	Points	% of Course
HW#1: Systems Thinking Homework	Individual	6	50	5%
WPR 1	Individual	11	125	12.5%
HW#2: Problem Definition Homework	Group	12	75	7.5%
HW#3: Solution Design Homework	Group	17	75	7.5%
WPR 2	Individual	22	125	12.5%
Project Presentation	Group	23	75	7.5%
HW#4: Decision Making Homework	Group	25	75	7.5%
Instructor Points	Individual	1-30	75	7.5%
Project Report	Group	30	125	12.5%
Term End Examination	Individual	TEE	200	20%
Course Total			1000	

5. ADMINISTRATIVE GUIDANCE.

- a. Minimum Requirements. Score at least 50% on the TEE and complete the course with at least 65%.
- b. Graded Events. Unless there are exceptional circumstances, cadets will take all WPRs and the TEE and complete all graded events during their primary scheduled times IAW the DPOM. Cadets who will miss an event must contact the instructor as soon as they know about the conflict. Cadets have the responsibility to coordinate with the instructor to make up any material/requirements missed.
- c. Late Submissions. Failure to submit a graded event on time for other than legitimate, compelling reasons will result in a reduced grade (5% per 24-hour period) and may result in disciplinary action. Cadets are required to submit all major graded events ($\geq 10\%$ of course grade) regardless of lateness. Any graded event not submitted prior to lesson 30 will receive a grade of zero. All graded events must be submitted to pass the course.
- d. Final Grades. Final grades will be awarded IAW reference 2.a, Department of Systems Engineering Admin Memo 1-1.
- e. Documentation. Cadets may use any available reference material in preparing course work but use of any references **MUST BE PROPERLY DOCUMENTED.** Documentation procedures are outlined in references 2b through 2d.

- f. Additional Instruction (AI). AI is available at mutually agreed upon times between the cadet and their instructor. Prior to visiting an instructor for additional instruction, cadets are expected to make a legitimate attempt to complete the daily assignments and practice problems to standard and bring those assignments with them to the AI session. Cadets are expected to have specific questions for any AI session. Cadets will bring their notes and course textbooks to any AI session. Cadets should primarily use their own instructor for AI; however, they are authorized to seek AI from other instructors teaching the course this semester. No AI is given after the first version of a WPR has been given, whether that version is the primary or a special make-ahead exam. The last AI session for the TEE is prior to 1600 on the last day of classes.
- g. Practice Problems. Practice Problems and reading assignments are assigned with the intent of enhancing the understanding of text material and assist the cadets in preparing for graded events. Instructors may choose to allocate some of their Instructor Points towards practice problems.
- h. Class. You are expected to bring your texts and laptop to each class. If you plan or anticipate being absent from class, let your instructor and section marcher know *prior to the class* in person or via email. If you are unable to attend class, you must inform your Instructor NLT 24hrs prior to the missed lesson.
- i. Teams. As mentioned above, the Microsoft Teams Page “SE AY22-1 SE301 – Engineering Design and Systems Management” will be the primary means of communication for this course. The course director will post all announcements and assignments to this page. It is your responsibility to consistently check this page for updated content and assignments. After you have read this sentence, send an email to your instructor acknowledging your understanding of the course standards.

FOR THE HEAD OF THE DEPARTMENT:

//ORIGINAL SIGNED//

RAYMOND T. VETTER
MAJ, FA49
Course Director

//ORIGINAL SIGNED//

VIKRAM MITTAL, PhD
Associate Professor
Program Director

DISTRIBUTION: 1 each instructor, SE301
 1 each cadet, SE301

Enclosure 3: SE301 Syllabus

Lsn	Day 1	Day 2	Title	Assignment (Complete Prior to Class)	Supplementary Reading (Optional)	Graded Event
Systems Thinking	1	8/16/2021	8/17/2021	Introduction to the Systems Decision Process (SDP)	Read: Parnell Ch 6 (p 185-193) Read: Parnell Ch 9.1-9.10 (p 275-290) To Do: Exercises 9.1, 9.3, & 9.9	
	2	8/18/2021	8/19/2021	Systems Thinking	Read: Parnell Ch 2 (p 27-40; 54-59) To Do: Exercise 2.1	Senge - Systems Thinking and Org. Learning Issue HW #1 Guidance
	3	8/20/2021	8/23/2021	DSRP Methodology	Read: Cabrera Ch. 3 (p. 52-69)	Watch: How Thinking Works (https://www.youtube.com/watch?v=dUqRTWCdXt4)
		8/24/2021		Day 1 Drop		
	4	8/26/2021	8/27/2021	Systems Thinking Workshop	Read: Systems Thinking Case Study	
Problem Definition	5	8/30/2021	8/31/2021	Introduction to Problem Definition	Read: Parnell Ch 10.1 (p 297-300) To Do: Exercise 10.1	Curtis - Doing the Right Problem
	6	9/1/2021	9/2/2021	Stakeholder Analysis	Read: Parnell Ch 1.3 (p 3-7) & Ch 10.2 (p 300-314) To Do: Exercise 10.2 Turn-In: Homework 1 Due NLT 1700 2SEP21	Reynolds - Stakeholder Theory and Managerial Decision-Making HW #1 Due (50pts) Issue HW#2 Guidance
		9/3/2021		Day 1 Drop		
	7	9/8/2021	9/7/2021	Functional & Requirements Analysis	Read: Parnell Ch 10.3 (p 314-325) To Do: Exercise 10.3	Carson - Implementing Structured Requirements to Improve Requirements Quality
	8	9/10/2021	9/9/2021	Value Modeling (Qualitative) I	Read: Parnell Ch 10.4 (p 326-331; 333-337) Read: Saaty - Decision Making with AHP To Do: Exercise 10.6 Turn-In: Homework 2 Part 1 Due NLT 1700 10SEP21	HW #2 Part 1 Due (15pts)
	9	9/13/2021	9/14/2021	Value Modeling (Quantitative) II	Read: Parnell Ch 10.4.3 (p 331-341) Read: SE301 HW 2 Guidance	
	10	9/16/2021	9/17/2021	Cabrera Systems Thinking Lecture (DROP)	Read: https://blog.cabreraaresearch.org/systems-thinking-in-a-7-images Watch: https://www.youtube.com/watch?v=-sfReUu3o0 Lecture: 1940-2040 15 September 2021, Teams	
	11	9/20/2021	9/21/2021	WPR 1	Review: Lessons 1-10	WPR 1 (125pts)
Solution Design	12	9/24/2021	9/23/2021	Introduction to Solution Design	Read: Parnell Ch 11.1-11.2 (p 353-365) To Do: Exercise 11.1 Turn-In: Homework 2 Due NLT 1700 24SEP21	Frey - Engineering Design Thinking HW #2 Due (60pts)
	13	9/29/2021	9/27/2021	Alternative Generation	Read: Parnell Ch 11.3-11.5.2 & 11.5.5-11.6 (p.365-370 & p.386-388) To Do: Exercises 11.2 & 11.3	Kelly - Creative Confidence Excerpt Issue HW #3 Guidance
	14	10/1/2021	9/30/2021	Alternative Generation II	Read: SE301 HW 3 Guidance	
	15	10/4/2021	10/5/2021	Design Methods	Read: Parnell Ch 3.3 (p 74-77) & Ch 7.5 (p 212-221) Read: Axiomatic Design Technology	Issue Project Presentation and Report Guidance
	16	10/7/2021	10/8/2021	Cost Analysis	Read: Parnell Ch 5.1-5.4 (p 137-172) To Do: Exercise 5.4 & 5.7	
	17	10/13/2021	10/12/2021	Introduction to Decision Making	Read: Parnell Ch 12.1-12.4 (p 395-409) To Do: Exercise 12.1 Turn-In: Homework 3 Due NLT 1700 13OCT21	Keeney - Creativity in Decision Making HW #3 Due (75pts)
Decision Making	18	10/15/2021	10/14/2021	Performance Modeling I	Read: Parnell Ch 4.1-4.5 (p 95-121) To Do: Exercises 4.5, 4.6, 4.7, & 4.8	Hakes - An Economic Evaluation of the Moneyball Hypothesis Issue HW #4 Guidance
	19	10/19/2021	10/18/2021	Performance Modeling II	Read: SE301 HW 4 Guidance	
	20	10/21/2021	10/22/2021	Sensitivity and Risk Analysis	Read: Parnell Ch 3.4-3.6 and 12.5-12.6 (p 409-417) To Do: Exercise 12.7	McManus - A Framework for Understanding Uncertainty
	21	10/25/2021	10/26/2021	Tradespace Analysis	Read: Parnell Ch 12.7-12.10 (p 417-432) To Do: Exercises 12.3, 12.4, 12.9 & 12.10	Ross - Multi-Attribute Trade space Exploration
Solution Implementation	22	10/27/2021	10/28/2021	WPR 2	Review: Lessons 12-21	WPR 2 (125pts)
		10/29/2021		Day 1 Drop		
	23	11/1/2021	11/3/2021	Introduction to Solution Implementation	Read: Parnell Ch 13.1-13.3 (p 447-453) To Do: 13.1 & 13.6 Turn-In: Project Presentation Due NLT 1700 3NOV21	Sauser - Why Projects Fail Project Presentation Due (75pts)
	24	11/4/2021	11/5/2021	Project Planning	Read: Parnell Ch 13.4 (p 453-456) To Do: 13.8 & 13.9	Lewis - One World Trade
	25	11/8/2021	11/9/2021	Execution, Monitoring, and Control	Read: Parnell Ch 13.5-13.8 (p 457-474) To Do: 13.13 & 13.14 Turn-In: Homework 4 Due NLT 1700 9NOV21	Sharon Project Management vs Systems Engineering Management HW #4 Due (75pts)
		11/10/2021		Day 1 Drop		
		11/12/2021		Day 1 Drop		
Course Summary	26	11/15/2021	11/16/2021	Introduction to Systems Thinking and Simulation	Read: Parnell Ch 4.6 (p 121-128) & Ch 4.1-4.5 (p 95-121) Read: System Dynamics (Bala et al) p. 3-22	
	27	11/19/2021	11/18/2021	Systems Thinking and Simulation Lab	Read: 7 Case Studies in Carbon and Climate	
	28	11/22/2021	11/23/2021	Project Report Working Session	To Do: Continue writing Project Report and Briefing	
		11/24/2021		Day 1 Drop		
		11/29/2021		Day 1 Drop		
	29	12/1/2021	12/2/2021	Project Presentation Feedback Session	To Do: Continue writing Project Report To Do: Watch assigned Project Group Presentation To Do: Come to Feedback Session with Questions	
		12/6/2021		Day 1 Drop		
	30	12/3/2021	12/7/2021	Course Summary	Read: Parnell Ch 14 (p 477-483) Turn-In: Project Report Due NLT 1700 7DEC21	Project Report (125pts)
		12/9/2021		Day 1 Drop		
		12/10/2021		Day 1 Drop		

Enclosure 4: SE301 Course Objective Assessment Table

SE301 Course Objectives		Cadet Assessment	Direct Assessment	Course Director's Assessment	Applicable Graded Events
1.	Apply systems thinking and the Systems Decision Process (SDP) to design an engineered solution to a large-scale, complex, multi-disciplinary engineering problem.		3.3	4.0	1) Homework 1 2) WPR 1 & 2 3) Project Pres. & Report 4) TEE
2.	Understand the four phases of the Systems Decision Process and perform the individual tasks in each phase to address an engineering problem.		3.0	4.0	1) Homework 2,3,4 2) WPR 1 & 2 3) Project Pres. & Report 4) TEE
3.	Perform stakeholder and requirement analysis to ensure alternatives achieve the desired outcome and account for the political, social, and economic dimensions of an engineering problem.		3.3	3.3	1) Homework 2 2) WPR 1 3) Project Pres. & Report 4) TEE
4.	Demonstrate creativity in the formulation of alternative solutions to an engineering problem.		3.0	3.5	1) Homework 3 2) WPR 1 & 2 3) Project Pres. & Report 4) TEE
5.	Apply Life Cycle Cost Analysis methods in comparing alternatives		3.3	2.5	1) Homework 4 2) WPR 1 3) Project Pres. & Report 4) TEE
6.	Apply mathematics, basic science, and engineering to analyze a physical system or process and apply the results to the solution of an engineering problem		3.3	3.5	1) Homework 3,4 2) WPR 2 3) Project Pres. & Report 4) TEE
7.	Understand the role of modeling and simulation in the development and analysis of an engineered solution to meet stakeholder requirements		3.0	2.5	1) Homework 4 2) WPR 2 3) TEE
8.	Communicate technical findings effectively, both orally and in writing		3.3	3.3	1) Homework 1-4 2) Project Presentation 3) Project Report
Overall Assessment			3.21	3.32	
NOTES: 1) Graded Events included: 2xWPR, 4xHomeworks, TEE, Project 2) Cadet Assessment is based on the End-of-Course Survey which is currently unavailable. 3) Course Director Assessment is: -- *Based on course average for "All graded events" -- All others based on individual questions on graded events Range: 1 = strongly disagree to 4 = strongly agree					

Enclosure 5: Course End Feedback
Tabular report aggregated for Course – SE301

Student surveys are unavailable at the time of course closeout. These will be added as soon as they become available.

Enclosure 6: Cadet Comments

Student comments are unavailable at the time of course closeout. These will be added as soon as they become available.



UNITED STATES MILITARY ACADEMY
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TAB F AY2021 COURSE ASSESSMENT



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996



MADN-SE

15 May 2021

MEMORANDUM FOR COL Rich Morales, Professor and Department Head, Department of Systems Engineering, United States Military Academy, West Point, New York 10996.

THRU LTC James Schreiner, Systems & Decision Sciences Program Director, Department of Systems Engineering, United States Military Academy, West Point, New York 10996.

SUBJECT: Academic Term (AT) 21-2 Course Assessment – SE301 Fundamentals of Engineering Design and Systems Management.

1. The attached report summarizes the SE301 assessment and curriculum change recommendations from AT 21-1.
2. EXSUM: This semester is the sixth consecutive semester that SE301 integrated CES cadets and DSE majors into a single course, which continues to be a success. Additionally, the course focused on the entire SDP, with five lessons on Systems Thinking and five lessons on each of the phases of the SDP. Overall performance is lower compared to previous semesters. Based on the AY 21-1 SE301 course assessment and requirements to deliver course content in a hybrid learning environment, several course improvement recommendations were implemented from the previous semester. These improvements focused on leveraging technology and pedagogical efficiencies which enabled flexible and redundant course material delivery and assessment.
3. The point of contact for this action is the undersigned.

Encls

1. Course Assessment
2. Course Instructional Memorandum
3. Course Syllabus
4. Course Objective Assessment Table
5. Course End Feedback
6. Cadet Course End Feedback Comments

ARTHUR J. MIDDLEBROOKS
MAJ, AR/FA49
Course Director

Enclosure 1: Course Assessment

Section I – Course Information and History

1. Historic Course Offerings and Enrollments.

a. Offered.

	AY17		AY18		AY19		AY20			AY 21-1	
	17-1	17-2	18-1	18-2	19-1	19-2	20-1	20-2	20-3	21-1	21-2
SE300	X	X	X	X	X						
SE301	X	X	X	X	X	X	X	X	X	X	X

b. Enrollments.

Academic Term	Enrollment	
	SE300	SE301
21-2	94	
21-1	121	
20-3	107	
20-2	91	
20-1	227	
19-2	111	
19-1	106	132
18-2	55	95
18-1	89	91
17-5		14
17-2	97	37
17-1	100	107

c. Historical Course Final Grade Averages.

Academic Term	Course Average	
	SE300	SE301
21-2	86.26	
21-1	86.26	
20-3	91.5%	
20-2	90.2%	
20-1	86.1%	
19-2	88.4%	
19-1	81.4%	86.5%
18-2	83.0%	88.0%
18-1	83.9%	89.0%
17-5		89.8%
17-2	82.8%	90.5%
17-1	84.5%	87.8%

2. Course Strategy.

SE301 provides the baseline approach to the fundamentals of systems thinking, systems engineering, and systems decision making. The course is designed to allow cadets the opportunity to learn engineering design, systems engineering, and engineering management processes on an individual and project team level so that each cadet will have the experience necessary to succeed in future Department of Systems Engineering courses. Cadets learn and apply the Systems Decision Process (SDP) throughout the 30-lesson course. The course includes one individual homework assignment that focuses on applying systems thinking to an emerging real-world initiative to integrate virtual reality in the Army's rotary-wing training curriculum. The course includes three group homework assignments where cadet teams implement the SDP to devise a strategy for various projects of their choosing. Lastly, the course includes two WPRs focused on course topics, a final project presentation, a final project report, and one Term End Exam (TEE). Individual work accounts for 67.5% of the course and the remaining 32.5% is group work.

Section II – Course Assessment

1. Achievement of Course Objectives.

- a. **Course Objective Assessment Table:** Enclosure 4
- b. **Results of Course End Feedback.** Enclosure 5
- c. **Results of Cadet Free-Form Comments.** Enclosure 6

2. Course Content.

- a. **Body of Knowledge.** The body of knowledge addressed across the five blocks of this course adequately cover the course objectives. The first block covers systems engineering and systems thinking and introduces the systems decision process. The remaining four blocks cover the four phases of the systems decision process in more detail. Each phase includes between 5-7 individual lessons.
- b. **Effective Assessment.** The department's major-graded-event content approval process provides an adequate assessment tool for the assessment of the course objectives.

3. Course Processes.

- a. **Textbooks.**
 - i. Parnell, Gregory S., Driscoll, Patrick J., Henderson, Dale L., Decision Making in Systems Engineering and Management, 2nd Edition, John Wiley & Sons, Inc., 2011.
 - ii. Cabrera, Derek and Cabrera, Laura, Systems Thinking Made Simple: New Hope for Solving Wicked Problems, 2nd Edition, Odyssean, 2015.
- b. **Software.** Cadets utilize Microsoft Excel, PowerPoint, and Word. SE301 provides an introduction to Microsoft Project and Vensim as well.
- c. **Graded Requirements.** The graded requirements in SE301 are appropriate for gauging student comprehension, extending fundamental ideas, and demonstrating the real-world relevance of the material. Across the five blocks of instruction, there were a total of nine

graded events. The course consisted of 67.5% individual work 32.5% group work with 4 to 5-person teams.

4. Course Pedagogy.

- a. **Lecture Materials:** The course balanced lectures with practical applications for each of the phases of the SDP. Generally, for each phase there were two - three lectures on the material followed by a “lab” type experience to build value models, implementation plans, cost analysis, etc. For some of the phases, the cadets were allowed one class to apply the tools to their homework assignments in class. Lectures were consistent with previous versions of SE301.
- b. **SE Case Studies:** This semester was the sixth time that SE Case Studies were introduced at the end of almost every class to reinforce topics in the lecture with real world applications of systems engineering. Cadets appeared to enjoy the discussions around the case studies as they provided relevance to the materials. Example case studies include: The Apollo Program, Ford Class Aircraft Carrier, and the Mars Climate Orbiter. These cases described both successes and failures in systems engineering. Consistent with each of the last four semesters, these case studies were often not discussed due to instructors running out of time. These case studies are too robust to add to the daily lesson prep. They should be truncated to focus on specific lesson objectives or removed from the course completely.
- c. **Homework Assignments:** Homework assignments followed a similar structure to past semesters, where the first homework assignment focuses on systems thinking and the subsequent homework assignments address each phase of the SDP. HW 1 focused on the applications of systems thinking, specifically, the DSRP methodology, to frame a problem/system of the cadet team’s choosing. Homework 2-4 focused on the application of each phase of the SDP to a group project. Each cadet presented a group project topic to their respective instructor immediately following the submission of Homework 1. Instructors selected 4-5 topics per section and assigned a team of 4-5 cadets to each of the selected topics. These topics remained the focus of each of the remaining homework assignments.
- d. **Integration of SE301/SE300:** This semester was the sixth semester in which SE301 cadets from D/SE majors and SE300 cadets from the CES were integrated into the same course and sections. Course evaluations are not available at this time, but cadets were asked to evaluate how they felt this combination of D/SE and CES contributed to their learning. Based on informal feedback received from instructors and cadets, it seems that integrating the course continues to be beneficial to everyone. Performance of CES versus D/SE cadets was tracked for every graded event except for the course end project, which included teams that were composed of CES and D/SE cadets.

Section III – Assessment of Recent Changes and Proposed Future Changes.

1. Assessment of Changes to Current Academic Term (based on previous Course Assessments).

- a. **Course Objectives.** No changes.
- b. **Textbook.** No changes.
- c. **Curriculum.** No changes.

d. Graded Requirements.

- i. Last semester, all graded events, including WPRs and the TEE were submitted on Blackboard. This semester, we continued with a similar strategy for all graded events, except for WPRs and the TEE, using Microsoft Teams. Regardless of the medium, using a digital submission process allows for instructors to easily use a standardized rubric, view each other's grades, and ensure equity across the course. The only downside to Teams is that it does not have SafeAssign. However, this semester's projects were substantively different from previous offerings, decreasing the likelihood of an honor issue.
- ii. Last semester, cadets were allowed to develop and pitch and topic of their own choosing for the course project. We sustained this approach during AY21-2 and recommend continuing to do so. Cadets were much more motivated to work on their project because they had agency over the topic.
- iii. Last semester, SE301 was taught using a hybrid approach. This semester, due to COVID19 protocols, we were forced to continue this practice. While Teams is an effective medium for online learning, I recommend SE301 go back to a live modality.
- iv. Last semester, due to the academic schedule adjustments, SE301 was unable to implement the Infantry Warrior Simulation (IWARS) lab. This semester, we changed the simulation lab to focus on System Dynamics. This change allowed cadets to experience a different approach to simulation. I recommend maintaining the use of System Dynamics, as it's applicability to social and technical problems surpasses that of iWARS.
- v. Last semester, the Written Partial Reviews and Term End Exam were administered and graded through Blackboard. This semester, all WPRs and the TEE were administered in-person, using paper-copies of the exams. If SE301 is taught in person, I recommend continuing with in-person, hand-written exams.

2. Proposed Change for Next Academic Term.

a. Course Objectives. None.

b. Textbook. Continue to make sure that the registrar's office knows that the Cabrera textbook should be the second edition. Also, cadets should not have to purchase the entire Cabrera text. Only three chapters of the text were required reading for the course this semester. I recommend trying to purchase only those chapters via an e-book at a reduced rate. Lastly, ensure the Green Book is the correct edition.

c. Curriculum.

- i. Reduce the number of Performance Modeling lessons to one in order to make room for an additional Quantitative Value Modeling lesson. During the semester, we found that while cadets conceptually understand value modeling, they need additional repetitions with it, especially when using Excel. As SE301 includes majors and non-majors, it is vitally important they understand how to conduct this analysis, especially for their homework and course project.
- ii. Reduce the number of Alternative Generation lessons to one in order to make room for an additional Cost Analysis. During the semester, we found that cadets can calculate Net Present Value; however, they struggle with including costs from across the system's lifecycle. This second lesson should focus on helping the cadets identify important costs during each phase of a system's lifecycle so that their cost profiles for the project are more complete.

- iii. Continue to allow cadets to choose their own project. This was something they cadets greatly enjoyed and allows them to see the universal applicability of the SDP. Additionally, continue to have the cadets record their presentations. This allows the instructor to view them at their leisure, forces the cadets to rehearse, and practice their public speaking skills. More importantly, this allows for the feedback sessions during lessons 28 and 29 to be more fruitful. A recommended COA is to have two groups per session in class, with the other two working remotely. For those in-person, print out the rubric and have them grade each other. Leave time for the cadets to provide each other with feedback. This is a great way to allow cadets to see other projects and give feedback to their peers.
- d. **Graded Requirements.** Consider revising Homework #4 so that it is less similar to the project presentation and report. The sequencing of this year's homework 4 resulted in a short window between the assignment's submission and that of the project presentation. If feedback was not provided on time, this led to cadets submitting the same work twice. Consider rewriting homework #4 or increasing the time between assignment submissions.



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6 August 2020

INSTRUCTIONAL MEMORANDUM
NO. 301-21-1

FUNDAMENTALS OF ENGINEERING DESIGN & SYSTEMS MANAGEMENT

1. **PURPOSE.** To provide each cadet with the administrative details necessary to successfully complete SE301, Fundamentals of Engineering Design and Systems Management.
2. **REFERENCES.**
 - a. Department of Systems Engineering Administrative Memo 1-1, "General Administrative Instructions," 8 July 2020.
 - b. Department of Systems Engineering Administrative Memo 1-2, "Standards for Written Requirements," 8 July 2020.
 - c. Department of Systems Engineering Administrative Memo 1-3, "Academic Documentation Policy," 8 July 2020.
 - d. USMA publication. "Documentation of Academic Work," Office of the Dean, July 2018.
 - e. "Communication Skills Handbook, A Basic Guide to Oral and Written Communications," Department of Systems Engineering, April 2000.
 - f. Course Website: Microsoft Teams – "SE AY21-2 SE301 – Engineering Design and Systems Management"
3. **GENERAL.** This memorandum provides each cadet with the course description, course objectives, course texts, lesson assignments, grading plan, and criteria for passing the course. Your instructor will inform you in advance of any deviation from the guidance provided in this memorandum.
4. **SPECIFIC.**
 - a. Course Scope. SE301 serves as the foundational course for all cadets taking the Engineering Management, Systems Engineering, or Systems and Decision Sciences

majors. This course presents the methodological framework and techniques for designing, implementing, managing, and re-engineering complex systems or processes. Cadets learn engineering design and engineering management processes and gain an appreciation for future environments and system life cycles. Cadets analyze real-world examples and complete practice problems to illustrate mastery of course topics. Cadets also use spreadsheet software and application design software for modeling and analyzing design alternatives. SE301 introduces the Systems Decision Process (SDP), incorporates material from courses in the USMA core curriculum, and previews the modeling and decision making tools that cadets will learn in follow-on Department of Systems Engineering courses. The course is designed to allow cadets the opportunity to learn engineering design and engineering management processes on an individual level so that each cadet will have the experience necessary to succeed in future Systems Engineering courses. Cadets will spend one to two lessons in a computer lab environment.

- b. Course Strategy. SE301 provides the baseline approach to the fundamentals of systems thinking, systems engineering, and systems decision making. The course is designed to allow cadets the opportunity to learn engineering design, systems engineering, and engineering management processes on an individual and project team level so that each cadet will have the experience necessary to succeed in future Department of Systems Engineering courses. Cadets learn and apply the Systems Decision Process (SDP) throughout the 30-lesson course. The course includes four homework assignments that focus on applying systems thinking and the SDP to a complex, engineering problem, two WPRs focused on course topics, a final project presentation, final project report, and one Term End Exam (TEE). Individual work accounts for 67.5% of the course and the remaining 32.5% is group work.
- c. Course Objectives.
 - 1) Apply systems thinking and the Systems Decision Process (SDP) to design an engineered solution to a large-scale, complex, multi-disciplinary engineering problem.
 - 2) Understand the four phases of the Systems Decision Process and perform the individual tasks in each phase to address an engineering problem.
 - 3) Perform stakeholder and requirement analysis to ensure alternatives achieve the desired outcome and account for the political, social, and economic dimensions of an engineering problem.
 - 4) Demonstrate creativity in the formulation of alternative solutions to an engineering problem.
 - 5) Apply Life Cycle Cost Analysis methods in comparing alternatives.
 - 6) Apply mathematics, basic science, and engineering to analyze a physical system or process and apply the results to the solution of an engineering problem.

- 7) Understand the role of modeling and simulation in the development and analysis of an engineered solution to meet stakeholder requirements.
 - 8) Communicate technical findings effectively, both orally and in writing.
- d. Course Texts.
- 1) Parnell, Gregory S., Driscoll, Patrick J., Henderson, Dale L., Decision Making In Systems Engineering and Management, 2nd Edition, John Wiley & Sons, Inc., 2011.
 - 2) Cabrera, Derek and Cabrera, Laura, Systems Thinking Made Simple: New Hope for Solving Wicked Problems, 2nd Edition, Odyssean, 2015.
- e. Additional References. Additional reading and course material will be posted on the course Teams page.
- f. Lesson Assignments. Syllabus is posted on the course Teams page.
- g. Performance Assessment.
- 1) The course consists of 1000 points. All graded work is individual work unless otherwise noted. The following is a breakdown of the graded work:
 - a) Homework Assignments (1x Individual, 50 points and 3x Group, 75 points each). The homework assignments will reinforce the phases of the SDP to include Systems Thinking (50 points), Problem Definition (75 points), Solution Design (75 points) and Decision Making (75 points). These individual homework assignments will support the semester long project to apply the SDP to a complex, engineering problem.
 - b) Written Partial Reviews (WPRs). (Individual, 2 x 125 points) The purpose is to reinforce concepts taught throughout the course. The WPRs will consist of interpretation, application, and explanation of concepts given scenario questions.
 - c) Project Report (Group, 1x 100 points) and Presentation (Group, 1 x 75 points). The purpose of the group project is to assess each cadet's ability to function as a contributing member of a team and apply the Systems Decision Process (SDP) to a complex, engineering problem and present a recommend solution and implementation plan to the client / decision-maker. The project will build upon the individual homework assignments that frame and define the problem, develop candidate solutions, and make a recommended decision on the candidate solutions. The group project will expand on these recommended solutions to develop an implementation plan for the decision maker. The report and presentation will evaluate the cadet's ability to communicate effectively in an oral and written manner.

- d) Term End Examination (TEE). (Individual, 175 points). The purpose of this exercise is to reinforce concepts taught during the entire course.
- e) Instructor Points (IP). (Individual, 75 points). The purpose of Instructor Points is to ensure cadets are prepared for class each day, complete assigned problems, and/or evaluate performance on in-class quizzes.

2) Student performance assessment summary:

Graded Event	Format	Lesson	Points	% of Course
HW#1: Systems Thinking Homework	Individual	6	50	5%
WPR 1	Individual	11	125	12.5%
HW#2: Problem Definition Homework	Individual/ Group	12	75	7.5%
HW#3: Solution Design Homework	Group	17	75	7.5%
WPR 2	Individual	22	125	12.5%
Project Presentation	Group	23	75	7.5%
HW#4: Decision Making Homework	Group	25	75	7.5%
Systems Thinking Presentation	Individual	30	50	5%
Instructor Points	Individual	1-30	75	7.5%
Term End Examination	Individual	TEE	175	17.5%
Project Report	Group	5 May	100	10%
Course Total			1000	

5. ADMINISTRATIVE GUIDANCE.

- a. Minimum Requirements. Score at least 50% on the TEE and complete the course with at least 65%.
- b. Graded Events. Unless there are exceptional circumstances, cadets will take all WPRs and the TEE and complete all graded events during their primary scheduled times IAW the DPOM. Cadets who will miss an event must contact the instructor as soon as they know about the conflict. Cadets have the responsibility to coordinate with the instructor to make up any material/requirements missed.
- c. Late Submissions. Failure to submit a graded event on time for other than legitimate, compelling reasons will result in a reduced grade (5% per 24-hour period) and may result in disciplinary action. Cadets are required to submit all major graded events ($\geq 10\%$ of course grade) regardless of lateness. Any graded event not submitted prior to lesson 30 will receive a grade of zero. All graded events must be submitted to pass the course.
- d. Final Grades. Final grades will be awarded IAW reference 2.a, Department of Systems Engineering Admin Memo 1-1.

- e. Documentation. Cadets may use any available reference material in preparing course work, but use of any references **MUST BE PROPERLY DOCUMENTED**. Documentation procedures are outlined in references 2b through 2d.
- f. Additional Instruction (AI). AI is available at mutually agreed upon times between the cadet and their instructor. Prior to visiting an instructor for additional instruction, cadets are expected to make a legitimate attempt to complete the daily assignments and practice problems to standard and bring those assignments with them to the AI session. Cadets are expected to have **specific questions for any AI session**. Cadets will bring their notes and course text books to any AI session. Cadets should primarily use their own instructor for AI, however they are authorized to seek AI from other instructors teaching the course this semester. No AI is given after the first version of a WPR has been given, whether that version is the primary or a special make-ahead exam. The last AI session for the TEE is prior to 1600 on the last day of classes.
- g. Practice Problems. Practice Problems and reading assignments are assigned with the intent of enhancing the understanding of text material and assist the cadets in preparing for graded events. Instructors may choose to allocate some of their Instructor Points towards practice problems.
- h. Class. You are expected to bring your texts and laptop to each class. If you plan or anticipate being absent from class, let your instructor and section marcher know *prior to the class* in person or via email. In order to comply with USMA and Department policy, no more than 12 Cadets will be allowed to attend class in person. Your Instructor will assign you specific lessons in which you will attend class in person. IAW COVID-19 mitigation measures, lessons 1-4 will be conducted remotely. Beginning with lesson 5, classes will be conducted using the hybrid approach, with some cadets in-person and the remaining attending virtually. You are responsible for ensuring you attend class in the appropriate manner. When attending class in a remote status, you must turn on your camera for the duration of the lesson. If you are quarantined due to COVID-19, you will (1) inform your Instructor and (2) attend class via the online platform. If you are unable to attend class, you must inform your Instructor NLT 24hrs prior to the missed lesson.
- i. Teams. As mentioned above, the Microsoft Teams Page “SE AY21-2 SE301 – Engineering Design and Systems Management” will be the primary means of communication for this course. The course director will post all announcements and assignments to this page. It is your responsibility to consistently check this page for updated content and assignments. After you have read this sentence, send an email to your instructor acknowledging your understanding of the course standards.

FOR THE HEAD OF THE DEPARTMENT:

// ORIGINAL SIGNED //

ARTHUR J. MIDDLEBROOKS
MAJ, AR/49
Course Director

// ORIGINAL SIGNED //

JAMES H. SCHREINER
LTC, EN / FA47
Program Director

DISTRIBUTION: 1 each instructor, SE301
 1 each cadet, SE301

Enclosure 3: SE301 Syllabus

Lsn	Day 1	Title	Assignment (Complete Prior to Class)	Supplementary Reading (Optional)	Graded Event	Significant Event
Systems Thinking	1/19/2021	Introduction to the Systems Decision Process (SDP)	Read: Parnell Ch 6 (p 185-193) Read: Parnell Ch 9.1-9.10 (p 275-290) To Do: Exercises 9.1, 9.3, & 9.9			Remote
	1/20/2021		DROP			Remote
	1/22/2021	Systems Thinking	Read: Parnell Ch 2 (p 27-40; 54-59) Read: Senge - Systems Thinking and Org. Learning To Do: Exercise 2.1		Issue HW #1 Guidance	Remote
	1/26/2021	DSRP Methodology	Read: Cabrera Ch. 3 (p. 52-69)	Watch: How Thinking Works (https://www.youtube.com/watch?v=dUqRTWCd504)		Remote
	1/27/2021		DROP			Remote
Problem Definition	1/29/2021	Systems Thinking Workshop	Read: Systems Thinking Case Study			Remote
	2/2/2021	Introduction to Problem Definition	Read: Parnell Ch 10.1 (p 297-300) To Do: Exercise 10.1	Curtis - Doing the Right Problem	Issue Systems Thinking Presentation Guidance	Study Day/Majors Open Hour 3 February
	2/4/2021	Stakeholder Analysis	Read: Parnell Ch 1.3 (p 3-7) & Ch 10.2 (p 300-314) To Do: Exercise 10.2 Turn-In: Homework 1 Due NLT 1700 4FEB21	Reynolds - Stakeholder Theory and Managerial Decision-Making	HW #1 Due (50pts)	
	2/8/2021	Functional & Requirements Analysis	Read: Parnell Ch 10.3 (p 314-325) To Do: Exercise 10.3	Carson - Implementing Structured Requirements to Improve Requirements Quality	Issue HW #2 Guidance	
	2/10/2021	Value Modeling (Qualitative)	Read: Parnell Ch 10.4 (p 326-331; 333-337) Read: Saaty - Decision Making with AHP To Do: Exercise 10.6			
	2/12/2021	Value Modeling (Quantitative)	Read: Parnell Ch 10.4.3 (p 331-341) Read: SE301 HW 2 Guidance			500th Night (12-14 Feb) Pres Day 15 Feb (No Class)
	2/17/2021	Cabrera Systems Thinking Lecture (DROP)	Read: https://blog.cabreraesearch.org/systems-thinking-in-a-7-images Watch: https://www.youtube.com/watch?v=-sfReUj3o0 Lecture: 1955-2105 17 February 2021, Teams			
Solution Design	2/19/2021		DROP			
	2/23/2021	WPR 1	Review: Lessons 1-10		WPR 1 (150pts)	
	2/25/2021	Introduction to Solution Design	Read: Parnell Ch 11.1-11.2 (p 353-365) To Do: Exercise 11.1 Turn-In: Homework 2 Due NLT 1700 25FEB21	Frey - Engineering Design Thinking	HW #2 Due (75pts)	Study Day 24 February
	3/1/2021	Alternative Generation I	Read: Parnell Ch 11.3-11.6 (p.365-388) To Do: Exercises 11.2 & 11.3	Kelly - Creative Confidence Excerpt	Issue HW #3 Guidance	
	3/3/2021	Alternative Generation II	Read: SE301 HW 3 Guidance			
	3/5/2021		DROP			No Classes 8 March
	3/11/2021	Design Methods	Read: Parnell Ch 3.3 (p 74-77) & Ch 7.5 (p 212-221) Read: Axiomatic Design Technology		Issue Project Presentation and Report Guidance	Study Day 10 March
Decision Making	3/15/2021	Cost Analysis	Read: Parnell Ch 5.1-5.4 (p 137-172) To Do: Exercise 5.4 & 5.7			
	3/17/2021	Introduction to Decision Making	Read: Parnell Ch 12.1-12.4 (p 395-409) To Do: Exercise 12.1 Turn-In: Homework 3 Due NLT 1700 17MAR21	Keeney - Creativity in Decision Making	HW 3 (75pts)	
	3/19/2021	Performance Modeling I	Read: Parnell Ch 4.1-4.5 (p 95-121) To Do: Exercises 4.5, 4.6, 4.7, & 4.8	Hakes - An Economic Evaluation of the Moneyball Hypothesis	Issue HW #4 Guidance	
	3/23/2021	Performance Modeling II	Read: SE301 HW 4 Guidance			Founder's Day
	3/25/2021	Sensitivity and Risk Analysis	Read: Parnell Ch 3.4-3.6 and 12.5-12.6 (p 409-417) To Do: Exercise 12.7	McManus - A Framework for Understanding Uncertainty		Study Day 24 March
	3/29/2021	Tradespace Analysis	Read: Parnell Ch 12.7-12.10 (p 417-432) To Do: Exercises 12.3, 12.4, 12.9 & 12.10	Ross - Multi-Attribute Trade space Exploration		Honorable Living Day 30 March
	4/1/2021		DROP			
Solution Implementation	4/6/2021	WPR 2	Review: Lessons 12-21		WPR 2 (125pts)	No Classes 5 April
	4/8/2021	Introduction to Solution Implementation	Read: Parnell Ch 13.1-13.3 (p 447-453) To Do: 13.1 & 13.6 Turn In: Project Presentation Due NLT 1700 8APR21	Sauser - Why Projects Fail	Project Presentation Due (100pts)	Study Day 7 April Mission Commnd Conference 8 April (Modified Day)
	4/12/2021	Project Planning	Read: Parnell Ch 13.4 (p 453-456) To Do: 13.8 & 13.9	Lewis - One World Trade		
	4/14/2021	Execution, Monitoring, and Control	Read: Parnell Ch 13.5-13.8 (p 457-474) To Do: 13.13 & 13.14 Turn-In: Homework 4 Due NLT 1700 14APR21	Sharon Project Management vs Systems Engineering Management	HW 4 (75pts)	
Course Summary	4/16/2021	Introduction to Systems Thinking and Simulation	Read: Parnell Ch 4.6 (p 121-128) & Ch 4.1-4.5 (p 95-121) Read: System Dynamics (Bala et al) p. 3-22			Modified Day 16 April Sandhurst 17 April
	4/20/2021	Systems Thinking and Simulation Lab	Read: 7 Case Studies in Carbon and Climate			Study Day 21 April
	4/23/2021	Project Presentation Feedback Session I	To Do: Continue writing Project Report To Do: Come to Feedback Session with Questions			
	4/28/2021	Project Presentation Feedback Session II	To Do: Continue writing Project Report To Do: Come to Feedback Session with Questions			Study Day 27 April Projects Day 29 April
	5/3/2021	Course Summary	Read: Parnell Ch 14 (p 477-483) Turn In: Systems Thinking Presentation Due NLT 1700 3MAY21		Systems Thinking Presentation Due (50pts)	
	5/5/2021		Turn In: Project Report Due NLT 1700 5MAY21		Project Report (150pts)	Study Day 6 May
	5/7/2021					
	5/11/2021					

Enclosure 4: SE301 Course Objective Assessment Table

SE301 Course Objectives		Cadet Assessment	Direct Assessment	Course Director's Assessment	Applicable Graded Events
1.	Apply systems thinking and the Systems Decision Process (SDP) to design an engineered solution to a large-scale, complex, multi-disciplinary engineering problem.		3.3	4.0	1) Homework 1 2) WPR 1 & 2 3) Project Pres. & Report 4) TEE
2.	Understand the four phases of the Systems Decision Process and perform the individual tasks in each phase to address an engineering problem.		3.0	4.0	1) Homework 2,3,4 2) WPR 1 & 2 3) Project Pres. & Report 4) TEE
3.	Perform stakeholder and requirement analysis to ensure alternatives achieve the desired outcome and account for the political, social, and economic dimensions of an engineering problem.		3.0	3.5	1) Homework 2 2) WPR 1 3) Project Pres. & Report 4) TEE
4.	Demonstrate creativity in the formulation of alternative solutions to an engineering problem.		3.0	3.5	1) Homework 3 2) WPR 1 & 2 3) Project Pres. & Report 4) TEE
5.	Apply Life Cycle Cost Analysis methods in comparing alternatives		3.7	2.5	1) Homework 4 2) WPR 1 3) Project Pres. & Report 4) TEE
6.	Apply mathematics, basic science, and engineering to analyze a physical system or process and apply the results to the solution of an engineering problem		3.3	3.0	1) Homework 3,4 2) WPR 2 3) Project Pres. & Report 4) TEE
7.	Understand the role of modeling and simulation in the development and analysis of an engineered solution to meet stakeholder requirements		2.3	2.5	1) Homework 4 2) WPR 2 3) TEE
8.	Communicate technical findings effectively, both orally and in writing		3.3	3.0	1) Homework 1-4 2) Project Presentation 3) Project Report
Overall Assessment			3.12	3.3125	
NOTES: 1) Graded Events included: 2xWPR, 4xHomeworks, TEE, Project 2) Cadet Assessment is based on the End-of-Course Survey which is currently unavailable. 3) Course Director Assessment is: -- *Based on course average for "All graded events" -- All others based on individual questions on graded events Range: 1 = strongly disagree to 5 = strongly agree					

Enclosure 5: Course End Feedback
Tabular report aggregated for Course – SE301

Student surveys are unavailable at the time of course closeout. These will be added as soon as they become available.

Enclosure 6: Cadet Comments

Student comments are unavailable at the time of course closeout. These will be added as soon as they become available.



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996



MADN-SE

11 December 2020

MEMORANDUM FOR COL Rich Morales, Professor and Department Head, Department of Systems Engineering, United States Military Academy, West Point, New York 10996.

THRU LTC James Schreiner, Systems & Decision Sciences Program Director, Department of Systems Engineering, United States Military Academy, West Point, New York 10996.

SUBJECT: Academic Term (AT) 21-1 Course Assessment – SE301 Fundamentals of Engineering Design and Systems Management.

1. The attached report summarizes the SE301 assessment and curriculum change recommendations from AT 21-1.
2. EXSUM: This semester is the fifth consecutive semester that SE301 integrated CES cadets and DSE majors into a single course, which continues to be a success. Additionally, the course focused on the entire SDP, with five lessons on Systems Thinking and five lessons on each of the phases of the SDP. Overall performance is lower compared to previous semesters. Based on the AY 20-3 SE301 course assessment and requirements to deliver course content in a hybrid learning environment, several course improvement recommendations were implemented from the previous semester. These improvements focused on leveraging technology and pedagogical efficiencies which enabled flexible and redundant course material delivery and assessment.
3. The point of contact for this action is the undersigned.

Encls

1. Course Assessment
2. Course Instructional Memorandum
3. Course Syllabus
4. Course Objective Assessment Table
5. Course End Feedback
6. Cadet Course End Feedback Comments

CLAY D. WOODY
AV/FA49
Course Director

Enclosure 1: Course Assessment

Section I – Course Information and History

1. Historic Course Offerings and Enrollments.

a. Offered.

	AY17		AY18		AY19		AY20			AY 21-1	
	17-1	17-2	18-1	18-2	19-1	19-2	20-1	20-2	20-3	21-1	21-2
SE300	X	X	X	X	X						
SE301	X	X	X	X	X	X	X	X	X	X	X

b. Enrollments.

Academic Term	Enrollment	
	SE300	SE301
20-1	121	
20-3	107	
20-2	91	
20-1	227	
19-2	111	
19-1	106	132
18-2	55	95
18-1	89	91
17-5		14
17-2	97	37
17-1	100	107

c. Historical Course Final Grade Averages.

Academic Term	Course Average	
	SE300	SE301
21-1	86.26	
20-3	91.5%	
20-2	90.2%	
20-1	86.1%	
19-2	88.4%	
19-1	81.4%	86.5%
18-2	83.0%	88.0%
18-1	83.9%	89.0%
17-5		89.8%
17-2	82.8%	90.5%
17-1	84.5%	87.8%

2. Course Strategy.

SE301 provides the baseline approach to the fundamentals of systems thinking, systems engineering, and systems decision making. The course is designed to allow cadets the opportunity to learn engineering design, systems engineering, and engineering management

processes on an individual and project team level so that each cadet will have the experience necessary to succeed in future Department of Systems Engineering courses. Cadets learn and apply the Systems Decision Process (SDP) throughout the 30-lesson course. The course includes one individual homework assignment that focuses on applying systems thinking to an emerging real-world initiative to integrate virtual reality in the Army's rotary-wing training curriculum. The course includes three group homework assignments where cadet teams implement the SDP to devise a strategy for various projects of their choosing. Lastly, the course includes two WPRs focused on course topics, a final project presentation, a final project report, and one Term End Exam (TEE). Individual work accounts for 57.5% of the course and the remaining 42.5% is group work.

Section II – Course Assessment

1. Achievement of Course Objectives.

- a. **Course Objective Assessment Table:** Enclosure 4
- b. **Results of Course End Feedback.** Enclosure 5
- c. **Results of Cadet Free-Form Comments.** Enclosure 6

2. Course Content.

- a. **Body of Knowledge.** The body of knowledge addressed across the five blocks of this course adequately cover the course objectives. The first block covers systems engineering and systems thinking and introduces the systems decision process. The remaining four blocks cover the four phases of the systems decision process in more detail. Each phase includes between 5-7 individual lessons.
- b. **Effective Assessment.** The department's major-graded-event content approval process provides an adequate assessment tool for the assessment of the course objectives.

3. Course Processes.

- a. **Textbooks.**
 - i. Parnell, Gregory S., Driscoll, Patrick J., Henderson, Dale L., Decision Making in Systems Engineering and Management, 2nd Edition, John Wiley & Sons, Inc., 2011.
 - ii. Cabrera, Derek and Cabrera, Laura, Systems Thinking Made Simple: New Hope for Solving Wicked Problems, 2nd Edition, Odyssean, 2015.
- b. **Software.** Cadets utilize Microsoft Excel, PowerPoint, and Word. SE301 provides an introduction to Microsoft Project as well.
- c. **Graded Requirements.** The graded requirements in SE301 are appropriate for gauging student comprehension, extending fundamental ideas, and demonstrating the real-world relevance of the material. Across the five blocks of instruction, there were a total of nine graded events. The course consisted of 57.5% individual work 42.5% group work with 4 to 5-person teams.

4. Course Pedagogy.

- a. **Lecture Materials:** The course balanced lectures with practical applications for each of the phases of the SDP. Generally, for each phase there were two - three lectures on the material followed by a “lab” type experience to build value models, implementation plans, cost analysis, etc. For some of the phases, the cadets were allowed one class to apply the tools to their homework assignments in class. Lectures were consistent with previous versions of SE301.
- b. **SE Case Studies:** This semester was the fifth time that SE Case Studies were introduced at the end of almost every class to reinforce topics in the lecture with real world applications of systems engineering. Cadets appeared to enjoy the discussions around the case studies as they provided relevance to the materials. Example case studies include: The Apollo Program, Ford Class Aircraft Carrier, and the Mars Climate Orbiter. These cases described both successes and failures in systems engineering. Consistent with each of the last four semesters, these case studies were often not discussed due to instructors running out of time. These case studies are too robust to add to the daily lesson prep. They should be truncated to focus on specific lesson objectives or removed from the course completely.
- c. **Homework Assignments:** Homework assignments followed a similar structure to past semesters, where the first homework assignment focuses on systems thinking and the subsequent homework assignments address each phase of the SDP. HW 1 focused on the applications of systems thinking, specifically, the DSRP methodology, to frame a current US Army initiative to implement commercial off-the-shelf virtual reality technology into the rotary wing training curriculum. Homework 2-4 focused on the application of each phase of the SDP to a group project. Each cadet presented a group project topic to their respective instructor immediately following the submission of Homework 1. Instructors selected 4-5 topics per section and assigned a team of 4-5 cadets to each of the selected topics. These topics remained the focus of each of the remaining homework assignments
- d. **Integration of SE301/SE300:** This semester was the fifth semester in which SE301 cadets from D/SE majors and SE300 cadets from the CES were integrated into the same course and sections. Course evaluations are not available at this time, but cadets were asked to evaluate how they felt this combination of D/SE and CES contributed to their learning. Based on informal feedback received from instructors and cadets, it seems that integrating the course continues to be beneficial to everyone. Performance of CES versus D/SE cadets was tracked for every graded event except for the course end project, which included teams that were composed of CES and D/SE cadets.

Section III – Assessment of Recent Changes and Proposed Future Changes.

1. Assessment of Changes to Current Academic Term (based on previous Course Assessments).

- a. **Course Objectives.** No changes.
- b. **Textbook.** No changes.
- c. **Curriculum.** No changes.
- d. **Graded Requirements.**
 - i. This semester, cadets were required to turn all graded events, to include WPRs and the TEE in via Blackboard. I recommend you continue to require cadets to turn their assignments in

on Blackboard for the SafeAssign capability. Assignments were also graded in Blackboard. Each assignment was accompanied by a standardized rubric which provided a basis for evaluation while still allowing appropriate grading flexibility given the unique nature of each project topic.

- ii. This semester, we continued to allow students to develop novel project topics due to the honor violations that occurred in the course one year prior. Moving forward and to ensure novelty in project ideas, I recommend allowing cadets to pitch project ideas. Based on cadet feedback during lesson 30 and through PDRs, the cadets really enjoyed picking their own project. As long as the instructor performs quality control on project selection and provides guidance on data collection efforts, this will work well in the future. You will also ensure cadet “buy-in.”
- iii. The SE301 instructor team and cadets did an excellent adjusting to the constrained learning environment which has resulted from the COVID-19 pandemic. The hybrid uses of synchronous lectures and asynchronous recordings proved effective and demonstrates a capability that may be capitalized on in the future. However, cadets lacked the traditional in-class collaborative opportunities due to constraints of learning in a hybrid environment. SE301 should return to a live modality at the earliest opportunity.
- iv. Due to the academic schedule adjustments, SE301 was unable to implement the Infantry Warrior Simulation (IWARS) lab.
- v. The Written Partial Reviews and Term End Exam were administered and graded through Blackboard during the semester. Paper copies of each exam were available to mitigate any risks associated with sustained/intermittent network connectivity. 118 out of 121 cadets completed each of the WPRs via Blackboard, and all 121 were able to complete the TEE via Blackboard. I recommend administering all exams through Blackboard during future iterations of the course. This modality provided maximum flexibility given current environmental constraints and significantly reduced the time and effort required for grading.

2. Proposed Change for Next Academic Term.

a. **Course Objectives.** None.

b. **Textbook.** Continue to make sure that the registrar’s office knows that the Cabrera textbook should be the second edition. Also, cadets should not have to purchase the entire Cabrera text. Only three chapters of the text were required reading for the course this semester. I recommend trying to purchase only those chapters via an e-book at a reduced rate. Lastly, ensure the Green Book is the correct edition.

c. **Curriculum.**

- i. I recommend allowing cadets to pitch project ideas. Based on cadet feedback during the course and through the PDRs, the cadets really enjoyed picking their own project. If the instructor performs quality control on project selection and assists with data collection efforts, this will work well in the future. You will also ensure cadet “buy-in.” This was an excellent mentorship opportunity from a research standpoint.
- ii. Consider replacing the DSRP methodology with MBSE and/or MIL STD 1521 to be more in-line with DoD engineering practices. At a minimum, lessons on MBSE should be included in this course.
- iii. If SE301 continues with DSRP, cover the SDP up front while including DSRP within the problem definition phase. Additionally, combine the readings covering Systems Thinking in the Green Book as the DSRP methodology adequately covers Systems Thinking.
- iv. Consider removing the supplemental readings for each lesson. Rather, focus more on the practical exercises and/or case studies within each lesson. The readings were assigned to

- support the Parnell text and introduce cadets to scholarly SE/EM sources beyond the SDP; however, were often difficult to comprehend (from the cadet perspective) and were not covered by graded events.
- v. Recommend scheduling the simulation block of instruction immediately following the modeling block.

d. Graded Requirements.

- i. This semester, 57.5% of the course points were based on individual work. I recommend changing this to 65%. While maintaining the same project design, the second homework assignment can be individual where the team forms after the problem definition phase is complete. This will also add to the substance of their work during the later phases of the SDP.
- ii. Consider adding additional individual technical writing assignment which is integrated with each Homework Assignment.



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996



MADN-SE

6 August 2020

INSTRUCTIONAL MEMORANDUM
NO. 301-21-1

FUNDAMENTALS OF ENGINEERING DESIGN & SYSTEMS MANAGEMENT

1. **PURPOSE.** To provide each cadet with the administrative details necessary to successfully complete SE301, Fundamentals of Engineering Design and Systems Management.
2. **REFERENCES.**
 - a. Department of Systems Engineering Administrative Memo 1-1, "General Administrative Instructions," 8 July 2020.
 - b. Department of Systems Engineering Administrative Memo 1-2, "Standards for Written Requirements," 8 July 2020.
 - c. Department of Systems Engineering Administrative Memo 1-3, "Academic Documentation Policy," 8 July 2020.
 - d. USMA publication. "Documentation of Academic Work," Office of the Dean, July 2018.
 - e. "Guidelines for Technical Communication," Departments of Systems Engineering and Civil and Mechanical Engineering, December 2019.
 - f. Course Website: <https://usma.blackboard.com>
3. **GENERAL.** This memorandum provides each cadet with the course description, objectives, texts, lesson assignments, grading plan, and criteria for passing the course. Your instructor will inform you in advance of any deviation from the guidance provided in this memorandum.
4. **SPECIFIC.**
 - a. Course Scope. SE301 serves as the foundational course for all cadets enrolled in the Engineering Management, Systems Engineering, or Systems and Decision Sciences majors. This course presents the methodological framework and techniques for

designing, implementing, managing, and re-engineering complex systems or processes. Cadets learn engineering design, engineering management processes, and gain an appreciation for future environments and system life cycles. Cadets analyze real-world examples and complete practice problems to illustrate mastery of course topics. Cadets also use spreadsheet software and application design software for modeling and analyzing design alternatives. SE301 introduces the Systems Decision Process (SDP), incorporates material from courses in the USMA core curriculum, and previews the modeling and decision-making tools that cadets will learn in follow-on courses. The course is designed to allow cadets the opportunity to learn engineering design and engineering management processes on an individual level so that each cadet will have the experience necessary to succeed in future Systems Engineering courses.

- b. **Course Strategy.** SE301 exposes cadets to the fundamentals of systems thinking, systems engineering, and systems decision making. The course enables cadets the opportunity to learn engineering design, systems engineering, and engineering management processes on an individual and project team level so that each cadet will have the experience necessary to succeed in future Department of Systems Engineering courses. Cadets learn and apply the Systems Decision Process (SDP) throughout the 30-lesson course. The course includes four homework assignments focused on the application of systems thinking and the SDP to a complex engineering problem chosen by each cadet team. Two WPRs will evaluate each individual cadet's mastery of the course and lesson objectives along with a final project presentation, final project report, and one Term End Exam (TEE).
- c. **Hybrid Learning Environment.** SE301 course content will be delivered in a hybrid learning environment during each lesson. In order to facilitate this hybrid approach within the prescribed limitations of classroom capacity, each section will be organized into 4 groups of 3-4 cadets each. Each group within a section will be given a unique letter designation (A-D) and scheduled for either a resident (In-class) or virtual (MS Teams) modality during each of the 30 lessons. While attending class virtually, cadets will be required to enable their camera for portions of each lesson. The modality schedule will rotate each lesson in order to ensure each student has an equitable learning experience throughout the semester. Groups will be assigned by each instructor, maintain section integrity and remain consistent throughout the semester. The modality schedule will be published on the course website prior to lesson 1.
- d. **Course Objectives.**
 - 1) Apply systems thinking and the Systems Decision Process (SDP) to design an engineered solution to a large-scale, complex, multi-disciplinary engineering problem.
 - 2) Understand the four phases of the Systems Decision Process and perform the individual tasks in each phase to address an engineering problem.
 - 3) Perform stakeholder and requirement analysis to ensure alternatives achieve the desired outcome and account for the political, social, and economic dimensions of an engineering problem.

- 4) Demonstrate creativity in the formulation of alternative solutions to an engineering problem.
- 5) Apply Life Cycle Cost Analysis methods in comparing alternatives.
- 6) Apply mathematics, basic science, and engineering to analyze a physical system or process and apply the results to the solution of an engineering problem.
- 7) Understand the role of modeling and simulation in the development and analysis of an engineered solution to meet stakeholder requirements.
- 8) Communicate technical findings effectively, both orally and in writing.

e. Course Texts.

- 1) Parnell, Gregory S., Driscoll, Patrick J., Henderson, Dale L., Decision Making In Systems Engineering and Management, 2nd Edition, John Wiley & Sons, Inc., 2011.
- 2) Cabrera, Derek and Cabrera, Laura, Systems Thinking Made Simple: New Hope for Solving Wicked Problems, 1st Edition, Odyssean, 2015.

f. Additional References. Additional reading and course material will be posted on the course Blackboard page.

g. Lesson Assignments. Syllabus is posted on the course Blackboard page.

h. Performance Assessment.

- 1) The course consists of 1000 points. All graded work is individual work unless otherwise noted. The following is a breakdown of the graded work:
 - a) Homework Assignments (Individual: 1 x 50 points; Individual/Group: 3 x 75 points). The homework assignments will reinforce the phases of the SDP to include Systems Thinking (50 points), Problem Definition (75 points), Solution Design (75 points), and Decision Making (75 points). These homework assignments will support the semester long project to apply the SDP to a complex, engineering problem.
 - b) Written Partial Reviews (WPRs). (Individual, 2 x 125 points) The purpose is to reinforce concepts taught throughout the course. The WPRs will consist of interpretation, application, and explanation of concepts given scenario questions.
 - c) Project Report (Group, 1x 100 points) and Presentation (Group, 1 x 100 points). The purpose of the group project is to assess each cadet's ability to function as a contributing member of a team and apply the Systems Decision Process (SDP) to a complex, engineering problem and present a recommend solution and

implementation plan to the client / decision-maker. The project will build upon the homework assignments that frame and define the problem, develop candidate solutions, and make a recommended decision on the candidate solutions. The group project will expand on these recommended solutions to develop an implementation plan for the decision maker. The report and presentation will evaluate the cadet's ability to communicate effectively in an oral and written manner.

- d) Term End Examination (TEE). (Individual, 200 points) The purpose of this exercise is to reinforce concepts taught during the entire course.
- e) Instructor Points (IP). (Individual, 75 points). The purpose of Instructor Points is to ensure cadets are prepared for and participate in class each day, complete assigned problems, and/or evaluate performance on in-class quizzes.

2) Student performance assessment summary:

Graded Event	Format	Lesson	Points	% of Course
Systems Thinking Homework	Individual	6	50	5%
WPR 1	Individual	10	125	12.5%
Problem Definition Homework	Group	13	75	7.5%
Instructor Points	Individual	15	25	2.5%
Solution Design Homework	Group	17	75	7.5%
WPR 2	Individual	22	125	12.5%
Decision Making Homework	Group	24	75	7.5%
Project Presentation	Group	28-29	100	10%
Project Report	Group	30	100	10%
Term End Examination	Individual	TEE	200	20%
Instructor Points	Individual	1-30	50	5%
Course Total			1000	

5. ADMINISTRATIVE GUIDANCE.

- a. Minimum Requirements. Score at least 50% on the TEE and complete the course with at least 65%.
- b. Graded Events. Unless there are exceptional circumstances, cadets will take all WPRs and the TEE and complete all graded events during their primary scheduled times IAW the DPOM. Cadets who will miss an event must contact the instructor as soon as they know about the conflict. Cadets have the responsibility to coordinate with the instructor to make up any material/requirements missed.
- c. Late Submissions. Failure to submit a graded event on time for other than legitimate, compelling reasons will result in a reduced grade (5% per 24-hour period) and may result in disciplinary action. Cadets are required to submit all major graded events (≥10% of course grade) regardless of lateness. Any graded event not submitted prior to

lesson 30 will receive a grade of zero. All graded events must be submitted to pass the course.

- d. Final Grades. Final grades will be awarded IAW reference 2.a, Department of Systems Engineering Admin Memo 1-1.
- e. Documentation. Cadets may use any available reference material in preparing course work, but use of any references **MUST BE PROPERLY DOCUMENTED**. Documentation procedures are outlined in references 2b through 2d.
- f. Additional Instruction (AI). AI is available at mutually agreed upon times between the cadet and their instructor. Prior to scheduling additional instruction, cadets are expected to make a legitimate attempt to complete the daily assignments and practice problems to standard and have those assignments available during the AI session. Cadets are expected to have **specific questions for any AI session**. Cadets will also have their notes and course textbooks readily available during any AI session. Cadets should primarily use their own instructor for AI. However, they are authorized to seek AI from other instructors teaching the course this semester. No AI is given after the first version of a WPR has been given, whether that version is the primary or a special make-ahead exam. The last AI session for the TEE is prior to 1600 on the last day of classes.
- g. Practice Problems. Practice Problems and reading assignments are assigned with the intent of enhancing the understanding of text material and assist the cadets in preparing for graded events. Instructors may choose to allocate some of their Instructor Points towards practice problems.
- h. Class. You are expected to have your texts and laptop available during each class. If you plan or anticipate being absent from class, let your instructor and section marcher know *prior to the class* in person or via email.

FOR THE HEAD OF THE DEPARTMENT:

// ORIGINAL SIGNED //

CLAY D. WOODY
MAJ, AV / FA49
Course Director

// ORIGINAL SIGNED //

JAMES H. SCHREINER
LTC, EN / FA47
Program Director

DISTRIBUTION: 1 each instructor, SE301
 1 each cadet, SE301

Enclosure 3: SE301 Syllabus

Ln	Day 1	Day 2	Title	Assignment (Complete Prior to Class)	SE Case	Graded Event
Systems Thinking	1	17-Aug	18-Aug	Introduction to the System Decision Process (SDP) Read: Pamell Ch 6 (p 185-193) Read: Pamell Ch 9.1-9.10 (p 275-290) To Do: Exercises 9.1, 9.3, & 9.9	Apollo Mission	
	2	19-Aug	20-Aug	Systems Thinking Read: Pamell Ch 2 (p 27-40; 54-59) Read: Senge - Systems Thinking and Org. Learning To Do: Exercise 2.1 Issue: Homework 1 Guidance	Onstar Case	
	3	21-Aug	24-Aug	DSRP Methodology Read: Cabrera Ch 3 (p 52-59)	Afghanistan COIN Model	
	4	25-Aug	28-Aug	Systems Thinking Workshop Read: Systems Thinking Case Study	Electrical Grid	
Problem Definition	5	31-Aug	1-Sep	Introduction to Problem Definition Read: Pamell Ch 10.1 (p 297-300) Read: Curtis - Doing the Right Problem To Do: Exercise 10.1	Soyuz 11 Accident	
	6	2-Sep	3-Sep	Cabrera Lecture (DROP) Read: Cabrera Ch 1-2 (p 12-50) Lecture: 1945 / 2045 2 September 2020, Arnold Aud Turn-In: Homework 1 Due NLT 1600 3SEP20	N/A	HW 1 (50pts)
	7	8-Sep	9-Sep	Stakeholder Analysis Read: Pamell Ch 1.3 (p 3-7) & Ch 10.2 (p 300-314) Read: Reynolds - Stakeholder Theory and Managerial Decision-Making To Do: Exercise 10.2 Issue: Homework 2 Guidance	CH-47F Chinook	
	8	10-Sep	11-Sep	Functional & Requirements Analysis Read: Pamell Ch 10.3 (p 314-325) Read: Carson - Implementing Structured Requirements to Improve Requirements Quality To Do: Exercise 10.3	Comanche Helicopter	
	9	14-Sep	16-Sep	Value Modeling (Qualitative) Read: Pamell Ch 10.4 (p 326-331; 333-337) Read: Saaty - Decision Making with AHP To Do: Exercise 10.6	GCV	
	10	17-Sep	18-Sep	WPR 1 Review: Lessons 1 - 9	N/A	WPR 1 (125pts) Dean's Hour 17SEP
	11	21-Sep	22-Sep	Value Modeling (Quantitative) Read: Pamell Ch 10.4.3 (p 331-341) Read: SE301 HW 2 Guidance	N/A	
	12	25-Sep	24-Sep	Introduction to Solution Design Read: Pamell Ch 11.1-11.2 (p 353-365) Read: Frey - Engineering Design Thinking To Do: Exercise 11.1	Mars Climate Orbiter	
	13	29-Sep	28-Sep	Alternative Generation I Read: Pamell Ch 11.3-11.6 (p.365-388) Read: Kelly - Creative Confidence Excerpt To Do: Exercises 11.2 & 11.3 Turn-In: Homework 2 Due NLT 1600 29SEP20	MRI Design	HW 2 (75pts)
	14	1-Oct	2-Oct	Alternative Generation II Issue: Homework 3 Guidance Read: SE301 HW 3 Guidance	N/A	
	15	5-Oct	8-Oct	Design Methods Read: Pamell Ch 3.3 (p 74-77) & Ch 7.5 (p 212-221) Read: Axiomatic Design Technology	B-52 Bomber	
Solution Design	16	9-Oct	13-Oct	Cost Analysis Read: Pamell Ch 5.1-5.4 (p 137-172) To Do: Exercise 5.4 & 5.7	N/A	
	17	14-Oct	15-Oct	Introduction to Decision Making Read: Pamell Ch 12.1-12.4 (p 395-409) Read: Keeney - Creativity in Decision Making To Do: Exercise 12.1 Turn-In: Homework 3 Due NLT 1600 15OCT20	SAOC	HW 3 (75pts)
	18	16-Oct	19-Oct	Performance Modeling I Read: Pamell Ch 4.1-4.5 (p 95-121) Read: Hakes - An Economic Evaluation of the Moneyball Hypothesis To Do: Exercises 4.5, 4.6, 4.7, & 4.8	AC-130J Accident	
	19	20-Oct	23-Oct	Performance Modeling II Read: SE301 HW 4 Guidance	N/A	
	20	26-Oct	27-Oct	Sensitivity and Risk Analysis Read: Pamell Ch 12.5-12.6 (p 409-417) Read: McManus - A Framework for Understanding Uncertainty To Do: Exercise 12.7	Hubble Telescope	
	21	28-Oct	29-Oct	Trade space Analysis Read: Pamell Ch 12.7-12.10 (p 417-432) Read: Ross - Multi-Attribute Trade space Exploration To Do: Exercises 12.3, 12.4, 12.9 & 12.10	Pamell UAV Study	
	22	2-Nov	3-Nov	WPR 2 Review: Lessons 12 - 21	N/A	WPR 2 (125pts) Dean's Hour 2NOV
	23	6-Nov	5-Nov	Introduction to Solution Implementation Read: Pamell Ch 13.1-13.3 (p 447-453) Read: Sauser - Why Projects Fail To Do: 13.1 & 13.6 Issue: Project Report / Presentation Guidance	Ford Rogue Manufacturing Plant	
	24	9-Nov	10-Nov	Project Planning Read: Pamell Ch 13.4 (p 453-456) Read: Lewis - One World Trade To Do: 13.8 & 13.9 Turn-In: Homework 4 Due NLT 1600 10NOV20	One World Trade Center	HW 4 (75pts)
	25	12-Nov	13-Nov	Execution, Monitoring, and Control Read: Pamell Ch 13.5-13.8 (p 457-474) Read: Sharon Project Management vs Systems Engineering Management To Do: 13.13 & 13.14	Ford Class Air Craft Carrier	
Decision Making	26	16-Nov	17-Nov	Introduction to Simulation Read: Pamell Ch 4.6 (p 121-128) & Ch 4.1-4.5 (p 95-121) Read: Ingalls Intro to Simulation OR Raychaudhuri Intro to Monte Carlo Sim To Do: Exercise 4.11, 4.18, & 4.19	N/A	
	27	18-Nov	20-Nov	Course Drop		
	28	23-Nov	24-Nov	Project Presentation Workshop Course Drop Turn In: Project Presentation Due NLT 2300 24NOV20	N/A	
	29	25-Nov	27-Nov	Project Presentations To Do: Rehearse Presentations	N/A	Project Presentation (100pts)
	30	30-Nov		Day 1 Drop		
	30	1-Dec	3-Dec	Course Summary Read: Pamell Ch 14 (p 477-483) Turn In: Project Report Due NLT 2300 04DEC20	N/A	
Course Project	31	4-Dec		Day 1 Drop		

Enclosure 4: SE301 Course Objective Assessment Table

SE301 Course Objectives		Cadet Assessment	Direct Assessment	Course Director's Assessment	Applicable Graded Events
1.	Apply systems thinking and the Systems Decision Process (SDP) to design an engineered solution to a large-scale, complex, multi-disciplinary engineering problem.	TBD	3.0	4.0	1) Homework 1 2) WPR 1 & 2 3) Project Pres. & Report 4) TEE
2.	Understand the four phases of the Systems Decision Process and perform the individual tasks in each phase to address an engineering problem.	TBD	3.0	4.0	1) Homework 2,3,4 2) WPR 1 & 2 3) Project Pres. & Report 4) TEE
3.	Perform stakeholder and requirement analysis to ensure alternatives achieve the desired outcome and account for the political, social, and economic dimensions of an engineering problem.	TBD	3.0	3.5	1) Homework 2 2) WPR 1 3) Project Pres. & Report 4) TEE
4.	Demonstrate creativity in the formulation of alternative solutions to an engineering problem.	TBD	3.0	4.0	1) Homework 3 2) WPR 1 & 2 3) Project Pres. & Report 4) TEE
5.	Apply Life Cycle Cost Analysis methods in comparing alternatives	TBD	3.0	2.5	1) Homework 4 2) WPR 1 3) Project Pres. & Report 4) TEE
6.	Apply mathematics, basic science, and engineering to analyze a physical system or process and apply the results to the solution of an engineering problem	TBD	3.0	3.0	1) Homework 3,4 2) WPR 2 3) Project Pres. & Report 4) TEE
7.	Understand the role of modeling and simulation in the development and analysis of an engineered solution to meet stakeholder requirements	TBD	3.0	2.5	1) Homework 4 2) WPR 2 3) TEE
8.	Communicate technical findings effectively, both orally and in writing	TBD	3.0	3.0	1) Homework 1-4 2) Project Presentation 3) Project Report
Overall Assessment		TBD	3.04	3.3125	
	NOTES: 1) Graded Events included: 2xWPR, 4xHomeworks, TEE, Project 2) Cadet Assessment is based on the End-of-Course Survey which is currently unavailable. 3) Course Director Assessment is: -- *Based on course average for "All graded events" -- All others based on individual questions on graded events Range: 1 = strongly disagree to 5 = strongly agree				

Enclosure 5: Course End Feedback
Tabular report aggregated for Course – SE301

Student surveys are unavailable at the time of course closeout. These will be added as soon as they become available.

Enclosure 6: Cadet Comments

Student comments are unavailable at the time of course closeout. These will be added as soon as they become available.



UNITED STATES MILITARY ACADEMY
WEST POINT.

SECTION II

EXAMINATIONS



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB G

WPR #1

Instructor Solution

Section _____

SE301 WPR #1 (Version 1)

AT 26-1

Name _____

Departure time _____

**DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996-1779**

SE 301 WPR #1

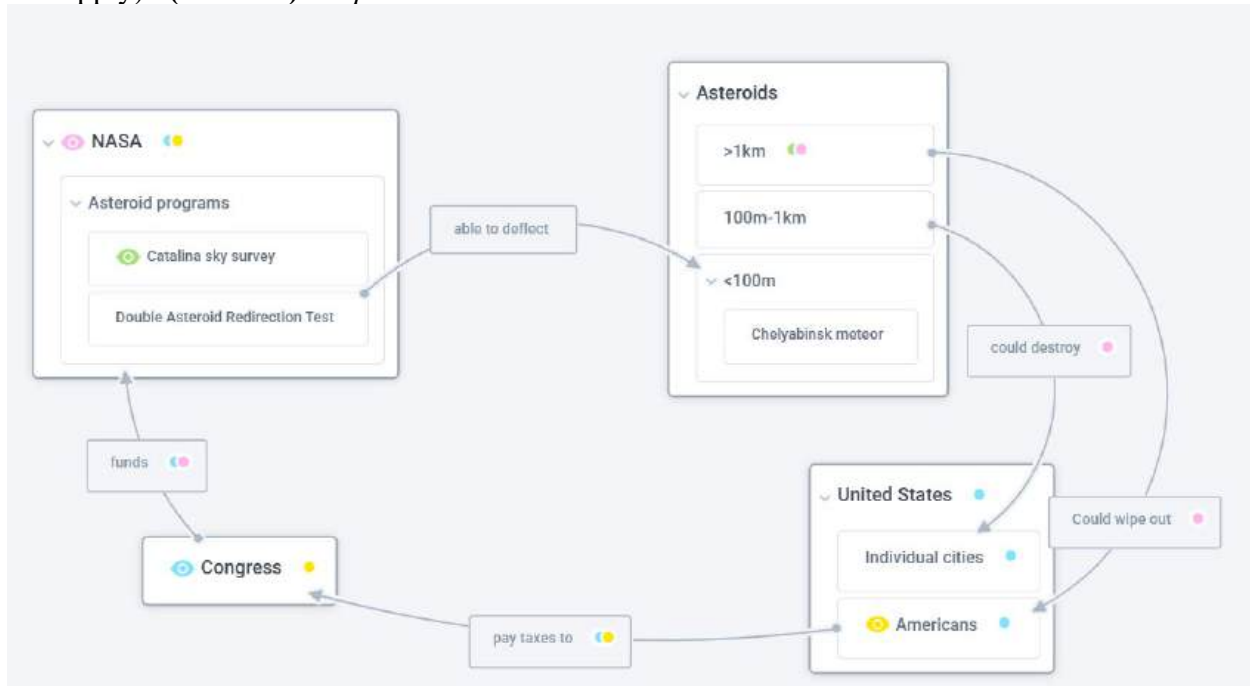
1. This is a **75-minute** exam **worth 125 points**. The exam consists of **12 pages**. Ensure that your exam is complete.
2. Place your name on each page in the space provided.
3. **AUTHORIZED REFERENCES:** You may use a pencil, straight edge (ruler, protractor, etc.), a **3 X 5 index note card (front and back)**, and your hand-held calculator.
4. **SHOW ALL YOUR WORK!** Unsupported or missing work earns little or no partial credit. Answer all questions to the fullest extent possible.
5. **EXTRA PAPER:** Use only the additional paper that is available at the front of the room. Attach the additional pages you use to your exam and turn them in.
6. Once you have completed the exam, fill in the departure time at the top of this page. Turn in your exam IAW your instructor's instructions.
7. This WPR will not be released from academic security until it has been returned. Before its return, cadets may not discuss any aspects of this exam with anyone except their SE301 instructor. **Initial here** _____

SECTION	POSSIBLE POINTS	SCORE
I. Systems Thinking	19	
II. Systems Decision Process & Systems Engineering	22	
III. Stakeholder Analysis	23	
IV. Functional and Requirements Analysis	26	
V. Value Modeling	35	
TOTAL	125	

Name _____

I. SYSTEMS THINKING (19 Points)

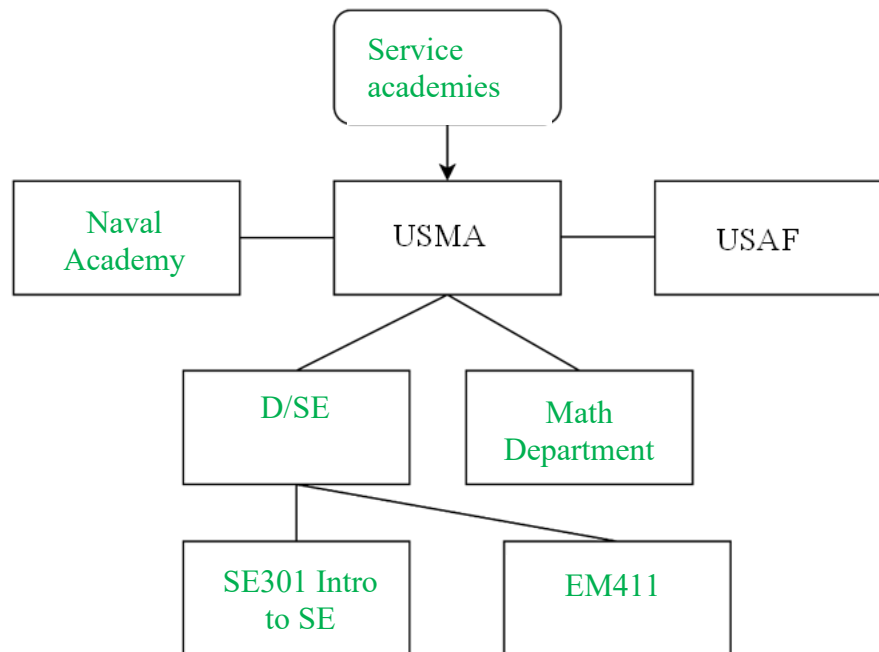
1. Using only the Plectica diagram below, which of the following statements are true (select all that apply). (4 Points) – 1 point each correct answer



- (a) NASA is a point
- (b) NASA is a view
- (c) The Catalina sky survey can see asteroids between 100m and 1km
- (d) NASA knows that asteroids between 100m and 1km could destroy American cities
- (e) The double asteroid redirection test could deflect asteroids between 100m and 1km
- (f) Individual cities pay taxes to Congress
- (g) Americans know that NASA exists
- (h) Americans know that asteroids larger than 1KM could wipe out Americans
- (i) Americans are part of individual cities
- (j) Congress knows about the Catalina Sky Survey

Name _____

2. NAME THE LEVEL OF HIERARCHY for the Spatial Arrangement **IN THE SPACES PROVIDED TO THE LEFT**, where the system of interest is the United States Military Academy at West Point. **FILL IN ALL OF THE BLANK BOXES** with the appropriate portion of the Spatial Arrangement, given the system indicated. **(9 Points)**

A) metasystemB) SystemC) subsystemD) element/component

3. Indicate each of the following descriptions with the appropriate system visibility **(6 Points)**

System	User understanding of system	System Visibility
Air Assault School	Complete transparency on a systems internal elements and processes	White Box
Cadet Branching Process	Offers partial knowledge of selected internal component interactions and processes	Grey Box
College Admissions	No visibility into exactly what internal elements are, how they are linked and behave.	Black Box

Name _____

II. Systems Decision Process and the Systems Engineer (22 Points).4. List the four phases of the Systems Decision Process (SDP) in the correct order: **(4 Points)**

1. __Problem Definition__

2. __Solution Design__

3. __Decision Making__

4. __Solution Implementation__

5. Describe the Systems Decision Process (SDP) and explain how it emphasizes systems thinking. **(7 Points)**

The SDP is a collaborative, iterative and value-based decision process that can be applied at any system life cycle stage. SDP follows a logical progression that embraces systems thinking. It considers the holistic view of the systems environment and emphasizes stakeholder value.

6. List 3 environmental considerations (from the SDP) and their types that Telsa should consider when developing the national rollout of robotaxis.. Do not use the same type of consideration more than once. **(9 points)**

Type of environmental consideration	Environmental consideration details for a Tesla Robotaxi rollout
Technological, Social, Historical, Health & Safety, Political	
Emotional, Moral/Ethical, Security, Ecological, Cultural	
Organizational, Legal, Economic	

Name _____

7. In the problem definition phase of the SDP, what two principal products are used to create the screening criteria? **(2 Points)**

Principal product 1: Requirements

Principal product 2: Value Functions

III. Stakeholder Analysis (23 Points).

8. Identify the type of survey error and correct the survey error. **(8 Points)**

- (a) How do you feel about the following statement? We should not remove study days from the West Point Schedule.

Strongly Agree

Agree

Disagree

Strongly Disagree

Type of survey error: Negative/Double Negative

Corrected question: How do you feel about the following statement? We should keep study days in the West Point schedule.

- (b) Calculus 2 is difficult. Does Calculus 1 help you feel prepared for Calculus 2?

Extremely Prepared

Prepared

Unprepared

Extremely Unprepared

Type of survey error: Leading

1. Corrected question: ~~Calculus 2 is difficult.~~ Does Calculus 1 help you feel prepared for Calculus 2?

Name _____

You are the Lead Systems Engineer on a project to develop a new floating bridge design for the Army Corps of Engineers. This bridge is designed to transport troops, vehicles, and equipment across rivers and other bodies of water.

Identify three stakeholders for this project by type and the stakeholder technique your team would utilize to elicit information from them. Be sure to explain why you would use the technique chosen. Use the table below for your answer. **(15 Points)**

Stakeholder Name	Stakeholder Type	Stakeholder Technique Used	Why?
Chief of Engineers	Decision Authority	Interview	There is one (1) Chief of Engineers
Engineer Commanders	Owner	Focus Group	Medium number of people, good at getting general ideal of stakeholders
Army Soldiers	Customer	Survey	Thousands of soldiers

Name _____

IV. Functional & Requirements Analysis (26 Points).

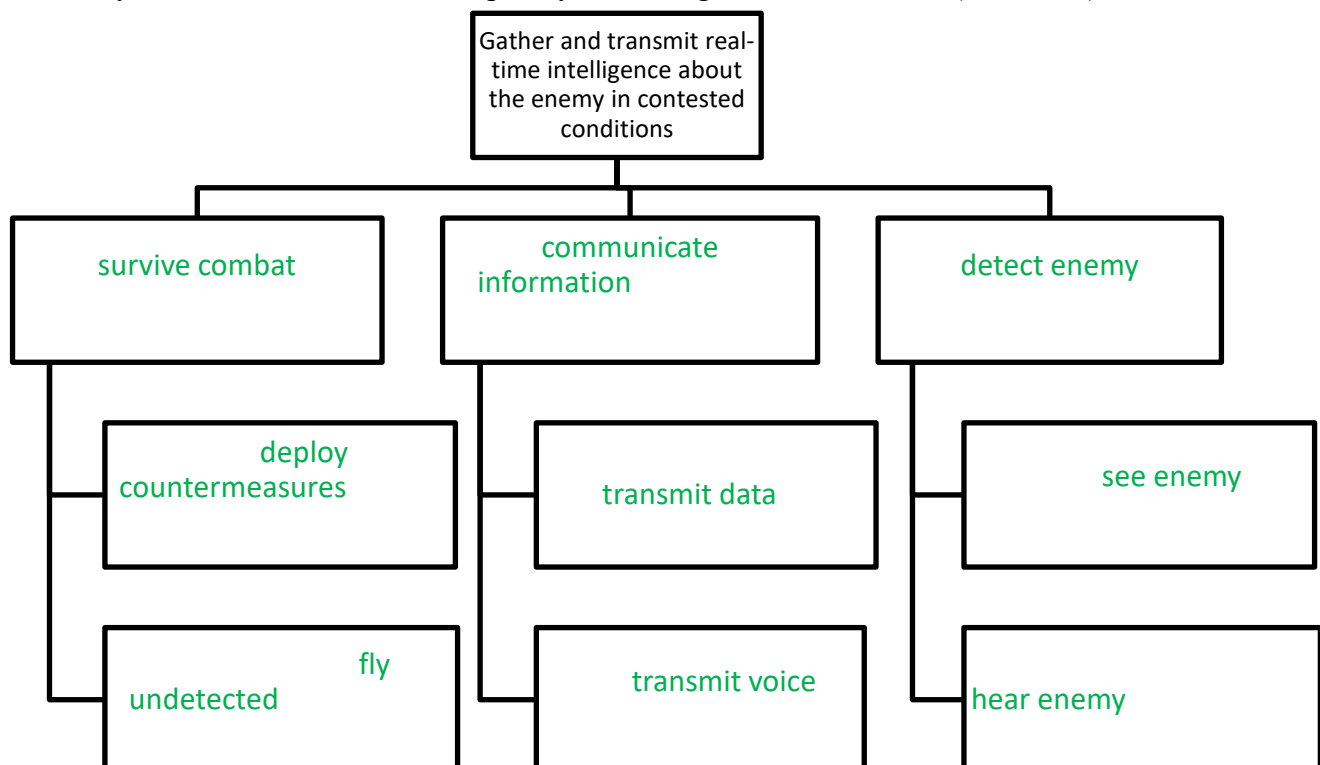
Carefully read the following answer from an Army engineer:

A reconnaissance helicopter is a vital intelligence-gathering asset, designed for stealth, speed, and advanced surveillance. As an army engineer, I focus on integrating sensor suites—electro-optical and infrared cameras, radar, and laser designators—to provide real-time battlefield intelligence. These lightweight, highly maneuverable aircraft allow forces to monitor enemy movements and terrain while minimizing detection.

Secure communication and data transmission are essential to their mission. Encrypted radio and satellite links ensure intelligence reaches commanders instantly, aiding tactical decisions. Engineers also equip these helicopters with networked systems that coordinate with other units, sometimes relaying targeting data to attack aircraft and artillery for precision strikes.

Survivability is a key design priority, as reconnaissance helicopters often operate in hostile environments. Stealth features, electronic warfare countermeasures, and missile defense systems help reduce threats. Additionally, redundant flight controls and lightweight armor enhance crew protection while maintaining agility. By combining advanced technology with strategic design, these helicopters play a crucial role in modern military operations.

9. Based on the engineer's answer and your understanding of the problem, create a functional hierarchy for a reconnaissance helicopter system using the boxes below. **(14 Points)**



Name _____

10. Rewrite the following functions for a reconnaissance helicopter (3 points):

Function 1: Main rotor

Rewritten function: _____ **Generate Lift** _____

Function 2: Uses gun

Rewritten function: _____ **Shoot Enemy** _____

Function 3: Provides transport

Rewritten function: _____ **Move Troops/Equipment** _____

11. Which of the following statements are well-written requirements? Choose all that apply (4 Points)

- (a) The JLTV shall be capable of destroying unmounted infantry at a minimum distance of 4,000 meters under typical weather conditions, with a kill probability not less than 80%.
- (b) The system shall respond in under 60 minutes 95% of the time.
- (c) The cadet shall complete the Fallen Comrades Half Marathon within a time of 3 hours in PTs, beginning and ending at Michie Stadium, in all weather conditions.
- (d) The Army Combat Boot shall weigh less than 1.5lbs and fit comfortably.
- (e) The system shall reach an altitude of at least 20 km.

12. Write two properly constructed requirement statements for an Army Aviation reconnaissance helicopter (8 Points)

1. ___ **The agent shall do what, how well, under what conditions.** _____

2. _____

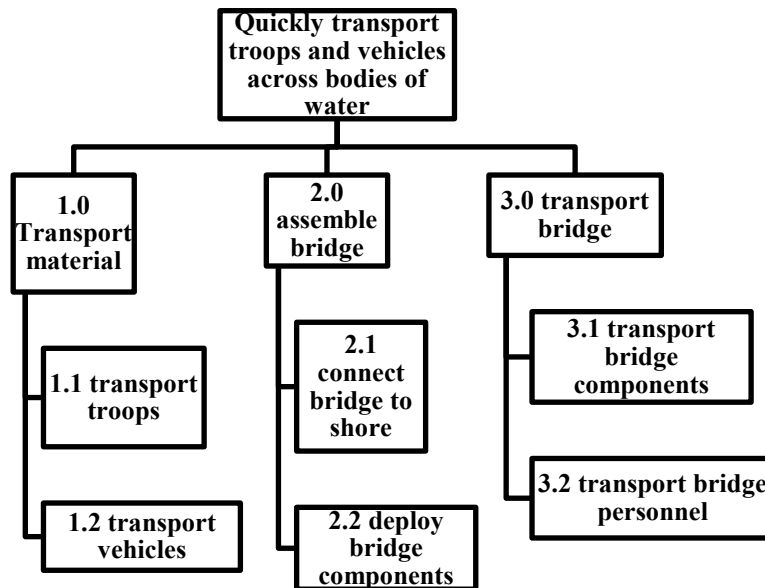
V. Value Modeling (35 Points).

13. What are the four levels of a Qualitative Value Model (Value Hierarchy)? (4 Points)

Level 1: ___ **Fundamental Objective** _____Level 2: ___ **Functions** _____Level 3: ___ **Objectives** _____Level 4: ___ **Value Measures** _____

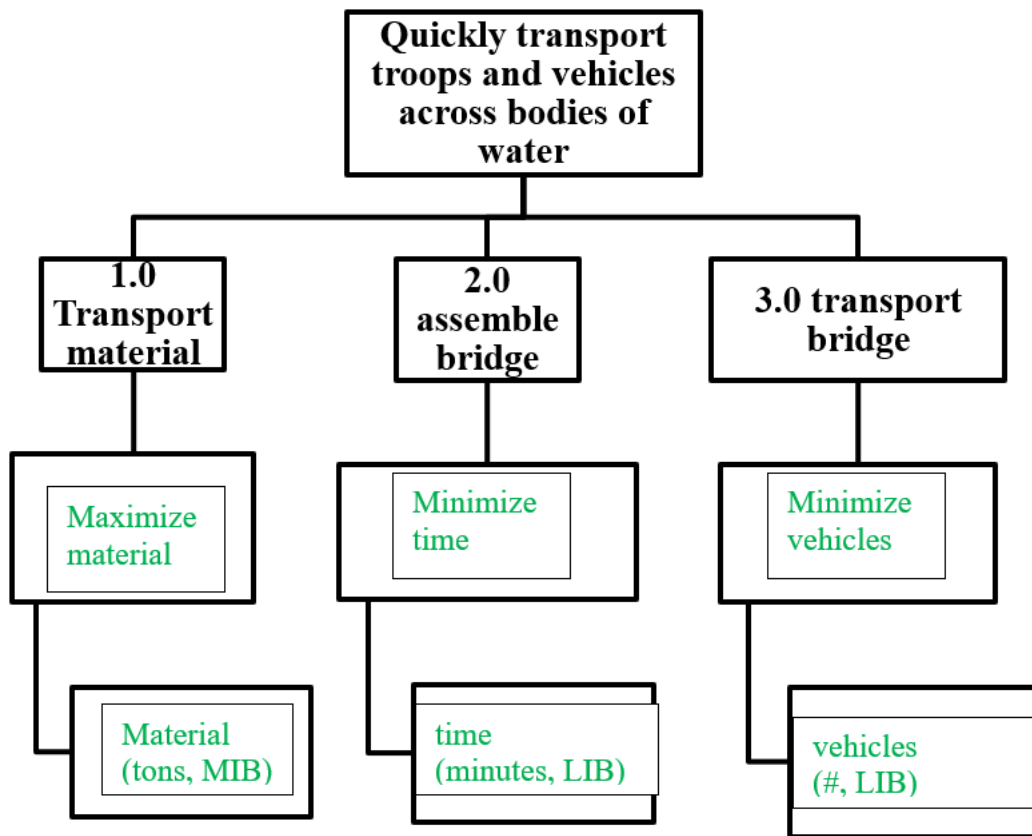
Name _____

14. Given the following Functional Hierarchy, create a Qualitative Value Model (Value Hierarchy) for a Army Corps of Engineers floating bridge: **(10 Points)**



Fill out the below template to create your Qualitative Value Model. You **ONLY NEED TO FILL OUT THE PROVIDED EMPTY BOXES** for your answer. Be sure to indicate the type and scale of each value measure.

Name _____



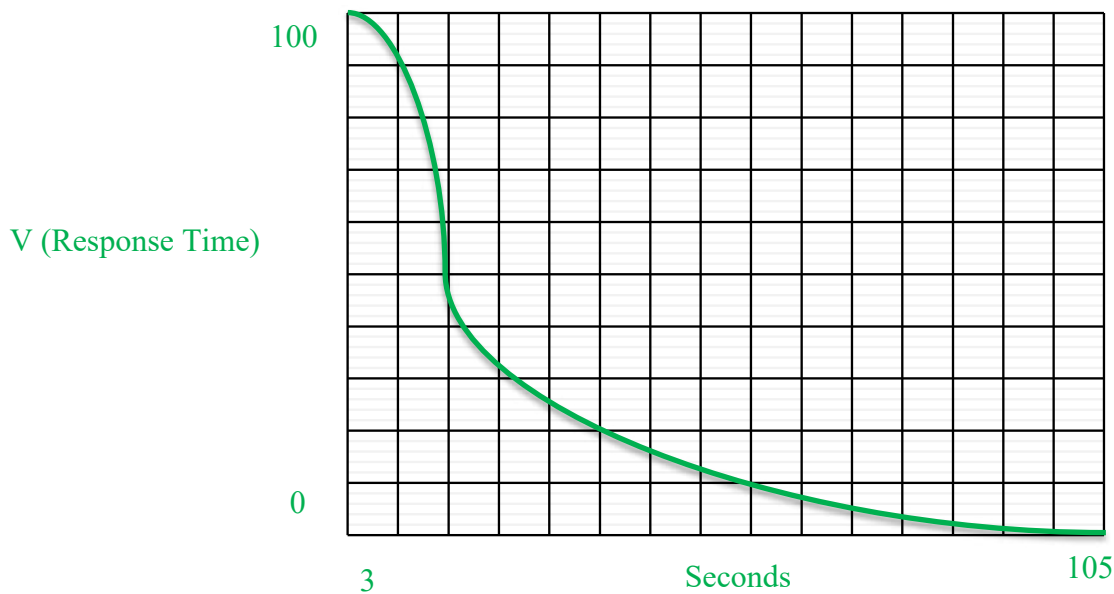
15. Create the value function for the following value measure, based on the following stakeholder analysis findings: **(6 Points)**

Even a perfect floating bridge system couldn't be expected to deploy in less than 3 minutes. Anything up to 10 minutes would be considered to be excellent but the Army would find anything slower than that increasingly frustrating and would not accept a deployment time longer than 105 minutes under any circumstances.

Fill in the blank spaces of the table below, based on the above information and create the value function in the grid below. Remember to format the value function correctly.

Value Measure	Scale	Worst	Ideal	Function Type
Response time	Seconds	105	3	S

Name _____



16. Now use your value function to determine how much of the available value from response time $V(x_{\text{response time},j})$ a stakeholder receives from system A that has a response time of 12 seconds and system B that has a response time of 3 seconds. Show how you derived these values on the value function you drew in the previous question. **(2 points)**

System A: _____ 5 (very low score) _____

System B: _____ 85 (very high score) _____

Name _____

17. With the help of your client, you determined the following bounds for three value measures.

Value measure	Worst	Ideal
Maximum weight capacity MIB	40 tons	100 tons
Number of engineers needed LIB	10	1
Lifetime deployments MIB	100	1000

Your client told you that moving from 40 ton to 100 ton maximum weight capacity would have largest impact on their decision. Moving from a number of engineers needed of 10 to 1 would have about a fourth as large an impact as moving from 40 ton to 100 ton maximum weight capacity. Increasing the number of lifetime deployments from 100 to 1,000 would have about three-quarters as large an impact as moving from 40 ton to 100 ton maximum weight capacity.

Based on the information provided, calculate the global weights in the table below.

Show your work and **be sure to round your answers to 2 decimal places.** (7 Points):

Show your work here:

<i>Value Measure</i>	<i>Swing Weight</i>	<i>Measure (Global) Weight</i>
Maximum weight capacity	100	.50
Number of engineers needed	25	.125
Lifetime deployments	75	.375

18. Unrelated to the previous questions, given the following data, determine the total value for

Name _____

each of the two alternatives (i.e. determine the total value for sneaker 1 and sneaker 2, respectively). You must show all your work. (6 Points)

Show your work here

Value measure	Global weight	$x_{sneaker\ 1,i}$	$x_{sneaker\ 2,i}$	$v_i(x_{sneaker\ 1,i})$	$v_i(x_{sneaker\ 2,i})$
Versatility	0.393	33	21	81	6
Durability	0.053	3/10	8/10	41	94
Performance	0.541	87.4	111.8	88	91
Stability	0.014	10	11	100	100

$$\text{Sneaker 1: } 81 \cdot .393 = 31.833$$

$$41 \cdot .053 = 2.173$$

$$88 \cdot .541 = 47.608$$

$$100 \cdot .014 = 1.4$$

$$\text{Total value of sneaker 1} = 83.014$$

$$\text{Sneaker 2: } 6 \cdot .393 = 2.358$$

$$94 \cdot .053 = 4.982$$

$$91 \cdot .541 = 49.231$$

$$100 \cdot .014 = 1.4$$

$$\text{Total value of sneaker 2} = 57.971$$

Sneaker 1 total value: 82.57

Sneaker 2 total value: 57.91

Outstanding Student Work

Section 12 Instructor LTC Cadell

SE301 WPR #1 (Version 1)
AT 26-1

Name Bendon Ingle

Departure time 1400

DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996-1779

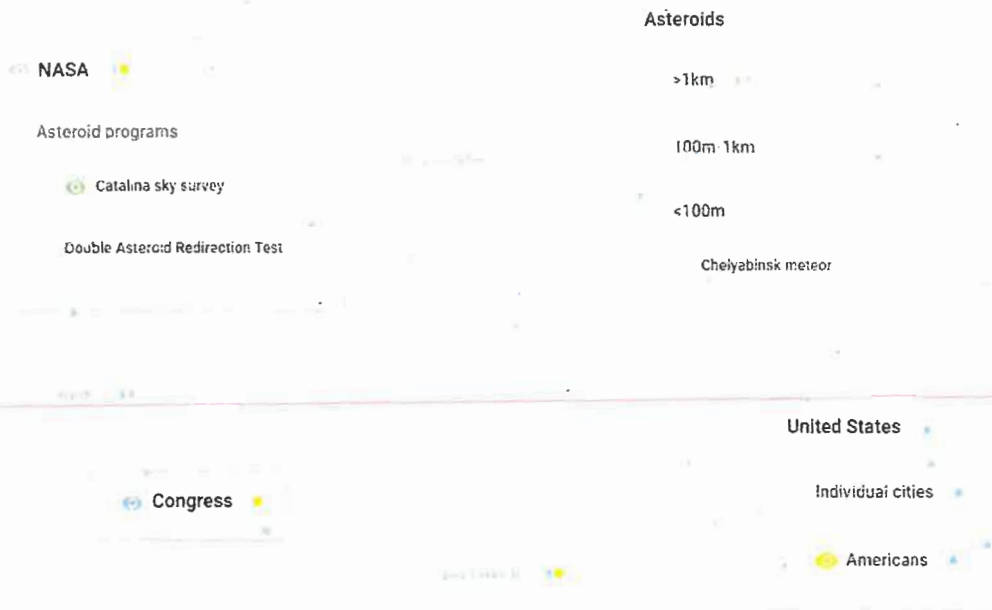
SE 301 WPR #1

1. This is a **75-minute** exam worth **125 points**. The exam consists of **12 pages**. Ensure that your exam is complete.
2. Place your name on each page in the space provided.
3. **AUTHORIZED REFERENCES:** You may use a pencil, straight edge (ruler, protractor, etc.), a **3 X 5 index note card (front and back)** during **last 20 minutes of exam period**, and your hand-held calculator.
4. **SHOW ALL YOUR WORK!** Unsupported or missing work earns little or no partial credit. Answer all questions to the fullest extent possible.
5. **EXTRA PAPER:** Use only the additional paper that is available at the front of the room. Attach the additional pages you use to your exam and turn them in.
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SECTION	POSSIBLE POINTS	SCORE
I. Systems Thinking	19	
II. System Decision Process & Systems Engineering	21	
III. Stakeholder Analysis	24	
IV. Functional and Requirements Analysis	26	
V. Value Modeling	35	
TOTAL	125	

Name Brandon Ingle**I. SYSTEMS THINKING (19 Points)**

1. Using only the Plectica diagram below, which of the following statements are true (select all that apply). (4 Points)



- ☒ (a) NASA is a point
- ☒ (b) NASA is a view
- ☒ (c) The Catalina sky survey can see asteroids between 100m and 1km
- ☒ (d) NASA knows that asteroids between 100m and 1km could destroy American cities
- ☒ (e) The double asteroid redirection test could deflect asteroids between 100m and 1km
- ☒ (f) Individual cities pay taxes to Congress
- ☒ (g) Americans know that NASA exists
- ☒ (h) Americans know that asteroids larger than 1KM could wipe out Americans
- ☒ (i) Americans are part of individual cities
- ☒ (j) Congress knows about the Catalina Sky Survey

Name Brandon Ingle

SE301 WPR #1 (Version 1)

AT 26-1

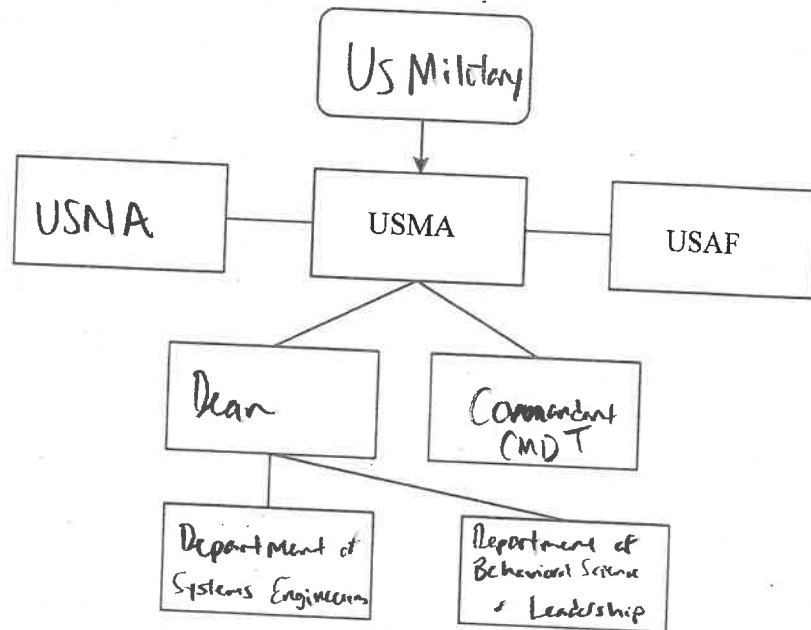
2. NAME THE LEVEL OF HIERARCHY for the Spatial Arrangement IN THE SPACES PROVIDED TO THE LEFT, where the system of interest is the United States Military Academy at West Point. FILL IN ALL OF THE BLANK BOXES with the appropriate portion of the Spatial Arrangement, given the system indicated. (9 Points)

A) Metasystem

B) System

C) Subsystem

D) Element



3. List and describe an example of the below levels of system visibility (6 Points)

System	User understanding of system	System Visibility
Air Assault School	offers full understanding of internal component interactions & processes	White Box
	Offers partial knowledge of selected internal component interactions and processes	Gray Box
Apple Security System	offers no knowledge of internal component interactions & processes	Black Box

Systems Decision Process and the Systems Engineer (24 Points).

Name Brandon Ingle

SE301 WPR #1 (Version 1)

AT 26-1

4. List the four phases of the Systems Decision Process (SDP) in the correct order: (4 Points)

1. Problem Definition
2. Solution Design
3. Decision Making
4. Solution Implementation

5. Describe the Systems Decision Process (SDP) and explain how it emphasizes systems thinking. (7 Points)

The SDP is an iterative process that considers stakeholders and what they value and clearly defines the problem to be solved. The SDP is built in a way that considers environmental factors & limitations so an optimal solution is met in the end. There is always information going into the solution design.

6. List 3 environmental considerations (from the SDP) and their types that Tesla should consider when developing the national rollout of robotaxis. Do not use the same type of consideration more than once. (9 points)

Type of environmental consideration	Environmental consideration details for a Tesla Robotaxi rollout
Economic	How will this impact jobs of taxidrivers?
Society	Will consumers be safe or trust robotaxis?
Legal	How will Tesla handle potential lawsuits for accidents from robotaxis?

7. In the problem definition phase of the SDP, what two principal products are used to create the screening criteria? (4 Points)

Principal product 1: Requirements

Principal product 2: Quantitative Value Model

II. Stakeholder Analysis (23 Points).

8. Identify the type of survey error and correct the survey error. (8 Points)

Name Brandon Ingle

SE301 WPR #1 (Version 1)

AT 26-1

- (a) How do you feel about the following statement? We should not remove study days from the West Point Schedule.

Extremely Prepared Prepared Unprepared Extremely Unprepared

Type of survey error: Double Negative

Corrected question: How do you feel about the following statement?

We should remove study days from the West Point Schedule.
Strongly Agree Agree Disagree Strongly Disagree

- (b) Calculus 2 is difficult. Does Calculus 1 help you feel prepared for Calculus 2?

Strongly Agree Agree Disagree Strongly Disagree

Type of survey error: Leading Question

Corrected question: How prepared does Calculus 1 help you feel for Calculus 2?

Extremely Prepared Prepared Unprepared Extremely Unprepared

Name Brandon Ingle

9. You are the Lead Systems Engineer on a project to develop a new floating bridge design for the Army Corps of Engineers. This bridge is designed to transport troops, vehicles, and equipment across rivers and other bodies of water.

Identify three stakeholders for this project by type and the stakeholder technique your team would utilize to elicit information from them. Be sure to explain why you would use the technique chosen. Use the table below for your answer. (15 Points)

Survey
Focus Groups

Stakeholder Name	Stakeholder Type	Stakeholder Technique Used	Why?
Company and lower soldiers building bridges	User	Survey	Scientific based way to extract feedback
Senior officers of EN Corps	Client	Focus Group	Agenda based, research moderated 6-12 stakeholders
Soldiers of other branches using the bridge	Consumer	Survey	Scientific based way to extract feedback for data on bridge use

Name Brandon Ingle

SE301 WPR #1 (Version 1)

AT 26-1

III. Functional & Requirements Analysis (26 Points).

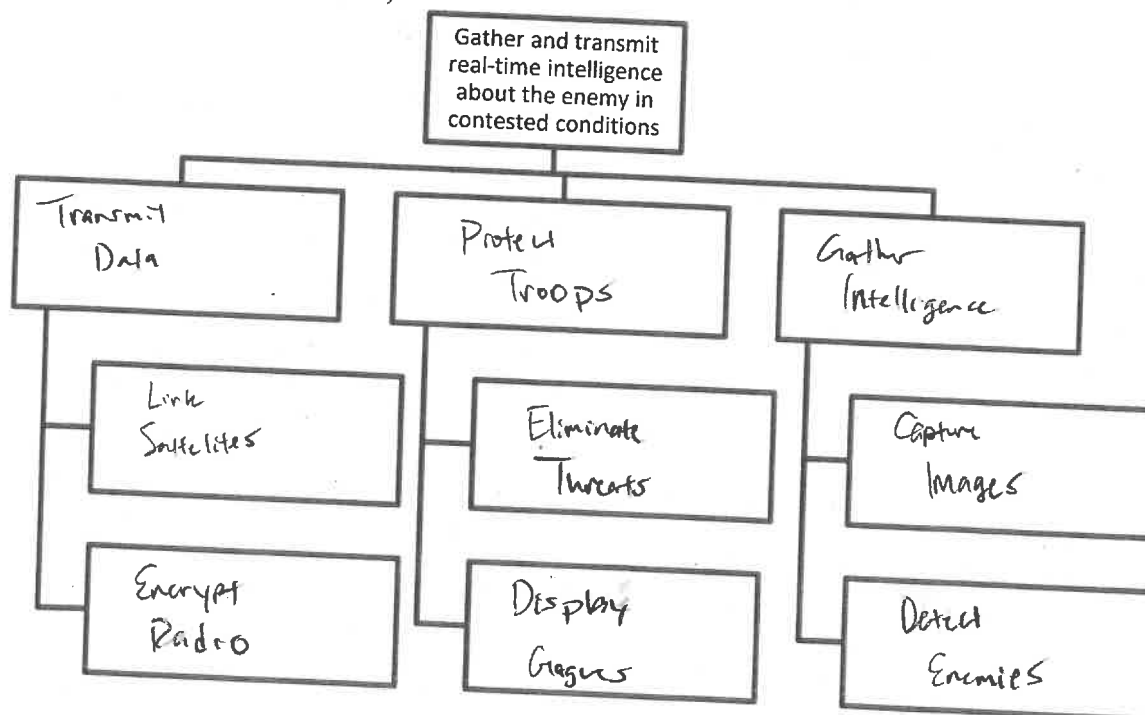
Carefully read the following answer from an Army engineer:

A reconnaissance helicopter is a vital intelligence-gathering asset, designed for stealth, speed, and advanced surveillance. As an army engineer, I focus on integrating sensor suites—electro-optical and infrared cameras, radar, and laser designators—to provide real-time battlefield intelligence. These lightweight, highly maneuverable aircraft allow forces to monitor enemy movements and terrain while minimizing detection.

Secure communication and data transmission are essential to their mission. Encrypted radio and satellite links ensure intelligence reaches commanders instantly, aiding tactical decisions. Engineers also equip these helicopters with networked systems that coordinate with other units, sometimes relaying targeting data to attack aircraft and artillery for precision strikes.

Survivability is a key design priority, as reconnaissance helicopters often operate in hostile environments. Stealth features, electronic warfare countermeasures, and missile defense systems help reduce threats. Additionally, redundant flight controls and lightweight armor enhance crew protection while maintaining agility. By combining advanced technology with strategic design, these helicopters play a crucial role in modern military operations.

10. Based on the engineer's answer and your understanding of the problem, create a functional hierarchy for a reconnaissance helicopter system using the boxes below. **FILL IN ALL EMPTY BOXES. (14 Points)**



Name Brandon Ingle

* 11. Rewrite the following functions (3 points):

Function 1: Main Rotor

Rewritten function: Spin rotor

Function 2: Uses gun

Rewritten function: Eliminate targets

Function 3: Provides transport

Rewritten function: Transports Troops

12. Which of the following statements are NOT well-written requirements? Choose all that apply (4 Points)

- (a) The JLTV shall be capable of destroying unmounted infantry at a minimum distance of 4,000 meters under typical weather conditions, with a kill probability not less than 80%.
 (b) The system shall respond in under 60 minutes 95% of the time.
 (c) The cadet shall complete the Fallen Comrades Half Marathon within a time of 3 hours in PTs, beginning and ending at Michie Stadium, in all weather conditions.
 (d) The Army Combat Boot shall weigh less than 1.5lbs and fit comfortably.
 (e) The system shall reach an altitude of at least 20 km.

13. Write two properly constructed requirement statements for an Army Aviation reconnaissance helicopter (6 Points)

- The helicopter shall be capable of flying for a minimum time of 10 hours under typical weather conditions with a full fuel tank and full crew.
 1. conditions with a full fuel tank and full crew.
 2. The helicopter shall be capable of transmitting at least 100 GB of data up to 100 km away under all weather conditions.

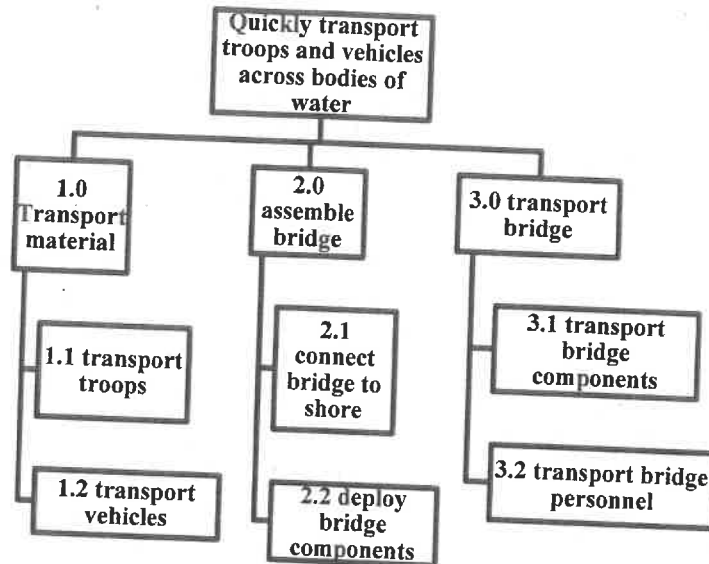
* IV. Value Modeling (35 Points).

14. What are the four levels of a Qualitative Value Model (Value Hierarchy)? (4 Points)

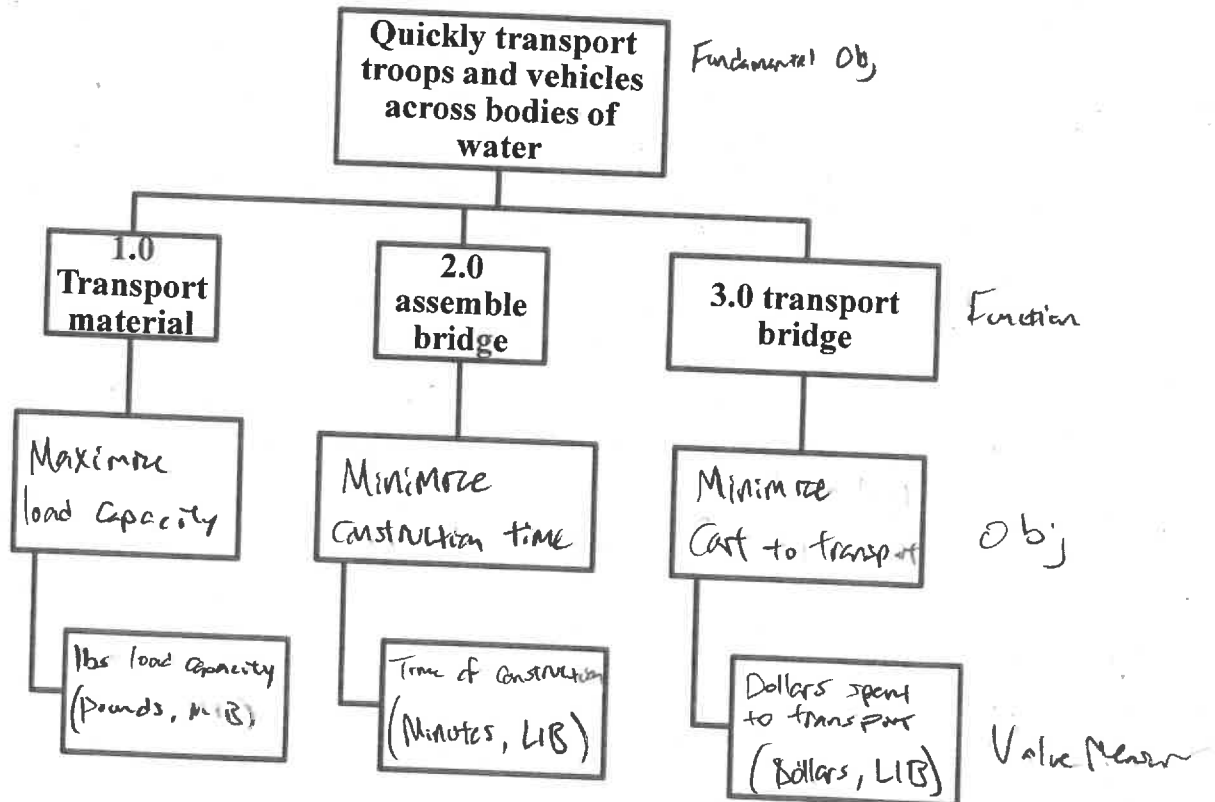
Level 1: Fundamental ObjectiveLevel 2: FunctionsLevel 3: ObjectivesLevel 4: Value Measures

15. Given the following Functional Hierarchy, create a Qualitative Value Model (Value Hierarchy) for a Army Corps of Engineers floating bridge: (10 Points)

Name Brandon Ingle



Fill out the below template to create your Qualitative Value Model. You **ONLY NEED TO FILL OUT THE PROVIDED EMPTY BOXES** for your answer. Be sure to indicate the type and scale of each value measure.



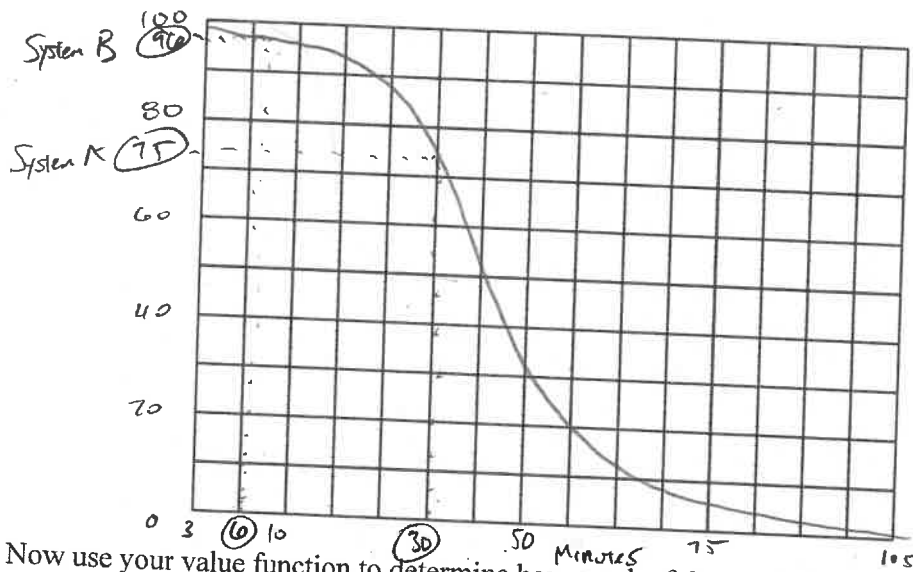
Name Brandon Ingle

16. Create the value function for the following value measure, based on the following stakeholder analysis findings: (6 Points)

Even a perfect floating bridge system couldn't be expected to deploy in less than 3 minutes. Anything up to 10 minutes would be considered to be excellent but the Army would find anything slower than that increasingly frustrating and would not accept a deployment time longer than 105 minutes under any circumstances.

Fill in the blank spaces of the table below, based on the above information and create the value function in the grid below. Remember to format the value function correctly.

Value Measure	Scale	Worst	Ideal	Function Type
Bridge deployment time	Minutes	105	3	S-Curve



17. Now use your value function to determine how much of the available value from bridge deployment time $V(x_{\text{response time},j})$ a stakeholder receives from system A that has a deployment time of 30 minutes ~~seconds~~ and system B that has a response time of 6 minutes. Show how you derived these values on the value function you drew in the previous question. (2 points)

System A: 75

System B: 96

Name Brandon Ingle

SE301 WPR #1 (Version 1)

AT 26-1

18. With the help of your client, you determined the following bounds for three value measures.

Value measure	Worst	Ideal
Maximum weight capacity MIB	40 tons	100 tons
Number of engineers needed LIB	10	1
Lifetime deployments MIB	100	1000

100
25
75

200

Your client told you that moving from 40 ton to 100 ton maximum weight capacity would have largest impact on their decision. Moving from a number of engineers needed of 10 to 1 would have about a fourth as large an impact as moving from 40 ton to 100 ton maximum weight capacity. Increasing the number of lifetime deployments from 100 to 1,000 would have about three-quarters as large an impact as moving from 40 ton to 100 ton maximum weight capacity.

Based on the information provided, calculate the global weights in the table below.
Show your work and **be sure to round your answers to 2 decimal places.** (7 Points):

Show your work here:

Weight Capacity

$$\frac{100}{200} = 0.5$$

Engineers

$$\frac{25}{200} = 0.125$$

Deployments

$$\frac{75}{200} = 0.375$$

Value Measure	Swing Weight	Measure (Global) Weight
Maximum weight capacity	100	0.5
Number of engineers needed	25	0.125
Lifetime deployments	75	0.375

= 1

Name Brandon hye

19. Unrelated to the previous questions, given the following data, determine the total value for each of the two alternatives (i.e. determine the total value for sneaker 1 and sneaker 2, respectively). You must show all your work. (6 Points)

Show your work here

$$GW \cdot V =$$

$$\begin{array}{rcl} \text{Sneaker 1:} & 0.393 \cdot 81 & = 31.833 \\ & 0.053 \cdot 41 & = 2.173 \\ & 0.541 \cdot 88 & = 47.608 \\ & 0.014 \cdot 100 & = 1.4 \\ & + & \\ & \hline & 83.014 \end{array}$$

$$\begin{array}{rcl} \text{Sneaker 2:} & 0.393 \cdot 6 & = 2.358 \\ & 0.053 \cdot 94 & = 4.982 \\ & 0.541 \cdot 91 & = 49.231 \\ & 0.014 \cdot 100 & = 1.4 \\ & + & \\ & \hline & 57.971 \end{array}$$

Sneaker 1 total value: 83.014

Sneaker 2 total value: 57.971

Fair-to-Good Student Work

Section 11 Instructor MAJ Jones

SE301 WPR #1 (Version 1)
AT 26-1

Name Jordan Hecht

Departure time 1511

**DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996-1779**

SE 301 WPR #1

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3. **AUTHORIZED REFERENCES:** You may use a pencil, straight edge (ruler, protractor, etc.), a **3 X 5 index note card (front and back)** during **last 20 minutes of exam period**, and your hand-held calculator.
4. **SHOW ALL YOUR WORK!** Unsupported or missing work earns little or no partial credit. Answer all questions to the fullest extent possible.
5. **EXTRA PAPER:** Use only the additional paper that is available at the front of the room. Attach the additional pages you use to your exam and turn them in.
6. Once you have completed the exam, fill in the departure time at the top of this page. Turn in your exam IAW your instructor's instructions.
7. This WPR will not be released from academic security until it has been returned. Before its return, cadets may not discuss any aspects of this exam with anyone except their SE301 instructor. Initial here JH

SECTION	POSSIBLE POINTS	SCORE
I. Systems Thinking	19	
II. System Decision Process & Systems Engineering	21	
III. Stakeholder Analysis	24	
IV. Functional and Requirements Analysis	26	
V. Value Modeling	35	
TOTAL	125	

Name Jordan Hecht**I. SYSTEMS THINKING (19 Points)**

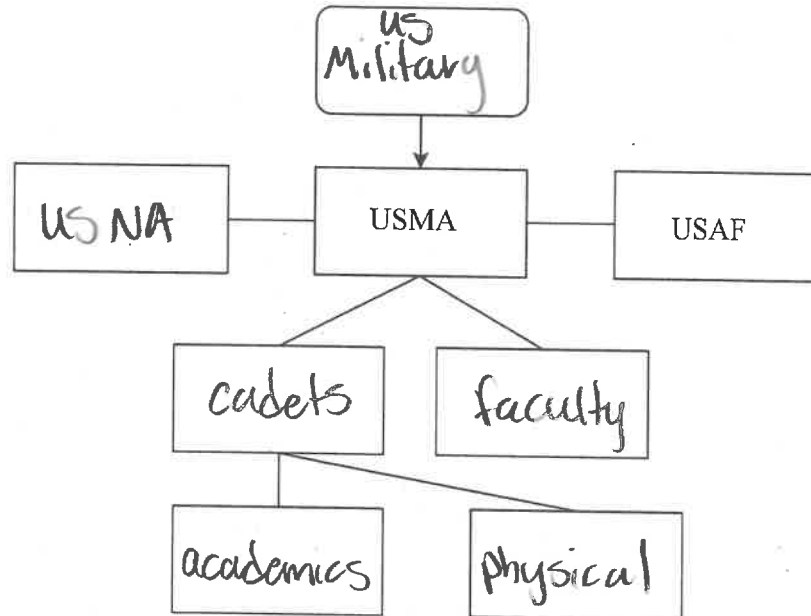
1. Using only the Plectica diagram below, which of the following statements are true (select all that apply). (4 Points)



- (a) NASA is a point
- (b) NASA is a view
- (c) The Catalina sky survey can see asteroids between 100m and 1km
- (d) NASA knows that asteroids between 100m and 1km could destroy American cities
- (e) The double asteroid redirection test could deflect asteroids between 100m and 1km
- (f) Individual cities pay taxes to Congress
- (g) Americans know that NASA exists
- (h) Americans know that asteroids larger than 1KM could wipe out Americans
- (i) Americans are part of individual cities
- (j) Congress knows about the Catalina Sky Survey

Name Jordan Hecht

2. NAME THE LEVEL OF HIERARCHY for the Spatial Arrangement **IN THE SPACES PROVIDED TO THE LEFT**, where the system of interest is the United States Military Academy at West Point. **FILL IN ALL OF THE BLANK BOXES** with the appropriate portion of the Spatial Arrangement, given the system indicated. (9 Points)

A) metasystemB) systemC) elementD) part

3. List and describe an example of the below levels of system visibility (6 Points)

System	User understanding of system	System Visibility
Air Assault School	offers all knowledge of selected internal component interactions and processes	white box
Branching process @ West Point	Offers partial knowledge of selected internal component interactions and processes	grey box
Delta force operations (to public)	offers no knowledge of selected internal component interactions and processes	Black Box

Systems Decision Process and the Systems Engineer (24 Points).

Name Jordan Hecht

4. List the four phases of the Systems Decision Process (SDP) in the correct order: (4 Points)

1. problem definition
2. solution consideration
3. solution implementation
4. solution evaluation

5. Describe the Systems Decision Process (SDP) and explain how it emphasizes systems thinking. (7 Points)

an iterative process which considers all things that influence or are affected by a system in order to solve an issue within the system.

6. List 3 environmental considerations (from the SDP) and their types that Tesla should consider when developing the national rollout of robotaxis. Do not use the same type of consideration more than once. (9 points)

Type of environmental consideration	Environmental consideration details for a Tesla Robotaxi rollout
moral/ethical	safety others on road with automated car
social	decrease in human-human interactions
political	decrease in jobs which should be protected by politicians

7. In the problem definition phase of the SDP, what two principal products are used to create the screening criteria? (4 Points)

Principal product 1: FCB matrixPrincipal product 2: value model**II. Stakeholder Analysis (23 Points).**

8. Identify the type of survey error and correct the survey error. (8 Points)

Name Jordan Hecht

- (a) How do you feel about the following statement? We should not remove study days from the West Point Schedule.

Extremely Prepared

Prepared

Unprepared

Extremely Unprepared

Type of survey error: double negativeCorrected question: How do you feel about the following statement?
We should have study days in the West Point schedule.

- (b) Calculus 2 is difficult. Does Calculus 1 help you feel prepared for Calculus 2?

Strongly Agree

Agree

Disagree

Strongly Disagree

Type of survey error: leading statementCorrected question: Did Calculus 1 help you feel prepared
for Calculus 2?

Name Jordan Hecht

9. You are the Lead Systems Engineer on a project to develop a new floating bridge design for the Army Corps of Engineers. This bridge is designed to transport troops, vehicles, and equipment across rivers and other bodies of water.

Identify three stakeholders for this project by type and the stakeholder technique your team would utilize to elicit information from them. Be sure to explain why you would use the technique chosen. Use the table below for your answer. (15 Points)

Stakeholder Name	Stakeholder Type	Stakeholder Technique Used	Why?
head of Army Corps of Engineers	Decision Authority	interview	large say on project and controls a lot of what will change
COL in charge of engineer brigade	User	interest group	Somewhat say on project, will implement new solution
Soldiers	Consumers	Survey	a lot of them, important information from lower levels

Name Jordan Hecht**III. Functional & Requirements Analysis (26 Points).**

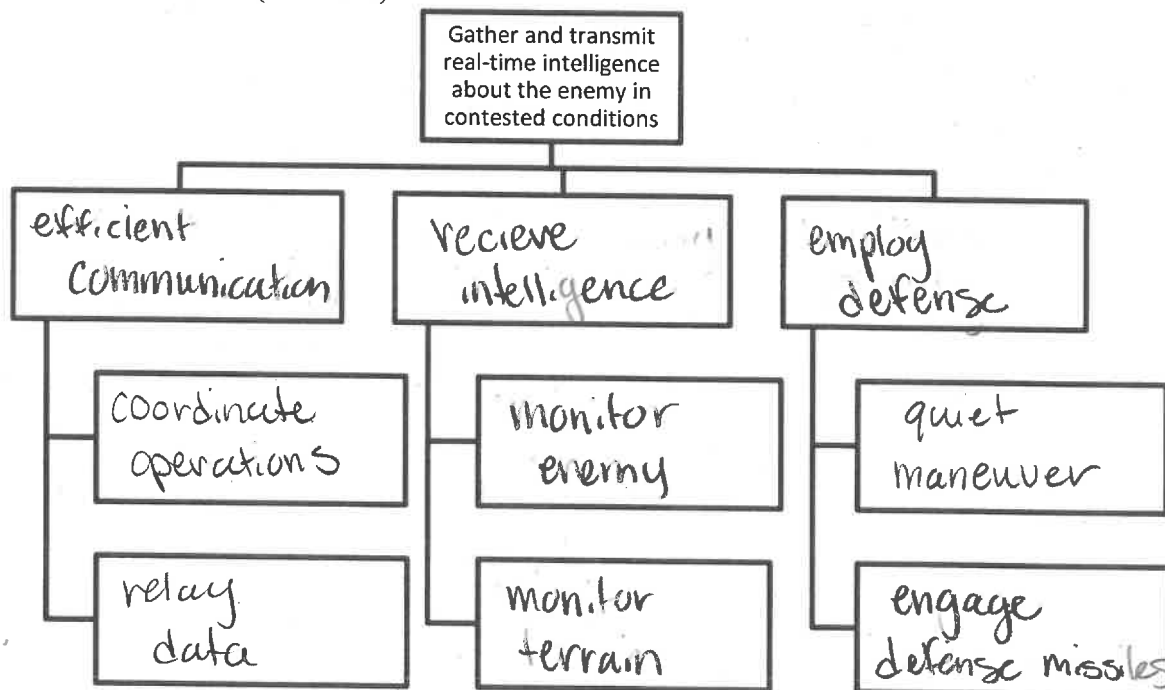
Carefully read the following answer from an Army engineer:

A reconnaissance helicopter is a vital intelligence-gathering asset, designed for stealth, speed, and advanced surveillance. As an army engineer, I focus on integrating sensor suites—electro-optical and infrared cameras, radar, and laser designators—to provide real-time battlefield intelligence. These lightweight, highly maneuverable aircraft allow forces to monitor enemy movements and terrain while minimizing detection.

Secure communication and data transmission are essential to their mission. Encrypted radio and satellite links ensure intelligence reaches commanders instantly, aiding tactical decisions. Engineers also equip these helicopters with networked systems that coordinate with other units, sometimes relaying targeting data to attack aircraft and artillery for precision strikes.

Survivability is a key design priority, as reconnaissance helicopters often operate in hostile environments. Stealth features, electronic warfare countermeasures, and missile defense systems help reduce threats. Additionally, redundant flight controls and lightweight armor enhance crew protection while maintaining agility. By combining advanced technology with strategic design, these helicopters play a crucial role in modern military operations.

10. Based on the engineer's answer and your understanding of the problem, create a functional hierarchy for a reconnaissance helicopter system using the boxes below. **FILL IN ALL EMPTY BOXES. (14 Points)**



Name Jordan Hecht

11. Rewrite the following functions (3 points):

Function 1: Main Rotor

Rewritten function: Controls movement

Function 2: Uses gun

Rewritten function: engages targets

Function 3: Provides transport

Rewritten function: relocates resources

12. Which of the following statements are NOT well-written requirements? Choose all that apply (4 Points)

shall, what, how well, in what conditions

- (a) The JLTV shall be capable of destroying unmounted infantry at a minimum distance of 4,000 meters under typical weather conditions, with a kill probability not less than 80%.
- ☒ (b) The system shall respond in under 60 minutes 95% of the time.
- (c) The cadet shall complete the Fallen Comrades Half Marathon within a time of 3 hours in PTs, beginning and ending at Michie Stadium, in all weather conditions.
- ☒ (d) The Army Combat Boot shall weigh less than 1.5lbs and fit comfortably.
- ☒ (e) The system shall reach an altitude of at least 20 km.

13. Write two properly constructed requirement statements for an Army Aviation reconnaissance helicopter (6 Points)

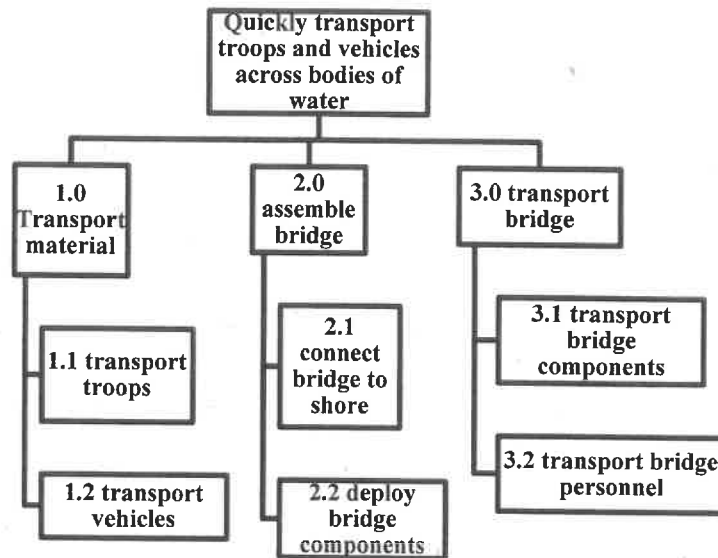
1. An Army Aviation Reconnaissance helicopter shall have integrated sensor suites, that detect enemy movement + terrain, 100% of the time, in clear skies
2. an Army Aviation Reconnaissance helicopter shall have missile defense systems that hit targets 9/10 times, under all weather conditions

IV. Value Modeling (35 Points).

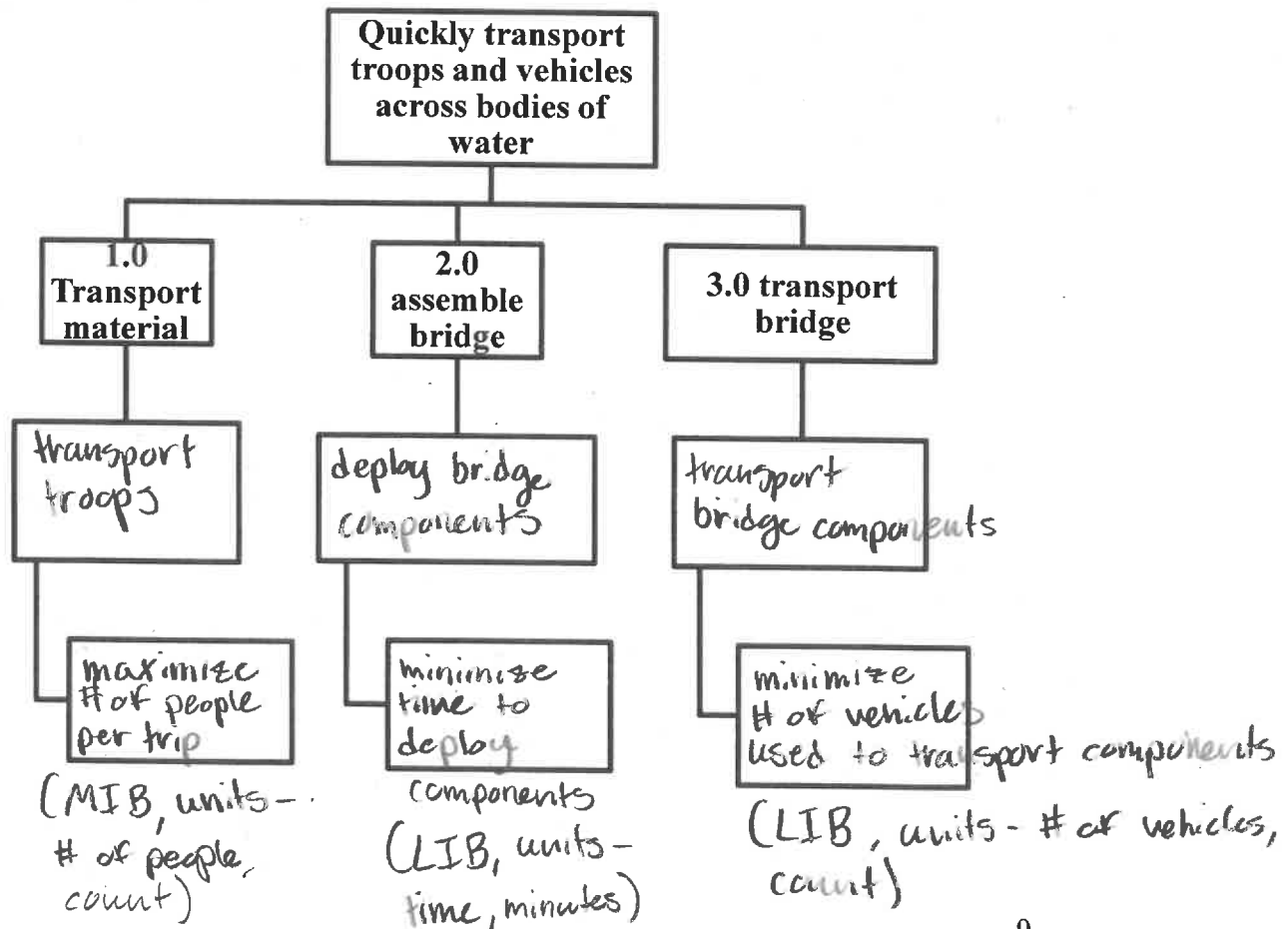
14. What are the four levels of a Qualitative Value Model (Value Hierarchy)? (4 Points)

Level 1: fundamental objectiveLevel 2: functionsLevel 3: objectivesLevel 4: value measures

15. Given the following Functional Hierarchy, create a Qualitative Value Model (Value Hierarchy) for a Army Corps of Engineers floating bridge: (10 Points)

Name Jordan Hecht

Fill out the below template to create your Qualitative Value Model. You **ONLY NEED TO FILL OUT THE PROVIDED EMPTY BOXES** for your answer. Be sure to indicate the type and scale of each value measure.



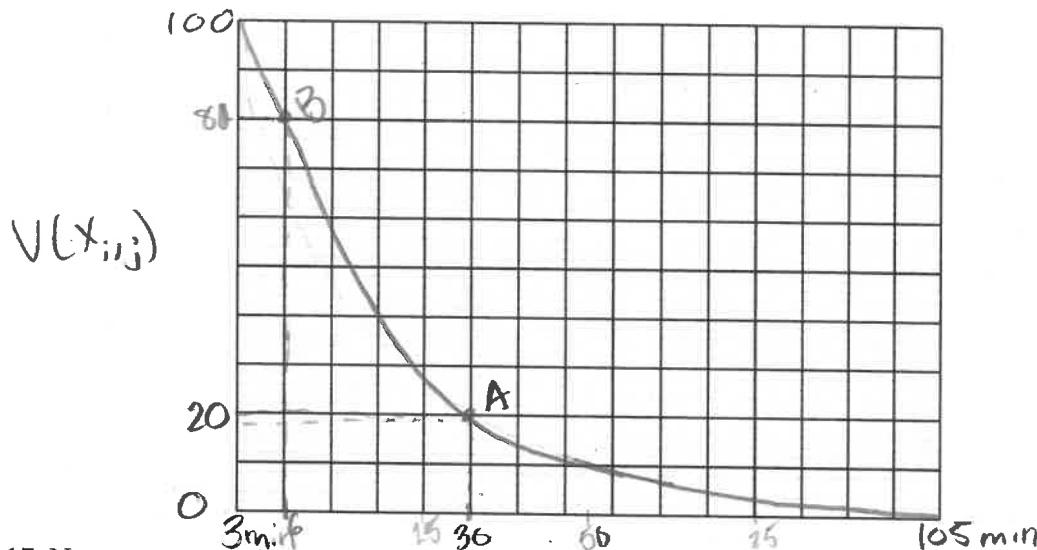
Name Jordan Hecht

16. Create the value function for the following value measure, based on the following stakeholder analysis findings: (6 Points)

Even a perfect floating bridge system couldn't be expected to deploy in less than 3 minutes. Anything up to 10 minutes would be considered to be excellent but the Army would find anything slower than that increasingly frustrating and would not accept a deployment time longer than 105 minutes under any circumstances.

Fill in the blank spaces of the table below, based on the above information and create the value function in the grid below. Remember to format the value function correctly.

Value Measure	Scale	Worst	Ideal	Function Type
Bridge deployment time	Minutes	105mins	3min	LIB



17. Now use your value function to determine how much of the available value from bridge deployment time $V(x_{\text{response time},j})$ a stakeholder receives from system A that has a deployment time of 30 minutes seconds and system B that has a response time of 6 minutes. Show how you derived these values on the value function you drew in the previous question. (2 points)

System A: 20

System B: 80

Name Jordan Hecht

18. With the help of your client, you determined the following bounds for three value measures.

Value measure	Worst	Ideal
Maximum weight capacity MIB	40 tons	100 tons
Number of engineers needed LIB	10	1
Lifetime deployments MIB	100	1000

Your client told you that moving from 40 ton to 100 ton maximum weight capacity would have largest impact on their decision. Moving from a number of engineers needed of 10 to 1 would have about a fourth as large an impact as moving from 40 ton to 100 ton maximum weight capacity. Increasing the number of lifetime deployments from 100 to 1,000 would have about three-quarters as large an impact as moving from 40 ton to 100 ton maximum weight capacity.

Based on the information provided, calculate the global weights in the table below.
Show your work and **be sure to round your answers to 2 decimal places.** (7 Points):

Show your work here:

$$100 + 25 + 75 = 200$$

$$\frac{100}{200} = 0.5$$

$$\frac{25}{200} = 0.13$$

$$\frac{75}{200} = 0.38$$

Value Measure	Swing Weight	Measure (Global) Weight
Maximum weight capacity	100	0.5
Number of engineers needed	25	0.13
Lifetime deployments	75	0.38

Sum

200

= 1.01

$$\frac{100}{200} = 0.5$$

$$\frac{25}{200} = 0.125$$

$$\frac{75}{200} = 0.375$$

Name Jordan Heant

19. Unrelated to the previous questions, given the following data, determine the total value for each of the two alternatives (i.e. determine the total value for sneaker 1 and sneaker 2, respectively). You must show all your work. (6 Points)

Show your work here

Sneaker 1

$$\begin{aligned}
 V: & 0.393(81) = 31.833 \\
 D: & 0.053(41) = 2.173 \\
 P: & 0.541(88) = 47.608 \\
 S: & 0.014(100) = 1.4 \\
 & \text{total} = 83.014
 \end{aligned}$$

Sneaker 1 total value: 83.014

Sneaker 2

$$\begin{aligned}
 V: & 0.393(6) = 2.358 \\
 V: & 0.053(94) = 4.982 \\
 P: & 0.541(91) = 49.231 \\
 S: & 0.014(100) = 1.4 \\
 & \text{total} = 57.971
 \end{aligned}$$

Sneaker 2 total value: 57.971

Poor Student Work

Section SI Instructor MAT JONES

SE301 WPR #1 (Version 1)
AT 26-1

Name London Laskey

Departure time 1525

DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996-1779

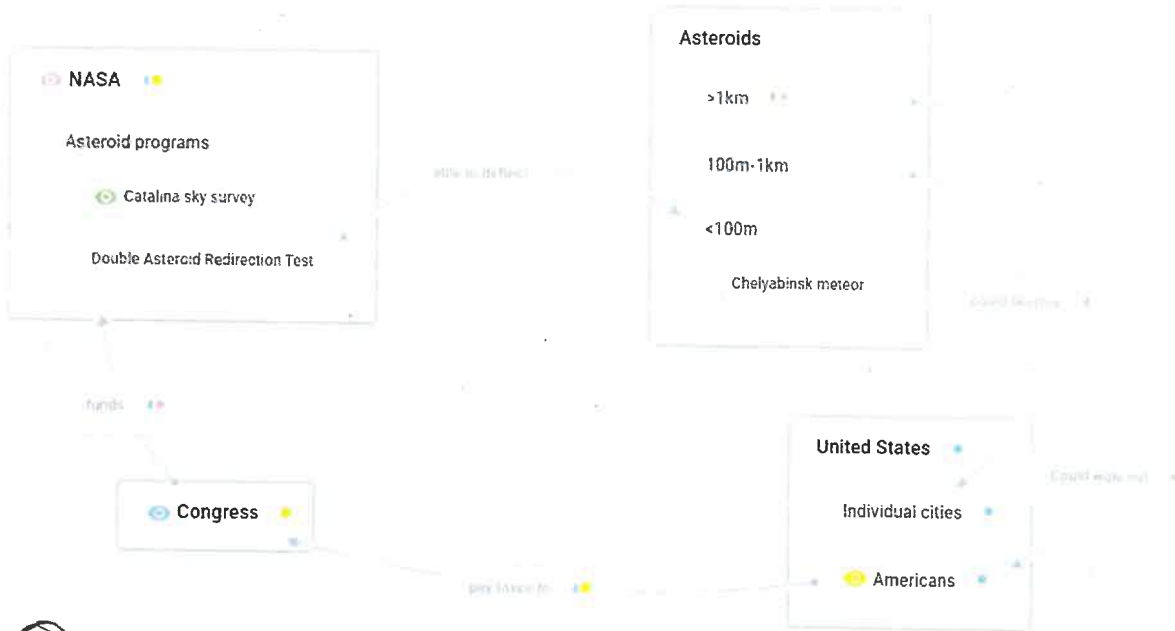
SE 301 WPR #1

1. This is a **75-minute** exam worth **125 points**. The exam consists of **12 pages**. Ensure that your exam is complete.
2. Place your name on each page in the space provided.
3. **AUTHORIZED REFERENCES:** You may use a pencil, straight edge (ruler, protractor, etc.), a **3 X 5 index note card (front and back)** during **last 20 minutes of exam period**, and your hand-held calculator.
4. **SHOW ALL YOUR WORK!** Unsupported or missing work earns little or no partial credit. Answer all questions to the fullest extent possible.
5. **EXTRA PAPER:** Use only the additional paper that is available at the front of the room. Attach the additional pages you use to your exam and turn them in.
6. Once you have completed the exam, fill in the departure time at the top of this page. Turn in your exam IAW your instructor's instructions.
7. This WPR will not be released from academic security until it has been returned. Before its return, cadets may not discuss any aspects of this exam with anyone except their SE301 instructor. Initial here LL

SECTION	POSSIBLE POINTS	SCORE
I. Systems Thinking	19	
II. System Decision Process & Systems Engineering	21	
III. Stakeholder Analysis	24	
IV. Functional and Requirements Analysis	26	
V. Value Modeling	35	
TOTAL	125	

Name Laundon Lasker**I. SYSTEMS THINKING (19 Points)**

1. Using only the Plectica diagram below, which of the following statements are true (select all that apply). (4 Points)



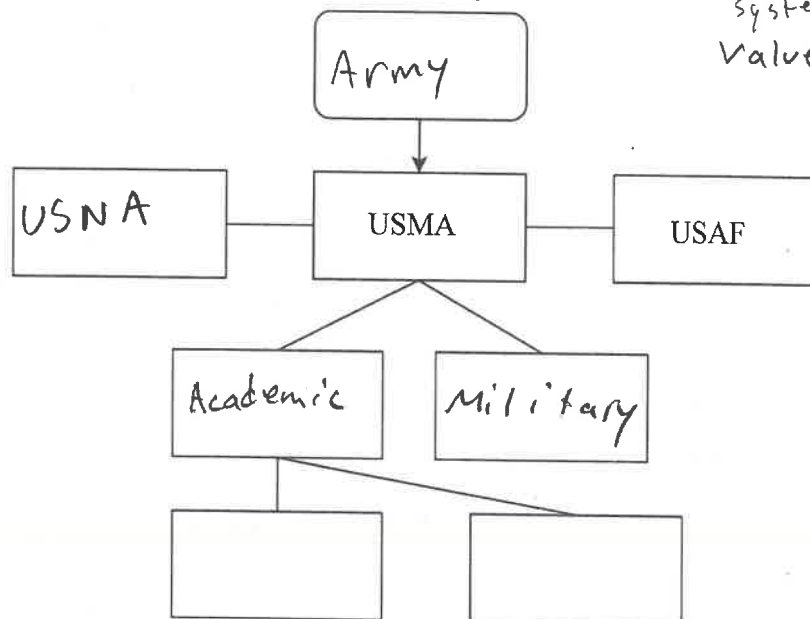
- (a) NASA is a point
- (b) NASA is a view
- (c) The Catalina sky survey can see asteroids between 100m and 1km
- (d) NASA knows that asteroids between 100m and 1km could destroy American cities
- (e) The double asteroid redirection test could deflect asteroids between 100m and 1km
- (f) Individual cities pay taxes to Congress
- (g) Americans know that NASA exists
- (h) Americans know that asteroids larger than 1KM could wipe out Americans
- (i) Americans are part of individual cities
- (j) Congress knows about the Catalina Sky Survey

Name Landon Laskey

2. NAME THE LEVEL OF HIERARCHY for the Spatial Arrangement **IN THE SPACES PROVIDED TO THE LEFT**, where the system of interest is the United States Military Academy at West Point. **FILL IN ALL OF THE BLANK BOXES** with the appropriate portion of the Spatial Arrangement, given the system indicated. (9 Points)

A) FunctionalB) SystemC) Value

D) _____



3. List and describe an example of the below levels of system visibility (6 Points)

System	User understanding of system	System Visibility
Air Assault School	offers training in air assault related operations requiring a score of 70% to qualify	Black Box
	Offers partial knowledge of selected internal component interactions and processes	Grey Box
		Black Box

Systems Decision Process and the Systems Engineer (24 Points).

Name Landon Laskey

4. List the four phases of the Systems Decision Process (SDP) in the correct order: (4 Points)

1. Problem definition
2. Solution design
3. Decision making
4. Solution implementation

5. Describe the Systems Decision Process (SDP) and explain how it emphasizes systems thinking. (7 Points)

SDP is a process used to navigate system related problems to accomplish your defined objective. It emphasizes systems thinking by considering an integrated set of elements.

6. List 3 environmental considerations (from the SDP) and their types that Tesla should consider when developing the national rollout of robotaxis. Do not use the same type of consideration more than once. (9 points)

Type of environmental consideration	Environmental consideration details for a Tesla Robotaxi rollout
economic	regular taxi drivers loosing jobs
legal	who is at fault if the robotaxis crashes
technology	How trustworthy is the tech used?

7. In the problem definition phase of the SDP, what two principal products are used to create the screening criteria? (4 Points)

Principal product 1: Decision

functional?

Principal product 2: Process**II. Stakeholder Analysis (23 Points).**

8. Identify the type of survey error and correct the survey error. (8 Points)

Name Landon Lasky

- (a) How do you feel about the following statement? We should not remove study days from the West Point Schedule.

Extremely Prepared

Prepared

Unprepared

Extremely Unprepared

Type of survey error: double barrelCorrected question: How do you feel about study days at WP?

<u>Strongly against</u>	<u>against</u>	<u>unsure</u>	<u>for</u>	<u>strongly for</u>
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- (b) Calculus 2 is difficult. Does Calculus 1 help you feel prepared for Calculus 2?

Strongly Agree

Agree

Disagree

Strongly Disagree

Type of survey error: Leading

Corrected question: Is CALC 1 a good prep for CALC 2? (0=not at all, 10=Yes, completely)

<u>0</u>	<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>	<u>5</u>	<u>6</u>	<u>7</u>	<u>8</u>	<u>9</u>	<u>10</u>
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Name Landon Laskey

9. You are the Lead Systems Engineer on a project to develop a new floating bridge design for the Army Corps of Engineers. This bridge is designed to transport troops, vehicles, and equipment across rivers and other bodies of water.

Identify three stakeholders for this project by type and the stakeholder technique your team would utilize to elicit information from them. Be sure to explain why you would use the technique chosen. Use the table below for your answer. (15 Points)

Stakeholder Name	Stakeholder Type	Stakeholder Technique Used	Why?
Commander	decision authority	interview	there is only one
Tax payers	consumer	survey	broad, collect the most data
Army	owner	interview	there are only a few people in army head quarters

Name Landon Laskey**III. Functional & Requirements Analysis (26 Points).**

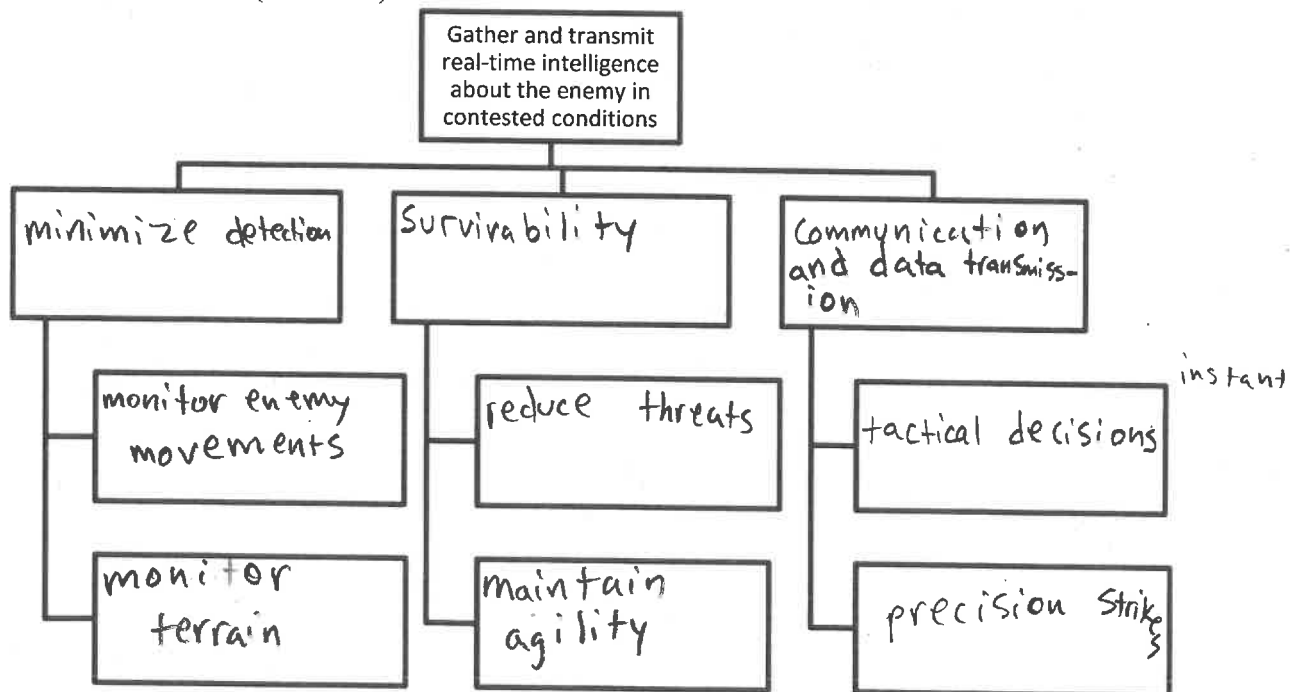
Carefully read the following answer from an Army engineer:

A reconnaissance helicopter is a vital intelligence-gathering asset, designed for stealth, speed, and advanced surveillance. As an army engineer, I focus on integrating sensor suites—electro-optical and infrared cameras, radar, and laser designators—to provide real-time battlefield intelligence. These lightweight, highly maneuverable aircraft allow forces to monitor enemy movements and terrain while minimizing detection.

Secure communication and data transmission are essential to their mission. Encrypted radio and satellite links ensure intelligence reaches commanders instantly, aiding tactical decisions. Engineers also equip these helicopters with networked systems that coordinate with other units, sometimes relaying targeting data to attack aircraft and artillery for precision strikes.

Survivability is a key design priority, as reconnaissance helicopters often operate in hostile environments. Stealth features, electronic warfare countermeasures, and missile defense systems help reduce threats. Additionally, redundant flight controls and lightweight armor enhance crew protection while maintaining agility. By combining advanced technology with strategic design, these helicopters play a crucial role in modern military operations.

10. Based on the engineer's answer and your understanding of the problem, create a functional hierarchy for a reconnaissance helicopter system using the boxes below. **FILL IN ALL EMPTY BOXES.** (14 Points)



Name Landon Laskey

11. Rewrite the following functions (3 points):

Function 1: Main Rotor

Rewritten function: ensure main rotor effectivenessFunction 2: ^{empty verb} ~~Uses~~ gunRewritten function: operates weaponFunction 3: ^{empty verb} ~~Provides~~ transportRewritten function: stage vehicles

12. Which of the following statements are NOT well-written requirements? Choose all that apply (4 Points)

- (a) The JLTV shall be capable of destroying unmounted infantry at a minimum distance of 4,000 meters under typical weather conditions, with a kill probability not less than 80%.
- ☒ (b) The system shall respond in under 60 minutes 95% of the time.
- (c) The cadet shall complete the Fallen Comrades Half Marathon within a time of 3 hours in PTs, beginning and ending at Michie Stadium, in all weather conditions.
- ☒ (d) The Army Combat Boot shall weigh less than 1.5lbs and fit comfortably.
- ☒ (e) The system shall reach an altitude of at least 20 km.

13. Write two properly constructed requirement statements for an Army Aviation reconnaissance helicopter (6 Points)

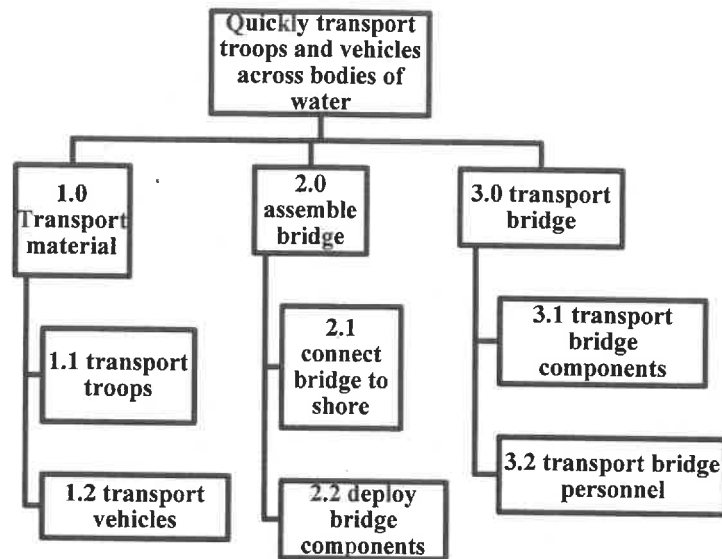
1. The Army Aviation recon helicopter shall pass army inspection in accordance to the standard in clear weather at FT Bragg.
2. The Army Aviation recon helicopter shall be capable of traveling a minimum distance of 200 miles under extreme weather conditions with 100% weight capacity.

IV. Value Modeling (35 Points).

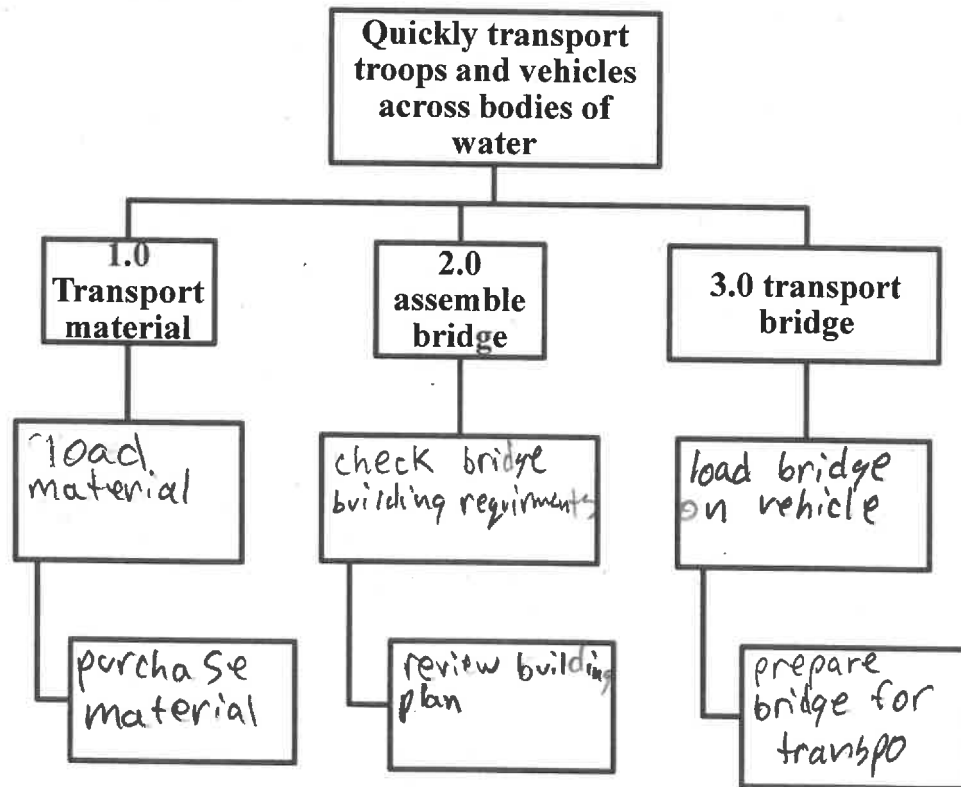
14. What are the four levels of a Qualitative Value Model (Value Hierarchy)? (4 Points)

Level 1: DistinctionsLevel 2: SystemLevel 3: RelationshipLevel 4: perspectives

15. Given the following Functional Hierarchy, create a Qualitative Value Model (Value Hierarchy) for a Army Corps of Engineers floating bridge: (10 Points)

Name Landon Laskey

Fill out the below template to create your Qualitative Value Model. You **ONLY NEED TO FILL OUT THE PROVIDED EMPTY BOXES** for your answer. Be sure to indicate the type and scale of each value measure.



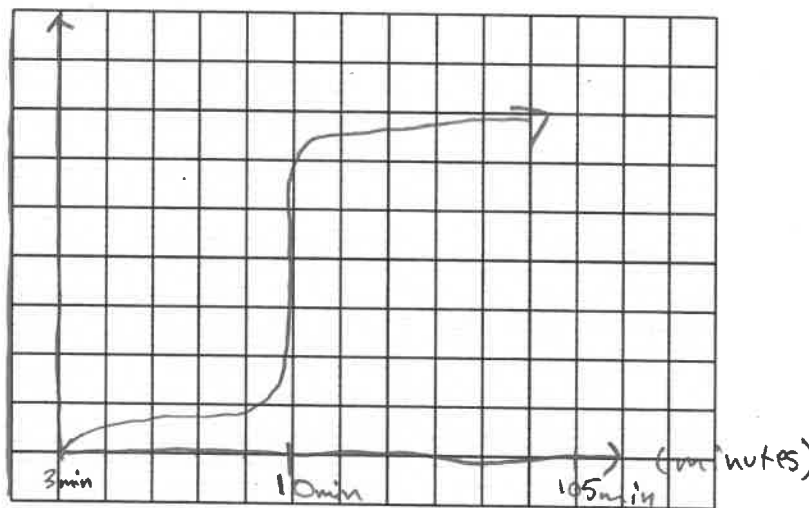
Name Landon Laskey

16. Create the value function for the following value measure, based on the following stakeholder analysis findings: (6 Points)

Even a perfect floating bridge system couldn't be expected to deploy in less than 3 minutes. Anything up to 10 minutes would be considered to be excellent but the Army would find anything slower than that increasingly frustrating and would not accept a deployment time longer than 105 minutes under any circumstances.

Fill in the blank spaces of the table below, based on the above information and create the value function in the grid below. Remember to format the value function correctly.

Value Measure	Scale	Worst	Ideal	Function Type
Bridge deployment time	Minutes	105 minutes	<10 minutes	S shape



17. Now use your value function to determine how much of the available value from bridge deployment time $V(x_{\text{response time},j})$ a stakeholder receives from system A that has a deployment time of 30 minutes seconds and system B that has a response time of 6 minutes. Show how you derived these values on the value function you drew in the previous question. (2 points)

System A: .2857

$$\rightarrow \frac{30 \text{ min}}{105 \text{ min}}$$

System B: .05714

$$\frac{6}{105}$$

Name Landon Laskey

18. With the help of your client, you determined the following bounds for three value measures.

Value measure	Worst	Ideal
Maximum weight capacity MIB	40 tons	100 tons
Number of engineers needed LIB	10	1
Lifetime deployments MIB	100	1000

Your client told you that moving from 40 ton to 100 ton maximum weight capacity would have largest impact on their decision. Moving from a number of engineers needed of 10 to 1 would have about a fourth as large an impact as moving from 40 ton to 100 ton maximum weight capacity. Increasing the number of lifetime deployments from 100 to 1,000 would have about three-quarters as large an impact as moving from 40 ton to 100 ton maximum weight capacity.

Based on the information provided, calculate the global weights in the table below.
Show your work and **be sure to round your answers to 2 decimal places.** (7 Points):

Show your work here:

$$GW = \frac{100}{200} = .5$$

$$\frac{25}{200} = .125$$

$$\frac{75}{200} = .375$$

✓

Value Measure	Swing Weight	Measure (Global) Weight
Maximum weight capacity	100	.5
Number of engineers needed	25	.125
Lifetime deployments	75	.375

total

200

Name Landon Lasky

19. Unrelated to the previous questions, given the following data, determine the total value for each of the two alternatives (i.e. determine the total value for sneaker 1 and sneaker 2, respectively). You must show all your work. (6 Points)

Show your work here

sneaker 1:

sneaker 2:

Sneaker 1 total value: _____

Sneaker 2 total value: _____



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB H

WPR #2

Instructor Solution

Section: _____

SE301 WPR #2 AT 26-1

Name: _____

Departure Time: _____

**DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996**

**SE301 WPR #2
26-1
VERSION (A)**

1. This is a **75-minute** exam **worth 90 pts.** The exam consists of **8 pages.** Ensure that your exam is complete.
2. Place your name on each page in the space provided.
3. **AUTHORIZED REFERENCES:** You may use a pencil, straight edge (ruler, protractor, etc.), a 3 X 5 index card (front and back), and your issued hand-held calculator.
4. **SHOW ALL YOUR WORK!** Unsupported or missing work earns little or no partial credit. Answer all questions to the fullest extent possible.
5. **EXTRA PAPER:** Use only the additional paper available at the front of the room. Attach the additional pages you use to your exam and turn them in.
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7. This WPR will not be released from academic security until it has been returned. Before its return, cadets may not discuss any aspects of this exam with anyone except their SE301 instructor. **Initial here** _____

SECTION	POSSIBLE POINTS	SCORE
I. Solution Design & Design Methods	20	
II. Alternative Generation	35	
III. Cost Analysis	35	
TOTAL	90	

Name _____

I. Solution Design & Design Methods [20 Pts]

1. True or False: [2 Pts] (check one)

The best design is almost always in the middle rather than an extreme option.

True ☐

False ☒

2. True or False: [2 Pts] (check one)

The first stage of the ideation process is to generate alternatives.

True ☐

False ☒

3. Which of the following are principles of good idea generation: [4 Pts]

Encourage wild ideas ☒

Omit early ideas ☐

Create numerous ideas ☒

Do not criticize ideas ☐

Build on ideas ☐

4. Which of the following are examples of idea generation methods (check all that apply)? [4 Pts]

Unstructured Brainwriting ☒

Oracle Method ☐

Structured Brainstorming ☒

Casino Royale Method ☐

All of the above ☐

5. Which of the following are types of errors to avoid in the ethics of solution design.

Check all that apply. [4 Pts]

Anecdotal evidence ☒

Vested interest ☒

Strawman ☒

Stochastic modeling ☐

Anachronistic argumentation ☐

None of the above ☐

6. Which of the following are approaches to alternative generation (check all that apply) [4 Pts.]

Integrated Benchmarking ☒

Quantitative Integration ☐

Functional Hierarchy ☐

Benchmarking ☒

Fundamental Objective ☐

Name _____

II. Alternative Generation [35 Pts]

You are part of the design team at Q-Branch responsible for developing a high-range rocket; your team has devised the following parameters conveyed in the table below.

Solution Design Parameters		
<i>Search Technique</i>	<i>Guidance</i>	<i>Nozzle Diameter (mm)</i>
Forward Looking Infrared	Inertial	>0.8
Laser	Global Positioning System	$0.81 \leq \text{Nozzle Diameter} < 1.6$
Audio	Wire	$1.6 \leq \text{Nozzle Diameter} \leq 1.8$
GSR	Optical	

7. Using Zwicky's Morphology, how many total possible combinations are there for the team to consider? Show your work. [3 Pts]

Answer: 4 * 4 * 3 = 48

8. Q You are designing a new electric scooter for urban commuting, and your team proposes a 15kg scooter with a 5,000 mAh battery. [5 Pts]

You have researched two similar scooters:

- Scooter X is 18kg and has a 4,000 mAh battery. It can travel 25 miles on a single charge.
- Scooter Y is 12kg and has a 5,500 mAh battery. It can travel 40 miles on a single charge.

Using benchmarking, what is a reasonable rough estimate for how far the new scooter will be able to travel?

Answer: 35 miles

Justification for your rough estimate: The range is estimated by interpolating battery capacity between Scooter X (4,000 mAh, 25 miles) and Scooter Y (5,500 mAh, 40 miles). The new scooter's 5,000 mAh battery is 1,000 mAh above Scooter X and 500 mAh below Scooter Y. The range per mAh is $(\frac{40 - 25}{5,500 - 4,000}) = 0.01$ miles/mAh. From Scooter X, $1,000 \text{ mAh} \times 0.01 = 10$ miles, so $25 + 10 = 35$ miles. From Scooter Y, $500 \text{ mAh} \times 0.01 = 5$ miles, so $40 - 5 = 35$ miles. The new scooter's weight (15kg) is halfway between Scooter X (18kg) and Scooter Y (12kg), balancing efficiency impacts, making 35 miles a reasonable estimate.

Name _____

9. MI6 is interested in acquiring a new Multi-Role Adaptive Rifle that can fire the 7.62mm, .300 Norma Magnum and .338 Norma Magnum. "M" insists, "the sniper rifle shall engage E-type silhouette targets at 1,500m with a greater than 50% hit rate during daylight, 0-5knt winds, and 24 degrees Celsius conditions." Without building a prototype, what are **two** ways MI6 could estimate a sniper rifle's ability to meet the requirement? What is one advantage and one disadvantage of each approach? [6 Pts]

Potential approaches are

Method 1: **Computer Model** Advantage: **low cost** Disadvantage: **do not actually fire weapon**

Method 2: **Subject Matter Expert Opinion** Advantage: **vast knowledge of weapons systems**
Disadvantage: **do not actually fire the weapon**

Method 3: **Benchmarking Against Similar Systems** Advantage: **several systems may be similar, lots of data available** Disadvantage: **systems may not be exact to proposed system**

10. Maj Boothroyd of Q-Branch created four alternatives for you to consider. The "must have" success criteria are search distance, guidance fidelity, and thrust. Based on the table below, which (if any) of the alternatives will be **screened out** and **why**? [5 Pts]

Alternative	Feasibility Screening Matrix			Go / NoGo
	Search Distance [m] (>1000 m)	Guidance Fidelity [%] (More than 75%)	Thrust [lbs] (>1000 lbs)	
Alt 1: Moonraker	1200 Go	95 Go	1010 Go	Go
Alt 2: Thunderball	999 No Go	77 Go	1120 Go	No Go
Alt 3: Risico	800 No Go	82 Go	900 No Go	No Go
Alt 4: Icebreaker	1150 Go	85 Go	1000 No Go	No Go

Why Go or NoGo: Alt : Risico because it does not meet the screening criteria for Search Distance and Thrust.

Name _____

11. Below is data collected from four (4) smartphone alternatives for battery life (hours), cost (\$), and charging time (minutes). Your engineering team has developed a new fast-charge adapter that can be added to the phones. The cost of the adapter is \$100 added to the price of the phone. Application of the adapter results in a 5% increase in battery life (rounded to the nearest whole number, do not round up) and a 15% decrease in charging time (rounded to two decimal places). If the fast-charge adapter is applied to the alternatives below, which alternative(s) meet the screening criteria (fill in the screening matrix and show all work)? [16 Pts]

iPhone 16: 20 hours, \$799, and 85 minutes
 Samsung Galaxy S23: 22 hours, \$849, and 70 minutes
 Google Pixel 9: 18 hours, \$599, and 90 minutes
 OnePlus 12: 24 hours, \$699, and 65 minute

	Battery Life	Cost	Charging time (min)	Go/NoGo
Screening Criteria	≥ 22 hours	$< \\$800$	≤ 70min	
iPhone 16	21 Hours	\$899	72.25	NO GO
Samsung Galay S24	23 Hours	\$949	59.50	NO GO
Google Pixel 9	18 Hours	\$699	76.50	NO GO
OnePlus 12	25 Hours	\$799	55.25	GO

Name _____

III. Cost Analysis [35 Pts]

12. Coach Q runs a professional soccer training academy. Which of the following are fixed costs and which are variable costs [4 Pts]:

- a) Cost of utilities at the training facility
Fixed ☐
Variable ☒
- b) Cost of renting the training field for the season
Fixed ☒
Variable ☐
- c) Cost of water and sports drinks provided to players during training
Fixed ☐
Variable ☒
- d) Cost of salaries for the coaching staff
Fixed ☒
Variable ☐

13. Coach Mezza manages a sports team that partners with Cava to provide Mediterranean-inspired meals for players during training camps and post-game events. Which of the following are recurring costs and which are non-recurring costs [4 Pts]:

- a) Cost to design a custom cava-branded food truck for event catering
Recurring ☐
Non-recurring ☒
- b) Cost of utilities for the Cava catering setup at event venues
Recurring ☒
Non-recurring ☐
- c) Cost to build a new Cava kitchen facility for preparing event meals
Recurring ☐
Non-recurring ☒
- d) Cost of annual training for cava staff on Mediterranean cuisine preparation.
Recurring ☒
Non-recurring ☐

Name _____

14. Cava Pita Co. operates a chain of Mediterranean fast-casual restaurants and provides catering services for corporate events. Are the following costs direct or indirect costs, and why?
[6 Pts]

a) Cost of a chef hired specifically to prepare Cava's signature harissa sauce for catering orders

Direct ☒

Indirect ☐

Reason: _____

b) Cost of utilities (electricity and water) for the Cava restaurant kitchen, fielding both catering orders and other restaurant operations

Direct ☐

Indirect ☒

Reason: _____

Name _____

15. Coach Falafel is the manager of a sports team that partners with Cava to provide Mediterranean meals for players during away games. To support the catering operations, Coach Falafel needs to transport Cava's catering equipment and staff from the Cava restaurant (located 50 miles from the team's stadium) to the game venue. He is considering two transportation options: Option 1 (Catering Van): Purchase a used catering van to transport equipment and staff; Option 2 (Rented Transport): Rent a shuttle service to transport equipment and staff, supplemented by a rideshare service for staff commuting to the shuttle pickup point. He completes a 4-year cost analysis to decide which option to select. Use the table below to calculate each option's Net Present Value (NPV). [20 pts]

	Catering van	Rented Transport
Initial Cost	\$30,000 (Used catering van)	\$500 (Specialized equipment cases)
Annual Maintenance/Operating Cost (The average cost you will pay for maintenance and operation)	\$3,000 (van upkeep, insurance)	\$1,200 (cleaning supplies, meal allowance)
Annual Transportation Cost	\$3,500 (gas, tolls)	\$7,000 (shuttle rental)
Resale Value	\$15,000	\$200
Interest Rate (<i>i</i>)	8%	

Round your answers to the nearest dollar and indicate the appropriate sign (positive or negative).

Catering Van NPV:

Catering Van NPV: -\$40,503

Answer: _____

Rented Transportation NPV:

Rented Transport NPV: -\$27,512

Answer: _____

16. If cost is the only concern in 14, which option should Coach Falafel pick? [1 Pt]

Catering Van ☐

Rented Transportation ☒

Outstanding Student Work

Section: 71

SE301 WPR #2 AT 26-1

Name: Garrett Dzurinko

Departure Time: 1446

**DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996**

**SE301 WPR #2
26-1
VERSION (A)**

1. This is a **75-minute** exam worth **90 pts**. The exam consists of **8 pages**. Ensure that your exam is complete.
2. Place your name on each page in the space provided.
3. **AUTHORIZED REFERENCES:** You may use a pencil, straight edge (ruler, protractor, etc.), a 3 X 5 index card (front and back), and your issued hand-held calculator. You are allowed to use the notecard for the final 20 minutes of the exam.
4. **SHOW ALL YOUR WORK!** Unsupported or missing work earns little or no partial credit. Answer all questions to the fullest extent possible.
5. **EXTRA PAPER:** Use only the additional paper available at the front of the room. Attach the additional pages you use to your exam and turn them in.
6. Once you have completed the exam, fill in the departure time at the top of this page. Turn in your exam IAW your instructor's instructions.
7. This WPR will not be released from academic security until it has been returned. Before its return, cadets may not discuss any aspects of this exam with anyone except their SE301 instructor. Initial here GD

SECTION	POSSIBLE POINTS	SCORE
I. Solution Design & Design Methods	20	
II. Alternative Generation	35	
III. Cost Analysis	35	
TOTAL	90	

Name Garrett Dzurinko**I. Solution Design & Design Methods [20 Pts]****1. True or False: [2 Pts] (check one)**

The best design is almost always in the middle rather than an extreme option.

True ☐False ☒**2. True or False: [2 Pts] (check one)**

The first stage of the ideation process is to generate alternatives.

True ☐False ☒**3. Which of the following are principles of good idea generation: [4 Pts]**Encourage wild ideas ☒Omit early ideas ☐Create numerous ideas ☒Do not criticize ideas ☒Build on ideas ☒**4. Which of the following are examples of idea generation methods (check all that apply)? [4 Pts]**Unstructured Brainwriting ☒Oracle Method ☐Structured Brainstorming ☒Casino Royale Method ☐All of the above ☐**5. Which of the following are types of errors to avoid in the ethics of solution design.**

Check all that apply. [4 Pts]

Anecdotal evidence ☒Vested interest ☒Strawman ☒Stochastic modeling ☐Anachronistic argumentation ☐None of the above ☐**6. Which of the following are approaches to alternative generation (check all that apply) [4 Pts.]**Integrated Benchmarking ☒Quantitative Integration ☐Functional Hierarchy ☐Benchmarking ☒Fundamental Objective ☐

Name Garett Dzulin**II. Alternative Generation [35 Pts]**

You are part of the design team responsible for developing a high-range rocket; your team has devised the following parameters conveyed in the table below.

Solution Design Parameters		
Search Technique	Guidance	Nozzle Diameter (mm)
Forward Looking Infrared	Inertial	>0.8
Laser	Global Positioning System	$0.81 \leq \text{Nozzle Diameter} < 1.6$
Audio	Wire	$1.6 \leq \text{Nozzle Diameter} \leq 1.8$
GSR		

7. Using Zwicky's Morphology, how many total possible combinations are there for the team to consider? Show your work. [3 Pts]

$$4 \times 3 \times 3 = 36$$

Answer: 36

8. You are designing a new electric scooter for urban commuting, and your team proposes a 15kg scooter with a 5,000 mAh battery. [5 Pts]

You have researched two similar scooters:

- Scooter X is 18kg and has a 4,000 mAh battery. It can travel 25 miles on a single charge.
- Scooter Y is 12kg and has a 5,500 mAh battery. It can travel 40 miles on a single charge.

Using benchmarking, what is a reasonable rough estimate for how far the new scooter will be able to travel?

Answer: 35 miles

Justification for your rough estimate: Our scooter's weight is 3kg less than Scooter X and 3kg more than Scooter Y and our battery is 1000mAh more than Scooter X and 500mAh less than Scooter Y. So the performance of our scooter is roughly between Scooter X and Y but closer to Scooter Y because of the battery. So our travel distance will be between 25 and 40 miles and closer to 40 miles.

Name Garrett Dzurinko

9. MI6 is interested in acquiring a new Multi-Role Adaptive Rifle that can fire the 7.62mm, .300 Norma Magnum and .338 Norma Magnum. "M" insists, "the sniper rifle shall engage E-type silhouette targets at 1,500m with a greater than 50% hit rate during daylight, 0-5knt winds, and 24 degrees Celsius conditions." Without building a prototype, what are **two** ways MI6 could estimate a sniper rifle's ability to meet the requirement? What is one advantage and one disadvantage of each approach? [6 Pts]

Method 1: Benchmarking against current rifles.

Advantage: Could have existing data from rifles that could provide accurate estimates.

Disadvantage: Might be a lack of Multi-Role Rifles to collect data from.

Method 2: Expert Opinion

Advantage: Low-cost method of estimating the rifles performance from SME.

Disadvantage: Could be biased against or for the rifle leading to inaccurate estimates.

10. Maj Boothroyd of Q-Branch created four alternatives for you to consider. The "must have" success criteria are search distance, guidance fidelity, and thrust. Based on the table below, which (if any) of the alternatives will be **screened out** and **why**? [5 Pts]

Feasibility Screening Matrix				
Alternative	Search Distance [m] (>1000 m)	Guidance Fidelity [%] (More than 75%)	Thrust [lbs] (>1000 lbs)	Go / NoGo
Alt 1: Moonraker	1200	95	1010	<u>GO</u>
Alt 2: Thunderball	999	77	1120	<u>NO GO</u>
Alt 3: Risico	800	82	900	<u>NO GO</u>
Alt 4: Icebreaker	1150	85	1000	<u>NO GO</u>

Why Go or NoGo:

Alt 1: GO - Meets all Screening Criteria

Alt 2: Screened out - Search distance isn't greater than 1000m

Alt 3: Screened Out - Search Distance not greater than 1000m and Thrust not greater than 1000 lbs.

Alt 4: Screened out - Thrust not greater than 1000 lbs

Name Garrett Dzurinko

11. Below is data collected from four (4) smartphone alternatives for battery life (hours), cost (\$), and charging time (minutes). Your engineering team has developed a new fast-charge adapter that can be added to the phones. The cost of the adapter is \$100 added to the price of the phone. Application of the adapter results in a 5% increase in battery life (rounded to the nearest whole number, do not round up) and a 15% decrease in charging time (rounded to two decimal places). If the fast-charge adapter is applied to the alternatives below, which alternative(s) meet the screening criteria (fill in the screening matrix and show all work)? [16 Pts]

iPhone 14: 20 hours, \$799, and 85 minutes

Samsung Galaxy S23: 22 hours, \$849, and 70 minutes

Google Pixel 7: 18 hours, \$599, and 90 minutes

OnePlus 11: 24 hours, \$699, and 65 minutes

	Battery Life	Cost	Charging time (min)	Go/NoGo
Screening Criteria	≥ 22 hours	$< \$800$	≤ 70 min	
1 iPhone 14	X 21 hours	X \$899	X 72.25 min	NO GO
2 Samsung Galay S23	23 hours	X \$949	59.50 min	NO GO
3 Google Pixel 7	X 18 hours	\$699	X 76.50 min	NO GO
4 OnePlus 11	25 hours	\$799	55.25 min	GO

1	$20 + 20(0.05) = 21$	$799 + 100 = 899$	$85 - 85(0.15) = 72.25$
2	$22 + 22(0.05) = 23$	$849 + 100 = 949$	$70 - 70(0.15) = 59.5$
3	$18 + 18(0.05) = 18$	$599 + 100 = 699$	$90 - 90(0.15) = 76.5$
4	$24 + 24(0.05) = 25$	$699 + 100 = 799$	$65 - 65(0.15) = 55.25$

Name Gariett Dzurko**III. Cost Analysis [35 Pts]**

12. Coach Q runs a professional soccer training academy. Which of the following are fixed costs and which are variable costs [4 Pts]:

- a) Cost of utilities at the training facility
Fixed ☐
Variable ☒
- b) Cost of renting the training facility for the season
Fixed ☒
Variable ☐
- c) Cost of water and sports drinks provided to players during training
Fixed ☐
Variable ☒
- d) Cost of salaries for the coaching staff
Fixed ☒
Variable ☐

13. Coach Mezza manages a sports team that partners with Cava to provide Mediterranean-inspired meals for players during training camps and post-game events. Which of the following are recurring costs and which are non-recurring costs [4 Pts]:

- a) Cost to design a custom cava-branded food truck for event catering
Recurring ☐
Non-recurring ☒
- b) Cost of utilities for the Cava catering setup at event venues
Recurring ☒
Non-recurring ☐
- c) Cost to build a new Cava kitchen facility for preparing event meals
Recurring ☐
Non-recurring ☒
- d) Cost of annual training for cava staff on Mediterranean cuisine preparation.
Recurring ☒
Non-recurring ☐

Name Garrett Dzurinko

14. Cava Pita Co. operates a chain of Mediterranean fast-casual restaurants and provides catering services for corporate events. Are the following costs direct or indirect costs, and why?
[6 Pts]

a) Cost of a chef hired specifically to prepare Cava's signature harissa sauce for catering orders

Direct ☒Indirect ☐Reason: The chef directly impacts the catering services by preparing the sauce

b) Cost of utilities (electricity and water) for the Cava restaurant kitchen, fielding both catering orders and other restaurant operations

Direct ☐Indirect ☒Reason: The utilities are not used exclusively for the catering service so they don't directly impact them.

Name Garrett Beninko

15. Coach Falafel is the manager of a sports team that partners with Cava to provide Mediterranean meals for players during away games. To support the catering operations, Coach Falafel needs to transport Cava's catering equipment and staff from the Cava restaurant (located 50 miles from the team's stadium) to the game venue. He is considering two transportation options: Option 1 (Catering Van): Purchase a used catering van to transport equipment and staff; Option 2 (Rented Transport): Rent a shuttle service to transport equipment and staff, supplemented by a rideshare service for staff commuting to the shuttle pickup point. He completes a 4-year cost analysis to decide which option to select. Use the table below to calculate each option's Net Present Value (NPV). [20 pts]

	Catering van	Rented Transport
Initial Cost	\$30,000 (Used catering van)	\$500 (Specialized equipment cases)
Annual Maintenance/Operating Cost (The average cost you will pay for maintenance and operation)	\$3,000 (van upkeep, insurance)	\$1,200 (cleaning supplies, meal allowance)
Annual Transportation Cost	\$3,500 (gas, tolls)	\$7,000 (shuttle rental)
Resale Value	\$15,000	\$200
Interest Rate (i)	8%	

Round your answers to the nearest dollar and indicate the appropriate sign (positive or negative).

	Year	Net Inc	Pre	Net Cash	Net present
Catering Van NPV: $P = F(1+i)^{-n}$	0	-30000		-30000	-30000
	1		-6500	-6500	-6018.52
	2		-6500	-6500	-5572.70
	3		-6500	-6500	-5159.91
	4	+15000	-6500	+8500	+6247.75
Answer: -40503.38					
Rented Transportation NPV:	0	-500		-500	-500
	1		-8200	-8200	-7592.59
	2		-8200	-8200	-7030.18
	3		-8200	-8200	-6504.42
	4	+2000	-8200	-6000	-5880.24
Answer: -27512.43					

16. If cost is the only concern in 14, which option should Coach Falafel pick? [1 Pt]

Catering Van ☐

Rented Transportation ☒

Fair-to-Good Student Work

Section: H1

SE301 WPR #2 AT 26-1

Name: Sam Olson

Departure Time: 0950

**DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996**

**SE301 WPR #2
26-1
VERSION (A)**

1. This is a **75-minute** exam worth **90 pts**. The exam consists of **8 pages**. Ensure that your exam is complete.
2. Place your name on each page in the space provided.
3. **AUTHORIZED REFERENCES:** You may use a pencil, straight edge (ruler, protractor, etc.), a 3 X 5 index card (front and back), and your issued hand-held calculator. You are allowed to use the notecard for the final 20 minutes of the exam.
4. **SHOW ALL YOUR WORK!** Unsupported or missing work earns little or no partial credit. Answer all questions to the fullest extent possible.
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6. Once you have completed the exam, fill in the departure time at the top of this page. Turn in your exam IAW your instructor's instructions.
7. This WPR will not be released from academic security until it has been returned. Before its return, cadets may not discuss any aspects of this exam with anyone except their SE301 instructor. Initial here SO

SECTION	POSSIBLE POINTS	SCORE
I. Solution Design & Design Methods	20	
II. Alternative Generation	35	
III. Cost Analysis	35	
TOTAL	90	

Name Sam Olson

I. Solution Design & Design Methods [20 Pts]

1. True or False: [2 Pts] (check one)

The best design is almost always in the middle rather than an extreme option.

True ☒

False ☐

2. True or False: [2 Pts] (check one)

The first stage of the ideation process is to generate alternatives.

True ☐

False ☒

3. Which of the following are principles of good idea generation: [4 Pts]

Encourage wild ideas ☒

Omit early ideas ☐

Create numerous ideas ☒

Do not criticize ideas ☒

Build on ideas ☒

4. Which of the following are examples of idea generation methods (check all that apply)? [4 Pts]

Unstructured Brainwriting ☒

Oracle Method ☐

Structured Brainstorming ☒

Casino Royale Method ☐

All of the above ☐

5. Which of the following are types of errors to avoid in the ethics of solution design.

Check all that apply. [4 Pts]

Anecdotal evidence ☒

Vested interest ☒

Strawman ☒

Stochastic modeling ☐

Anachronistic argumentation ☐

None of the above ☐

6. Which of the following are approaches to alternative generation (check all that apply) [4 Pts.]

Integrated Benchmarking ☒

Quantitative Integration ☐

Functional Hierarchy ☐

Benchmarking ☒

Fundamental Objective ☐

Name

Sam Olson

II. Alternative Generation [35 Pts]

You are part of the design team responsible for developing a high-range rocket; your team has devised the following parameters conveyed in the table below.

Solution Design Parameters		
Search Technique	Guidance	Nozzle Diameter (mm)
Forward Looking Infrared	Inertial	>0.8
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Audio	Wire	$1.6 \leq \text{Nozzle Diameter} \leq 1.8$
GSR		

7. Using Zwicky's Morphology, how many total possible combinations are there for the team to consider? Show your work. [3 Pts]

$$4 \times 3 \times 3$$

Answer: 36

8. You are designing a new electric scooter for urban commuting, and your team proposes a 15kg scooter with a 5,000 mAh battery. [5 Pts]

You have researched two similar scooters:

- Scooter X is 18kg and has a 4,000 mAh battery. It can travel 25 miles on a single charge.
- Scooter Y is 12kg and has a 5,500 mAh battery. It can travel 40 miles on a single charge.

Using benchmarking, what is a reasonable rough estimate for how far the new scooter will be able to travel?

Answer: 33 miles

Justification for your rough estimate: it is halfway between scooters X and Y in weight, and just over the midpoint in battery capacity. It is

reasonable that the new scooter has a range that is just over the midpoint of the other two scooters' ranges.

Name

Sam Olson

9. MI6 is interested in acquiring a new Multi-Role Adaptive Rifle that can fire the 7.62mm, .300 Norma Magnum and .338 Norma Magnum. "M" insists, "the sniper rifle shall engage E-type silhouette targets at 1,500m with a greater than 50% hit rate during daylight, 0-5knt winds, and 24 degrees Celsius conditions." Without building a prototype, what are **two** ways MI6 could estimate a sniper rifle's ability to meet the requirement? What is one advantage and one disadvantage of each approach? [6 Pts]

Method 1: Computer modeling

Advantage: Cheap, precise

Disadvantage: Does not account for human operator → ~~cannot~~ only accounts for what simulation allows

Method 2: (Integrated) benchmarking

Advantage: Takes known capabilities from all parts to create estimate

Disadvantage: Parts do not always work in tandem and retain the same capabilities

10. Maj Boothroyd of Q-Branch created four alternatives for you to consider. The "must have" success criteria are search distance, guidance fidelity, and thrust. Based on the table below, which (if any) of the alternatives will be **screened out** and **why**? [5 Pts]

Feasibility Screening Matrix				Go / NoGo
Alternative	Search Distance [m] (>1000 m)	Guidance Fidelity [%] (More than 75%)	Thrust [lbs] (>1000 lbs)	
Alt 1: Moonraker	1200	95	1010	Go
Alt 2: Thunderball	999	77	1120	margin No Go
Alt 3: Risico	800	82	900	No Go
Alt 4: Icebreaker	1150	85	1000	margin No Go

Why Go or NoGo: Alt 1 meets all requirements. GO

- Both Alt 2 and Alt 4 could be screened out for not meeting search distance and thrust criteria, respectively. However, the margin of error is very small, and to disqualify an otherwise promising alternative based on the difference of 2 meters or 1 pound is silly. Technically NoGo, but should be considered.
- Alt 3, on the other hand, significantly misses the mark on search distance and thrust. Clear NoGo

Name _____

11. Below is data collected from four (4) smartphone alternatives for battery life (hours), cost (\$), and charging time (minutes). Your engineering team has developed a new fast-charge adapter that can be added to the phones. The cost of the adapter is \$100 added to the price of the phone. Application of the adapter results in a 5% increase in battery life (rounded to the nearest whole number, do not round up) and a 15% decrease in charging time (rounded to two decimal places). If the fast-charge adapter is applied to the alternatives below, which alternative(s) meet the screening criteria (fill in the screening matrix and show all work)? [16 Pts]

iPhone 14: 20 hours, \$799, and 85 minutes

Samsung Galaxy S23: 22 hours, \$849, and 70 minutes

Google Pixel 7: 18 hours, \$599, and 90 minutes

OnePlus 11: 24 hours, \$699, and 65 minutes

	Battery Life	Cost	Charging time (min)	Go/NoGo
Screening Criteria	≥ 22 hours	$< \$800$	≤ 70 min	
iPhone 14	20 21 hours	\$899	85 72.25 min	No Go
Samsung Galay S23	23 hours	\$949	70 59.5 min	No Go
Google Pixel 7	18 18.9 hours	\$699	90 76.5 min	No Go
OnePlus 11	25 hours	\$799	55.25 min	Go

OnePlus 11 meets all screening criteria and is the logical choice

Name

Sam Olson**III. Cost Analysis [35 Pts]**

12. Coach Q runs a professional soccer training academy. Which of the following are fixed costs and which are variable costs [4 Pts]:

- a) Cost of utilities at the training facility
Fixed ☒
Variable ☐
- b) Cost of renting the training facility for the season
Fixed ☒
Variable ☐
- c) Cost of water and sports drinks provided to players during training
Fixed ☐
Variable ☒
- d) Cost of salaries for the coaching staff
Fixed ☒
Variable ☐

13. Coach Mezze manages a sports team that partners with Cava to provide Mediterranean-inspired meals for players during training camps and post-game events. Which of the following are recurring costs and which are non-recurring costs [4 Pts]:

- a) Cost to design a custom cava-branded food truck for event catering
Recurring ☐
Non-recurring ☒
- b) Cost of utilities for the Cava catering setup at event venues
Recurring ☒
Non-recurring ☐
- c) Cost to build a new Cava kitchen facility for preparing event meals
Recurring ☐
Non-recurring ☒
- d) Cost of annual training for cava staff on Mediterranean cuisine preparation.
Recurring ☒
Non-recurring ☐

Name Sam Olson

14. Cava Pita Co. operates a chain of Mediterranean fast-casual restaurants and provides catering services for corporate events. Are the following costs direct or indirect costs, and why?
[6 Pts]

a) Cost of a chef hired specifically to prepare Cava's signature harissa sauce for catering orders

Direct ☒Indirect ☐Reason: contributes directly to Cava Pita Co's output, which is food.

b) Cost of utilities (electricity and water) for the Cava restaurant kitchen, fielding both catering orders and other restaurant operations

Direct ☐Indirect ☒Reason: Does not directly contribute to output, but without electricity and water the company could not run.

Name _____

15. Coach Falafel is the manager of a sports team that partners with Cava to provide Mediterranean meals for players during away games. To support the catering operations, Coach Falafel needs to transport Cava's catering equipment and staff from the Cava restaurant (located 50 miles from the team's stadium) to the game venue. He is considering two transportation options: Option 1 (Catering Van): Purchase a used catering van to transport equipment and staff; Option 2 (Rented Transport): Rent a shuttle service to transport equipment and staff, supplemented by a rideshare service for staff commuting to the shuttle pickup point. He completes a 4-year cost analysis to decide which option to select. Use the table below to calculate each option's Net Present Value (NPV). [20 pts]

	Catering van	Rented Transport
Initial Cost	\$30,000 (Used catering van)	\$500 (Specialized equipment cases)
Annual Maintenance/Operating Cost (The average cost you will pay for maintenance and operation)	\$3,000 (van upkeep, insurance)	\$1,200 (cleaning supplies, meal allowance)
Annual Transportation Cost	\$3,500 (gas, tolls)	\$7,000 (shuttle rental)
Resale Value	\$15,000	\$200
Interest Rate (i)	8%	

Round your answers to the nearest dollar and indicate the appropriate sign (positive or negative).

Catering Van NPV:

$$(-30000 - 6944.44 - 6430.04 - 5953.74 + 5512.72)$$

$$= -43815.49$$

Year 0: -30000
 Year 1: -7500
 Year 2: -7500
 Year 3: -7500
 Year 4: -7500 + 15000

$$NPV = -30000 - \frac{7500}{1.08} - \frac{7500}{1.08^2} - \frac{7500}{1.08^3} + \frac{15000}{1.08^4}$$

$$= -43815.49$$

$$P = F(1+i)^{-N}$$

$$P^1 = -7500(1.08)^{-1}$$

$$P^4 = (-7500 + 15000)(1.08)^{-4}$$

Rented Transportation NPV:

$$(-500 - 7592.59 - 7030.18 - 6509.43 - 5880.74)$$

$$= -27512.43$$

Answer: -27512.43

Year	Cost	NPV
0	500	-500
1	8200	-7592.59
2	8200	-7030.18
3	8200	-6509.43
4	8200 + 200	-5880.74

$$P^1 = -8200(1.08)^{-1}$$

$$P^4 = (-8200 + 200)(1.08)^{-4}$$

16. If cost is the only concern in 14, which option should Coach Falafel pick? [1 Pt]

Catering Van ☐Rented Transportation ☒

Poor Student Work

Section: _____

SE301 WPR #2 AT 26-1

Name: Frank Frawley

Departure Time: 0818

**DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996**

**SE301 WPR #2
26-1
VERSION (A)**

1. This is a **75-minute** exam worth **90 pts.** The exam consists of **8 pages.** Ensure that your exam is complete.
2. Place your name on each page in the space provided.
3. **AUTHORIZED REFERENCES:** You may use a pencil, straight edge (ruler, protractor, etc.), a 3 X 5 index card (front and back), and your issued hand-held calculator. You are allowed to use the notecard for the final 20 minutes of the exam.
4. **SHOW ALL YOUR WORK!** Unsupported or missing work earns little or no partial credit. Answer all questions to the fullest extent possible.
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SECTION	POSSIBLE POINTS	SCORE
I. Solution Design & Design Methods	20	
II. Alternative Generation	35	
III. Cost Analysis	35	
TOTAL	90	

Name Fradeliric**I. Solution Design & Design Methods [20 Pts]****1. True or False: [2 Pts] (check one)**

The best design is almost always in the middle rather than an extreme option.

True ☒False ☐**2. True or False: [2 Pts] (check one)**

The first stage of the ideation process is to generate alternatives.

True ☐False ☒**3. Which of the following are principles of good idea generation: [4 Pts]**Encourage wild ideas ☐Omit early ideas ☐Create numerous ideas ☒Do not criticize ideas ☒Build on ideas ☒**4. Which of the following are examples of idea generation methods (check all that apply)? [4 Pts]**Unstructured Brainwriting ☒Oracle Method ☐Structured Brainstorming ☒Casino Royale Method ☐All of the above ☐**5. Which of the following are types of errors to avoid in the ethics of solution design.**

Check all that apply. [4 Pts]

Anecdotal evidence ☒Vested interest ☒Strawman ☒Stochastic modeling ☐Anachronistic argumentation ☐None of the above ☐**6. Which of the following are approaches to alternative generation (check all that apply) [4 Pts.]**Integrated Benchmarking ☒Quantitative Integration ☐Functional Hierarchy ☐Benchmarking ☒Fundamental Objective ☐

Name Frederic

II. Alternative Generation [35 Pts]

You are part of the design team responsible for developing a high-range rocket; your team has devised the following parameters conveyed in the table below.

Solution Design Parameters		
Search Technique	Guidance	Nozzle Diameter (mm)
Forward Looking Infrared	Inertial	>0.8
Laser	Global Positioning System	$0.81 \leq \text{Nozzle Diameter} < 1.6$
Audio	Wire	$1.6 \leq \text{Nozzle Diameter} \leq 1.8$
GSR		

7. Using Zwicky's Morphology, how many total possible combinations are there for the team to consider? Show your work. [3 Pts]

Search technique guidance nozzle diameter
4 3 3

Answer: 36

$$4 \times 3 \times 3 = 36$$

8. You are designing a new electric scooter for urban commuting, and your team proposes a 15kg scooter with a 5,000 mAh battery. [5 Pts]

You have researched two similar scooters:

- Scooter X is 18kg and has a 4,000 mAh battery. It can travel 25 miles on a single charge.
- Scooter Y is 12kg and has a 5,500 mAh battery. It can travel 40 miles on a single charge.

Using benchmarking, what is a reasonable rough estimate for how far the new scooter will be able to travel?

Answer: 32.5 miles

Justification for your rough estimate: given that the weight and battery conditions of our scooter are exactly in the middle of scooter X and Y, it is reasonable to believe the distance would proportionally adhere to the trend of weight and battery to distance.

Name Frankelizio

- * 9. MI6 is interested in acquiring a new Multi-Role Adaptive Rifle that can fire the 7.62mm, .300 Norma Magnum and .338 Norma Magnum. "M" insists, "the sniper rifle shall engage E-type silhouette targets at 1,500m with a greater than 50% hit rate during daylight, 0-5knt winds, and 24 degrees Celsius conditions." Without building a prototype, what are **two** ways MI6 could estimate a sniper rifle's ability to meet the requirement? What is one advantage and one disadvantage of each approach? [6 Pts]

Method 1: Existing IdeaAdvantage: Already functions, can be studiedDisadvantage: more costly,Method 2: Bench marking using a range of different rifles/calibers with respective proven capabilitiesAdvantage: Save money and timeDisadvantage: Could be very inaccurate

10. Maj Boothroyd of Q-Branch created four alternatives for you to consider. The "must have" success criteria are search distance, guidance fidelity, and thrust. Based on the table below, which (if any) of the alternatives will be **screened out** and **why**? [5 Pts]

Feasibility Screening Matrix				Go / NoGo
Alternative	Search Distance [m] (>1000 m)	Guidance Fidelity [%] (More than 75%)	Thrust [lbs] (>1000 lbs)	
Alt 1: Moonraker	✓ 1200	✓ 95	✓ 1010	Go
Alt 2: Thunderball	✗ 999	✓ 77	✓ 1120	No Go
Alt 3: Risico	✗ 800	✓ 82	✗ 900	No Go
Alt 4: Icebreaker	✓ 1150	✓ 85	✓ 1000	Go

Why Go or NoGo:

Thunderball and Risico were NoGos because they did not meet "must have" success criteria of one or more requirements

Moonraker and Icebreaker got Go's because they met all success criteria

Name Frankie

11. Below is data collected from four (4) smartphone alternatives for battery life (hours), cost (\$), and charging time (minutes). Your engineering team has developed a new fast-charge adapter that can be added to the phones. The cost of the adapter is \$100 added to the price of the phone. Application of the adapter results in a 5% increase in battery life (rounded to the nearest whole number, do not round up) and a 15% decrease in charging time (rounded to two decimal places). If the fast-charge adapter is applied to the alternatives below, which alternative(s) meet the screening criteria (fill in the screening matrix and show all work)? [16 Pts]

iPhone 14: 20 hours, \$799, and 85 minutes

Samsung Galaxy S23: 22 hours, \$849, and 70 minutes

Google Pixel 7: 18 hours, \$599, and 90 minutes

OnePlus 11: 24 hours, \$699, and 65 minutes

	Battery Life	Cost	Charging time (min)	Go/NoGo
Screening Criteria	≥ 22 hours	$< \$800$	≤ 70 min	
iPhone 14	21 hrs	\$899	72.25 min	No Go
Samsung Galaxy S23	23 hrs	\$949	59.5 min	No Go
Google Pixel 7	18 hrs	\$699	76.5 min	No Go
OnePlus 11	25 hrs	\$799	55.25 min	Go

Name Franelizio**III. Cost Analysis [35 Pts]**

12. Coach Q runs a professional soccer training academy. Which of the following are fixed costs and which are variable costs [4 Pts]:

- a) Cost of utilities at the training facility
Fixed ☐
Variable ☒
- b) Cost of renting the training facility for the season
Fixed ☒
Variable ☐
- c) Cost of water and sports drinks provided to players during training
Fixed ☐
Variable ☒
- d) Cost of salaries for the coaching staff
Fixed ☒
Variable ☐

13. Coach Mezze manages a sports team that partners with Cava to provide Mediterranean-inspired meals for players during training camps and post-game events. Which of the following are recurring costs and which are non-recurring costs [4 Pts]:

- a) Cost to design a custom cava-branded food truck for event catering
Recurring ☐
Non-recurring ☒
- b) Cost of utilities for the Cava catering setup at event venues
Recurring ☒
Non-recurring ☐
- c) Cost to build a new Cava kitchen facility for preparing event meals
Recurring ☐
Non-recurring ☒
- d) Cost of annual training for cava staff on Mediterranean cuisine preparation.
Recurring ☒
Non-recurring ☐

Name Foudehria

14. Cava Pita Co. operates a chain of Mediterranean fast-casual restaurants and provides catering services for corporate events. Are the following costs direct or indirect costs, and why?
[6 Pts]

a) Cost of a chef hired specifically to prepare Cava's signature harissa sauce for catering orders

Direct ☐Indirect ☒

Reason: The owner of Cava Pita Co. doesn't directly interact/physically pay this chef.

b) Cost of utilities (electricity and water) for the Cava restaurant kitchen, fielding both catering orders and other restaurant operations

Direct ☒Indirect ☐

Reason: This cost is directly handled by the owner.

Name Fradelizio

15. Coach Falafel is the manager of a sports team that partners with Cava to provide Mediterranean meals for players during away games. To support the catering operations, Coach Falafel needs to transport Cava's catering equipment and staff from the Cava restaurant (located 50 miles from the team's stadium) to the game venue. He is considering two transportation options: Option 1 (Catering Van): Purchase a used catering van to transport equipment and staff; Option 2 (Rented Transport): Rent a shuttle service to transport equipment and staff, supplemented by a rideshare service for staff commuting to the shuttle pickup point. He completes a 4-year cost analysis to decide which option to select. Use the table below to calculate each option's Net Present Value (NPV). [20 pts]

	Catering van	Rented Transport
Initial Cost	\$30,000 (Used catering van)	\$500 (Specialized equipment cases)
Annual Maintenance/Operating Cost (The average cost you will pay for maintenance and operation)	\$3,000 (van upkeep, insurance)	\$1,200 (cleaning supplies, meal allowance)
Annual Transportation Cost	\$3,500 (gas, tolls)	\$7,000 (shuttle rental)
Resale Value	\$15,000	\$200
Interest Rate (i)	8%	

Round your answers to the nearest dollar and indicate the appropriate sign (positive or negative).

Catering Van NPV:
Costs:
Initial: -30,000
Annual: -6,500
Resale: +15,000

$$FV = PV(1+i)^N$$

$$= -6,500(1+0.08)^4$$

$$FV = -8,843.18$$

After 4 years:

$$\begin{aligned} &-30,000 \\ &-8,843.18 \\ &+15,000 \\ &\hline &-23,843.18 \end{aligned}$$

\$ -23,843.18

Answer: _____

Rented Transportation NPV:

Costs:
Initial: -500
Annual: -8,200
Resale: +200

$$FV = -8,200(1+0.08)^4$$

$$FV = -11,156.01$$

After 4 years:

$$\begin{aligned} &-500 \\ &-11,156.01 \\ &+200 \\ &\hline &-11,456.01 \end{aligned}$$

Answer: -11,456.01

16. If cost is the only concern in 14, which option should Coach Falafel pick? [1 Pt]

Catering Van ☐

Rented Transportation ☒



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB I

WPR #3

Instructor Solution

Section: _____

SE301 WPR #3 AT 26-1

Name: _____

Departure Time: _____

**DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996**

**SE301 WPR #3
AT 26-1
VERSION (A)**

1. This is a **75-minute** exam **worth 90 pts**. The exam consists of **9 pages**. Ensure that your exam is complete.
2. Place your name on each page in the space provided.
3. **AUTHORIZED REFERENCES:** You may use a pencil, straight edge (ruler, protractor, etc.), a 3 X 5 index card (front and back), and your issued hand-held calculator. The notecard is accessible for last 20 minutes of exam period.
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SECTION	POSSIBLE POINTS	SCORE
I. Decision Making	29	
II. Performance Modeling	20	
III. Sensitivity Analysis	23	
IV. Risk Analysis	18	
TOTAL	90	

Name _____

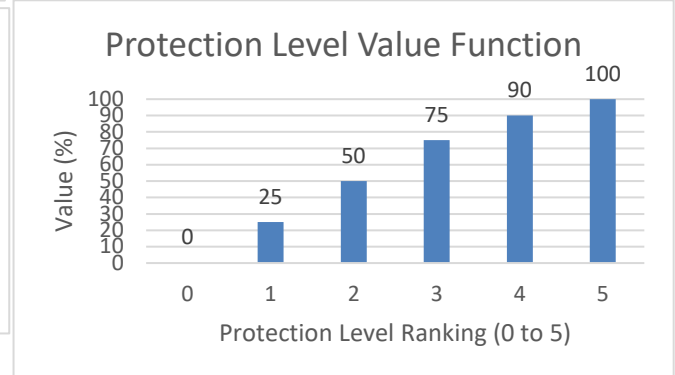
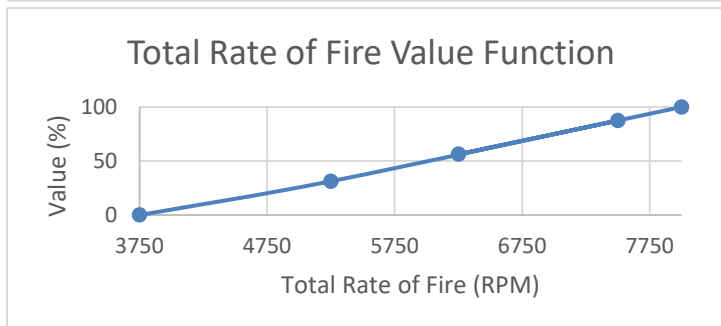
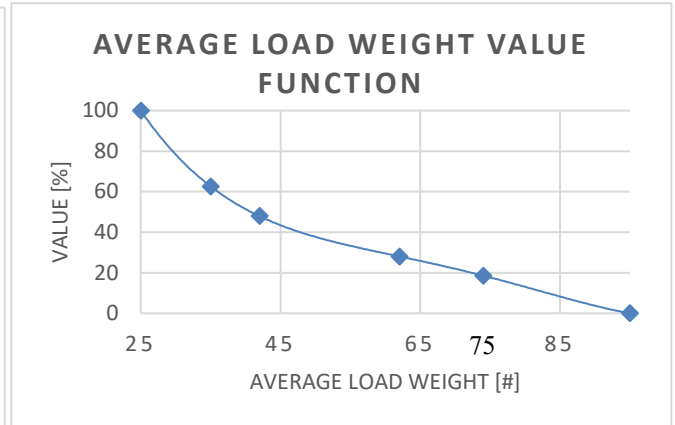
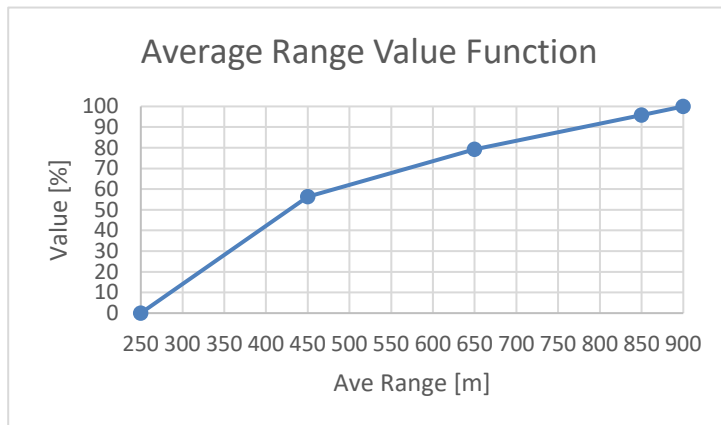
I. Decision Making [29 Pts]

1. West Point has decided to put together a Sandhurst Squad. The Academy hopes that this squad will improve Soldier readiness and training by assisting in range of coverage and firing capacities. You have been asked to assess the value of different possible options. You've collected the following information based on stakeholder values:

RAW DATA MATRIX

Candidate Solutions	Ave Range (m)	Ave Load Weight (#)	Total Rate of Fire (RPM)	Protection Level (0 – 5 Scale)
Ace of Base	450	42	5250	1
Heavy Metal	950	75	7750	4
Bad Blood	850	62	7500	3
Funky Town	650	35	6250	2
Ideal	900	25	8000	5

Determine the values for cells A, B, C, D, and E in the value matrix below using the following value functions: [10 Pts]



Name _____

VALUE MATRIX

Candidate Solutions	Ave Range (m)	Ave Load Weight (#)	Total Rate of Fire (RPM)	Protection Level (0 to 5 Scale)
Ace of Base	56	48	31	D
Heavy Metal	100	C	95	90
Bad Blood	A	28	90	E
Funky Town	B	60	70	50

A: 95 B: 80 C: 20 D: 25 E: 75

2. Using the unweighted values in Question 1's value matrix and the global weights below, calculate the values for cells A, B, C, D and E in the weighted value matrix (round up to nearest whole number). [10 Pts]

Value Measure	w _i	Value Measure	w _i
Ave Range	0.30	Total Rate of Fire	0.08
Ave Load Weight	0.20	Protection Level	0.42

WEIGHTED VALUE MATRIX

Candidate Solutions	Ave Range (m)	Ave Load Weight (#)	Total Rate of Fire (RPM)	Protection Level (0 to 5 Scale)
Ace of Base	17	10	3	D
Heavy Metal	30	C	8	38
Bad Blood	A	6	8	E
Funky Town	B	12	6	21

A: 29 B: 24 C: 4 D: 11 E: 32

3. How do you calculate the *total system value* for each candidate solution? [3 Pts]

- ☐ Multiply each value matrix column by corresponding global weights.
- ☒ Sum the row of the weighted value matrix for each candidate solution
- ☐ Use the value functions.
- ☐ Vary the swing weight for a value measure from 0 to 100.

4. Fill in the blanks. The three sub-tasks of the Decision Making phase of the SDP are Value Scoring/costing, sensitivity / risk analysis, and improvement / tradeoff analysis. [3 Pts]

Name _____

5. Describe the notation term w_i of the additive value model equation requested below. [3 Pts]

$$V(x_j) = \sum_{i=1}^n [w_i \times v_i(x_{j,i})]$$

w_i = Global weight, value of importance across value measures the decision maker considers

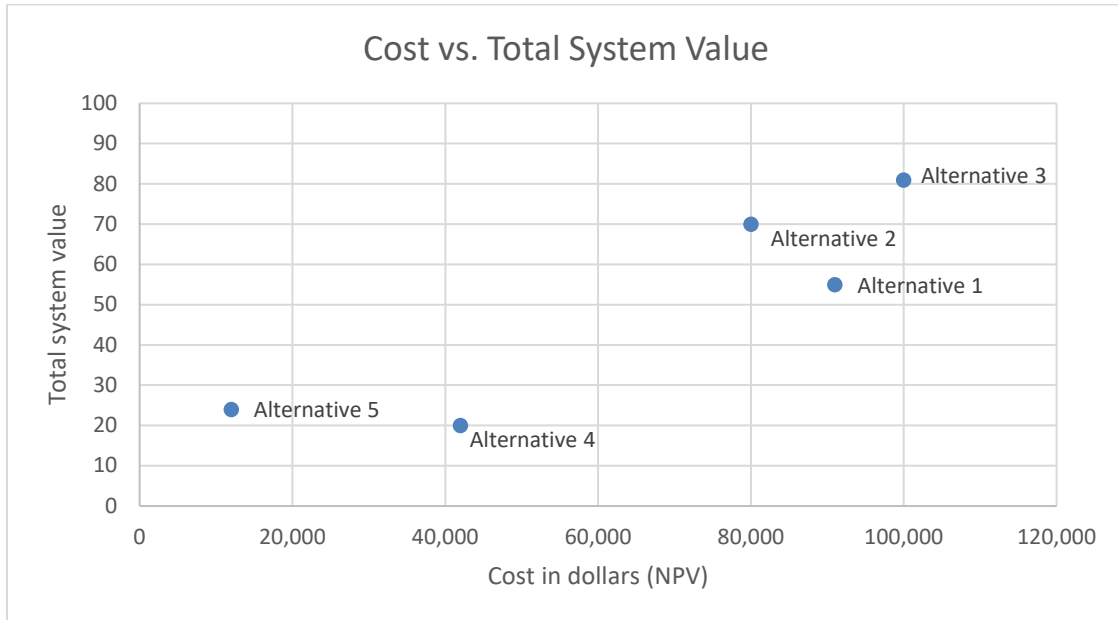
Performance Modeling. [20 Pts]

6. Explain why models are useful. [3 Pts]

Flexibility, time, money

Name _____

7. Assume the chart below reflects the Total Value and Cost of the final candidate solutions.



a. Identify the candidate solution(s) that are dominated. [2 pts]

Dominated solution: Alternative 4

Dominated solution: Alternative 1

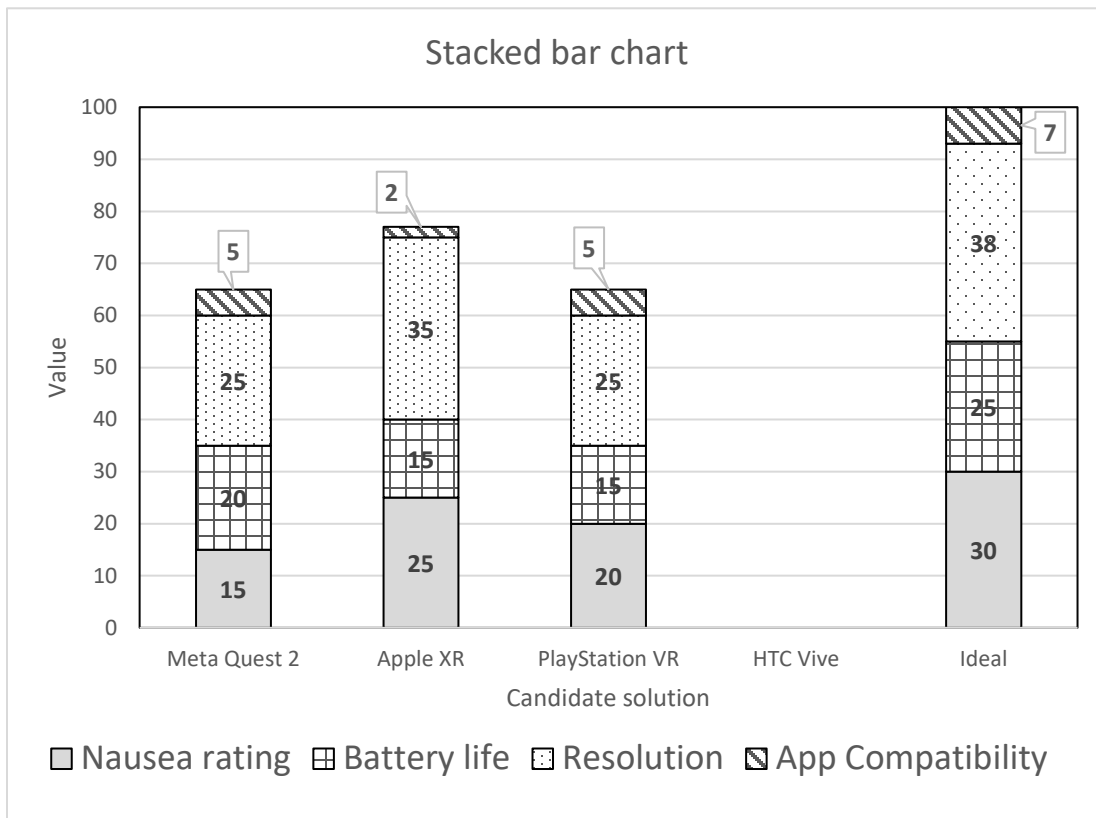
b. Explain what it means for a solution to be dominated and why the above alternatives are dominated. [4 pts]

More cost, less value

Name _____

8. Based on the Weighted Value Matrix of virtual reality headsets below, complete the stacked bar chart, by hand, for the **HTC Vive** candidate solution. Label the values for each segment: [4 Pts]

Candidate Solutions	Nausea rating	Battery life	Resolution	App Compatibility	Total Value
Meta Quest 2	15	20	25	5	65
Apple XR	25	15	35	2	77
PlayStation VR	20	15	25	5	65
HTC Vive	22	18	20	7	67
Ideal	30	25	38	7	100



9. Which VR headset should the decision maker pick? [2 Pts]

- ☐ Meta quest 2
- ☐ Apple XR
- ☐ PlayStation VR
- ☐ HTC Vive
- ☒ Not enough information to say

Name _____

10. Using the stacked bar chart, which value measure is the Apple XR losing the most value on compared to the ideal? [2 Pts]

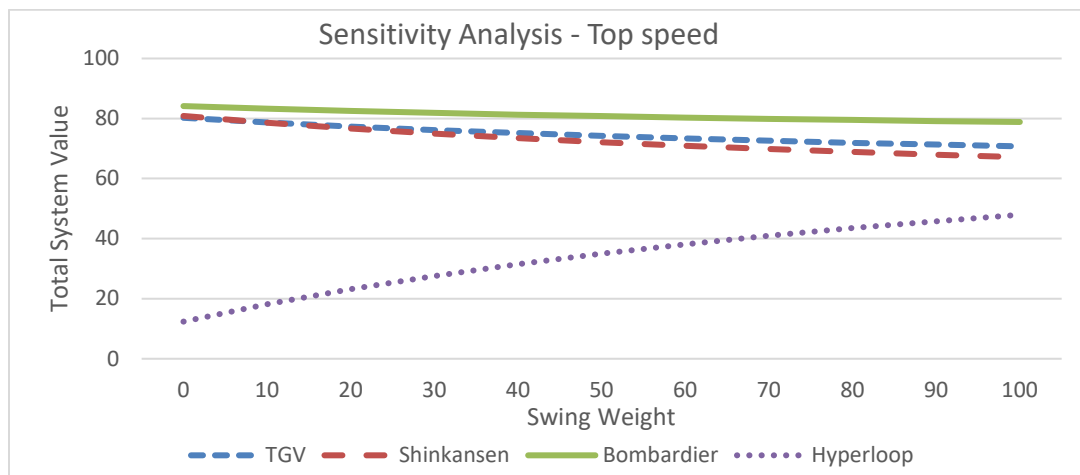
- ☐ Nausea rating
- ☒ Battery life
- ☐ Resolution
- ☐ App Compatibility
- ☐ Not enough information to say

11. What is a design change that the Apple XR could make that would improve its performance on that value measure? [3 Pts]

Increase battery life through updated design, etc.

II. Sensitivity Analysis [23 Pts]

12. You want to run a sensitivity analysis on the high-speed rail analysis, specifically the top speed measure, which had an original swing weight of 80. The results are in the graph below. [4 Pts]



Is your overall scoring process sensitive to the swing weight of the "Top speed" measure? And is it significant? (YES or NO)

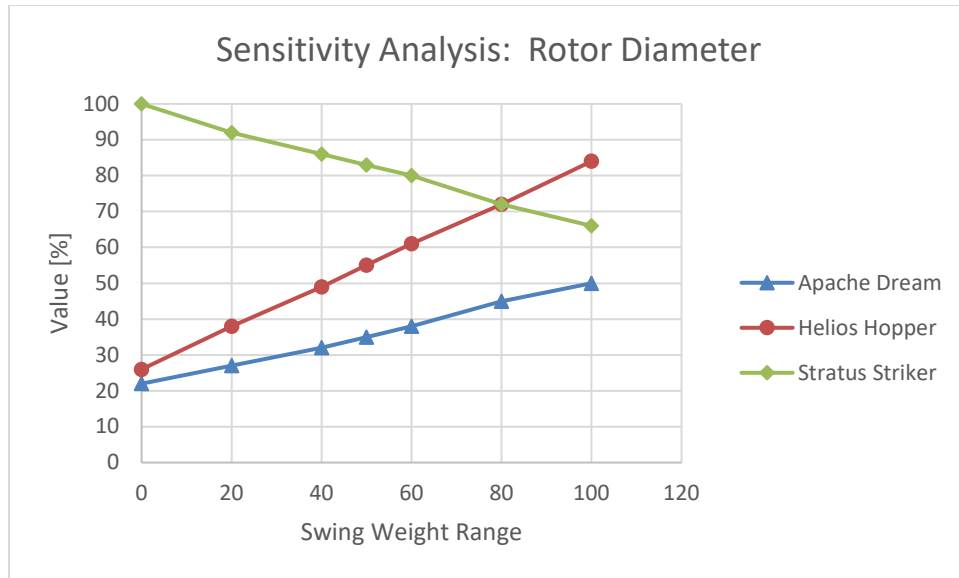
Sensitive: Y or ☒ N (Circle one)

Significant: Y or ☒ N (Circle one)

13. Assume that after you conducted Sensitivity Analysis on your value measures, none of them are sensitive to change. Explain what this means for your decision authority. [3 Pts]

14. You want to run a sensitivity analysis, specifically regarding rotor diameter, which had an original swing weight of 50. The results are in the graph below. [4 Pts]

Name _____



Sensitive: **Y** or N (Circle one)

Significant: Y or **N** (Circle one)

15. Explain why the process is sensitive and/or significant [4 Pts]

The highest value alternative changes across the swing weight = significant. If it is within +/- 10% of the swing weight = significant.

16. Explain why the design team conducts a sensitivity analysis in the SDP and what actions the design team would take if a value measure is considered sensitive/significant [8 Pts]



Name _____

III. Risk Analysis [18 Pts]

17. Which of the following are risk types in systems engineering? *Choose all that apply.* [2 Pts]

- ☒ Schedule
- ☐ Safety
- ☐ Environmental
- ☒ Programmatic
- ☐ Quality

18. Identify the risk mitigation methods in SE301. Choose all that apply. [2 Pts]

- ☒ Avoid
- ☐ Delay
- ☐ Buffering
- ☒ Reduce
- ☐ Investigation

19. What order do the Systems Engineering risk management activities take place? [2 Pts]

- ☐ Identification, Acceptance, Mitigation
- ☐ Observe, Assessment, Management
- ☒ Identification, Assessment, Mitigation
- ☐ Identification, Assessment, Management

20. Read the following description of various risks at the Red River Army Depot. Then categorize each risk. [12 Pts]

The systems engineering team at the U.S. Army's Red River Army Depot in Texarkana, Texas is developing risk mitigation plans for the depot commander. The team reviewed maintainer safety incidents, platform rework rates, production delays, and quality assurance protocols.

In reviewing incident reports over the past several years, the team identified a pattern of injuries associated with the hydraulic press used for Bradley track shoe assembly. One critical incident caused a maintainer to lose two fingers when the press cycled unexpectedly during a jam-clearing procedure. The team determined that this type of injury, while life-altering, occurred only rarely.

Vehicles failing Final Quality Gates result in moderate degradation of depot throughput.

Recurring issues include improperly seated turret ring gear bolts, mismatched electrical connectors in the fire-control system, and hull welds that fail non-destructive testing. The team confirmed these defects appeared regularly in nearly every production batch.

The team also analyzed the risk of a catastrophic failure in the depot's only large-scale X-ray inspection chamber used for detecting internal casting flaws in transmission housings. A total loss of this system would severely cripple overhaul schedules and downstream unit readiness.

After exhaustive review of all equipment logs and downtime records, the team found zero documented instances of such a failure.

Name _____

Fill in the following risks using this risk assessment matrix:

Risk Assessment Matrix		Probability (expected frequency)				
		Frequent: Continuous, regular, or inevitable occurrences	Likely: Several or numerous occurrences	Occasional: Sporadic or intermittent occurrences	Seldom: Infrequent occurrences	Unlikely: Possible occurrences but improbable
Severity (expected consequence)		A	B	C	D	E
Catastrophic: Mission failure, unit readiness eliminated; death, unacceptable loss or damage	I	EH	EH	H	H	M
Critical: Significantly degraded unit readiness or mission capability; severe injury, illness, loss or damage	II	EH	H	H	M	L
Moderate: Somewhat degraded unit readiness or mission capability; minor injury, illness, loss, or damage	III	H	M	M	L	L
Negligible: Little or no impact to unit readiness or mission capability; minimal injury, loss, or damage	IV	M	L	L	L	L
Legend: EH - Extremely High Risk H - High Risk M - Medium Risk L - Low Risk						

Risk 1 description: _____ Hydraulic Press _____

Probability: Unlikely/Seldom Severity: Catastrophic/Critical Risk level: M _____

Risk 2 description: Quality Gates _____

Probability: Frequent Severity: Negligible Risk level: M _____

Risk 3 description: X-Ray Failure _____

Probability: Unlikely Severity: Catastrophic Risk level: M _____

Outstanding Student Work

Section: I 1

SE301 WPR #3 AT 26-1

Name: Abigail Araya

Departure Time: 1106 02 DEC 25

**DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996**

**SE301 WPR #3
AT 26-1
VERSION (A)**

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Name _____

I. Decision Making [29 Pts]

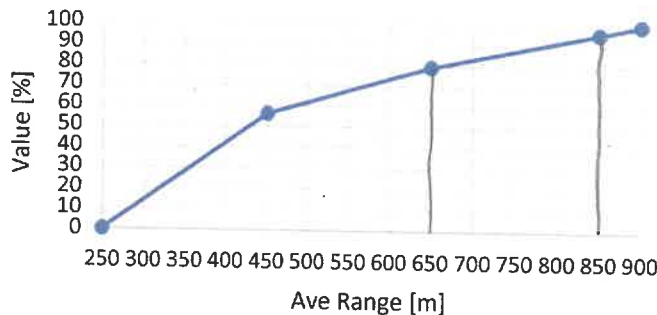
1. West Point has decided to put together a Sandhurst Squad. The Academy hopes that this squad will improve Soldier readiness and training by assisting in range of coverage and firing capacities. You have been asked to assess the value of different possible options. You've collected the following information based on stakeholder values:

RAW DATA MATRIX

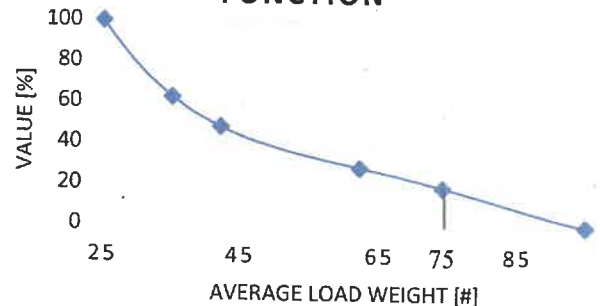
Candidate Solutions	Ave Range (m)	Ave Load Weight (#)	Total Rate of Fire (RPM)	Protection Level (0 – 5 Scale)
Ace of Base	450	42	5250	1
Heavy Metal	950	75	7750	4
Bad Blood	850	62	7500	3
Funky Town	650	35	6250	2
Ideal	900	25	8000	5

Determine the values for cells A, B, C, D, and E in the value matrix below using the following value functions: [10 Pts]

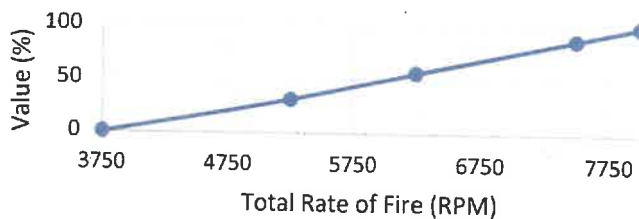
Average Range Value Function



AVERAGE LOAD WEIGHT VALUE FUNCTION



Total Rate of Fire Value Function



Protection Level Value Function



Name _____

VALUE MATRIX

Candidate Solutions	Ave Range (m)	Ave Load Weight (#)	Total Rate of Fire (RPM)	Protection Level (0 to 5 Scale)
Ace of Base	56	48	31	D
Heavy Metal	100	C	95	90
Bad Blood	A	28	90	E
Funky Town	B	60	70	50

A: 97 B: 80 C: 20 D: 25 E: 75

2. Using the unweighted values in Question 1's value matrix and the global weights below, calculate the values for cells A, B, C, D and E in the weighted value matrix (round up to nearest whole number). [10 Pts]

Value Measure	w _i	Value Measure	w _i
Ave Range	0.30	Total Rate of Fire	0.08
Ave Load Weight	0.20	Protection Level	0.42

WEIGHTED VALUE MATRIX

Candidate Solutions	Ave Range (m)	Ave Load Weight (#)	Total Rate of Fire (RPM)	Protection Level (0 to 5 Scale)
Ace of Base	17	10	3	D
Heavy Metal	30	C	8	38
Bad Blood	A	6	8	E
Funky Town	B	12	6	21

A: 29 B: 24 C: 4 D: 11 E: 32

3. How do you calculate the **total system value** for each candidate solution? [3 Pts]

- ☐ Multiply each value matrix column by corresponding global weights.
- ☒ Sum the row of the weighted value matrix for each candidate solution
- ☐ Use the value functions.
- ☐ Vary the swing weight for a value measure from 0 to 100.

4. Fill in the blanks. The three sub-tasks of the Decision Making phase of the SDP are value scoring + costing, sensitivity + risk analyses, and improvement + tradeoff analysis. [3 Pts]

Name _____

5. Describe the notation term W_i of the additive value model equation requested below. [3 Pts]

$$V(x_j) = \sum_{i=1}^n [w_i \times v_i(x_{j,i})]$$

$W_i =$ global weight of value measure i

The global weight is a metric used to show the relative importance of the value measures to each other, & the total sum of the global weight is equal to 1.

For example, for AIT J, w/ 3 value measures, it could be:

$$0.4(x_{j,1}) + 0.5(x_{j,2}) + 0.1(x_{j,3})$$

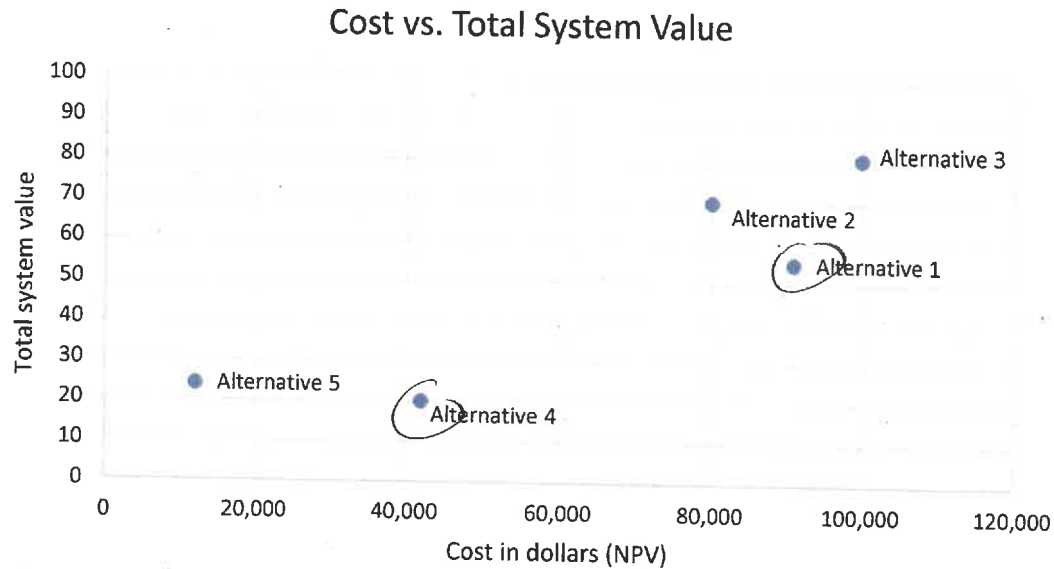
II. Performance Modeling. [20 Pts]

6. Explain why models are useful. [3 Pts]

They give us an abstract representation of a system that saves time & money & is flexible.

Name _____

7. Assume the chart below reflects the Total Value and Cost of the final candidate solutions.



a. Identify the candidate solution(s) that are dominated. [2 pts]

Dominated solution: Alternative 4

Dominated solution: Alternative 1

b. Explain what it means for a solution to be dominated and why the above alternatives are dominated. [4 pts]

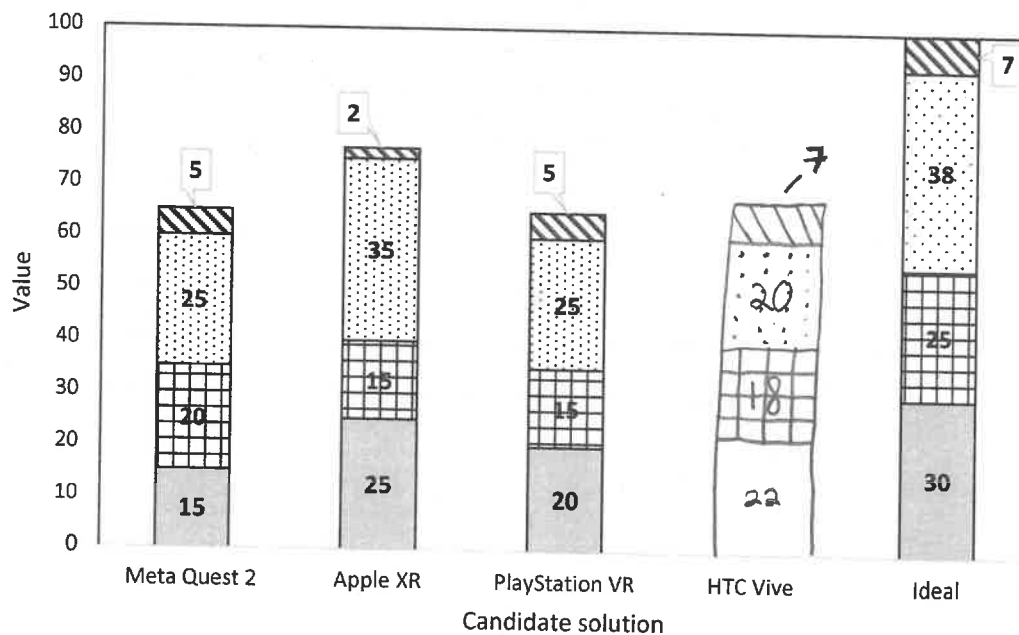
Dominated solns have a higher cost & lower value than other solns. Alt 4 is more expensive & has a lower total value than Alt 5, and the same for Alt 1 in comparison to Alt. 2.

Name _____

8. Based on the Weighted Value Matrix of virtual reality headsets below, complete the stacked bar chart, by hand, for the **HTC Vive** candidate solution. Label the values for each segment: [4 Pts]

Candidate Solutions	Nausea rating	Battery life	Resolution	App Compatibility	Total Value
Meta Quest 2	15	20	25	5	65
Apple XR	25	15	35	2	77
PlayStation VR	20	15	25	5	65
HTC Vive	22	18	20	7	67
Ideal	30	25	38	7	100

Stacked bar chart



☐ Nausea rating
 ☐ Battery life
 ☐ Resolution
 ☐ App Compatibility

9. Which VR headset should the decision maker pick? [2 Pts]

- ☐ Meta quest 2
- ☐ Apple XR
- ☐ PlayStation VR
- ☐ HTC Vive
- ☒ Not enough information to say

Name _____

10. Using the stacked bar chart, which value measure is the Apple XR losing the most value on compared to the ideal? [2 Pts]

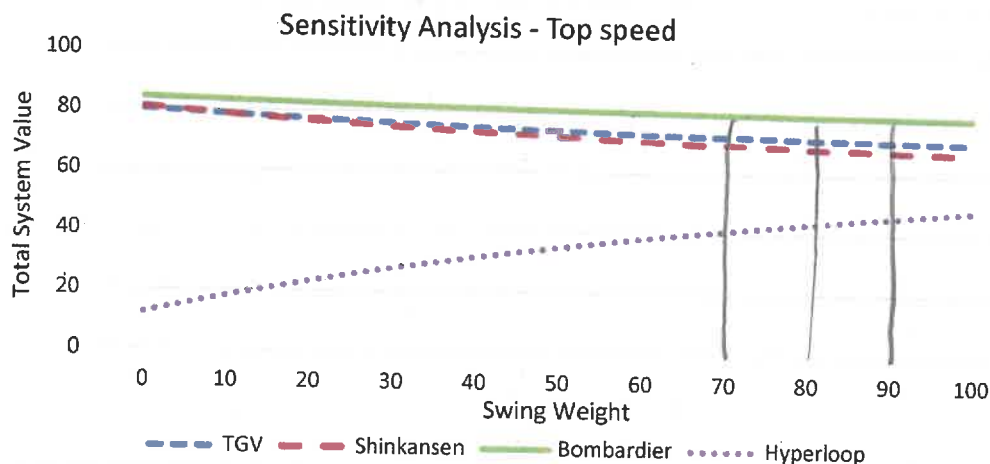
- ☐ Nausea rating
- ☒ Battery life
- ☐ Resolution
- ☐ App Compatibility
- ☐ Not enough information to say

11. What is a design change that the Apple XR could make that would improve its performance on that value measure? [3 Pts]

Assuming that the high resolution is likely a primary cause of battery drain, the designers could reduce the resolution slightly, which would prolong the battery life.

III. Sensitivity Analysis [23 Pts]

12. You want to run a sensitivity analysis on the high-speed rail analysis, specifically the top speed measure, which had an original swing weight of 80. The results are in the graph below. [4 Pts]



Is your overall scoring process sensitive to the swing weight of the "Top speed" measure? And is it significant? (YES or NO)

Sensitive: Y or **N (Circle one)**

Significant: Y or **N (Circle one)**

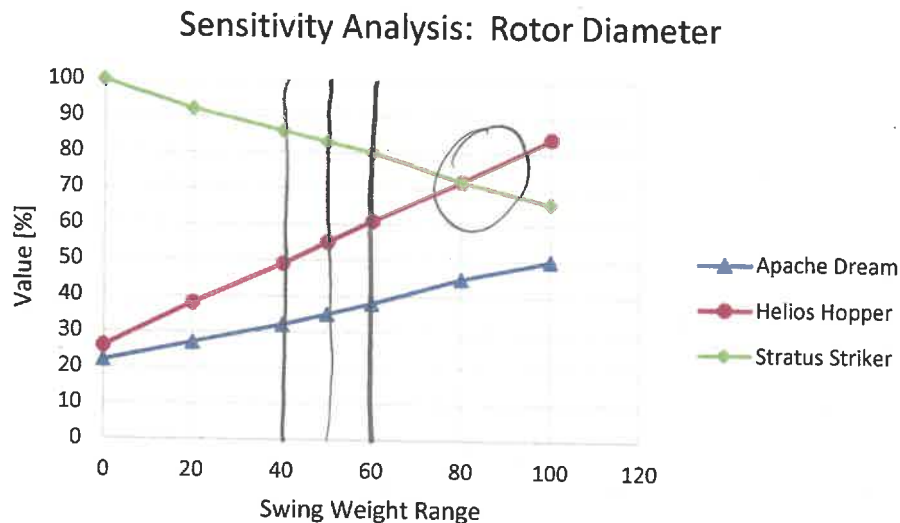
13. Assume that after you conducted Sensitivity Analysis on your value measures, none of them are sensitive to change. Explain what this means for your decision authority. [3 Pts]

This means that, despite changing the relative importance of the value of top speed for each of the candidate solutions, there was no change to which of the alternatives had the best value.

This means that ~~it~~ is the best option for that value measure.
that top alternative

Name _____

14. You want to run a sensitivity analysis, specifically regarding rotor diameter, which had an original swing weight of 50. The results are in the graph below. [4 Pts]



Sensitive Y or N (Circle one)

Significant: Y or N (Circle one)

15. Explain why the process is sensitive and/or significant [4 Pts]

IT IS SENSITIVE BECAUSE, AT SWING WEIGHT 80, THE BEST OPTION CHANGES FROM STRATUS STRIKER TO HELIOS HOPPER.

IT IS NOT SIGNIFICANT BECAUSE, WITHIN $\pm 10\%$ OF THE ORIGINAL SWING WEIGHT (MEANING, BETWEEN 40 & 60), THERE WAS NO CHANGE IN THE BEST OPTION.

16. Explain why the design team conducts a sensitivity analysis in the SDP and what actions the design team would take if a value measure is considered sensitive/significant [8 Pts]

a) They conduct this analysis to see whether/to what extent the relative importance placed upon value measures (which contribute to the tot. system value) affect the resulting best option. This helps stakeholders get a more complete picture of the strengths/weaknesses of each option.

b) what actions they would take:

- 1) tell stakeholders
- 2) possibly re-evaluate the swing weights of certain value measures
- 3) re-calculate, present new findings, go from there

Name _____

IV. Risk Analysis [18 Pts]

17. Which of the following are risk types in systems engineering? *Choose all that apply.* [2 Pts]

- ☒ Schedule
- ☐ Safety
- ☐ Environmental
- ☒ Programmatic
- ☐ Quality

18. Identify the risk mitigation methods in SE301. Choose all that apply. [2 Pts]

- ☒ Avoid
- ☐ Delay
- ☐ Buffering
- ☒ Reduce
- ☐ Investigation

*for avoid
accept
reduce
transfer*

19. What order do the Systems Engineering risk management activities take place? [2 Pts]

- ☐ Identification, Acceptance, Mitigation
- ☐ Observe, Assessment, Management
- ☒ Identification, Assessment, Mitigation
- ☐ Identification, Assessment, Management

id, assess, mitigation

20. Read the following description of various risks at the Red River Army Depot. Then categorize each risk. [12 Pts]

The systems engineering team at the U.S. Army's Red River Army Depot in Texarkana, Texas is developing risk mitigation plans for the depot commander. The team reviewed maintainer safety incidents, platform rework rates, production delays, and quality assurance protocols.

In reviewing incident reports over the past several years, the team identified a pattern of injuries associated with the hydraulic press used for Bradley track shoe assembly. One critical incident caused a maintainer to lose two fingers when the press cycled unexpectedly during a jam-clearing procedure. The team determined that this type of injury, while life-altering, occurred only rarely.

Vehicles failing Final Quality Gates result in moderate degradation of depot throughput.

Recurring issues include improperly seated turret ring gear bolts, mismatched electrical connectors in the fire-control system, and hull welds that fail non-destructive testing. The team confirmed these defects appeared regularly in nearly every production batch.

The team also analyzed the risk of a catastrophic failure in the depot's only large-scale X-ray inspection chamber used for detecting internal casting flaws in transmission housings. A total loss of this system would severely cripple overhaul schedules and downstream unit readiness. After exhaustive review of all equipment logs and downtime records, the team found zero documented instances of such a failure.

Name _____

Fill in the following risks using this risk assessment matrix:

Risk Assessment Matrix		Probability (expected frequency)				
		Frequent: Continuous, regular, or inevitable occurrences	Likely: Several or numerous occurrences	Occasional: Sporadic or intermittent occurrences	Seldom: Infrequent occurrences	Unlikely: Possible occurrences but improbable
Severity (expected consequence)		A	B	C	D	E
Catastrophic: Mission failure, unit readiness eliminated, death, unacceptable loss or damage	I	EH	EH	H	H	M
Critical: Significantly degraded unit readiness or mission capability, severe injury, illness, loss or damage	II	EH	H	H	M	L
Moderate: Somewhat degraded unit readiness or mission capability, minor injury, illness, loss, or damage	III	H	M	M	L	L
Negligible: Little or no impact to unit readiness or mission capability, minimal injury, loss, or damage	IV	M	L	L	L	L
Legend: EH - Extremely High Risk H - High Risk M - Medium Risk L - Low Risk						

Risk 1 description: loss of limb (2 fingers)

Probability: critical Severity: seldom Risk level: Medium

Risk 2 description: vehicle quality defects

Probability: moderate Severity: frequent Risk level: High

Risk 3 description: total failure/loss of inspection chamber

Probability: unlikely Severity: catastrophic Risk level: Medium

Fair-to-Good Student Work

Section: I - 1

SE301 WPR #3 AT 26-1

Name: Thomas Oakley

Departure Time: _____

**DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996**

**SE301 WPR #3
AT 26-1
VERSION (A)**

1. This is a **75-minute** exam worth **90 pts**. The exam consists of **9 pages**. Ensure that your exam is complete.
2. Place your name on each page in the space provided.
3. **AUTHORIZED REFERENCES:** You may use a pencil, straight edge (ruler, protractor, etc.), a 3 X 5 index card (front and back), and your issued hand-held calculator. The notecard is accessible for last 20 minutes of exam period.
4. **SHOW ALL YOUR WORK!** Unsupported or missing work earns little or no partial credit. Answer all questions to the fullest extent possible.
5. **EXTRA PAPER:** Use only the additional paper available at the front of the room. Attach the additional pages you use to your exam and turn them in.
6. Once you have completed the exam, fill in the departure time at the top of this page. Turn in your exam IAW your instructor's instructions.
7. This WPR will not be released from academic security until it has been returned. Before its return, cadets may not discuss any aspects of this exam with anyone except their SE301 instructor. Initial here TO

SECTION	POSSIBLE POINTS	SCORE
I. Decision Making	29	
II. Performance Modeling	20	
III. Sensitivity Analysis	23	
IV. Risk Analysis	18	
TOTAL	90	

Name _____

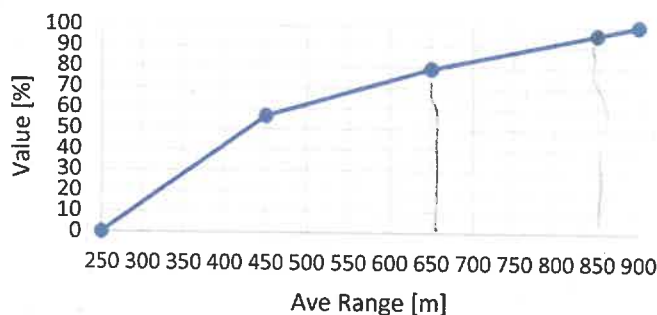
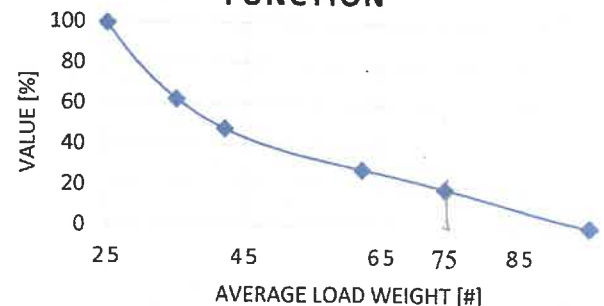
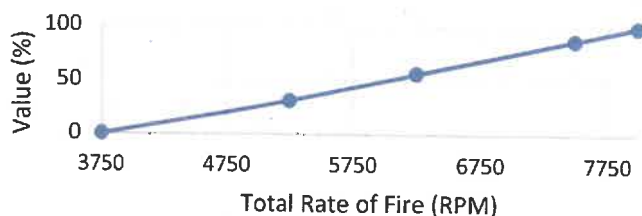
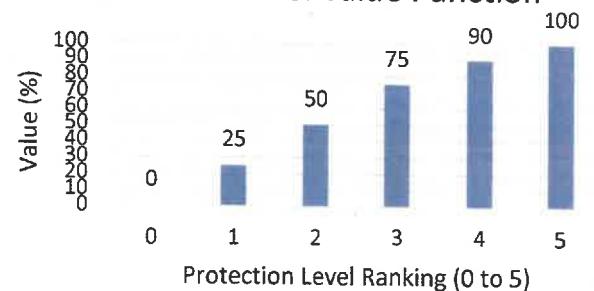
I. Decision Making [29 Pts]

1. West Point has decided to put together a Sandhurst Squad. The Academy hopes that this squad will improve Soldier readiness and training by assisting in range of coverage and firing capacities. You have been asked to assess the value of different possible options. You've collected the following information based on stakeholder values:

RAW DATA MATRIX

Candidate Solutions	Ave Range (m)	Ave Load Weight (#)	Total Rate of Fire (RPM)	Protection Level (0 – 5 Scale)
Ace of Base	450	42	5250	1
Heavy Metal	950	75	7750	4
Bad Blood	850	62	7500	3
Funky Town	650	35	6250	2
Ideal	900	25	8000	5

Determine the values for cells A, B, C, D, and E in the value matrix below using the following value functions: [10 Pts]

Average Range Value Function**AVERAGE LOAD WEIGHT VALUE FUNCTION****Total Rate of Fire Value Function****Protection Level Value Function**

Name _____

VALUE MATRIX

Candidate Solutions	Ave Range (m)	Ave Load Weight (#)	Total Rate of Fire (RPM)	Protection Level (0 to 5 Scale)
Ace of Base	56	48	31	D
Heavy Metal	100	C	95	90
Bad Blood	A	28	90	E
Funky Town	B	60	70	50

A: 96 B: 80 C: 20 D: 25 E: 75

2. Using the unweighted values in Question 1's value matrix and the global weights below, calculate the values for cells A, B, C, D and E in the weighted value matrix (round up to nearest whole number). [10 Pts]

Value Measure	w _i	Value Measure	w _i
Ave Range	0.30	Total Rate of Fire	0.08
Ave Load Weight	0.20	Protection Level	0.42

WEIGHTED VALUE MATRIX

Candidate Solutions	Ave Range (m)	Ave Load Weight (#)	Total Rate of Fire (RPM)	Protection Level (0 to 5 Scale)
Ace of Base	17	10	3	D
Heavy Metal	30	C	8	38
Bad Blood	A	6	8	E
Funky Town	B	12	6	21

A: 29 B: 24 C: 4 D: 11 E: 32

3. How do you calculate the **total system value** for each candidate solution? [3 Pts]

- ☐ Multiply each value matrix column by corresponding global weights.
☒ Sum the row of the weighted value matrix for each candidate solution
☐ Use the value functions.
☐ Vary the swing weight for a value measure from 0 to 100.

4. Fill in the blanks. The three sub-tasks of the Decision Making phase of the SDP are Value scoring and costing, sensitivity and risk analysis, and improvement + trade off analysis. [3 Pts]

Name _____

5. Describe the notation term w_i of the additive value model equation requested below. [3 Pts]

$$V(x_j) = \sum_{i=1}^n [w_i \times v_i(x_{j,i})]$$

$w_i =$ w_i is the global weight

The global weight is used to change specific values from a value matrix into the weighted value matrix which can then be used to find the total system value.

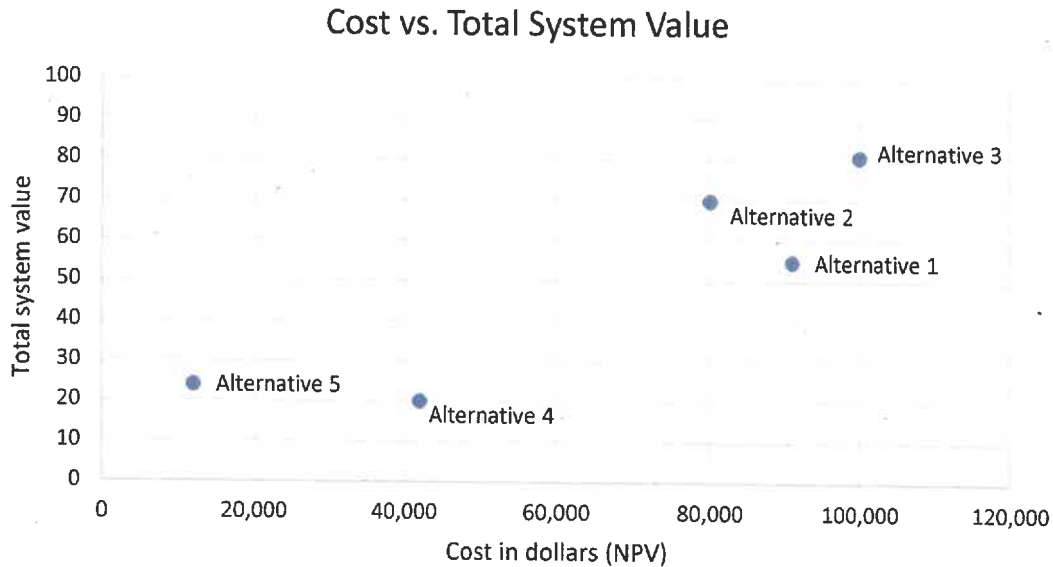
II. Performance Modeling. [20 Pts]

6. Explain why models are useful. [3 Pts]

Models save time and money and are flexible.

Name _____

7. Assume the chart below reflects the Total Value and Cost of the final candidate solutions.



a. Identify the candidate solution(s) that are dominated. [2 pts]

Dominated solution: Alternative 1

Dominated solution: Alternative 4

b. Explain what it means for a solution to be dominated and why the above alternatives are dominated. [4 pts]

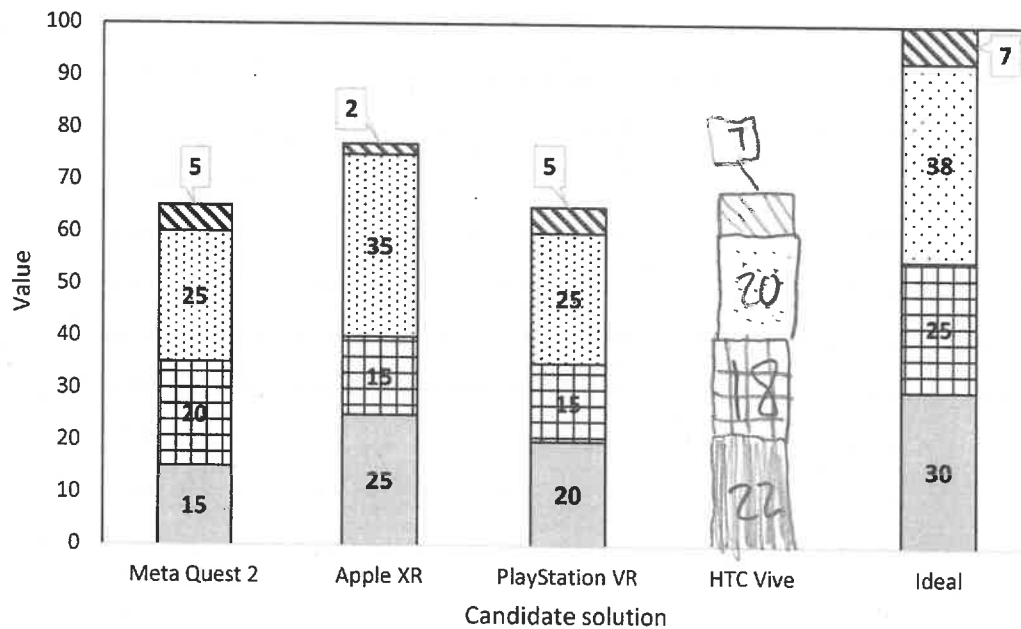
A dominated solution has a cheaper and higher valued alternative. Alternative 2 is cheaper and higher valued than alternative 1. Alternative 5 is cheaper and higher valued than alternative 4.

Name _____

8. Based on the Weighted Value Matrix of virtual reality headsets below, complete the stacked bar chart, by hand, for the **HTC Vive** candidate solution. Label the values for each segment: [4 Pts]

Candidate Solutions	Nausea rating	Battery life	Resolution	App Compatibility	Total Value
Meta Quest 2	15	20	25	5	65
Apple XR	25	15	35	2	77
PlayStation VR	20	15	25	5	65
HTC Vive	22	18	20	7	67
Ideal	30	25	38	7	100

Stacked bar chart



☐ Nausea rating
 ☐ Battery life
 ☐ Resolution
 ☐ App Compatibility

9. Which VR headset should the decision maker pick? [2 Pts]

- ☐ Meta quest 2
- ☐ Apple XR
- ☐ PlayStation VR
- ☐ HTC Vive
- ☒ Not enough information to say

Name _____

10. Using the stacked bar chart, which value measure is the Apple XR losing the most value on compared to the ideal? [2 Pts]

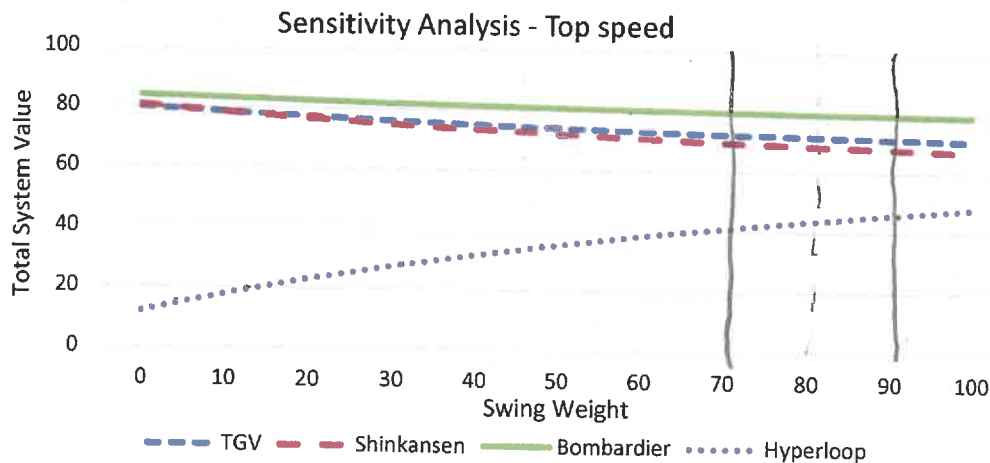
- ☐ Nausea rating
- ☒ Battery life
- ☐ Resolution
- ☐ App Compatibility
- ☐ Not enough information to say

11. What is a design change that the Apple XR could make that would improve its performance on that value measure? [3 Pts]

Apple XR should try to modify the design to include a better battery life.

III. Sensitivity Analysis [23 Pts]

12. You want to run a sensitivity analysis on the high-speed rail analysis, specifically the top speed measure, which had an original swing weight of 80. The results are in the graph below. [4 Pts]



Is your overall scoring process sensitive to the swing weight of the "Top speed" measure? And is it significant? (YES or NO)

Sensitive: Y or N (Circle one)

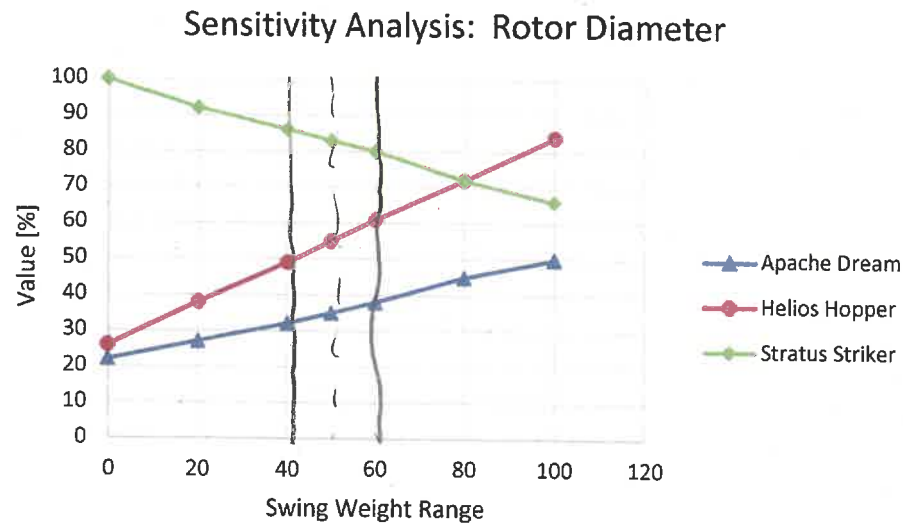
Significant: Y or N (Circle one)

13. Assume that after you conducted Sensitivity Analysis on your value measures, none of them are sensitive to change. Explain what this means for your decision authority. [3 Pts]

It means that a change to each of the value measure will not change the outcome.

Name _____

14. You want to run a sensitivity analysis, specifically regarding rotor diameter, which had an original swing weight of 50. The results are in the graph below. [4 Pts]



Sensitive: Y or N (Circle one)

Significant: Y or N (Circle one)

15. Explain why the process is sensitive and/or significant [4 Pts]

The process is sensitive because the value of the status striker becomes less than the helios hopper as you increase the swing weight. It is not significant because this change does not occur ± 10 of the original swing weight.

16. Explain why the design team conducts a sensitivity analysis in the SDP and what actions the design team would take if a value measure is considered sensitive/significant [8 Pts]

A design team conducts sensitivity analysis to determine how a change in swing weight would affect the outcome through the tested value measures. If the value measures are sensitive, changes in them will affect the outcome. If a value is significant, the design team would likely make a design change to accommodate for the value measure that is significant. This would change the preferred solution.

Name _____

IV. Risk Analysis [18 Pts]

17. Which of the following are risk types in systems engineering? *Choose all that apply.* [2 Pts]

- ☒ Schedule
- ☐ Safety
- ☐ Environmental
- ☒ Programmatic
- ☐ Quality

18. Identify the risk mitigation methods in SE301. Choose all that apply. [2 Pts]

- ☒ Avoid
- ☐ Delay
- ☐ Buffering
- ☒ Reduce
- ☐ Investigation

19. What order do the Systems Engineering risk management activities take place? [2 Pts]

- ☐ Identification, Acceptance, Mitigation
- ☐ Observe, Assessment, Management
- ☒ Identification, Assessment, Mitigation
- ☐ Identification, Assessment, Management

20. Read the following description of various risks at the Red River Army Depot. Then categorize each risk. [12 Pts]

The systems engineering team at the U.S. Army's Red River Army Depot in Texarkana, Texas is developing risk mitigation plans for the depot commander. The team reviewed maintainer safety incidents, platform rework rates, production delays, and quality assurance protocols.

In reviewing incident reports over the past several years, the team identified a pattern of injuries associated with the hydraulic press used for Bradley track shoe assembly. One critical incident caused a maintainer to lose two fingers when the press cycled unexpectedly during a jam-clearing procedure. The team determined that this type of injury, while life-altering, occurred only rarely.

Vehicles failing Final Quality Gates result in moderate degradation of depot throughput. Recurring issues include improperly seated turret ring gear bolts, mismatched electrical connectors in the fire-control system, and hull welds that fail non-destructive testing. The team confirmed these defects appeared regularly in nearly every production batch.

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Name _____

Fill in the following risks using this risk assessment matrix:

Risk Assessment Matrix		Probability (expected frequency)				
		Frequent: Continuous, regular, or inevitable occurrences	Likely: Several or numerous occurrences	Occasional: Sporadic or intermittent occurrences	Seldom: Infrequent occurrences <i>Seldom</i>	Unlikely: Possible occurrences but improbable
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Catastrophic: Mission failure, unit readiness eliminated, death, unacceptable loss or damage	I	EH	EH	H	H	M
Critical: Significantly degraded unit readiness or mission capability, severe injury, illness, loss or damage	II	EH	H	H	M	L
Moderate: Somewhat degraded unit readiness or mission capability, minor injury, illness, loss, or damage	III	H	M	M	L	L
Negligible: Little or no impact to unit readiness or mission capability; minimal injury, loss, or damage	IV	M	L	L	L	L
Legend: EH - Extremely High Risk H - High Risk M - Medium Risk L - Low Risk						

Risk 1 description: hydraulic press injuries (Finger loss due to press cycling unexpectedly during jam procedure)

Probability: Seldom Severity: critical (severe injury) Risk level: medium

Risk 2 description: Vehicles failing Final Quality Gates

Probability: Frequent Severity: Moderate Risk level: High

Risk 3 description: Failure in depots only large scale X-ray

Probability: Unlikely Severity: Critical Risk level: Low

Poor Student Work

Section: J1

SE301 WPR #3 AT 26-1

Name: Timur Kim

Departure Time: _____

**DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996**

**SE301 WPR #3
AT 26-1
VERSION (A)**

1. This is a **75-minute** exam worth **90 pts**. The exam consists of **9 pages**. Ensure that your exam is complete.
2. Place your name on each page in the space provided.
3. **AUTHORIZED REFERENCES:** You may use a pencil, straight edge (ruler, protractor, etc.), a 3 X 5 index card (front and back), and your issued hand-held calculator. The notecard is accessible for last 20 minutes of exam period.
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Name _____

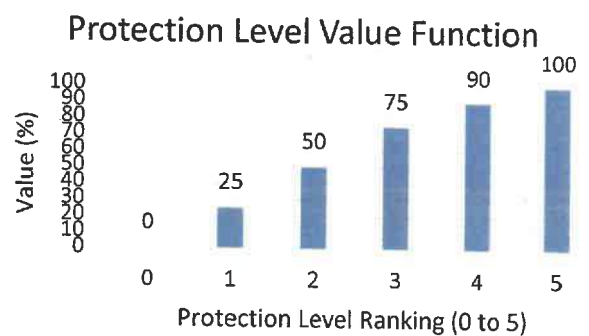
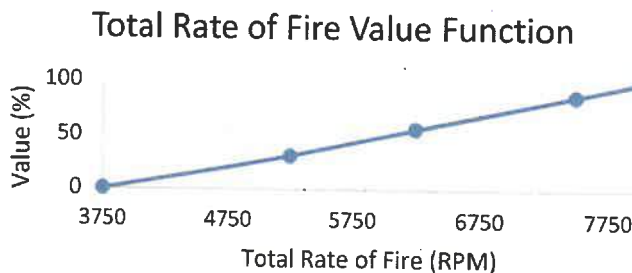
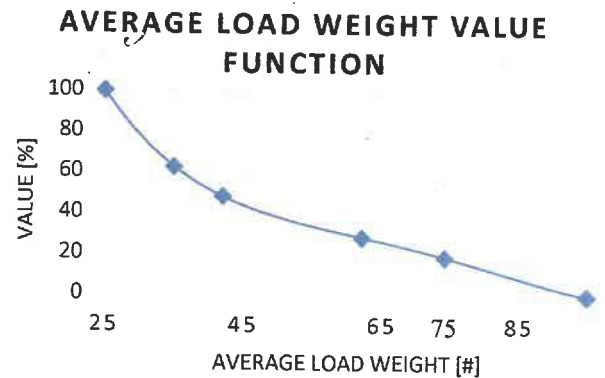
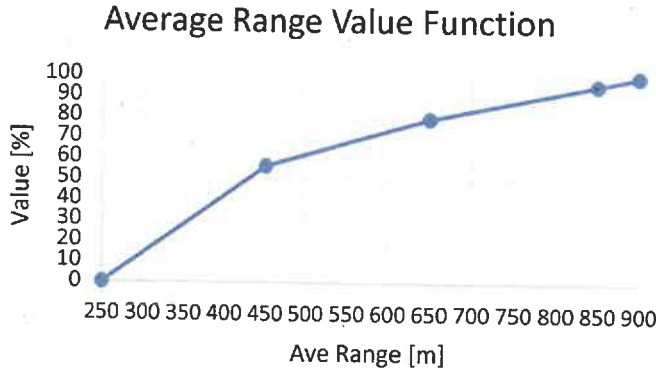
I. Decision Making [29 Pts]

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Determine the values for cells A, B, C, D, and E in the value matrix below using the following value functions: [10 Pts]



Name _____

VALUE MATRIX

Candidate Solutions	Ave Range (m)	Ave Load Weight (#)	Total Rate of Fire (RPM)	Protection Level (0 to 5 Scale)
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Bad Blood	A	28	90	E
Funky Town	B	60	70	50

A: 95% B: 100% C: 20% D: 25% E: 75%

2. Using the unweighted values in Question 1's value matrix and the global weights below, calculate the values for cells A, B, C, D and E in the weighted value matrix (round up to nearest whole number). [10 Pts]

Value Measure	w _i	Value Measure	w _i
Ave Range	0.30	Total Rate of Fire	0.08
Ave Load Weight	0.20	Protection Level	0.42

WEIGHTED VALUE MATRIX

Candidate Solutions	Ave Range (m)	Ave Load Weight (#)	Total Rate of Fire (RPM)	Protection Level (0 to 5 Scale)
Ace of Base	17	10	3	D
Heavy Metal	30	C	8	38
Bad Blood	A	6	8	E
Funky Town	B	12	6	21

A: 26 B: 30 C: 4 D: 10.5 E: 31.5

3. How do you calculate the *total system value* for each candidate solution? [3 Pts]

- ☐ Multiply each value matrix column by corresponding global weights.
- ☒ Sum the row of the weighted value matrix for each candidate solution
- ☐ Use the value functions.
- ☐ Vary the swing weight for a value measure from 0 to 100.

4. Fill in the blanks. The three sub-tasks of the Decision Making phase of the SDP are

Assessment, Value Weighting, and Solution. [3 Pts]

Name _____

5. Describe the notation term w_i of the additive value model equation requested below. [3 Pts]

$$V(x_j) = \sum_{i=1}^n [w_i \times v_i(x_{j,i})]$$

w_i = The weight assigned to the values. ~~then~~ its multiplied by v_i as it add weight. ~~to be~~
↳ value

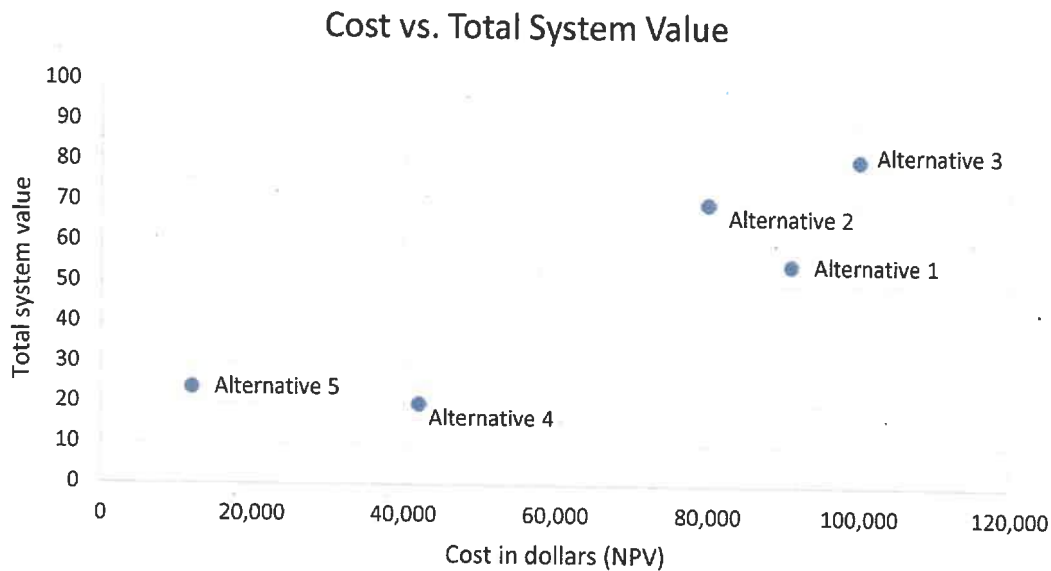
II. Performance Modeling. [20 Pts]

6. Explain why models are useful. [3 Pts]

Models are useful as they organize data in a digestible way which allows you to perceive/think of different outcomes more efficiently.

Name _____

7. Assume the chart below reflects the Total Value and Cost of the final candidate solutions.



a. Identify the candidate solution(s) that are dominated. [2 pts]

Dominated solution: Alternate 4

Dominated solution: Alternate 2

b. Explain what it means for a solution to be dominated and why the above solution has alternatives are dominated. [4 pts]

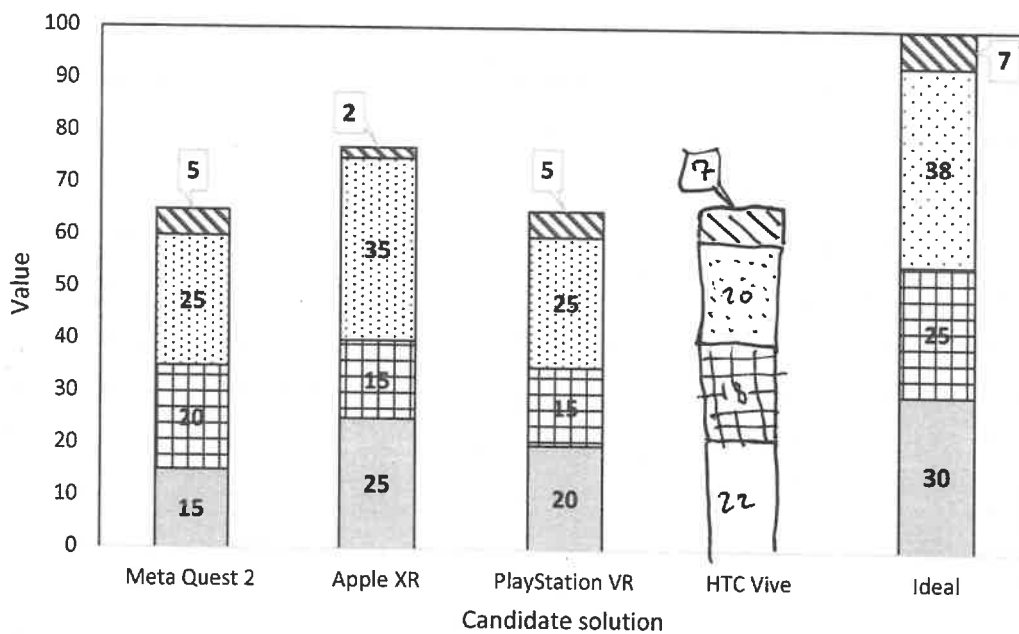
another ✓ Less cost and higher value. The above
alts are dominated by 3 and 2.

Name _____

8. Based on the Weighted Value Matrix of virtual reality headsets below, complete the stacked bar chart, by hand, for the **HTC Vive** candidate solution. Label the values for each segment: [4 Pts]

Candidate Solutions	Nausea rating	Battery life	Resolution	App Compatibility	Total Value
Meta Quest 2	15	20	25	5	65
Apple XR	25	15	35	2	77
PlayStation VR	20	15	25	5	65
HTC Vive	22	18	20	7	67
Ideal	30	25	38	7	100

Stacked bar chart



☐ Nausea rating
 ☐ Battery life
 ☐ Resolution
 ☐ App Compatibility

9. Which VR headset should the decision maker pick? [2 Pts]

- ☐ Meta quest 2
- ☒ Apple XR
- ☐ PlayStation VR
- ☐ HTC Vive
- ☐ Not enough information to say

Name _____

10. Using the stacked bar chart, which value measure is the Apple XR losing the most value on compared to the ideal? [2 Pts]

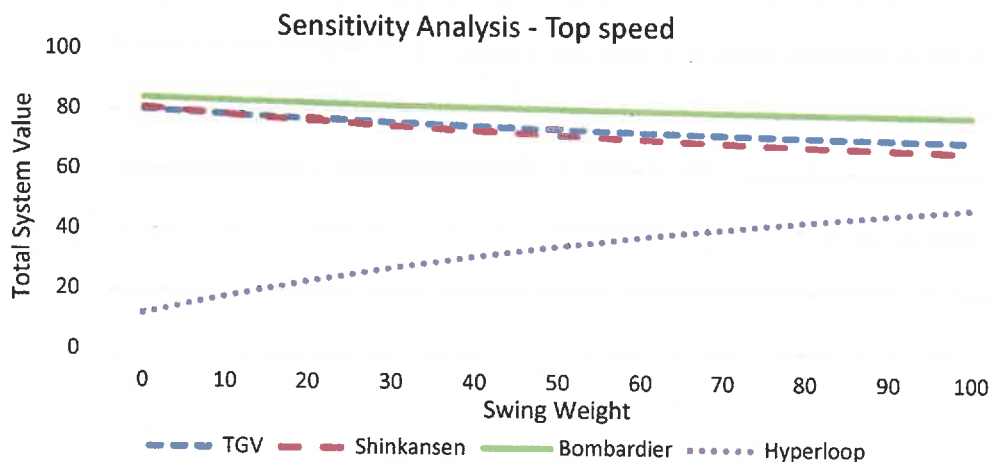
- ☐ Nausea rating
- ☒ Battery life
- ☐ Resolution
- ☒ App Compatibility
- ☐ Not enough information to say

11. What is a design change that the Apple XR could make that would improve its performance on that value measure? [3 Pts]

Try to invest in more efficient battery/ power systems for the headset - if not possible try make a compromise on some of the resolution if it will significantly increase batt. life.

III. Sensitivity Analysis [23 Pts]

12. You want to run a sensitivity analysis on the high-speed rail analysis, specifically the top speed measure, which had an original swing weight of 80. The results are in the graph below. [4 Pts]



Is your overall scoring process sensitive to the swing weight of the "Top speed" measure? And is it significant? (YES or NO)

Sensitive: ☒ YES or ☐ NO (Circle one)

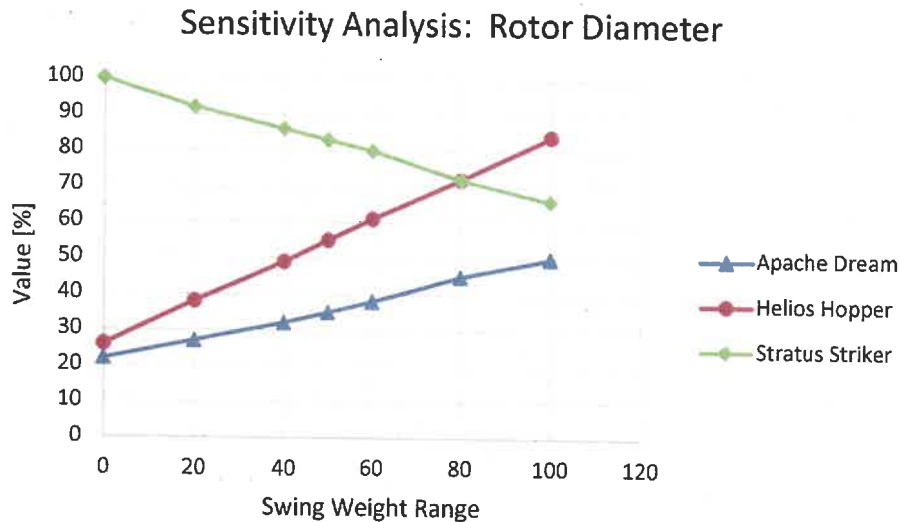
Significant: ☒ YES or ☐ NO (Circle one)

13. Assume that after you conducted Sensitivity Analysis on your value measures, none of them are sensitive to change. Explain what this means for your decision authority. [3 Pts]

If none of them are sensitive to change then that implies your decision authority has little effect.

Name _____

14. You want to run a sensitivity analysis, specifically regarding rotor diameter, which had an original swing weight of 50. The results are in the graph below. [4 Pts]



Sensitive: ☒ or N (Circle one)

Significant: ☒ or N (Circle one)

15. Explain why the process is sensitive and/or significant [4 Pts]

The process is sensitive as it reacts highly to changes in the swing weights and it is also significant as it shows the interactions between the variables.

16. Explain why the design team conducts a sensitivity analysis in the SDP and what actions the design team would take if a value measure is considered sensitive/significant [8 Pts]

The design team conducts a sensitivity analysis in order to see how the value measures interact with and respond to change. Actions the design team could take is to generate new alternatives or change their values.

Name _____

IV. Risk Analysis [18 Pts]17. Which of the following are risk types in systems engineering? *Choose all that apply.* [2 Pts]

- ☒ Schedule
- ☒ Safety
- ☒ Environmental
- ☐ Programmatic
- ☒ Quality

18. Identify the risk mitigation methods in SE301. Choose all that apply. [2 Pts]

- ☐ Avoid
- ☒ Delay
- ☒ Buffering
- ☒ Reduce
- ☒ Investigation

19. What order do the Systems Engineering risk management activities take place? [2 Pts]

- ☐ Identification, Acceptance, Mitigation
- ☐ Observe, Assessment, Management
- ☒ Identification, Assessment, Mitigation
- ☐ Identification, Assessment, Management

20. Read the following description of various risks at the Red River Army Depot. Then categorize each risk. [12 Pts]

The systems engineering team at the U.S. Army's Red River Army Depot in Texarkana, Texas is developing risk mitigation plans for the depot commander. The team reviewed maintainer safety incidents, platform rework rates, production delays, and quality assurance protocols.

In reviewing incident reports over the past several years, the team identified a pattern of injuries associated with the hydraulic press used for Bradley track shoe assembly. One critical incident caused a maintainer to lose two fingers when the press cycled unexpectedly during a jam-clearing procedure. The team determined that this type of injury, while life-altering, occurred only rarely.

Vehicles failing Final Quality Gates result in moderate degradation of depot throughput. Recurring issues include improperly seated turret ring gear bolts, mismatched electrical connectors in the fire-control system, and hull welds that fail non-destructive testing. The team confirmed these defects appeared regularly in nearly every production batch.

The team also analyzed the risk of a catastrophic failure in the depot's only large-scale X-ray inspection chamber used for detecting internal casting flaws in transmission housings. A total loss of this system would severely cripple overhaul schedules and downstream unit readiness.

After exhaustive review of all equipment logs and downtime records, the team found zero documented instances of such a failure.

Name _____

Fill in the following risks using this risk assessment matrix:

Risk Assessment Matrix		Probability (expected frequency)				
		Frequent: Continuous, regular, or inevitable occurrences	Likely: Several or numerous occurrences	Occasional: Sporadic or intermittent occurrences	Seldom: Infrequent occurrences	Unlikely: Possible occurrences but improbable
Severity (expected consequence)		A	B	C	D	E
Catastrophic: Mission failure, unit readiness eliminated, death, unacceptable loss or damage	I	EH	EH	H	H	M
Critical: Significantly degraded unit readiness or mission capability, severe injury, illness, loss or damage	II	EH	H	H	M	L
Moderate: Somewhat degraded unit readiness or mission capability, minor injury, illness, loss, or damage	III	H	M	M	L	L
Negligible: Little or no impact to unit readiness or mission capability, minimal injury, loss, or damage	IV	M	L	L	L	L
Legend: EH - Extremely High Risk H - High Risk M - Medium Risk L - Low Risk						

Risk 1 description: Safety

Probability: Seldom Severity: Critical Risk level: Medium Risk

Risk 2 description: Quality

Probability: Frequent Severity: Moderate Risk level: High Risk

Risk 3 description: Schedule

Probability: Unlikely Severity: Critical Risk level: Low





UNITED STATES MILITARY ACADEMY
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TAB J TERM END EXAMINATION

Term End Examinations are kept in a secure location and can be accessed
by request to the Program Director.



UNITED STATES MILITARY ACADEMY
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TAB L FINAL COURSE GRADES



DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996



MADN-SE

20 December 2025

MEMORANDUM FOR LTC John Caddell, Systems and Decision Sciences Program Director,
Department of Systems Engineering, West Point, NY 10996

SUBJECT: TEE and Final Grade Approval for Academic Year Term (AYT) 26-1 – SE301:
Fundamentals of Engineering Design & Systems Management

1. **RECOMMENDATION.** I recommend the approval of the TEE with moving the cutline below as shown in the Appendix C. The exam was comprehensive, straightforward, covered wide range of lesson objectives. The exam resulted in a course average of **88.8%** (down from 91.5% in 25-5, 92.1% in 25-2). The TEE was nearly identical to that administered in 25-2 to facilitate longitudinal analysis.
2. **ADMINISTRATIVE DATA.**
 - a. Primary Exam: total of 154 cadets have taken their exam during their regular scheduled day.
 - (1) Wednesday, December 17 2025 (1800-2130): 123 cadets tested.
 - (2) Thursday, December 18 2025 (1500-1830): 26 cadets tested.
 - (3) Friday, December 19 2025 (0730-1100): 5 cadets tested.
 - b. Weight of Exam: 200 points, which equates to 20% of the course grade.
 - c. Material Covered: The TEE was comprehensive and covered topics from across the course. It specifically covered systems engineering, systems thinking, and the four phases of the Systems Decision Process (SDP): problem definition, solution design, decision making, and solution implementation.
3. **SUMMARY STATISTICS AND ANALYSIS:**
 - a. Cadets scored an average of 175.35 points on the TEE (87.7%). This is somewhat lower than 25-5 (90.4%) and 25-2 (88.9%) but close to 24-2 (88.2%).
 - b. The use of gradescope allowed a detailed analysis of performance on TEE sections. Cadets performed well across the vast majority of questions. Low scores were seen on the following questions:
 - (1) Q13 where cadets create value functions (72%). Points were lost for precise formatting errors and general understanding of the value function.
 - (2) Q22 where cadets are asked to describe the purpose of a value function, over 30% of the cadets left it blank and overlooked the question
 - (3) Q25 asking for an analysis of design improvements (80%). This question may need to be rewritten to emphasize that we want design changes and not just an identification of the value measure.

- (4) Q32 the questions on verification and validation (87%, 62%, 64%, 72%). This indicates a need for more clarity in how we teach this.
- c. Gradescope grading continued to give much greater insight into how each part of the exam was graded.
- d. Historical Course Averages: Table 2 presents SE301/SE300 Course averages back to AYT 16-1. The average for this year is in line with previous semesters.

Table 2: Historical Course Averages

Academic Term	SE301 Average
AYT 26-1	87.7%
AYT 25-2	91.5%
AYT 25-2	92.1%
AYT 25-1	90.8%
AYT 24-2	89.7%
AYT 24-1	88.3%
AYT 23-2	84.6%
AYT 23-1	85.3%
AYT 22-2	83.5%
AYT 22-1	86.8%
AYT 21-2	83.5%
AYT 21-1	85.9%
AYT 20-2	93.0% ¹
AYT 20-1	85.9%

4. **TEE Summary:** Similar to SE301's WPRs, the only authorized references for the TEE was a 8.5x11 handwritten piece of paper, no time restrictions. The exam focused on fundamental concepts from the course and the interpretation and analysis of results. Where possible, students demonstrated their abilities by hand or were provided with the correct "answer" as a starting point for analysis.
5. **Failures:** There were no failures on the TEE.

¹ During AYT 20-2, the TEE was administered under the following conditions: If a cadet was failing, the TEE was mandatory and presented an opportunity to pass the course. If a cadet was passing, it was optional. Cadets who took the TEE had 24 hours to complete it. The TEE was open-book and open-notes, but no collaboration was authorized with other individuals. Moreover, the TEE could only help a cadet's grade. Specifically, if a cadet's TEE grade was lower than his grade at Lesson 30, then his "Lesson 30 grade" was used as his final grade.

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6. **Current Course Grades:** For course OML see Appendix B.
7. The point of contact for this memorandum is the undersigned at joseph.kearney@westpoint.edu.

Joseph P. Kearney
MAJ, EN/AC
SE301 Course Director

Appendix A: Detailed Analysis

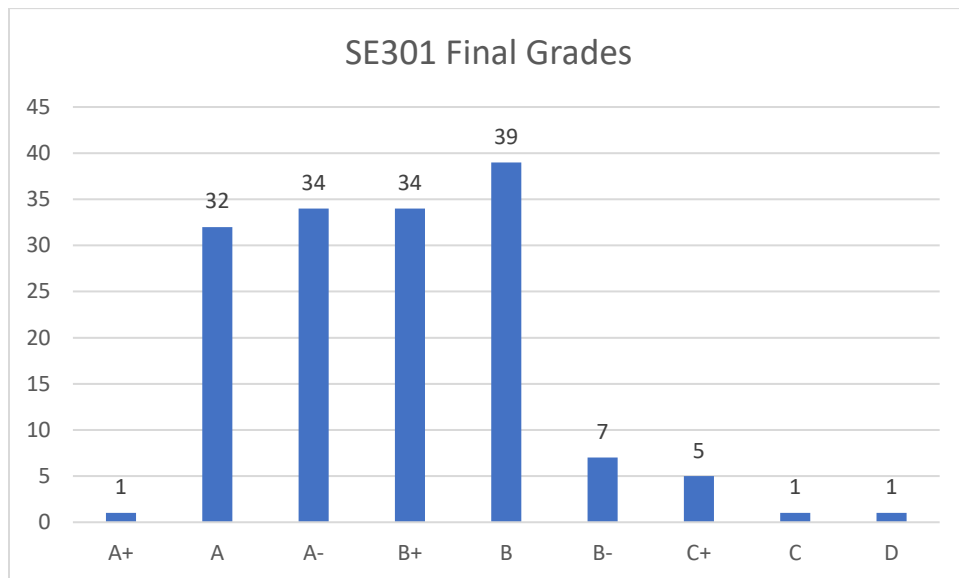


Figure 1: Course-wide performance by letter grade

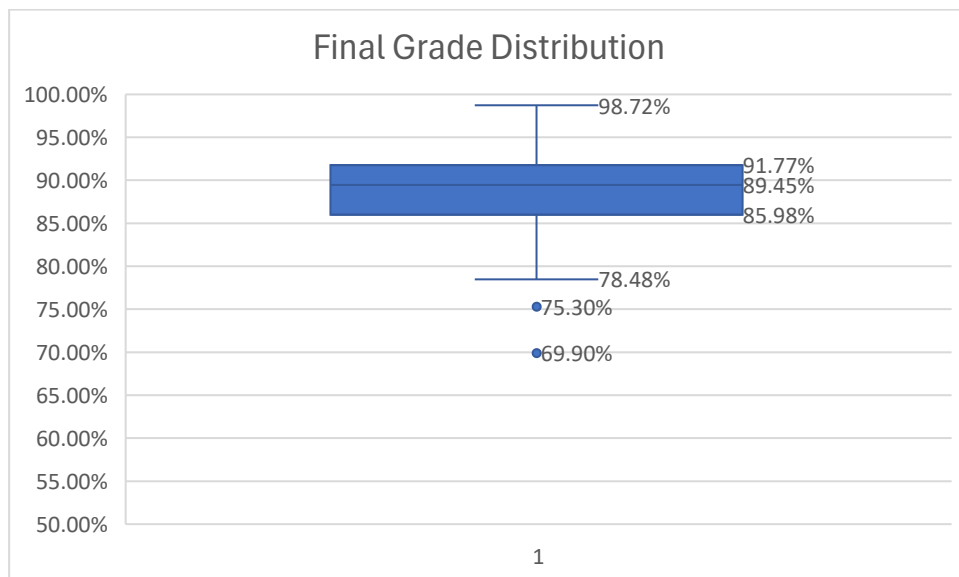


Figure 2: SE301 Final Grade Distribution

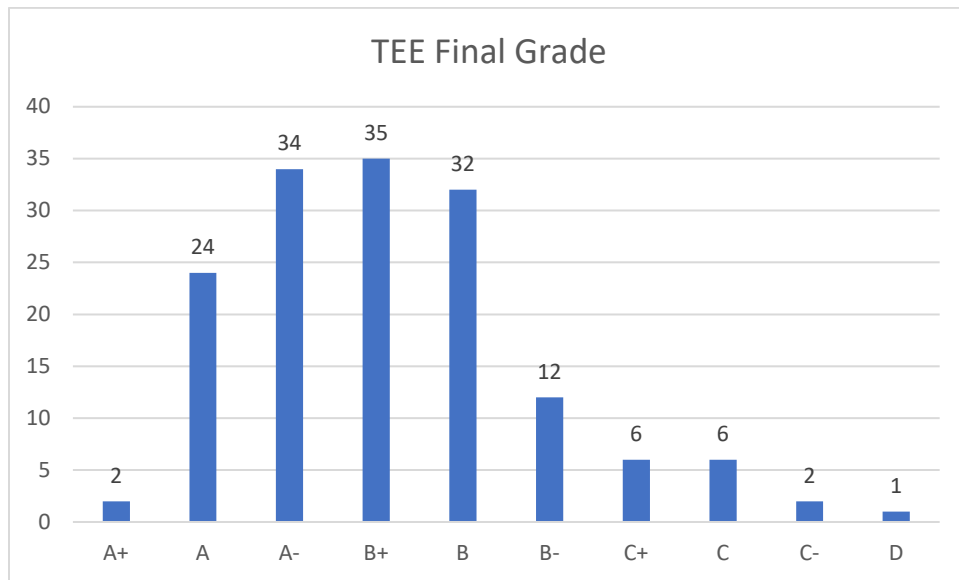


Figure 3: SE301 TEE Grades

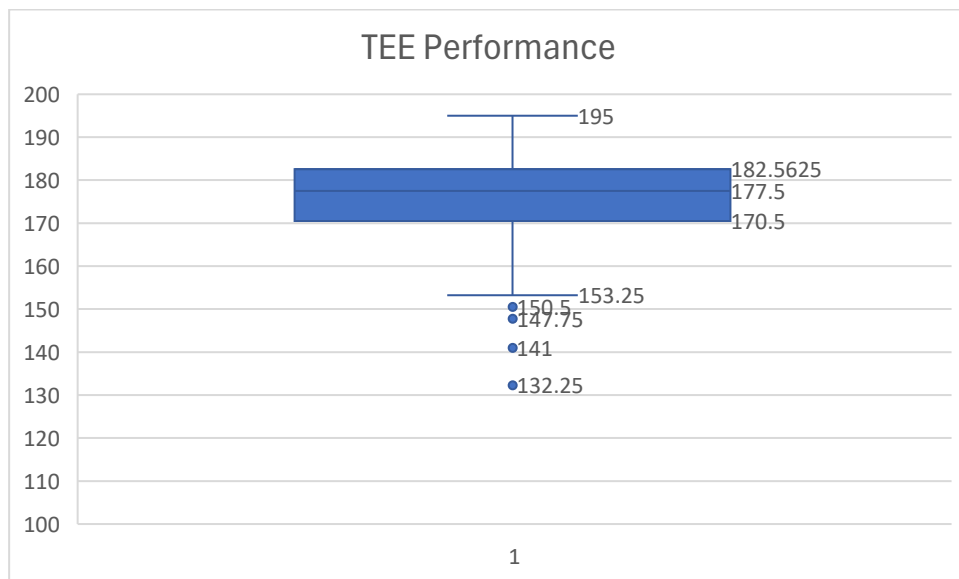


Figure 4: SE301 TEE Distribution (out of 200 pts)

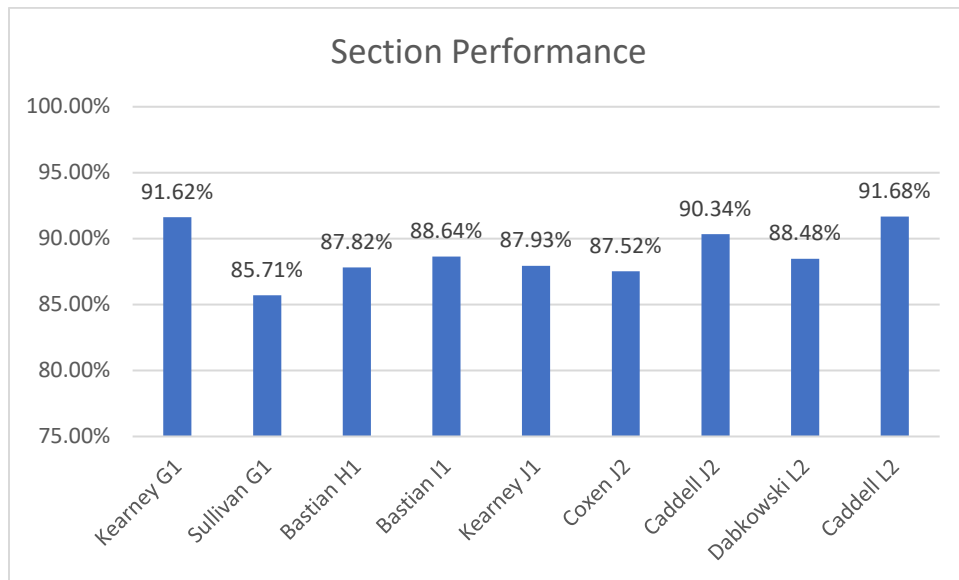


Figure 5: Section Performance

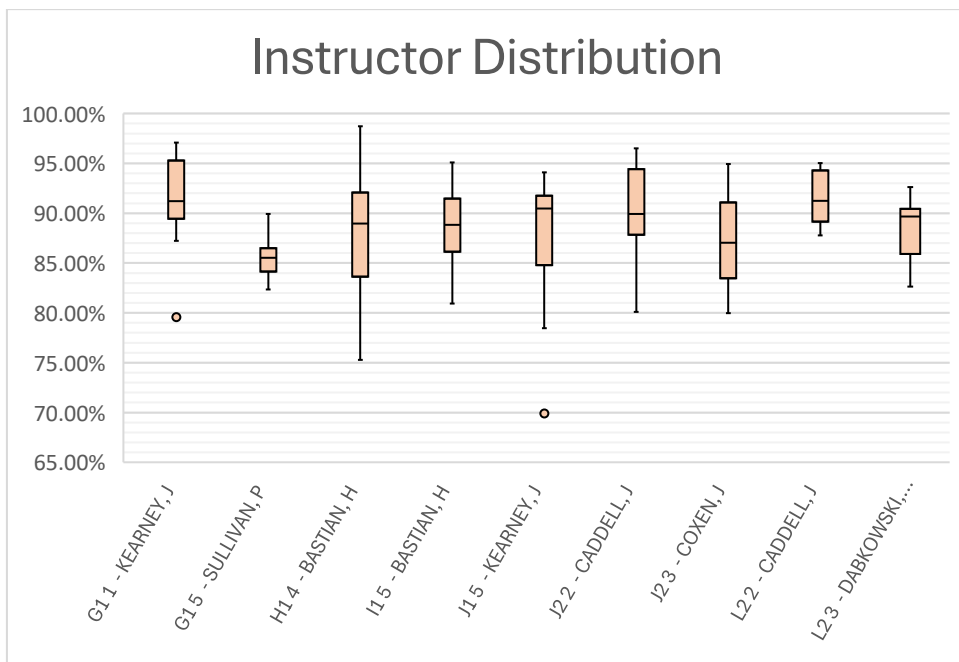


Figure 6: Section Distribution

Appendix B: Course OML (with TEE Performance)

Student	Course Points	Percent	Final Grade	Recommended	TEE%
Clark, Zachary	987.2	98.72%	A+	A+	97.00%
Kempinski, Chester	970.75	97.08%	A	A	93.75%
Clemons, Mark	968.5	96.85%	A	A	97.50%
Fillion, Jack	967	96.70%	A	A	96.50%
Lu, Jerry	965	96.50%	A	A	92.25%
Ahn, Cheyunne	959.75	95.98%	A	A	95.25%
Ingle, Brandon	959.75	95.98%	A	A	90.75%
Grdinich, Noah	957	95.70%	A	A	92.00%
Lester, Nicholas	956.3	95.63%	A	A	97.00%
Brenner, Jacob	951	95.10%	A	A	93.00%
Woodring, Barron	950.5	95.05%	A	A	94.38%
Thorley, Ethan	950.25	95.03%	A	A	97.50%
Flynn, Michael	949.25	94.93%	A	A	93.50%
Smith, Dylan	948.5	94.85%	A	A	90.00%
Mavoides, Gavin	948.25	94.83%	A	A	90.50%
Satpute, Chinmay	944.9	94.49%	A	A	93.38%
Smith, Ruby	944.3	94.43%	A	A	90.38%
Sirimaturos, Wyatt	943.5	94.35%	A	A	93.38%
Geczik, Kristin	942.25	94.23%	A	A	88.63%
Russell, Finlay	941.75	94.18%	A	A	91.75%
Gonzalez, John	941	94.10%	A	A	93.50%
McDaniel, John	941	94.10%	A	A	89.00%
Remaly, Ryan	941	94.10%	A	A	92.25%
Botchwey, Edward	940.5	94.05%	A	A	93.00%
Allert, Erin	934.2	93.42%	A	A	89.75%
Johnson, William	931.85	93.19%	A	A	95.25%
Roberts, Alexander	930.25	93.03%	A	A	90.75%
Dewan, Zev	928.4	92.84%	A	A	93.50%
Laczkowski, Tate	927.25	92.73%	A	A	93.25%
Mason, Weston	926.6	92.66%	A	A	88.25%
Hecht, Jordan	926.5	92.65%	A	A	92.75%
Johnson, Jacob	926.35	92.64%	A	A	88.50%
Congdon, Matthew	923.65	92.37%	A	A	88.50%
Edwards, Madeline	922	92.20%	A-	A-	88.75%

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Perine, Alexander	920.95	92.10%	A-	A-	92.38%
Kim, Sean	920.3	92.03%	A-	A-	87.25%
Miller, Andrew	919.5	91.95%	A-	A-	91.50%
Schrader, Colt	918.2	91.82%	A-	A-	90.25%
Dzurinko, Garrett	917.5	91.75%	A-	A-	91.50%
Veletto, Quintin	916.6	91.66%	A-	A-	89.63%
Wagner, Dustin	916.6	91.66%	A-	A-	87.25%
Kinton, Hayden	916.15	91.62%	A-	A-	91.00%
Garlock, Brennan	915.75	91.58%	A-	A-	94.50%
Haldezos, Ethan	915.25	91.53%	A-	A-	89.75%
Kennedy, Ryan	915.25	91.53%	A-	A-	91.00%
Gamboa, Isaac	914.95	91.50%	A-	A-	90.25%
Chun, Jake	914.8	91.48%	A-	A-	93.50%
Lechner, Steele	912.58	91.26%	A-	A-	82.00%
Johnson, Adam	912.55	91.26%	A-	A-	90.75%
Streich, Brayden	910.75	91.08%	A-	A-	92.25%
Simpson, Alexander	910.65	91.07%	A-	A-	91.70%
Kraus, Jackson	909.3	90.93%	A-	A-	86.13%
Crenshaw, Mathias	909.01	90.90%	A-	A-	90.50%
Amaya, Abigail	909	90.90%	A-	A-	90.25%
Finnessy, Kiera	908	90.80%	A-	A-	88.75%
Schrichte, Kolbet	906.65	90.67%	A-	A-	89.50%
Waldron, Andrew	905.95	90.60%	A-	A-	86.75%
Monahan, William	904.7	90.47%	A-	A-	88.50%
McGlone, Maxwell	903.2	90.32%	A-	A-	85.75%
Strange, Anderson	902.68	90.27%	A-	A-	89.00%
Jensen, Megan	902.45	90.25%	A-	A-	90.50%
Dunfee, Timothy	902.3	90.23%	A-	A-	81.50%
Brooks, Chancellor	901.5	90.15%	A-	A-	90.00%
McLaughlin, James	901.45	90.15%	A-	A-	86.00%
Masterson, Brody	901.35	90.14%	A-	A-	83.50%
Mathews, Ayden	901.25	90.13%	A-	A-	89.75%
Faherty, Patrick	901.2	90.12%	A-	A-	87.25%
Goetz, Thomas	899.35	89.94%	B+	B+	93.25%
Wood, Atticus	899.25	89.93%	B+	B+	84.75%
McGettigan, Matthew	899.2	89.92%	B+	B+	85.25%
Murphy, Zachary	898.65	89.87%	B+	B+	88.75%
Nopper, Coleman	898.18	89.82%	B+	B+	89.50%
Counts, Kaia	897.1	89.71%	B+	B+	85.38%
Dudlar, Gunnar	897	89.70%	B+	B+	89.75%
Schwartz, Nicole	895.65	89.57%	B+	B+	92.13%

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Singleton, Corey	895.15	89.52%	B+	B+	88.63%
Tabaska, Nollenn	894.93	89.49%	B+	B+	88.13%
Olubamowo, Similolu	894	89.40%	B+	B+	91.25%
Filipowicz, Gunner	893.25	89.33%	B+	B+	89.00%
Concannon, Hailey	892.78	89.28%	B+	B+	91.25%
Roseme, Zachary	891.25	89.13%	B+	B+	90.00%
Al-Droubi, Sammy	891	89.10%	B+	B+	87.88%
Haney, William	890.08	89.01%	B+	B+	88.50%
Bamgbala, Jubril	889.7	88.97%	B+	B+	78.75%
Fradelizio, Frank	889.65	88.97%	B+	B+	85.25%
Ferris, Christopher	888.25	88.83%	B+	B+	87.00%
Melendez-Eyster, Elijah	888.25	88.83%	B+	B+	89.38%
Kim, Danny	888	88.80%	B+	B+	93.00%
St. Louis, Ella	887.35	88.74%	B+	B+	85.25%
Orcutt, Samuel	887.06	88.71%	B+	B+	92.50%
James, Gatlin	882.43	88.24%	B+	B+	82.13%
Leone, Nicholas	882.35	88.24%	B+	B+	88.50%
Weikel, Jonathan	882.15	88.22%	B+	B+	92.25%
De Santos, Christopher	881.93	88.19%	B+	B+	90.00%
Kemp, Benjamin	881.15	88.12%	B+	B+	85.00%
Oakley, Thomas	880.2	88.02%	B+	B+	89.00%
Quackenbush, Alexander	880.13	88.01%	B+	B+	81.25%
Haynia, Maximus	878.56	87.86%	B+	B+	88.13%
Jeong, Yewon	878.5	87.85%	B+	B+	81.38%
Summers, Quintin	878.33	87.83%	B+	B+	90.00%
Meyer, Kathryn	877.65	87.77%	B+	B+	94.75%
Yun, Brian	872.25	87.23%	B	B	85.25%
Rose, Isaiah	870.5	87.05%	B	B	85.75%
Cesoni, Geneva	868.53	86.85%	B	B	84.00%
Dickey, Alexis	867.33	86.73%	B	B	89.13%
Smith, Cooper	867.13	86.71%	B	B	80.00%
Rockey, Dylan	866.7	86.67%	B	B	82.50%
Cunningham, Austin	866.25	86.63%	B	B	86.25%
Choi, Nicole	865.03	86.50%	B	B	91.25%
Spaulding, Aedan	864.75	86.48%	B	B	85.75%
Moon, Johan	864.35	86.44%	B	B	87.00%
Hahn, Calvin	862.95	86.30%	B	B	89.25%
Indukuri, Medha	861.63	86.16%	B	B	92.50%
Ralston, Samuel	861.4	86.14%	B	B	77.75%
Nack, Truman	860.55	86.06%	B	B	88.25%
White, Chad	859.95	86.00%	B	B	79.50%

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Grabowski, Joshua	859.5	85.95%	B	B	88.00%
Fitts, John	859.45	85.95%	B	B	79.75%
Kim, Minwook	858.68	85.87%	B	B	77.88%
Hicke, Zane	858.25	85.83%	B	B	82.25%
Romo, Joncarlo	858.05	85.81%	B	B	89.50%
Thabit, Sarah	856.1	85.61%	B	B	91.38%
Duran, Noah	852.6	85.26%	B	B	91.00%
Keltner, Asa	851.25	85.13%	B	B	77.38%
Woolley, Kylie	851.06	85.11%	B	B	82.00%
Mat Sahari, Muhammad Faidh	850.35	85.04%	B	B	90.75%
Kim, Dylan	849.7	84.97%	B	B	76.63%
Johnson, Foster	847.83	84.78%	B	B	84.75%
Kumar, Rohan	845.03	84.50%	B	B	81.13%
Levindoski, Blake	840.28	84.03%	B	B	88.50%
Jeffords, Jonathan	838.86	83.89%	B	B	83.75%
Riddle, Kade	837.45	83.75%	B	B	85.00%
Spencer, Daniel	836.76	83.68%	B	B	86.25%
Jones, Olivia	835.45	83.55%	B	B	85.50%
Prieto, Andrew	834.75	83.48%	B	B	88.00%
Newsom, Baylor	832.3	83.23%	B	B	75.25%
Yoo, Jonathan	827.23	82.72%	B	B	81.13%
Deforest, Adam	826.65	82.67%	B	B	73.88%
Wu, Jeffrey	826.43	82.64%	B	B	87.50%
Williams, Christopher	826	82.60%	B	B	86.00%
Clerk, David	824.75	82.48%	B-	B-	81.25%
De La Torre, David	823.63	82.36%	B-	B-	94.25%
Cullum, Riley	821.75	82.18%	B-	B-	82.25%
Marks, Landon	816.41	81.64%	B-	B-	88.25%
Mobley, Duke	815.65	81.57%	B-	B-	85.75%
Perciavalle, Trieste	809.33	80.93%	B-	B-	74.25%
Diamond, Timothy	801	80.10%	B-	B-	70.50%
Demeritt, Luke	799.75	79.98%	C+	C+	76.88%
Eremita, Nicholas	795.63	79.56%	C+	C+	82.75%
Andrade, Elena	793.95	79.40%	C+	C+	83.75%
Izzo, Breanna	790.15	79.02%	C+	C+	85.50%
Laskey, Landon	784.75	78.48%	C+	C+	70.50%
Olson, Samuel	752.95	75.30%	C	C	78.88%
Kim, Timur	699	69.90%	D	D	66.13%



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SECTION III

ASSIGNMENTS



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TAB M HOMEWORK (1-4)



UNITED STATES MILITARY ACADEMY
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HOMEWORK #1

Fundamentals of Engineering Design and System Management

SE 301 – AT26-1

Homework 1: Systems Thinking and Research & Stakeholder Analysis

Assigned: Lesson 1

Due: **09/15/2025** NLT 2300 via Canvas (75 Points)

This is an **individual** assignment. Submit your work on this worksheet Canvas along with a digitally signed coversheet IAW the Dean's Documentation and Acknowledgement of Academic Work.

BACKGROUND:

The last mile problem in logistics describes the problem of getting packages to consumers from the last hub. This final delivery step has always suffered from high costs and low efficiency. Companies have experimented with many solutions including using delivery bicycles and customer pickup from hubs. Several companies are experimenting with a new solution: drone delivery of packages up to 15 miles away from a hub.

Proponents argue that drone delivery will reduce congestion and allow for faster delivery times for customers, while critics argue that drones do little to improve customer experience and come with significant risks.



Amazon has trialed drone delivery in cities including College Station, Texas and Phoenix, Arizona but they have plans to expand substantially. Many other companies including Flytrex, DroneUp, Zipline, and Google's Wing have begun to [operate drone delivery services](#) for retailers including Walmart. The Federal Aviation Administration (FAA) has responsibility for regulating commercial drone usage and has been considering the best way to regulate the usage.

Fundamentals of Engineering Design and System Management

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You have been hired by the City of Chattanooga to investigate the different perspectives on drone delivery. You are having an on-line discussion with some stakeholders and should also make use of scholarly articles and government websites to support your research, your tasks are to:

1.1. Conduct an online interview with the stakeholders listed below (note that the specific scenario and characters have been fictionalized).

[Jay Merkel- Executive Director of the UAS Integration Office at the FAA](#)

[Jerry Larkin - Mayor of Chattanooga](#)

[Paul Misener- Amazon's Vice President for Global Public Policy](#)

[Laura Timmons - Head of the College Station Against Drones \(CSAD\) Action Group.](#)

1.2 Read the following survey report on public opinion towards drone delivery

[Opinion Survey to Reduce Uncertainty in Public and Stakeholder Perception of Unmanned Aircraft](#)

1.3 Identify **at least two** other scholarly or government sources relevant to the problem.

1.4. Use the interviews, survey and your additional sources to complete the stakeholder analysis and systems thinking worksheet below.

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1. (10 points) Stakeholders: Identify at least five stakeholder groups for your project (don't just list the interview subjects) and state why they are stakeholders; identify their types and the technique you would use to gather their feedback.

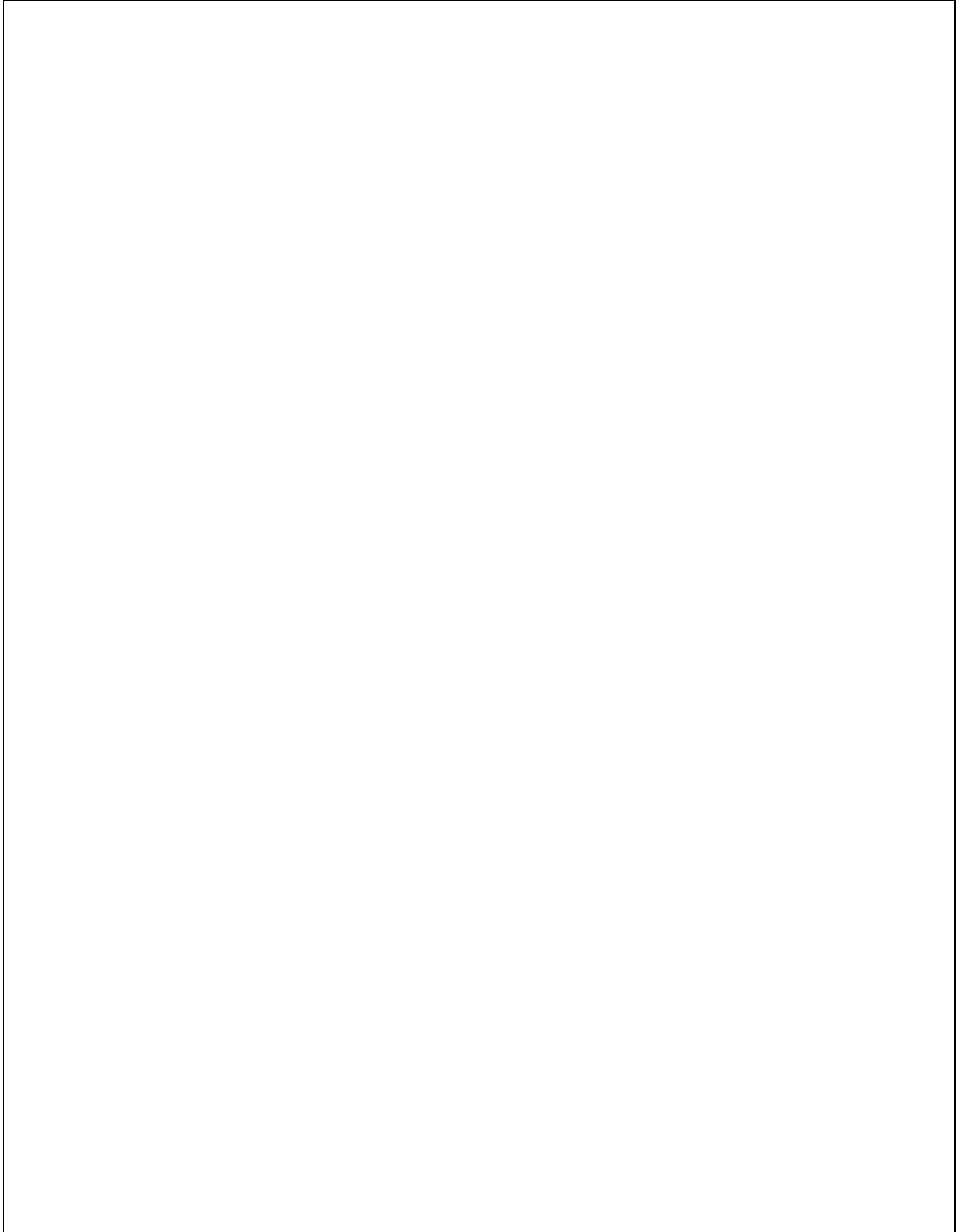
Stakeholder	Why are they a stakeholder?	What type of stakeholder are they and why?

Fundamentals of Engineering Design and System Management

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2. FCR Matrix (20 points): Summarize what you learned from your research and stakeholder analysis in a Findings, Conclusions, and Recommendations (FCR) matrix that shows the perspective of the stakeholder groups you identified. Cite sources used in the table. Use 10 point font and limit your table to 2 pages.

Fundamentals of Engineering Design and System Management
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DSRP Diagram using Plectica

Apply the DSRP methodology to better understand the current system affecting drone delivery in Chattanooga. Show your analysis using a Plectica diagram following the DSRP rules. Make sure that your diagram:

- 1) Distinction (thing/other): shows what is part of the system and what is not;
- 2) System (part/whole): Decomposes the system as you see fit to identify the various parts that make up the whole system;
- 3) Relationship (action/reaction): Shows the potential relationships inside and outside of the system that may impact your recommendation;
- 4) Perspective (point/view) shows how the perspective of at least three stakeholders differ from each other

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3. Plectica Diagram (15 points):

URL to diagram: _____

Image:

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4. Interpretation of DSRP diagram (8 points)

Explain the key insights that your Plectica diagram illustrates about the problems of Drone Delivery in Chattanooga.

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5 Use of DSRP concepts (7 points)

Identify and explain each dimension of DSRP and two elements of each dimension. For example, for the dimension Distinction, identify two Things that the system is (e.g., T1 and T2) and two Things that it is not (e.g., O1 and O2). Your examples must be present on your DSRP diagram.

SIMPLE RULE OR PATTERN			
(D)_____	(S)_____	(R)_____	(P))_____
ELEMENT 1			
(T)_____	(P)_____	(A)_____	(P)_____
1.	1.	1.	1.
2.	2.	2.	2.
ELEMENT 2			
(O)_____	(W)_____	(R)_____	(V)_____
1.	1.	1.	1.
2.	2.	2.	2.

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6. Assessment of Research (15 points) (this is graded based on your use of research throughout the rest of this homework and use of appropriate citations)

References (list your sources here using APA 7th Edition format)



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HOMEWORK #2

SE301 Course Project



Homework 2: Problem Definition

Assigned: Lesson 10

Due: October 9th at 2359

(70 Points)

You will complete the Problem Definition Phase of the SDP in a group of 3-5, assigned by your instructor. Only one team member will submit IAW your instructor's guidance. Submit your work with a coversheet IAW the Dean's Documentation of Academic Work.

The homework in the rest of SE301 will follow the phases of the SDP to solve an important engineering problem. Read all the instructions and **submit your answers in the form of the 1) the problem definition worksheet** in this document and **2) an excel worksheet containing your value functions**.

You should pick **ONE** problem from the options listed in the next section. When picking a problem statement think about:

- 1) How interesting is the problem?
- 2) Is this an area you will be able to understand well enough to make meaningful recommendations (e.g. if you need to understand quantum mechanics to understand solutions, consider picking a different topic)?
- 3) Is there likely to be data available to help estimate costs and assess performance of alternatives? If everything related to the problem is classified, it may be sensible to pick a different topic.

Remember that the initial problem statement is never the final problem statement. The problem definition phase should leave you with a more refined problem statement following the format explained in the problem definition phase slides. **You will likely tackle a narrower problem than the initial problem statement covers.**

You should also pick a sensible client for the problem you have picked. Pick an individual or organization who would be interested in systems engineering solutions that tackle your initial problem statement.

Homework overview

Your task is to apply the steps of the problem definition phase of the SDP to the problem you picked from the list in order to help the client you identified. In later homeworks, your group will apply the solution design and decision making phases to the same problem.

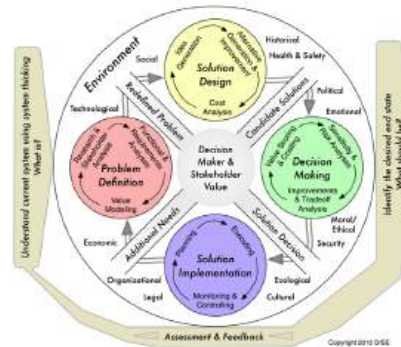


Figure 2: System Decision Process

You will present your analysis in the worksheet below. You should write clearly and professionally in the style of a brief to a senior leader, equivalent to a corporation's CEO.

The points are divided up as shown in the following table:

Section	Task	Points
Inputs	Initial problem statement/client	2
Research & Stakeholder analysis	Sources used	4
	Stakeholder engagement	bonus 4 pts
	Stakeholders	3
	FCR matrix	9
Functional & Requirements analysis	Functional hierarchy	8
	Functional hierarchy explanation	3
	Requirements	4
Value modeling	Qualitative value model	6
	Value measure weights	4
	Value functions	10
Outputs	Redefined problem statement	2
	Screening criteria	4
	Conclusion	3
Additional	Professionalism	4
	Coherency	4
	Total	70 + bonus

Possible Problems:

Electric Car Makers Facing Lithium Shortage

Initial problem statement: The global shift toward electric vehicles (EVs) has created an unprecedented demand for lithium, a key component in EV batteries. However, the supply of lithium is struggling to keep pace. How can electric car manufacturers mitigate the risks associated with lithium shortages?

Background: <https://apnews.com/article/china-ev-lithium-united-states-battery-87eb9382a0181bb7ee64e835efe7b170>

Building Enough Compute for AI Needs

Initial problem statement: The rapid development and deployment of artificial intelligence technologies have created an insatiable demand for high-performance computing resources, particularly GPUs. This surge has led to supply shortages, increased costs, and bottlenecks in AI research and product development. US companies are scrambling to secure these resources, but how can they build compute infrastructure to meet future AI demands?

Background: <https://www.nytimes.com/2023/08/16/technology/ai-gpu-chips-shortage.html>

Taiwan Energy Import Dependence

Initial problem statement: Taiwan relies on imports for approximately 97% of its energy needs, making it highly vulnerable to disruptions in maritime supply chains. This dependency poses significant risks to Taiwan's national security. How can Taiwan reduce its reliance on imported energy and enhance its energy security?

Background: <https://www.atlanticcouncil.org/blogs/new-atlanticist/does-taiwans-massive-reliance-on-energy-imports-put-its-security-at-risk/>

Protecting U.S. Fleets from Sea Drones

Initial problem statement: The rise of unmanned sea drones, as demonstrated in the ongoing conflict in Ukraine, presents new challenges for naval security. How can the U.S. Navy counter this emerging threat?

Background: <https://cepa.org/article/ukraines-marauding-sea-drones-bewilder-russia/>

Impact of neighborhood location and its influence on access of healthy foods

Initial problem statement: There is a growing problem of obesity in the United States, especially in urban areas where accessibility of supermarkets and grocery stores is an issue. In addition, access to fast-food outlets and convenience stores has grown and may contribute to this problem.

Background: [How cities can have an impact on healthy food options | World Economic Forum](#)

SE301: Fundamentals of Engineering Design and System Management – AY 26-1

Homework 2: Problem Definition

Use of special operations forces to deter enemy threats

Initial problem statement: How can US Special Operations Forces be used to counter cooperative threats, deter and avoid war, and operate under a budget constrained/recruitment problematic War department?

Background: [How the Pentagon can use special operations forces to better compete](#)

Countering the use of drones by our enemies

Initial Problem Statement: Drones are a low cost, potentially high payoff delivery systems. They are being used on the battlefield today in places like Ukraine. What types of countermeasures can the US develop to deal with the growing threat of drone use in warfare?

Background: <https://rmcglobal.com/small-unmanned-aerial-systems-suas-and-the-force-protection-threat-to-dod/>

California's water shortage problem

Initial Problem Statement: California has a large population and increased farming demands for water. How does California address the increased demand for its limited water supply?

Background: <https://www.nytimes.com/interactive/2023/12/14/climate/california-water-crisis-drought.html>

Bridging the Marine Corps' Digital Chasm

Initial problem statement: In order for the Marines to leverage data and information to enhance the service's warfighting capabilities, it must equip tactical-level units to develop technological solutions to the problems they encounter rather than relying on enterprise-level innovation. How might this capability be created in a zero-growth manning and budgeting environment?

Background: <https://mwi.westpoint.edu/bridging-the-marine-corps-digital-chasm/>

Military Dining Facilities Nutrition Efforts

Initial Problem Statement: The Department of War is struggling to support food service operations across its bases while prioritizing nutrition and overall health improvements for service members. As of 2023, 68% of active-duty service members are overweight or obese. Improvements are ongoing but the DoW needs additional efforts to address this issue that further impacts personnel readiness.

Background: https://www.gao.gov/products/gao-24-106155?utm_campaign=usgao_email&utm_content=topic_natldefense&utm_medium=email&utm_source=govdelivery

DoW Childcare Struggles

Initial Problem Statement: The Department of War is struggling to support its active-duty stakeholders to meet demand of childcare across installations. A shortage of childcare providers has left around 9,000 children waiting months for spots at one of the military's childcare centers. Improvements are ongoing but DoW needs additional efforts to address this issue that impacts family readiness.

Background: <https://mwi.westpoint.edu/caring-for-children-and-retaining-families-the-gaps-in-military-child-care/>

Input to the Problem Definition Phase

You should start by outlining your initial problem statement and the client you are working for.

Initial Problem Statement: (1)

--

Client (1):

--

Research and stakeholder analysis

Conduct research. Wikipedia is not considered a scholarly resource, but it can often point you in the right direction. The links provided above could be helpful but do not limit yourself to these when trying to understand the problem. You should conduct sufficient research (document your sources) to justify your analysis. It is strongly recommended that you begin your research first and early.

- ☐ Research approach (4): What sources did you consult to conduct your research? How did you find those sources? What systems did you investigate and why? You should use at least **four** outside references on similar systems to the one you are examining.

- ☐ Stakeholder engagement (4 bonus points): Bonus points are available for any engagement with stakeholders identified as experts in the field. For instance, if any department at West Point has some Subject Matter Expert (SME), reach out and schedule a time to interview and survey to get some perspective. Describe any stakeholder analysis (interview, focus group, survey) you conducted.

Homework 2: Problem Definition

- ☐ Stakeholders (3): Identify at least three stakeholder groups for your project and state why they are stakeholders; identify their types and the technique you would use to gather their feedback.

Stakeholder	Why are they a stakeholder?	What type of stakeholder are they and why?	What technique and why?

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Homework 2: Problem Definition

- FCR Matrix (9): Summarize what you learned from your research and stakeholder analysis in a Findings, Conclusions, and Recommendations (FCR) matrix that shows the perspective of the stakeholder groups you identified. The FCR matrix should contribute to your future analysis, especially in developing your value model.

Functional and Requirements Analysis

Develop a functional hierarchy for your system: identify the fundamental objective and at least four supporting functions. In this section, you should present the diagram and describe the functions and how you developed the hierarchy.

- ☐ Functional hierarchy (8): Present a functional hierarchy for your system. This should include at least **four** functions that are based on the stakeholder analysis. Each function should have at least two subfunctions. Draw the functional hierarchy using PowerPoint or draw.io.

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Homework 2: Problem Definition

- ☐ Functional hierarchy explanation (3): Explain your functional hierarchy with reference to your research and stakeholder analysis.

- ☐ Requirements analysis (3): List and justify (with reference to your research and stakeholder analysis) 6 requirements that a solution to your problem must meet. Make sure to format requirements correctly.

Requirement	Justification

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Homework 2: Problem Definition

Requirement	Justification

Value Modeling

A major part of value-focused thinking is developing a model to **evaluate** alternatives before generating them. In this portion of the assignment, use your stakeholder analysis, research, and functional analysis to create a value model for your system.

- ☐ Qualitative value model (6): Present a qualitative value model for the system. The Qualitative Value Model should be an extension of the functional hierarchy. Draw the functional hierarchy using PowerPoint or draw.io.

At a minimum, you should have **two** objectives per function and at least **one** value measure for each objective. This means you should have a minimum of 8 (4 top-level functions * 2 objectives * 1 value measure) value measures. Make sure that you follow the rules about formatting the qualitative value model.

Homework 2: Problem Definition

- Value measure weights (4): Develop a set of value measure weights for each of your value measures **using the swing weights approach** discussed in the first quantitative value modeling lesson. Show your calculations below.

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Homework 2: Problem Definition

- ☐ **Value functions (10):** Develop and present value functions for each of your value measures. Include a graphical representation of each value function and a description of why you chose that scale for your value function. (Add additional rows to the table if your value model includes more than 8 value measures).
- ☐ **Draw your value functions using excel. Submit the excel worksheet containing the tables and plotted value functions with this homework.**

Hint: Your value functions should range from a threshold value (0 value score) to an ideal value (100 value score). Value functions are not restricted to straight lines; you must ensure that your value function is logical and captures the understanding of the stakeholder range of acceptable values.

Value measure	Value function graph	Explanation of value function

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Homework 2: Problem Definition

Value measure	Value function graph	Explanation of value function

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Homework 2: Problem Definition

Value measure	Value function graph	Explanation of value function

Outputs of the Problem Definition Phase

In the course of problem definition, we often realize the initial problem statement was not an accurate summary of the true problem.

- ☐ Redefined problem statement (2): What is your redefined problem statement and what led you to make any changes?

- ☐ Screening criteria (4): List all of the screening criteria you have identified during problem definition. Screening criteria are developed from the value model (the minimum feasible values for each value measure) and the requirements analysis.

The final output of the problem definition phase is the value model. You have already specified this in detail in the earlier section, so no additional work is needed here.

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Homework 2: Problem Definition

- ☐ Conclusion (3): Provide a written summary that describes what you learned from the problem definition phase and how it will help you with the project going forward.

References

Insert your sources here. Make sure that the sources are appropriately cited throughout the worksheet.



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HOMEWORK #3

Fundamentals of Engineering Design and Systems Management

SE 301 – AY 26-1

SE301 Course Project



Homework 3: Solution Design

Assigned: Lesson 16

Due: 07 2359 November 2025 IAW your instructor's guidance
(70 Points)

This is a group assignment. Submit your work as a PDF with a coversheet IAW the Dean's Documentation of Academic Work.

This portion of the project builds on the work that you performed on your previous homework assignment. In this project phase, you will execute the tasks in the Solution Design phase of the SDP, including idea generation, alternative generation and improvement, and cost analysis. This document outlines the requirements for each task in the Solution Design phase.

Submission instructions

You should document all references in accordance with APA format. Include at least four outside references to support your analysis.

Submission documents:

- Revised HW2 worksheet
- HW3 worksheet
- Excel sheet with cost analysis for each alternative

Point summary

Question	points
Resubmitted worksheet	1
Summary of changes	2
Idea generation method	1
Explanation of use of method	1
Ideas generated	4
Basic alternative generation	2
Design parameters	3
Alternative generation visualization	5
Explanation of alternatives	10
Raw data on alternatives	10
Feasibility screening matrix	6
Initial cost analysis	20
Consistency across sections and HW2/HW3	5
Total	70

Updates to Problem Definition

In addition to this worksheet, you must also submit the HW2 (problem definition) worksheet with any changes you have made in response to feedback and your experience completing the Solution Design phase.

☐ *Please tick to confirm that you have attached your revised HW2 worksheet (1)*

Briefly summarize the changes you have made to the HW2 worksheet: (2)

Inputs to the Solution Design Phase

The inputs to the Solution Design phase are the following parts of your revised HW2 worksheet:

- Redefined problem statement
- Screening criteria
- Value model

Idea generation

You should first generate many ideas using one or more idea generation techniques covered in SE301. Remember to follow the principles of good idea generation:

- No Criticism
- Encourage Wild Ideas
- Numerous Ideas
- Build on Ideas

SE301: Fundamentals of Engineering Design and Systems Management – AY 26-1

Homework 3: Solution Design

Idea generation method(s) used (1):

--

Explain how you used the method(s) (1):

--

List all the ideas your generated (4): You should give enough detail so that a reader can understand what the idea is, but you do not need to flesh out every idea in depth.

--

Alternative generation

Develop one alternative using one of the following alternative generation techniques:

- Existing Idea
- Benchmarking
- Integrated Benchmarking

Basic alternative generation: Explain the alternative and how you used the chosen technique to develop it (2):

Now develop **five additional** potential alternative systems for your project using the alternative design approach covered in SE301: Zwicky's morphology.

Your design approach should make use of the ideas you generated in the previous section and should follow the steps explained in the alternative generation lesson.

You must identify **at least five design parameters** for your system. Use these parameters with Zwicky's Morphological Box and make use of your Functional Hierarchy to generate the five potential alternatives of the system. Describe the process and themes used to generate these alternative designs. Present your version of Zwicky's Morphological Box to visualize how you developed the various alternative designs for the system. *Hint: Cost is never a design parameter.*

You must show how you derive the design parameters from your functional hierarchy.

Remember: design parameters are ways in which the thing is built (**e.g. battery size, materials used, wing shape etc.**) not ways in which the thing performs (**e.g. battery life, thrust generated, average time between repairs etc.**).

Design parameters from functional hierarchy (3)

Zwicky's morphology: (5)

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Homework 3: Solution Design

Alternatives: Summary of the 5 additional alternatives you generated (10)

Alternative name	Explanation of the alternative

Generate raw data on alternatives for feasibility screening

Raw data collection: For each screening criteria you developed in your revised HW2 worksheet, explain 1) what the screening criteria is (e.g. the system shall...), 2) what variable is measured for this criteria (e.g. time to cover 5,000m in muddy conditions), and 3) what the unit of the measure is (e.g. seconds) (10).

Screening criteria	Measure	Unit

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Homework 3: Solution Design

In this homework you will use a large language model to score your alternatives on these criteria. Go to this link:

<https://thayerai.azurewebsites.net/interface.html?botid=1599>

Then enter:

- 1) System type: a description of the type of system you're building
- 2) alternatives description: a description of each of your alternatives with sufficient detail that the AI can sensibly estimate scores for them
- 3) the measure/unit for each screening criteria

System type: replacement for the M1 Abrams Alternative description: alternative 1: The proposed replacement for the M1 Abrams features a hybrid-electric powertrain for improved fuel efficiency and silent running, modular composite armor for easy upgrades, and an active protection system to intercept incoming threats. It is armed with a 130mm smoothbore cannon, enhanced by an autoloader for a higher rate of fire and reduced crew size, along with a digital fire-control system that can track multiple targets and integrate with other battlefield assets. The design includes a lower-profile hull and turret for reduced detectability, while its advanced suspension and track system provide superior off-road mobility. alternative 2: The proposed replacement for the M1 Abrams features a hybrid-electric powertrain for improved	fuel efficiency and silent running, modular composite armor for easy upgrades, and an active protection system to intercept incoming threats. It is armed with a 105mm smoothbore cannon, enhanced by an autoloader for a higher rate of fire and reduced crew size, along with a digital fire-control system that can track multiple targets and integrate with other battlefield assets. The design includes a lower-profile hull and turret for reduced detectability, while its advanced suspension and track system provide superior off-road mobility. Performance measures: 1) maximum speed on muddy ground (kph) 2) maximum speed on roads (kph) 3) kill rate on T72 tank at 1km range over open ground (%) 4) kill rate on T72 tank at 2km range over open ground (%) 5) average number of enemy 70mm shells that new tank is hit by before it is immobilized (#)
---	---

The AI may tell you where your input is not sufficiently detailed. Make any changes needed until it will give you a score. The AI will then give reasoning for a scoring and a raw data point for that measure. Repeat this process for each alternative. If the AI's reasoning reflects a misunderstanding of your system or a performance measure, change the description of the alternative/measure so that it correctly understands it.

Only ever enter all your alternatives at once: do not enter one at a time.

Tell the AI to "Continue" when it reaches its limit, until it has scored all of the alternatives.

Important: this technique is being used for educational purposes in this course and is not a reliable technique for estimating performance for real systems.

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Feasibility Screening Matrix: Using the LLM method described above, gather data on each of your 6 alternatives to execute feasibility screening for all of the screening criteria you developed in the revised HW2 worksheet. Present your findings in a feasibility screening matrix. You must have **at least four** candidate solutions to progress into the Decision Making phase. *Hint: If you have screened out too many alternatives, return to the alternative generation step and develop additional feasible alternatives and rescore it along with all other alternatives using the LLM method. (6)*

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Initial Cost Analysis (20)

Lifecycle cost analysis: Conduct a lifecycle cost analysis for each of your remaining candidate solutions. As part of this analysis, you will need to identify the lifespan of your system. Calculate the total initial cost (acquisition cost) for each alternative and the net present value of the lifecycle costs for each alternative. (20)

Show your calculations, assumptions, and sources **in an excel file** submitted with this homework. A reader should be able to easily understand where each number in your excel sheet comes from.

Clearly state your assumptions—and justify the costs you choose to use. Use the 4.03% Federal Funds rate as your interest rate. **Do not** consider inflation in your calculations. Support your cost analysis with research.

Fill in the following table with the name of each remaining candidate solution, the total initial cost (acquisition cost), and the net present value of its lifecycle costs. Do not include calculations in this table. **Calculations must be shown in the excel file submitted with this homework.**

[illegible]

SE301: Fundamentals of Engineering Design and Systems Management – AY 26-1
Homework 3: Solution Design

References

Insert your sources here. Make sure that the sources are appropriately cited throughout the worksheet.



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HOMEWORK #4

Fundamentals of Engineering Design and Systems Management

SE 301 – AY 26-1

SE301 Course Project

Homework 4: Decision Making

Assigned: Lesson 24

Due: **05 DEC at 2359** – IAW your instructor's guidance
(70 Points)



This is a group assignment. Submit your work as a PDF with a coversheet IAW the Dean's Documentation of Academic Work.

This portion of the project builds on the work that you performed on your previous two homework assignments. In this project phase, you will execute the tasks in the Decision Making phase of the SDP, including value scoring & costing, sensitivity & risk analysis, and improvements & tradeoff analysis. This document outlines the requirements for each task in the Decision Making phase.

Submission instructions

You should document all references in accordance with APA format. Include at least four outside references to support your analysis.

Submission documents:

- Cover sheet
- HW4 worksheet
- HW4 value Decision Making Excel spreadsheet
- Revised HW2 worksheet
- Revised HW3 worksheet
- Revised HW3 cost analysis Excel spreadsheet

Question	Points
Attached updated inputs	1
Summarized changes	3
Inputted raw scores	3
Value scoring	15
Stacked bar chart	4
Sensitivity analysis	10
Sensitivity interpretation	2
Risk analysis	9
Improved alternatives	5
Trade space analysis	10
Solution decision	4
Consistency across sections	4
Total	70

Updates to Problem Definition and Solution Design

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- In addition to this worksheet, you must also submit the following documents with any changes you have made in response to feedback and your experience completing the Decision Making phase.
 - HW2 (problem definition) worksheet
 - HW3 (solution design) worksheet
 - HW3 (solution design) excel cost analysis spreadsheet

☐ *Please tick to confirm that you have attached your revised HW2 and HW3 worksheets and the cost analysis spreadsheet (1)*

Briefly summarize the changes you have made to the documents above (only discuss changes made since submitting HW3): (3)

Inputs to Decision Making

The inputs to the Decision Making phase are the following parts of your revised HW3 worksheet:

- Candidate solutions

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Value scoring

You will submit the Excel spreadsheet model you used for this analysis as an attachment. The Excel model should be a standalone document that someone who is familiar with your work up to this point could understand. You may use comments within the Excel file if your model requires additional explanation. Most of the points for value scoring will reflect the quality of your Excel Model.

Input the raw scores for each value measure for each of the candidate solutions developed in Homework 3 into your Excel Model. (3)

Note that you should have generated raw scores for these measures in HW3 when you conducted feasibility screening.

Calculate the overall value (in the range 0 – 100) of the candidate solutions. Include the name of each candidate solution and its total value in the table below (15)

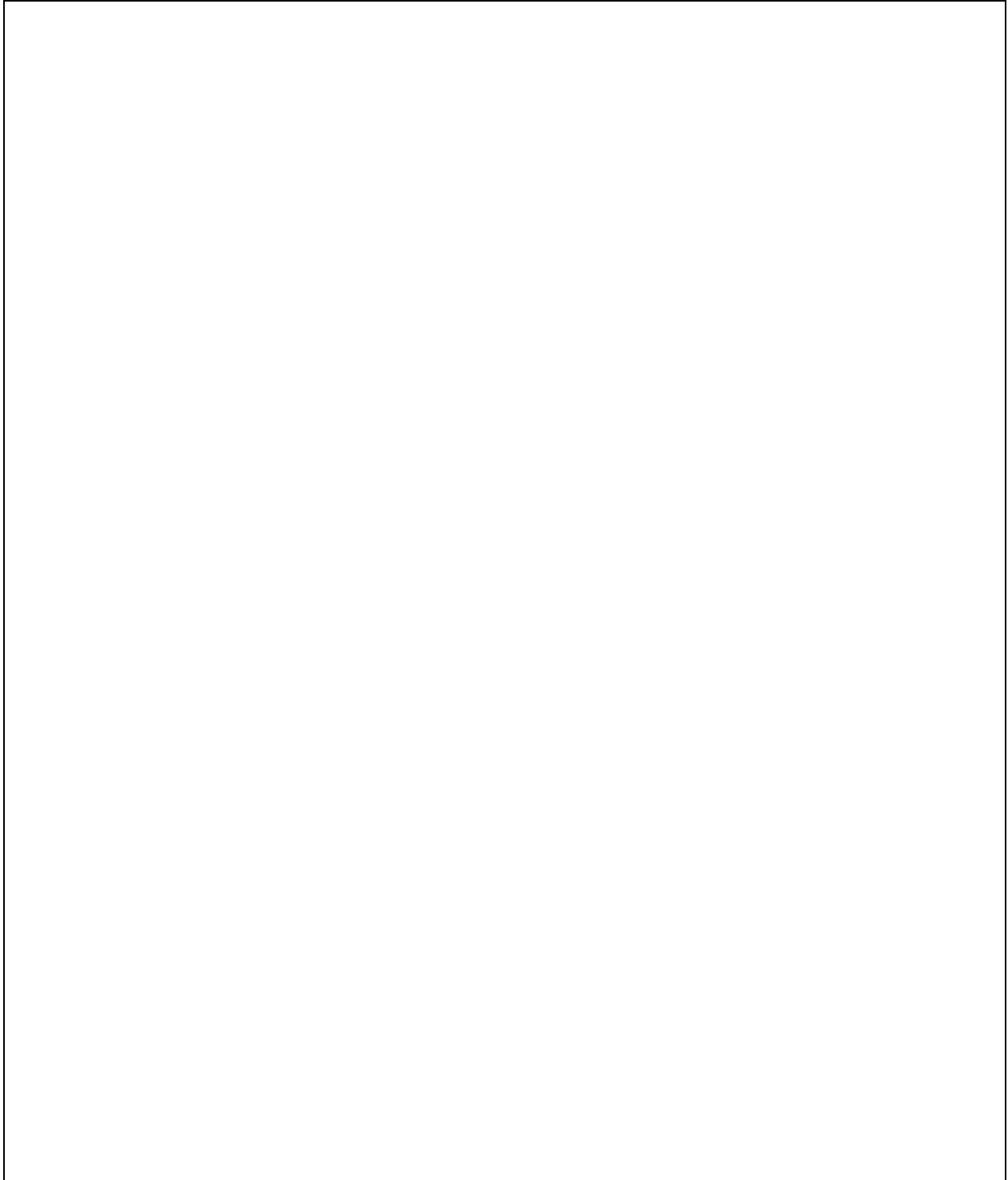
[illegible]

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Plot a stacked bar chart of the alternatives (4):

Your stacked bar chart should also be included as part of your attached Excel Model.



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Sensitivity analysis

Conduct sensitivity analysis on two of the individual weights for your value measures (10).

Use a Data Table to determine if the recommended alternative solution changes and, if so, to what degree (sensitive/significantly sensitive). Your sensitivity analysis should be included as part of your attached Excel Model. Include your two sensitivity analysis plots below (they should visually indicate the significance range):

Value measure name:	Sensitive:	Significant:

Value measure name:	Sensitive:	Significant:

Interpret your sensitivity results (2):

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Risk analysis

Conduct a risk analysis. Identify and describe three risk factors and why they are important, their types and methods of mitigation. Describe the likelihood and severity of impact on your system and explain how the chosen mitigation method changes the likelihood and severity of impact on your system (9)

Description of risk	Likelihood	Severity	Mitigation approach explanation	Mitigation approach type	Effect of mitigation approach on likelihood/severity

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Improvement and tradeoff analysis

Based on your initial analysis of your alternatives (including your stacked bar chart), provide recommendations to improve your best alternative. As part of this recommendation, identify value gaps and propose a way to close these gaps to develop your improved solution. Improvements must be explained in how the design changes lead to the value measure improvements (5).

Your analysis of improved alternatives should be included as part of your attached Excel Model.

Explain your improved alternative here:

Show a new stacked bar chart including your improved alternative here:

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Graphically depict the trade space (i.e., the cost versus value) to identify the best candidate solutions. Identify any dominated solutions (if present) (10).

Your trade space analysis should be included as part of your attached Excel Model.

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Identify your proposed solution to the decision maker and discuss how you traded off cost and value for that solution. State why you are recommending this solution. (4)